STUDY SCRIPT · OCULAR ANATOMY · PRINT ME

Ocular Anatomy: the study script

Every chapter in one place: what the exam asks, the anatomy in tables and pathways, pictures to label, and a self-check.

HOW TO READ THIS SCRIPT

Each fact carries its source: Lecture handout or slides · TB textbook (Remington & Goodwin, chapter · page) · Your notes found only in your own lecture notes, so check it · Exam named in Prof. Goodwin's exam guidelines or flagged in the lecture.

Cover the right-hand columns and recall them before you read them. Label the blank figures before you open the answer.

Contents

1. Introduction: building an eye
2. Eyelids & lacrimal system
3. Cornea, conjunctiva, sclera & limbus
4. Crystalline lens
5. Uvea
6. IOP & aqueous chambers
7. Retina
8. Orbit
9. Extraocular muscles
10. Cranial nerve innervation
11. Pupillary pathway
12. Visual pathway

CHAPTER I · LECTURE I "BUILDING AN EYE" · TEXTBOOK CH. I

Introduction: building an eye

The map and the vocabulary every later chapter uses.

WHAT THE EXAM ASKS

The exam guidelines have no objective of their own for this lecture: it is the groundwork for all the others. Two things from it are tested directly:

- Exam **Label the globe.** The handout marks the anterior chamber "(LABEL)": be able to name every structure on the horizontal section (Fig. 1.1). § Handout I · p. 3
- **Use the direction words correctly**: anterior/posterior, superior/inferior, inner/outer, vitread/sclerad. Every later objective ("location", "origin and insertion") is written in them. § Handout I · p. 5

1 · The globe is built in three coats

The eye (the **globe**) is a sphere of three coats, or tunics, from outside in: § TB I · p. I

COAT	MADE OF	JOB
Outer · fibrous	**cornea** (front, transparent) + **sclera** (white, opaque); they meet at the **limbus**	protects, keeps the globe's shape, resists the pressure of the fluids inside
Middle · vascular = *uvea*	**iris** + **ciliary body** + **choroid**	blood supply; the iris controls light, the ciliary body makes aqueous and moves the lens
Inner · neural	**retina**	turns light into a nerve signal

The transparent **conjunctiva** covers the sclera. § TB I · p. I

2 · Structure by structure

The handout's eleven points in its own order. The middle column is what the textbook says; the right column is what you wrote in the lecture.

STRUCTURE	WHAT IT IS AND DOES	YOUR NOTES
Retina	Innermost, neural coat. Changes light energy into a signal; the signal leaves the eye through the **optic nerve** to the brain. § TB I · p. I	back of the eye, inner to the choroid · photoreceptors (rods + cones) → ganglion cells → optic nerve
Choroid	Posterior part of the uvea: a network of vessels with a dense capillary bed. Surrounds the retina and **feeds the outer retinal layers**. § TB I · p. I	blood vessels, nutrients ("our energy") · outer to the retina
Sclera	Opaque white outer coat of dense connective tissue: protects and keeps the shape. § TB I · p. I	connective tissue, the "skeleton" of the eye

Cornea	Transparent front of the outer coat. Lets light in and, by refraction, helps focus it on the retina. § TB I · p. I	connective tissue, transparent, to get light to the retina
Iris / pupil	Most anterior part of the uvea. A diaphragm that controls how much light enters the pupil; two iris muscles, supplied by the autonomic nervous system, set the pupil's size. § TB I · p. I	the coloured part · has muscles inside · pupil large in dim light, small in bright Your notes
Lens	The *crystalline* lens, behind the iris, adds refractive power to focus images on the retina; it changes shape to see near objects: **accommodation**. § TB I · p. I	focuses the light · behind the iris
Ciliary body	Continuous with the iris at its root. Produces the aqueous humour and contains the muscle that changes the lens shape. § TB I · p. I	muscular · relaxed: distance; contracts: accommodation Your notes
Uvea	Iris + ciliary body + choroid: the vascular middle coat. § Handout I · p. 3 · TB I · p. I	"a lot of blood flow"
Vitreous chamber	Largest space, behind the lens, against the inner retina. Holds the **gel-like** vitreous humour. § TB I · p. I	clear inside · gives nutrition and structure Your notes
Anterior chamber	Between the cornea (front) and the iris + front of the lens (back). Holds aqueous humour, which nourishes the cornea and lens. § TB I · p. I Label	aqueous · feeds the cornea and lens
Extraocular muscles	Outside the globe, attached to its outer coat; move the eye and work with the other eye's muscles for binocular vision. **Six** per eye (below). § TB I · p. I · Slides I · #3	outside the eye (6) · coordinate eye movement
Ocular adnexa	The supporting structures around the globe: **conjunctiva**, **eyelids** (cover and protect the front of the eye; their glands make the tear film), **tear film** (lubricates). § Handout I · p. 3–4 · TB I · p. I	eyelids: protection · tear film: supplies oxygen to the cornea, keeps it moist Your notes

ONLY IN YOUR NOTES · CHECK IN LATER CHAPTERS

"No blood vessels in the lens and cornea" Your notes: the textbook's chapter 1 does not say it; chapters 3 (cornea) and 4 (lens) will.

3 · Three spaces inside the globe

SPACE	IN FRONT	BEHIND / AROUND	CONTAINS
Anterior chamber	cornea	iris and the front of the lens	aqueous humour

Posterior chamber	iris	lens lies in it; its outer border is the ciliary body	aqueous humour
Vitreous chamber	lens	inner retina	vitreous humour (gel)

The anterior and posterior chambers connect through the **pupil**. The ciliary body makes the aqueous for both. § TB 1 · p. 1

The path of light, then of the signal

cornea → anterior chamber → pupil → lens → vitreous → retina → optic nerve → **brain**

§ TB 1 · p. 1 (light is focused by the cornea and lens onto the retina; the signal leaves through the optic nerve)

4 · Label the globe

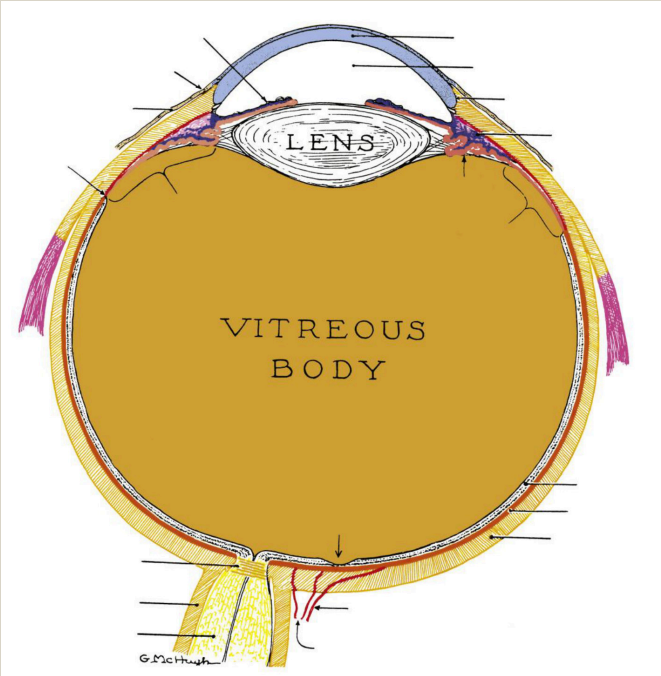
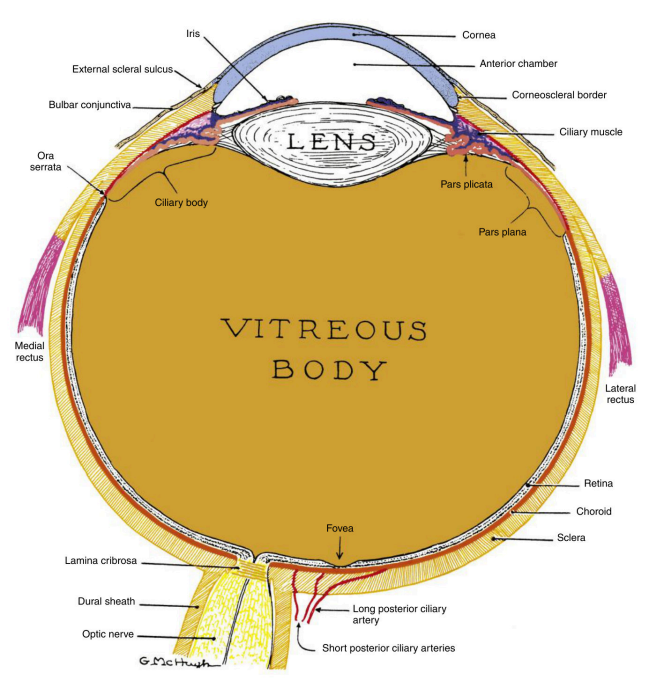


Fig. 1.1, labels removed. Name all 22 lines, then open the answer. § TB 1 · p. 2 · Slides 1 · #2

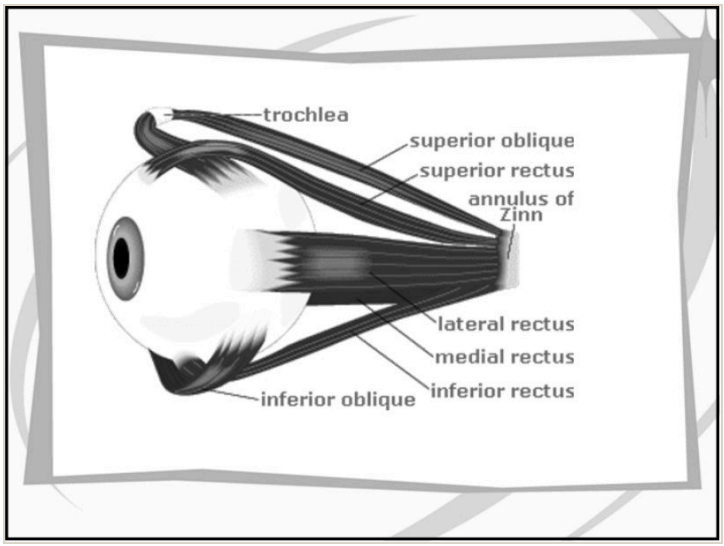
▼ Answer: Fig. 1.1 with labels



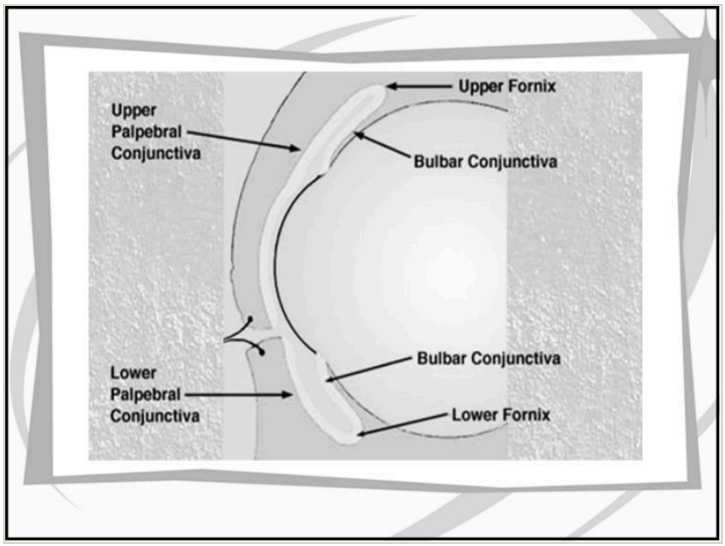
Clockwise from the top: cornea · anterior chamber · corneoscleral border · ciliary muscle · pars plicata · pars plana · lateral rectus · retina · choroid · sclera · fovea · long and short posterior ciliary arteries · optic nerve · dural sheath · lamina cribrosa · medial rectus · ora serrata · ciliary body · bulbar conjunctiva · external scleral sulcus · iris.

Many of these names (pars plana, ora serrata, lamina cribrosa) are only defined in later chapters. Here, learn *where* they are.

5 · Muscles and conjunctiva at a glance



Six extraocular muscles: four recti (superior, inferior, medial, lateral) and two obliques (superior, inferior). The recti start at the annulus of Zinn; the superior oblique runs through the trochlea. § Slides 1 · #3



Conjunctiva: **palpebral** (lines the lids) · **bulbar** (covers the globe) · upper and lower **fornix** where they fold into each other. § Slides 1 · #4

Both return in detail: muscles in chapter 9, conjunctiva in chapters 2 and 3.

6 · Directions and planes

TERM	ALSO CALLED	MEANS
Anterior	ventral	toward the front
Posterior	dorsal	toward the back
Superior	rostral Lecture · cranial TB	toward the head
Inferior	caudal	away from the head § your mnemonic: caudal → "cuddle, foot massage" (toward the feet)
Medial / lateral		toward / away from the midline
Proximal / distal		near / away from the point of origin

§ Handout 1 · p. 5 · TB 1 · p. 1 · Fig. 1.2 (Slides 1 · #5). The handout says *rostral* and the textbook *cranial* for superior: know both.

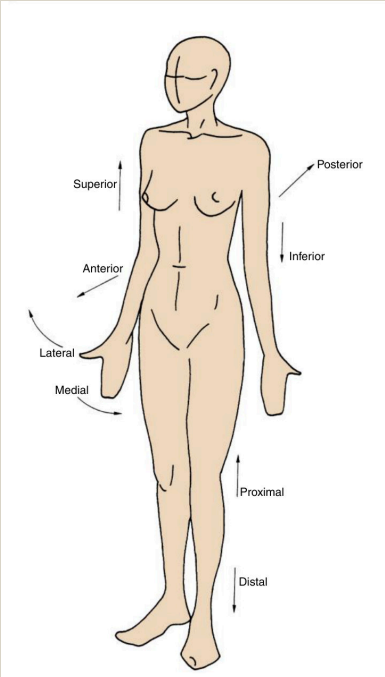


Fig. 1.2 Anatomic directions. § TB 1 · p. 3

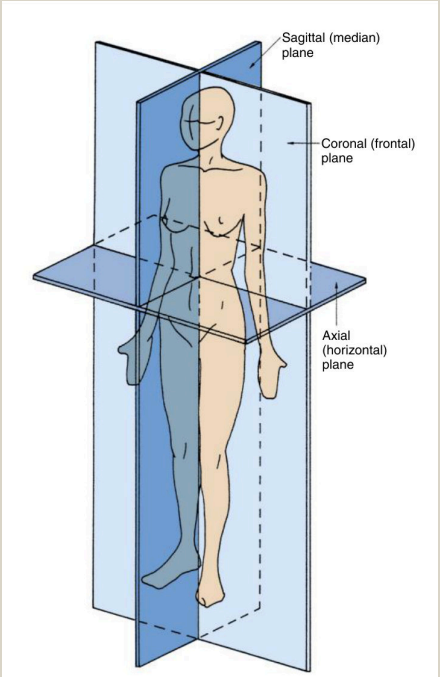


Fig. 1.3 Anatomic planes. § TB 1 · p. 3

PLANE	RUNS	DIVIDES INTO
Sagittal	vertical, front to back	right and left sides
Midsagittal	sagittal, through the midline	right and left *halves*
Coronal (frontal)	vertical, side to side	anterior and posterior parts
Axial (transverse, horizontal)	horizontal	superior and inferior parts

§ TB 1 · p. 2. Fig. 1.1 is a horizontal (axial) section.

Directions on the globe

- **Anterior / posterior** are measured from the **anterior pole**, the centre of the cornea: the pupil is anterior to the ciliary body.
- **Inner / outer** are measured from the **centre of the globe**, which lies in the vitreous: the retina is inner to the sclera.
- **Vitread** = toward the vitreous (inward) · **sclerad** = toward the sclera (outward).

§ Handout 1 · p. 5 · TB 1 · p. 2

NOT TAUGHT FROM TEXTBOOK CHAPTER 1

The textbook's chapter 1 goes on to refractive errors, eye instruments, basic histology, cell physiology and cell junctions. Lecture 1 does not cover them, so they are left out here. Where a later lecture needs a piece of them (e.g. junctions for the blood–retinal barrier), that chapter brings it in.

Key facts

3 coats: fibrous (cornea + sclera) · vascular (uvea = iris + ciliary body + choroid) · neural (retina)
3 spaces: anterior chamber · posterior chamber (both aqueous, joined at the pupil) · vitreous chamber (gel)
6 extraocular muscles: 4 recti + 2 obliques
Reference points: anterior pole for front/back, centre of the globe for inner/outer

Self-check

▼ Name the three coats of the globe and what each is made of.

Fibrous: cornea + sclera (meeting at the limbus). Vascular (uvea): iris, ciliary body, choroid. Neural: retina. § TB 1 · p. 1

▼ Which structure produces the aqueous humour, and which structures does the aqueous nourish?

The ciliary body. It nourishes the surrounding structures, especially the cornea and lens. § TB 1 · p. 1

▼ What are the boundaries of the anterior chamber? Of the posterior chamber?

Anterior: cornea in front, iris and front of the lens behind. Posterior: behind the iris, lens within it, ciliary body as its outer border. They connect through the pupil. § TB 1 · p. 1

▼ Which part of the retina does the choroid feed?

The outer retinal layers. § TB I · p. 1

▼ "The retina is ___ to the sclera." Fill in inner or outer, and say what the reference point is.

Inner. Inner/outer refer to the centre of the globe, in the vitreous. § TB I · p. 2

▼ What do vitread and sclerad mean?

Vitread: toward the vitreous (inward). Sclerad: toward the sclera (outward). § Handout I · p. 5 · TB I · p. 2

▼ A coronal plane divides the body into which parts? An axial plane?

Coronal: anterior and posterior. Axial: superior and inferior. § TB I · p. 2

▼ Give two other words for "superior" and one for "inferior".

Superior: rostral (handout), cranial (textbook). Inferior: caudal. § Handout I · p. 5 · TB I · p. 1

§ Sources for this chapter: Handout I "Building an Eye" (with your notes) · Slides I "Introduction" · Textbook ch. 1, Introduction to the Visual System, pp. 1–3.

Eyelids and lacrimal system

What the lids are made of, which muscle moves them, which gland makes which tear layer, and the route the tears take to the nose.

WHAT THE EXAM ASKS

Prof. Goodwin's guideline objectives for "Ocular Adnexa", word for word: § Review · p. 1

- Exam "Be able to identify the external features of the eyelids" → §1 (label Fig. 2.2)
- Exam "Describe the functions of each structures that make up the eyelids" → §2
- Exam "Describe the function of each of the muscles of the eyelids" → §3
- Exam "Describe the difference of ptosis appearance depending on the muscle involved" → §3, Ptosis
- Exam "Describe how the glands of the eyelid contribute to the different tear layers" → §4 and §5
- Exam "Describe how the tears drain from the orbit" → §6

Your typed notes beside these objectives: "picture / tarsal plate / orbital plate / levator muller helps keep eye up / aquaous and acessory / goblet cells, muzin / ablauf von tränendrüse bis nase" Your notes. Read as hints: identify from a **picture**; tarsal plate; orbital *septum* (probably meant by "orbital plate"); levator and Müller hold the lid up; aqueous layer from the lacrimal and accessory glands; goblet cells make the mucin; and the **route from the lacrimal gland to the nose** ("Ablauf von Tränendrüse bis Nase"), so learn §6 as a sequence.

1 · External features of the eyelids

The eyelids (**palpebrae**) are folds of skin and tissue that cover the globe when closed. Four jobs: (1) cover and protect the globe, (2) hold the structures that make the tear film, (3) spread the tears over the eye on opening, (4) push the tears toward the drainage at the medial canthus on closing. On closure the upper lid moves down over the cornea; the lower lid rises only slightly. § TB 2 · p. 10

FEATURE	WHAT AND WHERE	FUNCTION / FACT	YOUR NOTES
Palpebral fissure	the opening between the open lids; about **11 mm** high in Caucasians, **8.5 mm** in Asians § TB 2 · p. 10	the upper lid covers the superior limbus by **1.5–2 mm**; the lower lid lies within 1 mm of the inferior limbus § TB 2 · p. 10	sketch: upper lid over the cornea "1,5–2 mm"
Lateral canthus	lateral corner where the lids meet, 5–7 mm medial to the bony orbital rim	in contact with the globe § TB 2 · p. 10	

Medial canthus	medial corner, at the medial orbital rim	separated from the globe by the **lacrimal lake**, a pool of tears § TB 2 · p. 10	
Plica semilunaris	narrow crescent fold of conjunctiva in the medial canthus, the floor of the lacrimal lake	lets the eye move laterally without stretching the bulbar conjunctiva § TB 2 · p. 10	"raised tissue"; "it enlarged when we look lateral" Your notes
Caruncle	small pink mass of modified skin just medial to the plica	epithelium with goblet cells, fine hairs, sweat and sebaceous glands § TB 2 · p. 10	"pink tissue", in the middle (of the canthus)
Eyelid margin	rests against the globe; holds the lashes and the meibomian pores § TB 2 · p. 11	medial 1/6 = **lacrimal portion** (no lashes, no pores) · lateral 5/6 = **ciliary portion**; they divide at the **lacrimal papilla** with the punctum § TB 2 · p. 12	"part / cili portion"
Cilia (lashes)	double or triple row; about **150** upper, **75** lower; curl up on the upper lid, down on the lower	richly innervated: the slightest touch triggers a protective blink § TB 2 · p. 11	= eyelashes
Pores of the meibomian glands	a row on the margin **posterior to the cilia**; the **gray line** runs between lashes and pores	release the lipid for the outermost tear layer § TB 2 · p. 11–12, 16	"outermost tear"

§ Handout 2 · p. 1 · Slides 2 · #2, #3, #11

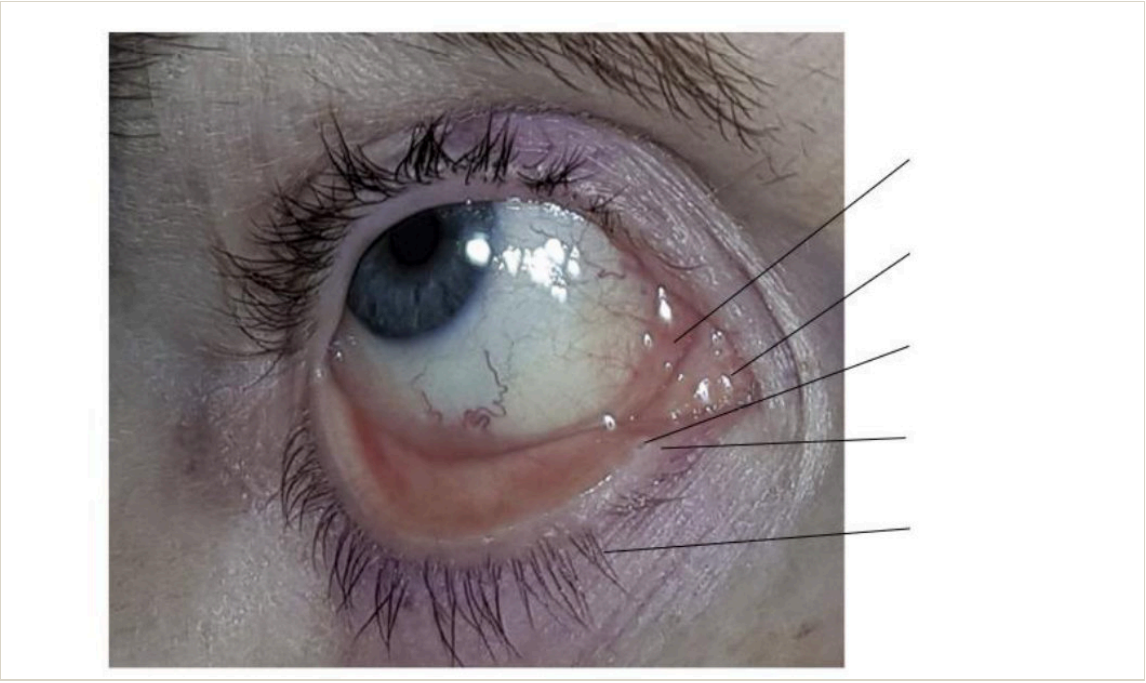
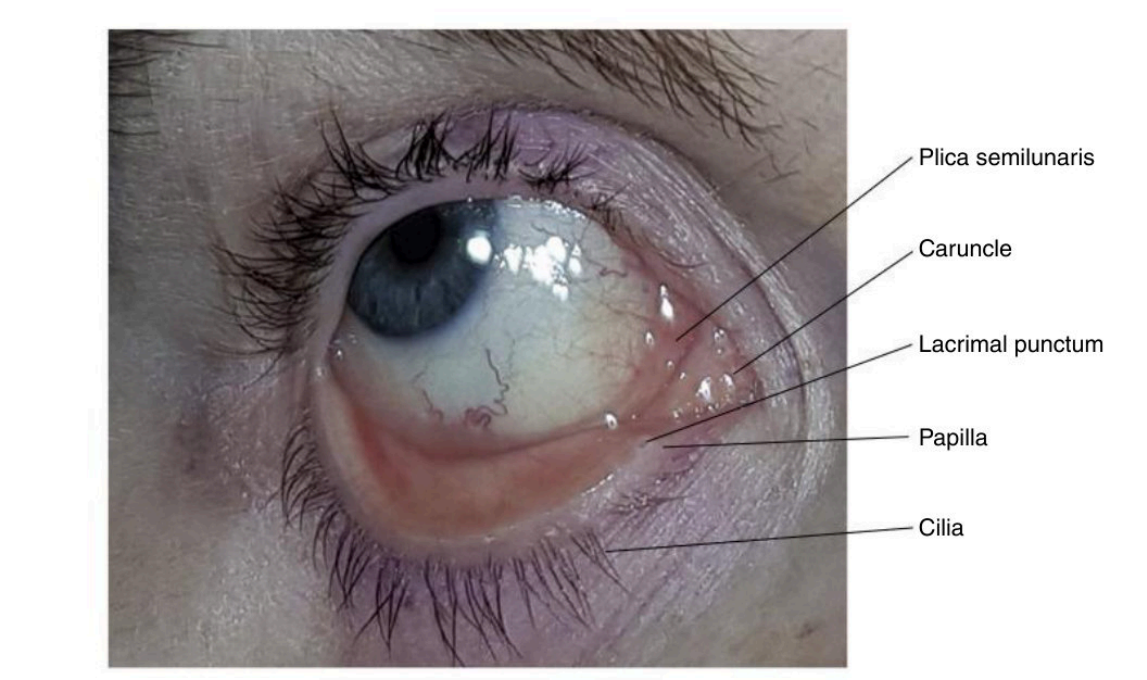


Fig. 2.2 Left medial canthus, labels removed. Name the five lines, then open the answer. § TB 2 · p. 11 · Slides 2 · #2 Label

▼ Answer: Fig. 2.2 with labels



Top to bottom: plica semilunaris · caruncle · lacrimal punctum · (lacrimal) papilla · cilia.

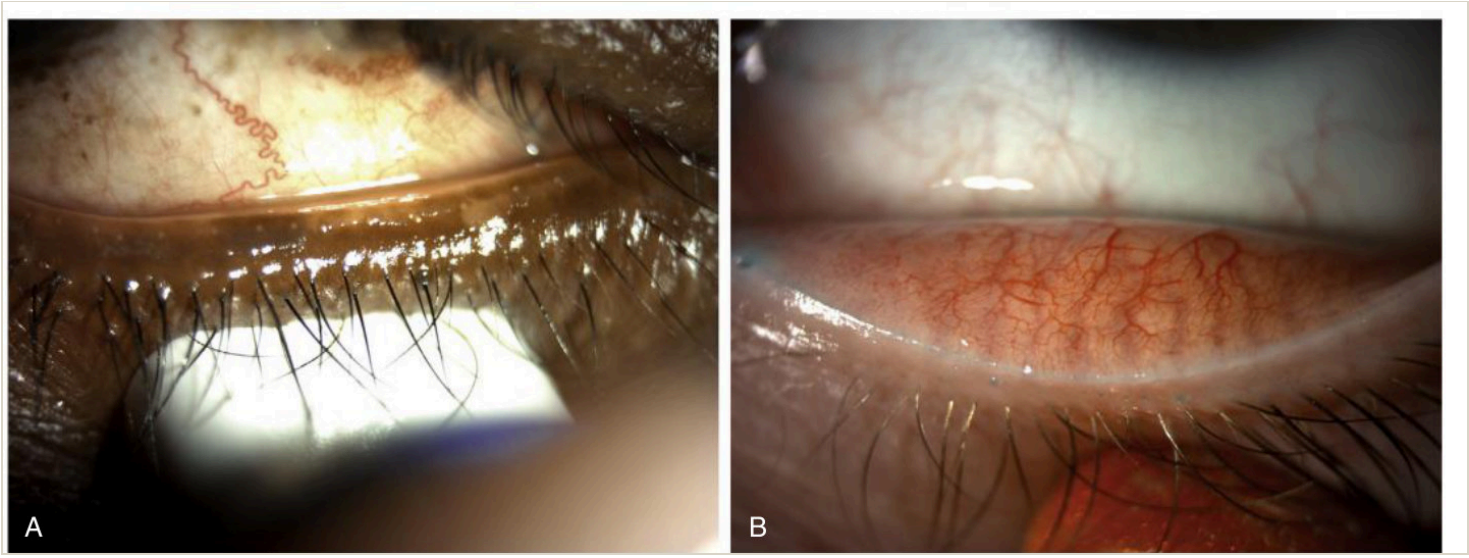


Fig. 2.5 Eyelid margin. **A** the row of meibomian gland pores behind the lashes · **B** the mucocutaneous junction, stained green, just behind the pores. § TB 2 · p. 12 · Slides 2 · #3

2 · The structures that make up the eyelid

The handout's list, front to back where it applies. Fig. 2.10 below shows them in a sagittal section.

STRUCTURE	WHAT AND WHERE	FUNCTION	CLINICAL / YOUR NOTES
Skin	the **thinnest skin in the body**; fine hairs, sebaceous and sweat glands; no fat beneath it in the tarsal part § TB 2 · p. 16	covers the lid; folds easily	loose tissue under it fills with blood or oedema (black eye) § TB 2 · p. 18
Palpebral conjunctiva	lines the inner surface of the lid; at the **fornix** it turns into the **bulbar** conjunctiva, which covers the sclera § TB 2 · p. 18	its **goblet cells** make the innermost, **mucin** layer of the tears; most numerous in the plica semilunaris, then inferonasal tarsal conjunctiva § TB 2 · p. 19	"goblet cells: innermost tear layer"; "fornix: bulbar conjunctiva is continuous" Your notes · pale conjunctiva can be a sign of anaemia § TB 2 · p. 19
Fornix	the fold where palpebral meets bulbar conjunctiva, upper and lower § TB 2 · p. 18 · Slides 1 · #4	holds the accessory lacrimal glands of Krause; the main lacrimal gland's ducts open into the upper fornix § TB 2 · p. 16, 24	

Mucocutaneous junction	on the lid margin, just behind the meibomian pores (**line of Marx**): skin epithelium becomes conjunctival epithelium § TB 2 · p. 11–12, 18	the granular and keratinised layers of the skin stop and a stratified columnar mucous epithelium begins; may hold stem cells for the palpebral conjunctiva § TB 2 · p. 18	
Tarsal plate (tarsus)	**dense connective tissue** in each lid; upper **10 mm** high in Caucasians, **8 mm** in Asians; lower **5 mm**. Front: against the submuscular tissue; back: stuck to the palpebral conjunctiva § Handout 2 · p. 3 · TB 2 · p. 16	gives the lid **rigidity and structure** and shapes it to the globe; holds the meibomian glands; the canthal tendons tie it to the orbital rim § TB 2 · p. 16	"can't bend" · clinical: eyelid eversion (box below)
Orbital septum	sheet of connective tissue from the **orbital rim** to the **levator aponeurosis** above the upper tarsus (3.7–4.4 mm above it), and to the **lower tarsal plate** § Handout 2 · p. 3 · TB 2 · p. 16	(a) separates the lid structures from the orbital contents · (b) a barrier: keeps **eyelid infections out of the orbit**, and so away from a direct route to the brain § Handout 2 · p. 3 Exam	"People can die if it comes in!" · "Thick" (see the notes box) Your notes

The muscles and glands, also part of the lid, follow in §3 and §4.

> CLINICAL CORRELATE · EYELID EVERSION
>
> To evert the upper lid, place the cotton-tipped applicator or fingertip **above the superior edge of the tarsal plate**; pressing on the middle of the plate does not work, because the plate is rigid. Once everted, the meibomian glands can show as yellow vertical streaks and the lacrimal gland may be seen above the tarsus.
>
> § TB 2 · p. 16, 24 · Handout 2 · p. 3

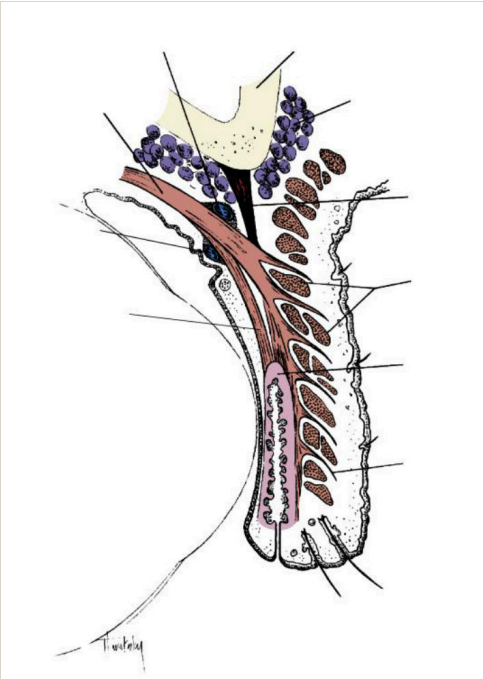
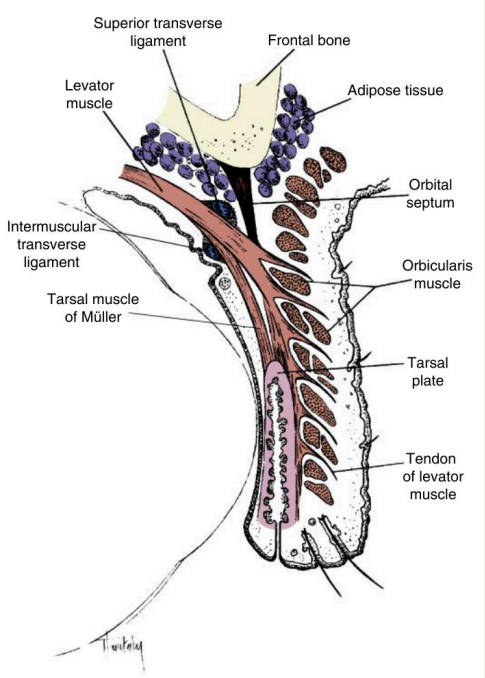


Fig. 2.10 Sagittal section of the upper eyelid, labels removed (globe on the left, skin on the right). Name the ten labels (the orbicularis has two lines). § TB 2 · p. 14 · Slides 2 · #4, #7 Label

▼ Answer: Fig. 2.10 with labels



Left side, top to bottom: superior transverse ligament · levator muscle · intermuscular transverse ligament · tarsal muscle of Müller. Right side, top to bottom: frontal bone · adipose tissue · orbital septum · orbicularis muscle · tarsal plate · tendon of the levator muscle (aponeurosis).

3 · The eyelid muscles

MUSCLE	TYPE · COURSE	FUNCTION	INNERVATION
Orbicularis oculi	striated; **encircles the palpebral fissure**, under the skin, from the lid margin out onto the orbital rim § TB 2 · p. 12–13	(1) closes the lids · (2) protects the eye · (3) renews the tear film · (4) keeps the lid margin near the globe · (5) helps meibomian secretion § Handout 2 · p. 4	**CN VII**, facial (temporal and zygomatic branches) § TB 2 · p. 13, 22
Superior palpebral levator	striated; starts in the **posterior orbit** (lesser wing of the sphenoid, above the optic foramen); runs forward **with the superior rectus**, its sheath joined to the rectus sheath; ends as the **levator aponeurosis** § Handout 2 · p. 5 · TB 2 · p. 14	**elevates the upper lid**; the link to the superior rectus raises the lid further when the eye looks up § Handout 2 · p. 5 · TB 2 · p. 14	**CN III**, oculomotor (superior division) § TB 2 · p. 14
Müller muscle (superior tarsal)	**smooth** muscle; arises from the underside of the levator where it turns into aponeurosis; **inserts into the upper edge of the tarsal plate** § Handout 2 · p. 5 · TB 2 · p. 15	additional elevation, about **2 mm**; widens the fissure in fear or surprise § TB 2 · p. 15	**sympathetic** (from the superior cervical ganglion) § Handout 2 · p. 5 · TB 2 · p. 22

Levator aponeurosis = the tendinous, fan-shaped expansion of the levator § Handout 2 · p. 5. It pierces the orbital septum and spreads across the whole lid: posterior fibres into the lower third of the front of the tarsal plate, anterior fibres between the orbicularis bundles into the skin, which makes the **upper lid crease**. Its lateral horn divides the lacrimal gland into orbital and palpebral lobes. § TB 2 · p. 14 Your notes: "not a m(uscle)", "spread out through the whole lid".

A smaller **inferior tarsal muscle** (smooth, sympathetic) lies in the lower lid; the lower lid's retractor is the capsulopalpebral fascia from the inferior rectus sheath. § TB 2 · p. 14–15

The orbicularis in parts

PART	WHERE	DOES
Palpebral portion	the part of the lid on the globe, nearest the margin (pretarsal + preseptal)	**gentle closure**, the involuntary blink, the voluntary wink; blinking renews the precorneal tear film § TB 2 · p. 13–14

· **Horner muscle**	deep palpebral fibres from the posterior lacrimal crest, **around the canaliculi**	moves tears through the canaliculi into the drainage system (§6) § TB 2 · p. 13
· **Riolan muscle**	at the lid margin, on both sides of the meibomian openings	**keeps the margin against the globe**; compresses the tarsus in a blink to express meibum § TB 2 · p. 13, 20
Orbital portion	concentric outer ring on the orbital rim	**tight closure** (forehead, temple, cheek join in): protection against pain or injury, reflex blepharospasm § TB 2 · p. 13–14

Antagonists: palpebral orbicularis ↔ levator · orbital orbicularis ↔ **frontalis** (the forehead muscle that raises the brow; Fig. 2.1, Slides 2 · #5). § TB 2 · p. 10, 14 The lid is held open by levator tone with the orbicularis relaxed; in a blink levator tone stops and the orbicularis fires. § TB 2 · p. 14

Your notes on the orbicularis: "in circle, help us blink · closes the eyelid → protects the eye · moves tear film through the eye".

CLINICAL CORRELATE · ECTROPION AND ENTROPION

	ECTROPION	ENTROPION
Margin	turns **outward**, away from the globe (Fig. 2.8)	turns **inward** (Fig. 2.9)
Cause	loss of orbicularis tone, normal in ageing	spasm of the orbicularis; or scarring after trauma or disease
Result	the punctum leaves the lacrimal lake → tears cannot drain → **epiphora** (tears overflow onto the cheek)	lashes rub the cornea → irritation, **corneal abrasion**

Both are more common in the lower lid and can be repaired surgically. § TB 2 · p. 13 · Handout 2 · p. 4 · Slides 2 · #6 Your notes: "outward: weakness of orbicularis m. · inwards (scarring): over-action of orbicularis m." — matches the textbook.

Ptosis: what it looks like depends on the muscle

Ptosis = drooping of the upper eyelid, from weakness or paralysis of the levator or of Müller muscle. § Handout 2 · p. 6 · TB 2 · p. 15 Exam

MUSCLE AFFECTED	APPEARANCE	EXAMPLE (FIG. 2.12)
Müller alone (sympathetic)	**mild**, less noticeable ptosis (Müller adds only ~2 mm)	A: Horner syndrome
Levator (CN III)	**severe**, obvious ptosis	B: cranial nerve III palsy
Both cases	the patient raises the lid with the **frontalis**: the brow on that side is raised and the forehead wrinkled	

§ TB 2 · p. 15 · Slides 2 · #8. Your notes: "Müller → subtle / levator → dramatic".

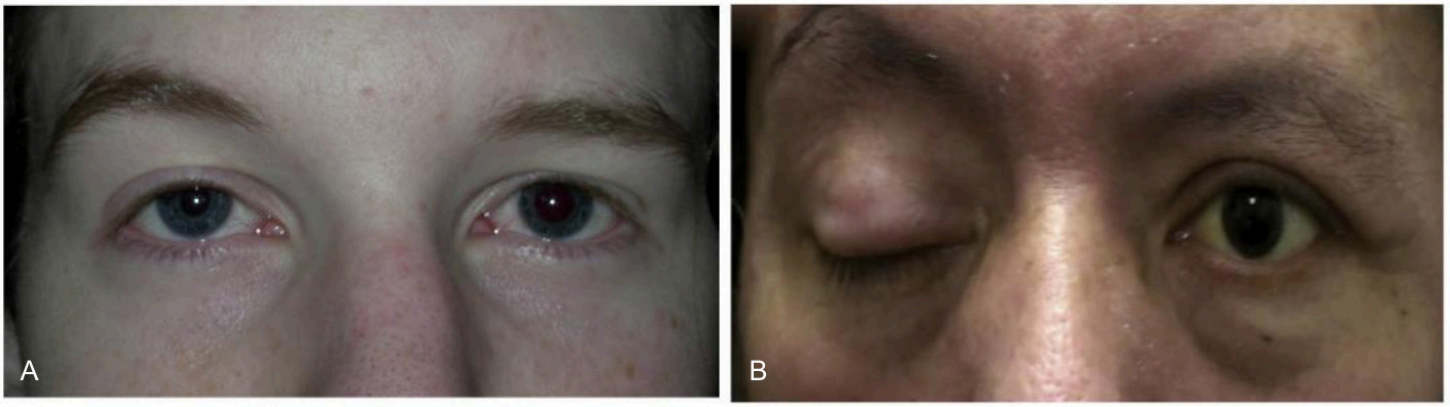


Fig. 2.12 **A** mild ptosis of the right lid, Horner syndrome (Müller) · **B** severe ptosis of the right lid after CN III palsy (levator). Both raise the right brow with the frontalis. § TB 2 · p. 15 · Slides 2 · #8

4 · The glands of the eyelids

Groundwork: three ways a gland secretes

MODE	WHAT LEAVES THE CELL	EYELID EXAMPLE
Holocrine	the **whole cell**, filled with secretion, breaks down	meibomian glands § TB 2 · p. 20
Apocrine	**part of the cytoplasm** is pinched off	glands of Moll § TB 2 · p. 21
Merocrine	only the **product**; the cell stays intact	accessory lacrimal glands (and the main lacrimal gland, which has the same make-up) § TB 2 · p. 22

§ TB 1 · p. 5, Fig. 1.6. Glands are also named by what they secrete: mucous, serous or **sebaceous** (oily).

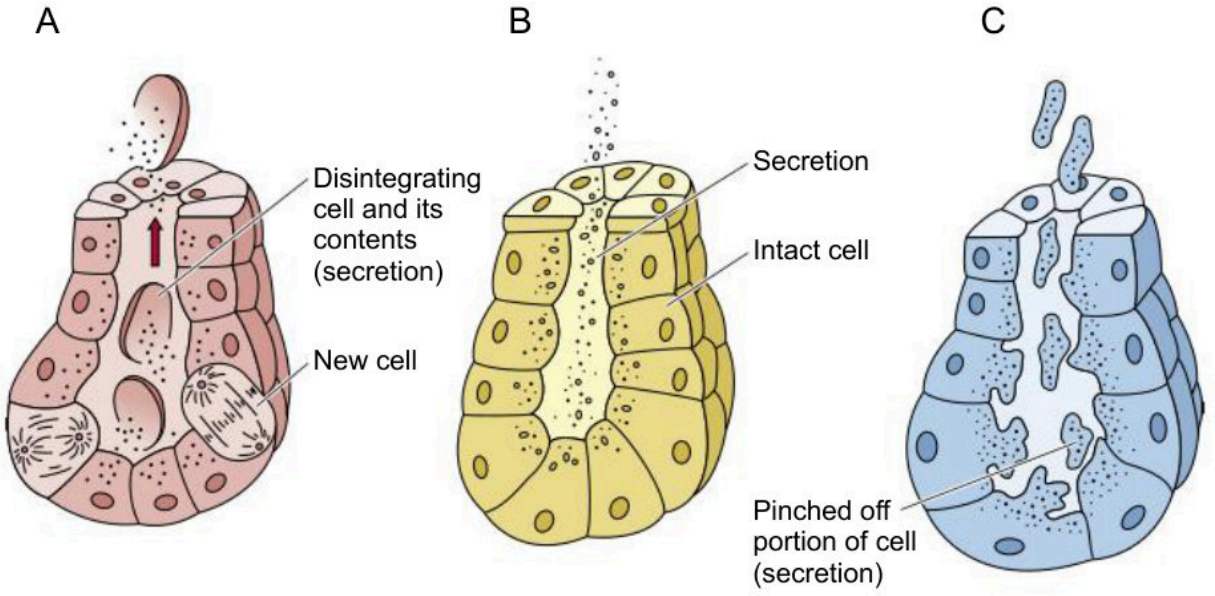


Fig. 1.6 **A** holocrine · **B** merocrine · **C** apocrine. § TB 1 · p. 5

The glands, and the tear layer each makes

GLAND	WHERE	KIND	SECRETES → FOR
Meibomian (tarsal) glands	**embedded in the tarsal plate**, vertical, pores on the margin behind the lashes; 25–40 upper, 20–30 lower § TB 2 · p. 16	sebaceous, **holocrine** § TB 2 · p. 20	meibum → the outer **lipid layer** of the tear film § Handout 2 · p. 6 · TB 2 · p. 16
Zeis glands	at the lash follicle, about two per follicle § TB 2 · p. 20	sebaceous, built like the meibomian glands but with only 1–2 acini § TB 2 · p. 20	sebum into the hair follicle → keeps the **lash from becoming brittle**; not a tear layer § Handout 2 · p. 6 · TB 2 · p. 16
Moll glands TB	near the lash follicle; open into the follicle, a Zeis duct or the margin	**apocrine**	antimicrobial proteins, defence of the lash and ocular surface; not a tear layer § TB 2 · p. 16, 20–21
Accessory lacrimal glands	upper and lower **conjunctival fornix** (glands of **Krause**: 20–40 upper, 6–8 lower) and the **orbital border of the tarsal plate** (glands of **Wolfring**, fewer) § Handout 2 · p. 6 · TB 2 · p. 16	**merocrine** § TB 2 · p. 22	→ the **aqueous layer** § Handout 2 · p. 6

Main lacrimal gland	in the lacrimal fossa of the frontal bone, **superotemporal** orbit; split by the levator aponeurosis into a larger orbital and a smaller palpebral lobe; ~12 ducts into the upper fornix § TB 2 · p. 24	merocrine (same make-up as the accessory glands) § TB 2 · p. 22, 24	water, electrolytes, antibacterial agents → the **aqueous layer** § Handout 2 · p. 7 · TB 2 · p. 24
Goblet cells	in the conjunctival epithelium (not a gland of the lid itself)	the cell swells with mucin and its surface ruptures	→ the innermost **mucin layer** § TB 2 · p. 19

Your notes: meibomian "in tarsal plate", "important in dry eye"; accessory lacrimal "to do with tears"; main lacrimal gland "lateral & superior, help to produce tears".

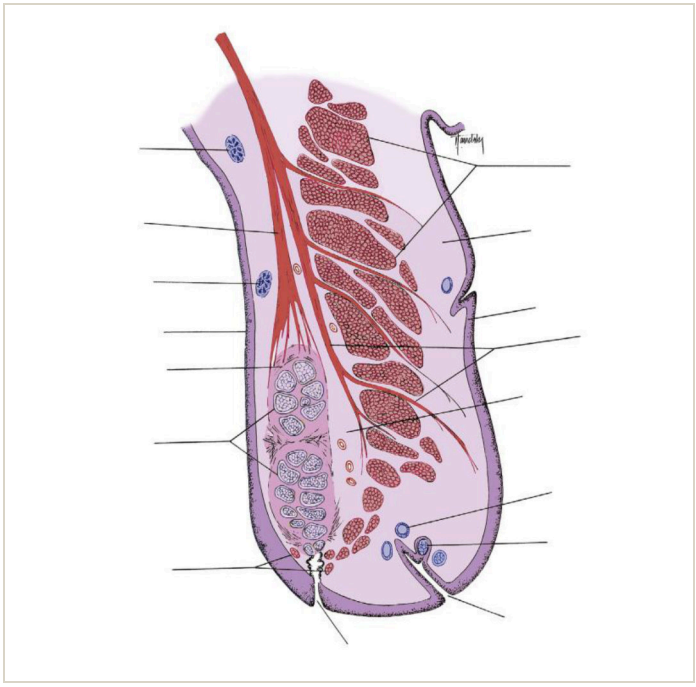
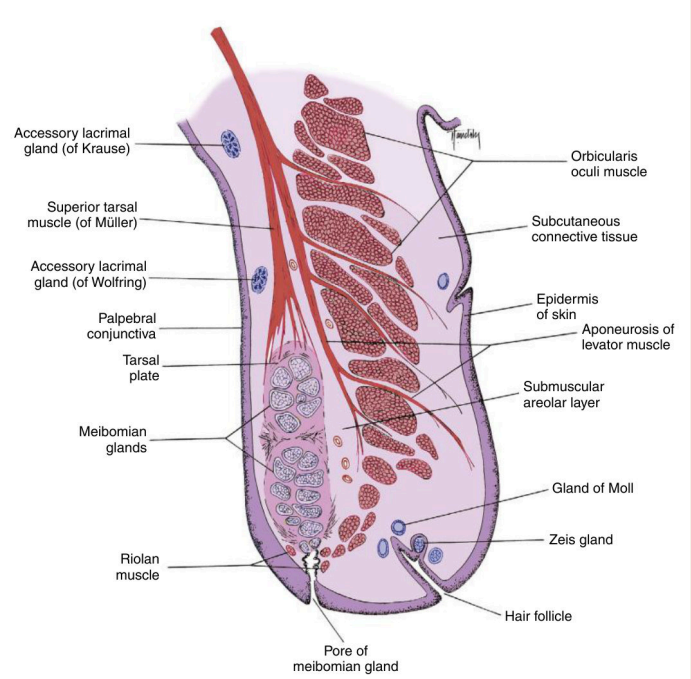


Fig. 2.13 Sagittal section of the eyelid, labels removed (conjunctiva on the left, skin on the right). Name the 16 labels: the glands first. § TB 2 · p. 17 Label

▼ Answer: Fig. 2.13 with labels



Left, top to bottom: accessory lacrimal gland (of Krause) · superior tarsal muscle (of Müller) · accessory lacrimal gland (of Wolfring) · palpebral conjunctiva · tarsal plate · meibomian glands · Riolan muscle. Bottom: pore of meibomian gland. Right, top to bottom: orbicularis oculi muscle · subcutaneous connective tissue · epidermis of skin · aponeurosis of levator muscle · submuscular areolar layer · gland of Moll · Zeis gland · hair follicle.

5 · The tear film

Functions

HANDOUT	TEXTBOOK ADDS	YOUR NOTES
a · lubricant between globe and eyelids	keeps the surface moist	
b · helps remove debris	traps debris and sloughed epithelial cells	(dirt …)
c · source of oxygen	the **primary** source of atmospheric oxygen for the **cornea**	(cornea! no vessels)
d · antibacterial substances	lysozyme, beta-lysin, lactoferrin, immunoglobulins	(protect from bacteria)
e · maintains corneal hydration	through changes in tonicity as tears evaporate	
f · regulates ocular-surface wound repair	growth factors and peptides	

not in the handout	a **smooth refractive surface** for good optics

§ Handout 2 · p. 7 · TB 2 · p. 23. Handout: six functions; textbook: seven. Learn the handout's six and know the seventh.

Three layers, three sources

LAYER	MADE BY	CONTAINS	FUNCTION
Lipid · outermost	**meibomian glands**	waxy esters, cholesterol, free fatty acids; ~53 nm thick	(i) **retards evaporation** · (ii) lubricates lid movement · (iii) **stabilises** the film by lowering surface tension, so tears do not overflow onto the cheeks
Aqueous · middle	**main and accessory lacrimal glands** (Krause, Wolfring)	salts, glucose, urea, enzymes, proteins, antibacterial substances; ~90% of the film	the watery bulk: moisture, oxygen, antibacterial defence
Mucin · innermost	**goblet cells** + the **glycocalyx** of the corneal and conjunctival surface cells	hydrophilic glycocalyx on the cells' **microvilli**, plus goblet-cell mucin	(i) makes the aqueous **adhere** to the ocular surface · (ii) lets it **spread evenly**; also reduces lid friction and traps microbes

§ Handout 2 · p. 7–8 · TB 2 · p. 23–24. Whole film 4–8 µm. The textbook adds that the aqueous and mucin layers have no sharp border. Your notes: "watery layer"; mucin "closest to the cornea, tracks the water and sticks it to cornea".

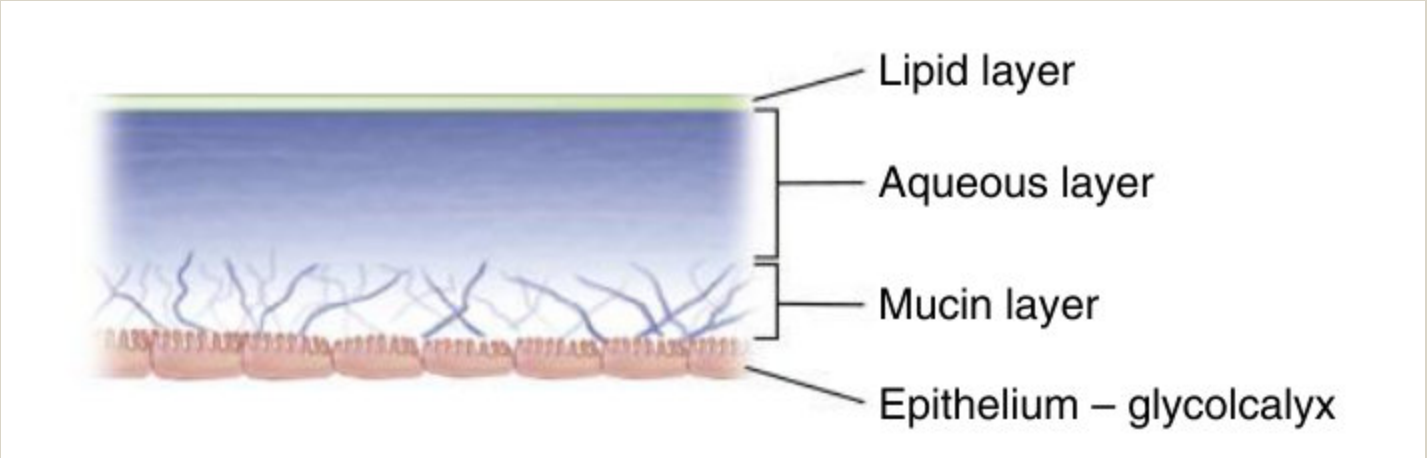


Fig. 2.29 The tear film: lipid · aqueous · mucin · epithelium with glycocalyx. § TB 2 · p. 24

Gland → layer, as one chain each

meibomian glands (tarsal plate) → pores behind the lashes → **lipid layer**

main lacrimal gland + Krause + Wolfring → upper fornix / conjunctiva → **aqueous layer**

conjunctival goblet cells + surface glycocalyx → **mucin layer**

§ Handout 2 · p. 6–8 · TB 2 · p. 16, 19, 24

> CLINICAL LINK · DRY EYE
>
> A deficit in any layer can cause dry eye: **aqueous** (ageing, Sjögren syndrome and other autoimmune disease affecting the lacrimal gland) · **lipid** (meibomian gland dysfunction → faster evaporation: evaporative dry eye) · **mucin** (loss of goblet cells, e.g. chemical burns). § TB 2 · p. 25 This is why you noted the meibomian glands as "important in dry eye". Contact lens wear is linked to a loss of meibomian glands (Fig. 2.14). § TB 2 · p. 16 · Slides 2 · #9

6 · How the tears drain

About **75%** of the tears leave through the nasolacrimal drainage system; the rest evaporate or are absorbed by the conjunctiva. § TB 2 · p. 25

From the lacrimal gland to the nose

lacrimal gland (superotemporal) → lateral upper fornix → across the globe (spread by the blink) → tear menisci on the lid margins → lacrimal lake (medial canthus) → upper + lower **puncta** → **canaliculi** (2 mm vertical, 8 mm horizontal) → common canaliculus → **lacrimal sac** → **nasolacrimal duct** → **valve of Hasner** → **inferior meatus of the nose**

§ TB 2 · p. 24–26 · Handout 2 · p. 8 · Slides 2 · #10. The handout lists the five drainage parts (puncta → canaliculi → lacrimal sac → nasolacrimal duct → valve of Hasner); the first boxes are how the tears get there. Exam your hint "Ablauf von Tränendrüse bis Nase".

PART	WHERE · SIZE	FACT TO KNOW
Puncta	one per lid, on the **lacrimal papilla** at the junction of the lacrimal and ciliary portions of the margin; lower punctum 0.1–0.9 mm wide	face the globe: seen only if the lid edge is turned out slightly § TB 2 · p. 25
Canaliculi	in each lid: **vertical ~2 mm** (with a dilation, the ampulla), then **horizontal ~8 mm** along the margin; join as the **common canaliculus**, which enters the side of the sac	elastic walls, wrapped by **Horner muscle**; the angle of entry into the sac is a valve against reflux § TB 2 · p. 25–26
Lacrimal sac	in the **lacrimal fossa** of the medial orbital wall (frontal process of the maxilla + lacrimal bone); ~10 mm (Fig. 2.31)	medial canthal tendon in front, Horner muscle behind § TB 2 · p. 26
Nasolacrimal duct	in the nasolacrimal canal of the maxilla; ends in the **inferior meatus** of the nose	the textbook text says **~15 mm**, its Fig. 2.31 says **12 mm**: know both. Most tears are absorbed by its lining before the nose § TB 2 · p. 25–26

Valve of Hasner	at the **end of the nasolacrimal duct** § Handout 2 · p. 8	a mucosal fold that stops fluid flowing back up from the nose § TB 2 · p. 26

What moves the tears

- **The blink:** the lids close from the **lateral canthus toward the medial**, pushing the menisci to the lacrimal lake and the puncta. § TB 2 · p. 10, 26
- **Capillary attraction** draws tears into the puncta and canaliculi between blinks. § TB 2 · p. 26
- **Horner muscle pump** (one theory): lid closure contracts it, shortening the canaliculi and expanding the sac, which pulls tears in; opening relaxes it and pushes them on into the duct. The mechanism is not fully known. § TB 2 · p. 26

Your notes: puncta "where the tears drain, upper and lower lid"; your sketch marks the canaliculi "2 mm" and "8 mm", as in Fig. 2.31.

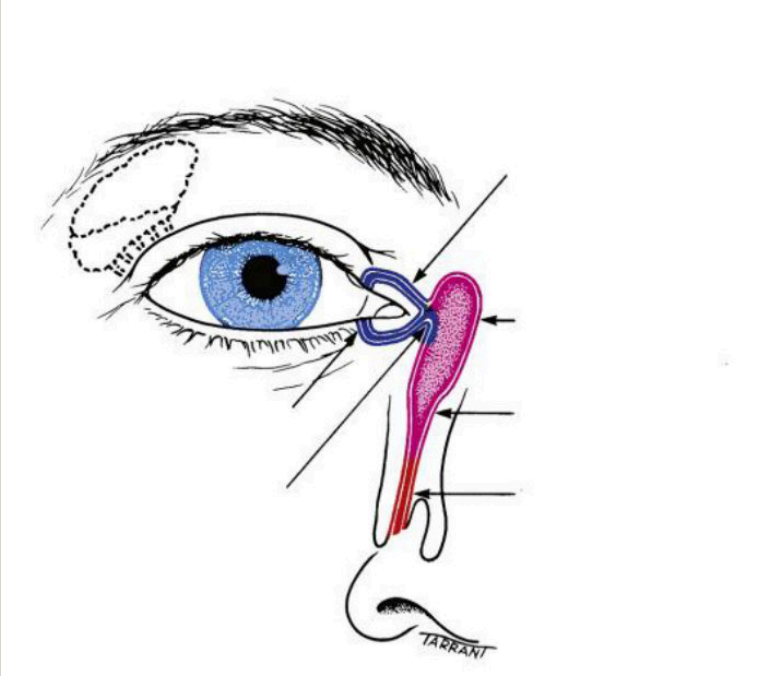
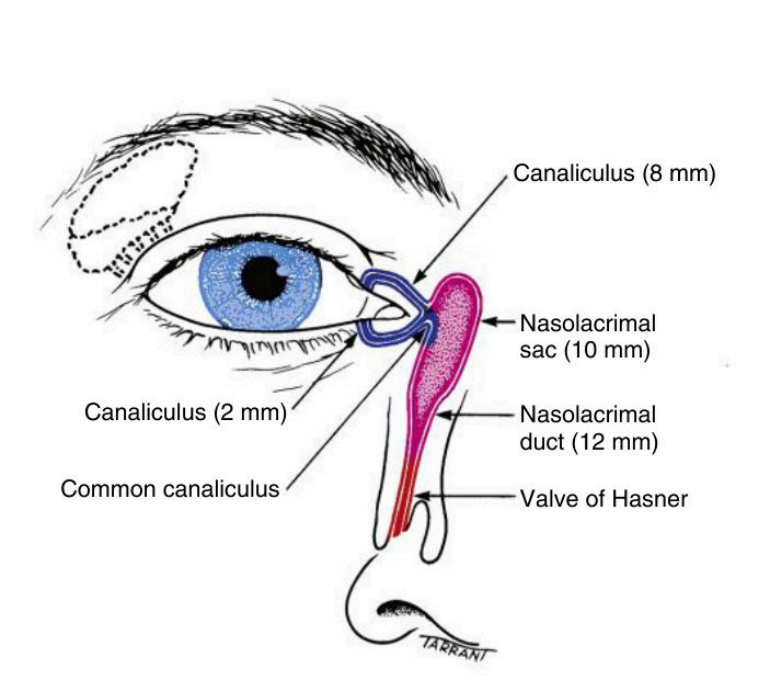


Fig. 2.31 Lacrimal drainage system, labels removed. Name the six lines and give the lengths. § TB 2 · p. 25 · Slides 2 · #10 Label

▼ Answer: Fig. 2.31 with labels



Canaliculus, horizontal part (8 mm) · nasolacrimal sac (10 mm) · nasolacrimal duct (12 mm) · valve of Hasner · canaliculus, vertical part (2 mm) · common canaliculus.

ONLY IN YOUR NOTES · CHECK THEM

- Orbital septum "**Thick**" Your notes: the textbook calls it a **thin** sheet of fibrous connective tissue; the handout just says "a sheet". Go with the textbook unless the lecturer said otherwise. § TB 2 · p. 16 · Handout 2 · p. 3
- Fornix: "bulbar conjunctiva is continuous (so tears won't go to the brain)" Your notes: the textbook confirms that palpebral and bulbar conjunctiva are continuous at the fornix, but gives no such reason. The barrier to the orbit and brain that the handout names is the **orbital septum**. § TB 2 · p. 18 · Handout 2 · p. 3
- Plica "enlarged when we look lateral" Your notes: the textbook says only that the fold lets the eye move laterally without stretching the bulbar conjunctiva. § TB 2 · p. 10

NOT TAUGHT FROM TEXTBOOK CHAPTER 2

The handout and the guidelines do not cover these, so they are left out: the eyebrows and the forehead muscles beyond the frontalis; lid topography (sulci, creases), epicanthus, lagophthalmos; lash disorders and prostaglandin lash growth; the lower-lid retractors and canthal tendons in detail; the microscopic layers of skin and conjunctival stroma, crypts of Henle, the lid wiper; hordeolum, chalazion, blepharitis, concretions; the sensory nerves and the **blood supply** of the lids (palpebral arcades); the nerve pathway of reflex tearing; tear break-up time and dry-eye treatment; the rival theories of tear drainage; ageing changes; references.

Key facts

11 mm (Caucasian) / **8.5 mm** (Asian): palpebral fissure height · upper lid covers the limbus by **1.5–2 mm**
10 mm (Caucasian) / **8 mm** (Asian): upper tarsal plate · **5 mm**: lower tarsal plate
~2 mm: extra lift from Müller muscle
~150 / 75 lashes upper / lower · **25–40 / 20–30** meibomian glands upper / lower
Innervation: orbicularis **CN VII** · levator **CN III** · Müller **sympathetic**
3 tear layers: lipid (meibomian) · aqueous (lacrimal + accessory) · mucin (goblet cells + glycocalyx) · film 4–8 µm, aqueous 90%
75% of tears drain via the nasolacrimal system · canaliculus **2 + 8 mm** · sac **10 mm** · duct **12–15 mm**
Margin: lacrimal portion = medial **1/6**, ciliary portion = lateral **5/6**

Self-check

▼ Name the structures you see in the medial canthus, from the globe outward.

Plica semilunaris (conjunctival fold, floor of the lacrimal lake) · caruncle (pink modified skin, just medial to the plica) · on the lid margin the lacrimal papilla with the lacrimal punctum · cilia. § TB 2 · p. 10–11 · Fig. 2.2

▼ Where on the lid margin are the meibomian pores, the mucocutaneous junction and the lashes, front to back?

Cilia in front, then the gray line, then the meibomian pores, then the mucocutaneous junction (line of Marx) just behind them. § TB 2 · p. 11–12

▼ What does the tarsal plate do, and how high is it?

Dense connective tissue that gives the lid rigidity and structure and holds the meibomian glands. Upper ~10 mm (Caucasian) or ~8 mm (Asian); lower ~5 mm. § Handout 2 · p. 3 · TB 2 · p. 16

▼ Where does the orbital septum run, and what are its two functions?

From the orbital rim to the levator aponeurosis above the upper tarsus, and to the lower tarsal plate. It separates the lid structures from the orbital contents and stops eyelid infections from entering the orbit, and from there the brain. § Handout 2 · p. 3

▼ Give the five functions of the orbicularis oculi and its nerve.

Closes the lids, protects the eye, renews the tear film, keeps the lid margin near the globe, helps meibomian secretion. Facial nerve, CN VII. § Handout 2 · p. 4

▼ Levator vs Müller: type of muscle, insertion, function, innervation.

Levator: striated, from the posterior orbit with the superior rectus, ends as the aponeurosis into the tarsus and lid skin; elevates the lid (more on upgaze); CN III. Müller: smooth, from the underside of the levator into the upper edge of the tarsal plate; ~2 mm extra elevation; sympathetic. § Handout 2 · p. 5 · TB 2 · p. 14–15

▼ A patient has a slight droop of one upper lid; another has a lid that almost closes the eye. Which muscle is involved in each, and what else do both show?

Slight: Müller muscle (sympathetic), as in Horner syndrome. Severe: levator (CN III palsy). Both raise the brow on that side with the frontalis, wrinkling the forehead. § TB 2 · p. 15 · Fig. 2.12

▼ Ectropion and entropion: direction, cause and consequence.

Ectropion: margin turns out, from loss of orbicularis tone with age; the punctum leaves the lacrimal lake → epiphora. Entropion: margin turns in, from orbicularis spasm or scarring; lashes rub the cornea → abrasion. § TB 2 · p. 13

▼ Which gland makes which tear layer? Name the secretion mode where the textbook gives it.

Lipid: meibomian glands (holocrine). Aqueous: main and accessory lacrimal glands of Krause and Wolfring (merocrine). Mucin: conjunctival goblet cells plus the surface glycocalyx. Zeis (sebaceous) and Moll (apocrine) glands serve the lashes, not a tear layer. § Handout 2 · p. 6–8 · TB 2 · p. 16, 19–22, 24 · TB 1 · p. 5

▼ What are the three functions of the lipid layer, and the two of the mucin layer?

Lipid: retards evaporation, lubricates, stabilises the film by lowering surface tension (no overflow onto the cheeks). Mucin: makes the aqueous adhere to the ocular surface and spread evenly. § Handout 2 · p. 7–8

▼ Trace a tear from the lacrimal gland to the nose.

Lacrimal gland (superotemporal) → lateral upper fornix → over the globe with each blink → menisci → lacrimal lake → upper and lower puncta → canaliculi (2 mm vertical, 8 mm horizontal) → common canaliculus → lacrimal sac → nasolacrimal duct → valve of Hasner → inferior meatus of the nose. § TB 2 · p. 24–26 · Handout 2 · p. 8

▼ What does Horner muscle do, and what does the valve of Hasner do?

Horner muscle (deep palpebral orbicularis around the canaliculi) helps pump tears through the canaliculi into the sac. The valve of Hasner, a mucosal fold at the end of the nasolacrimal duct, stops fluid flowing back up from the nose. § TB 2 · p. 13, 26

§ Sources for this chapter: Review slide "Ocular Adnexa" objectives (with your typed notes) · Handout 2 "Ocular Adnexa" (with your notes) · Slides 2 "Ocular Adnexa" · Textbook ch. 2, Ocular Adnexa and Lacrimal System, pp. 10–26 · Textbook ch. 1, p. 5 (modes of secretion).

Cornea, conjunctiva, sclera & limbus

The outer coat of the globe: five corneal layers, how the cornea is fed and kept clear, and the tissues that wrap the sclera.

WHAT THE EXAM ASKS

Prof. Goodwin's objectives for "Cornea, Conjunctiva and Sclera": § Review 14 · p. 1

- Exam "Describe the average central corneal thickness" → §1. Your review note: "535–555 im Zentrum". Your notes
- Exam "Describe the clinical significance of the five corneal layers" → §1 table, last column.
- Exam "Describe the layers and structures involved with a recurrent corneal erosion" → §2.
- Exam "Describe where the cornea obtains oxygen and nutrients" → §4.
- Exam "Describe the functions of the sclera, episclera, Tenon capsule, and conjunctiva" → §5–7.

The handout adds innervation (§3) and the limbus (§8). Its four named clinical correlates are in the green boxes: fluorescein dye, recurrent corneal erosion, corneal opacification, corneal neovascularization. § Handout 3 · p. 1–6

1 · Five corneal layers

The cornea is the main refracting part of the eye. It is transparent and has no blood vessels. In front of it is the tear film; behind it is the aqueous of the anterior chamber. **Central corneal thickness: 535–555 µm**. The periphery is thicker: 640–670 µm. § TB 3 · p. 30

From front to back: **epithelium · Bowman layer · stroma · Descemet membrane · endothelium**. § TB 3 · p. 30 · Handout 3 · p. 1–3

LAYER	STRUCTURE	CLINICAL SIGNIFICANCE

1 · Epithelium	Stratified, **5–7 cells** thick, about 50 μm. Three cell types: surface squamous cells (2 cells thick), wing cells (2–3 layers), basal columnar cells (1 layer). **Surface cells** secrete a **glycocalyx** that adjoins the mucin layer of the tear film. Their microvilli and microplicae enlarge the surface and make the tear film more stable. **Tight junctions (zonula occludens)** join the lateral walls of the surface cells, near the apex. **Basal cells** secrete the basement membrane and are attached to it by **hemidesmosomes**. § TB 3 · p. 30–31 · Handout 3 · p. 1–2	The tight junctions are a barrier: substances pass *through* the cells, not between them, and the cornea takes up no excess fluid from the tears. This is why **fluorescein** stains only where the epithelium is damaged. Weak hemidesmosomes cause **recurrent corneal erosion** (§2). Superficial wounds heal quickly and without a scar. § TB 3 · p. 30, 46 Your notes: "keep the water out" Your notes
2 · Bowman layer	8–19 μm. A dense sheet of **randomly interwoven collagen fibrils**. **Acellular**. It is a transition into the stroma rather than a true membrane. It ends at the corneal periphery. § TB 3 · p. 32–33 · Handout 3 · p. 2	Made before birth by the epithelium; it **does not regenerate**. If injured, it is replaced by epithelial cells or by stromal scar tissue. It is very resistant to shearing, penetration and infection. § TB 3 · p. 33, 46 Your notes: "hard to see, so thin" · "never comes back … → scar" Your notes
3 · Stroma	450–500 μm, about **90%** of the corneal thickness. Made of **collagen fibrils, keratocytes and extracellular ground substance**. The fibrils have a uniform diameter (25–35 nm) and lie parallel in 200–300 **lamellae** parallel to the surface; each lamella runs from limbus to limbus. Keratocytes (corneal fibroblasts) lie between the lamellae and maintain them. § TB 3 · p. 33, 35 · Handout 3 · p. 2	**Transparency** comes from the very regular fibril arrangement and diameter and the restricted distance between fibrils. Fibrils closer than half a wavelength of light cancel each other's scatter (destructive interference), so less than 1% of light is scattered. If the spacing is disturbed (edema) or the fibrils are disordered (scar), the cornea becomes opaque: **corneal opacification**. Repaired stroma has thicker, less ordered fibrils, so stromal wounds scar. § TB 3 · p. 37, 46 Your notes: "if not, it will be a white scar; fibers not regular" Your notes
4 · Descemet membrane	The **basement membrane of the endothelium**. It is produced throughout life, so it **thickens with age**: about 5 μm in children, about 15 μm over a lifetime, doubled by age 40. Elastic: if torn, it curls into the anterior chamber. § TB 3 · p. 37–38 · Handout 3 · p. 3	Very **resistant** to trauma and enzymes, and it **can be regenerated** if damaged (by keratocytes and endothelium). Its end at the limbus is a thickened ring, the **Schwalbe line**, a landmark of the angle (chapter 6). Mounds of extra basement membrane: *Hassall–Henle bodies* in the periphery (common with age), *guttata* in the centre (a sign of endothelial dysfunction). § TB 3 · p. 38–39, 47

5 · Endothelium	A **single layer of flattened cells**, 5 μm thick; 70–80% are **hexagonal** (the "endothelial mosaic"). Their junctions are a slightly *leaky* barrier that lets nutrients in from the aqueous. **Ionic pumps** move ions, and water follows, out of the stroma into the aqueous: this keeps the cornea at its right hydration. § TB 3 · p. 38–39, 44 · Handout 3 · p. 3	The cells **do not divide**. When cells are lost, the neighbours enlarge and spread out to cover the gap, so cell density falls with age: 3000–4000 cells/mm² in children, 1000–2000 at 80. Below about 400–500 cells/mm² the layer fails and the stroma swells (corneal **edema**); in Fuchs dystrophy the pumps fail. § TB 3 · p. 39, 45 Your notes: "pump water out of cornea into [the aqueous]" · "they spread out and get bigger to fill empty space" Your notes

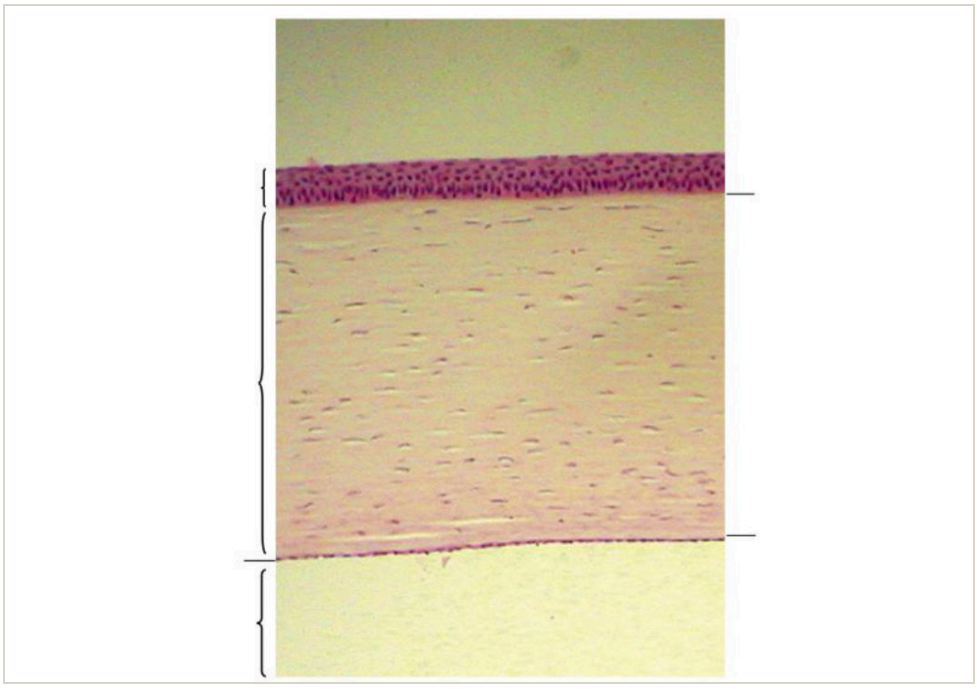
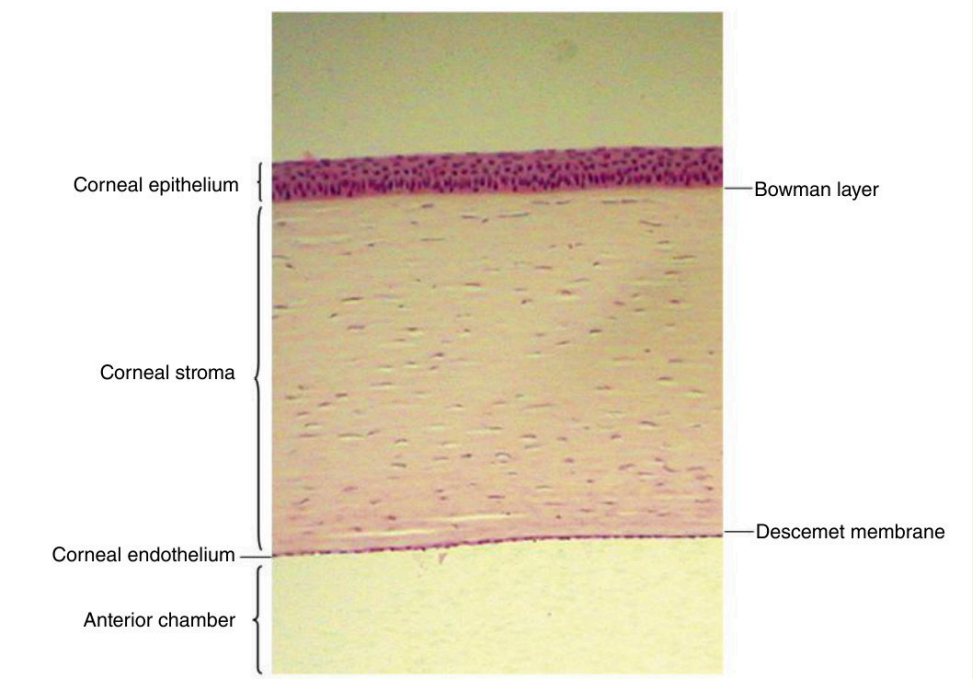


Fig. 3.3, labels removed. Name the five layers and the space behind them (6 lines), then open the answer. § TB 3 · p. 32 · Slides 3 · #2

▼ Answer: Fig. 3.3 with labels



Left, top to bottom: corneal epithelium · corneal stroma · corneal endothelium · anterior chamber. Right: Bowman layer (under the epithelium) · Descemet membrane (above the endothelium).

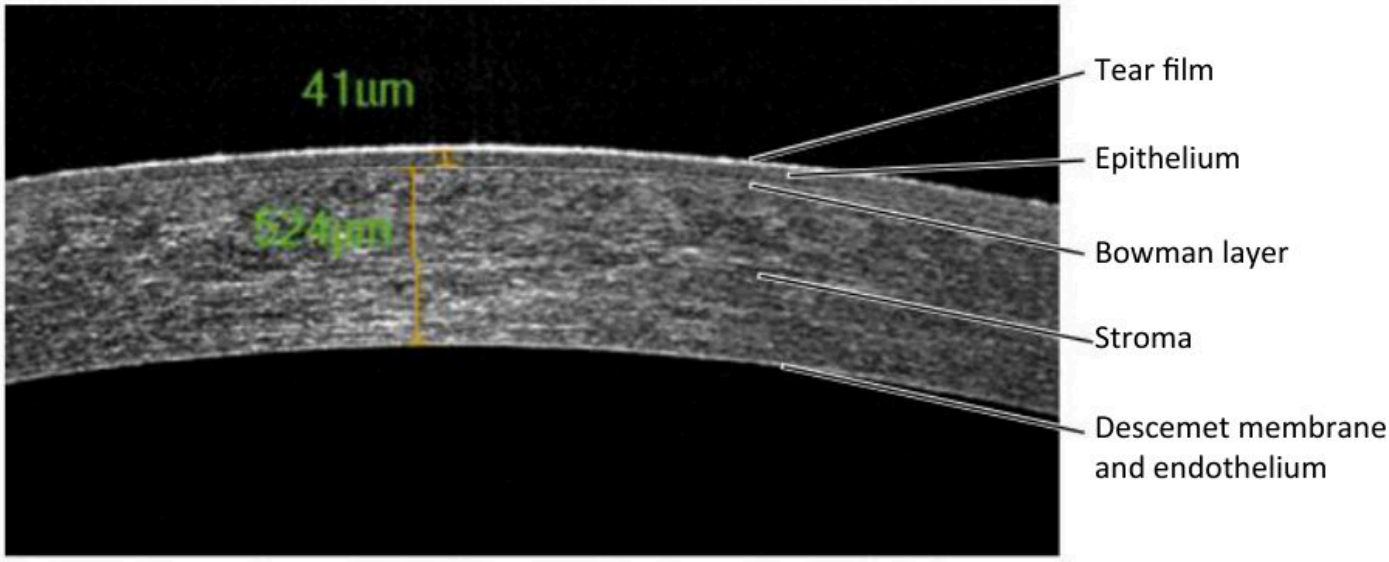


Fig. 3.9 Anterior segment OCT: the same layers in a living eye. The orange calliper across the full cornea reads 524 µm. § TB 3 · p. 35 · Slides 3 · #6 Your notes: "OCT = cross-section of the cornea · tear layer really bright". Your notes

CLINICAL CORRELATE · FLUORESCEIN DYE

Fluorescein in the tear film does not enter the epithelium while the **zonula occludens** are intact. Where the tight junctions are broken, the dye passes through Bowman layer into the anterior stroma. Through the slit lamp's **cobalt blue filter** the defect glows bright green (Fig. 3.5). § TB 3 · p. 30 · Handout 3 · p. 1 · Slides 3 · #4

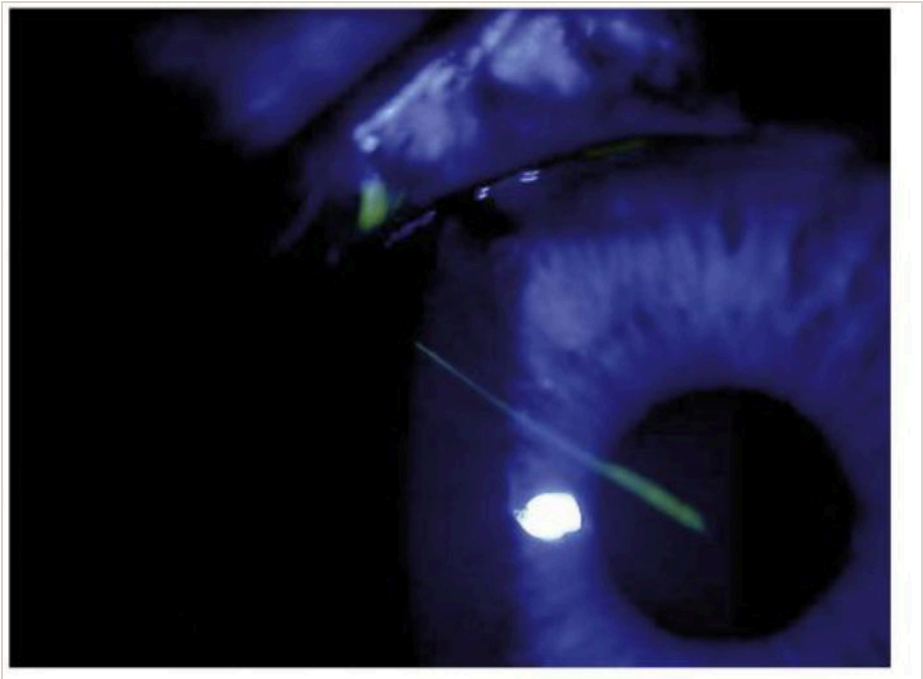


Fig. 3.5 A paper cut to the cornea: the epithelial defect stains green under cobalt blue light. § TB 3 · p. 33 · Slides 3 · #4

CLINICAL CORRELATE · CORNEAL OPACIFICATION

The cornea loses transparency when the stromal order is lost. **Edema** changes the refractive index and disturbs the fibril spacing, so destructive interference no longer cancels the scatter. A **scar** forms when fibrils are rebuilt wide and disordered. Active keratocytes during wound healing can cause a temporary haze. § TB 3 · p. 37 · Handout 3 · p. 2

Stromal swelling goes backward, because the anterior stroma is stiffer and Bowman layer is fixed. Descemet membrane then folds (striae). Endothelial failure causes much more edema than a damaged epithelium. § TB 3 · p. 34, 45

2 · Recurrent corneal erosion

Groundwork from chapter 1: a **hemidesmosome** is a strong junction that joins the *basal* side of a cell to its basement membrane and the connective tissue beneath. It is built like half a desmosome: a plaque inside the cell, with filaments running from it into the underlying connective tissue. § TB 1 · p. 8–9 (Fig. 1.9)

What holds the epithelium on

basal epithelial cell → hemidesmosome → basement membrane → anchoring fibrils → through Bowman layer → **anterior stroma**

§ TB 3 · p. 31 (the basal cells secrete the basement membrane and attach to it by hemidesmosomes; anchoring fibrils run from the hemidesmosomes through Bowman layer into the anterior stroma) · Handout 3 · p. 2

CLINICAL CORRELATE · RECURRENT CORNEAL EROSION

What fails: the hemidesmosomes or anchoring fibrils are abnormal. The epithelium is poorly attached to its basement membrane, or the basement membrane to the stroma beneath, so the epithelium **sloughs off again and again**. §TB 3 · p. 31 · Handout 3 · p. 2

- **Layers involved:** epithelium (basal cells) · its basement membrane · Bowman layer, crossed by the anchoring fibrils · anterior stroma.
- **Causes:** a superficial abrasion that did not heal completely, or an **epithelial basement membrane dystrophy**.
- **Age:** the basement membrane doubles in thickness by age 60. When it becomes thicker than the anchoring fibrils are long, the epithelium can slough.
- **Pain:** very painful, because the dense nerve endings in the epithelium are torn (§3).

§TB 3 · p. 31

Healing: a minor abrasion heals in 24–48 h with the hemidesmosomes rebuilt. If the basement membrane is damaged, healing takes much longer. The textbook says "up to 6 weeks" on p. 31 and "months" on p. 46: know that it is slow. §TB 3 · p. 31, 46

Your notes: "after an abrasion the epithelium grows back, but the hemidesmosomes do NOT grow back correctly" · review: "hemidesmosome problem in dystrophy". Your notes

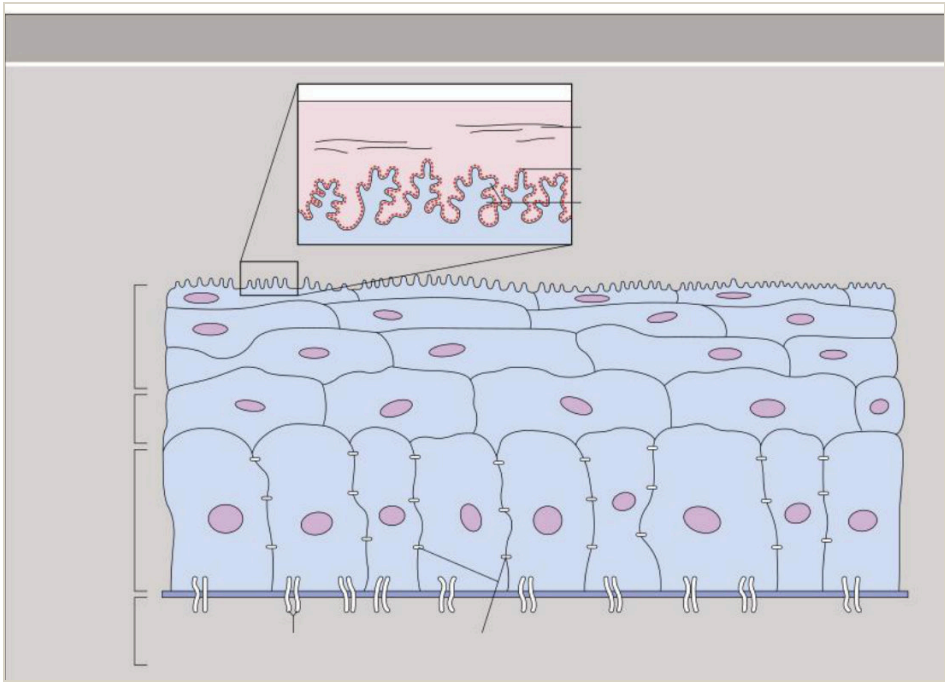
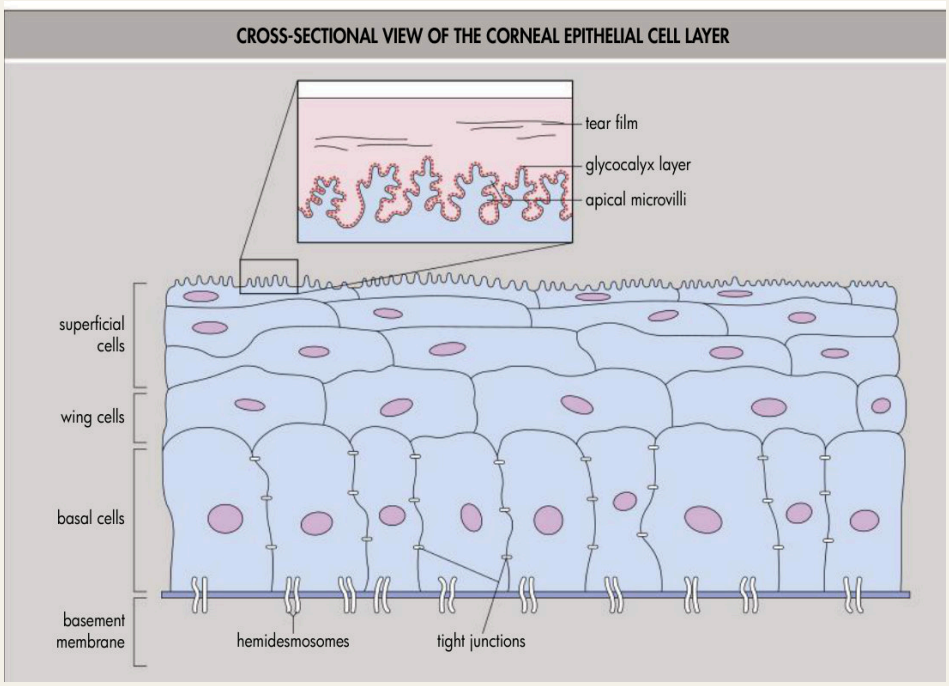


Fig. 3.4, labels removed. Name the 9 lines: the three items in the inset, the three cell layers, and the three structures at the base. Then point to the structures that fail in recurrent erosion. §TB 3 · p. 33 · Slides 3 · #3

▼ Answer: Fig. 3.4 with labels



Inset: tear film · glycocalyx layer · apical microvilli. Left: superficial cells · wing cells · basal cells · basement membrane. Bottom: hemidesmosomes · tight junctions. Recurrent erosion: the hemidesmosomes (and the anchoring fibrils beneath the basement membrane) fail.

3 · Corneal innervation

trigeminal nerve (V) → ophthalmic division (V1) → nasociliary nerve → long + short ciliary nerves → peripheral stroma → **free nerve endings in the epithelium**

§ TB 3 · p. 42 · Handout 3 · p. 4

- Some **70–80** large nerves enter the peripheral stroma. About 1 mm in, they lose their myelin and branch into three plexuses (midstromal, subepithelial, subbasal). Passing through Bowman layer, they lose their Schwann cell covering. § TB 3 · p. 42
- The epithelium has about 400 times the nerve-ending density of skin, about 7000 nociceptors/mm². Even a light touch is felt as pain. **Descemet membrane and endothelium have no nerve endings.** § TB 3 · p. 42
- **Role in structure and function**: the nerves are *neurotrophic*; they help maintain the tissue. Without them the epithelium becomes more permeable, divides less, adheres less and heals poorly (neurotrophic keratitis; one cause is **diabetes**). § TB 3 · p. 42–43 · Handout 3 · p. 4

Your notes: "highly sensitive, one of the most [one word unreadable] in the body" · "diabetes damages the nerve → staining in the eye". The textbook confirms both. Your notes

4 · Where the cornea gets oxygen and nutrients

The cornea is **avascular**; it is fed by diffusion. Having no vessels is an important part of its transparency. § TB 3 · p. 43

NEED	MAIN SOURCE	SMALLER SOURCES
Oxygen	**Atmospheric oxygen dissolved in the tear film**	aqueous humour · limbal capillaries. With the eye closed, about two-thirds comes from capillaries and the rest from the aqueous.
Nutrients (glucose, amino acids, vitamins)	**Aqueous humour**, through the leaky endothelium	conjunctival and episcleral capillaries in the **limbus**

§ TB 3 · p. 43, 45 · Handout 3 · p. 4. (The textbook says the closed-eye oxygen comes from "capillaries"; it does not name which ones.)

air → tear film → epithelium → **cornea: O_2**

ciliary body → aqueous → leaky endothelium → **cornea: nutrients**

If oxygen falls (e.g. a tight contact lens with poor tear exchange), the cornea switches to anaerobic glycolysis. Lactate builds up, draws water into the stroma, and the cornea swells. § TB 3 · p. 45

> CLINICAL CORRELATE · CORNEAL NEOVASCULARIZATION
>
> When the cornea is short of oxygen, new vessels **sprout from the perilimbal capillaries** and grow into the cornea (Fig. 3.20A, B). In a contact lens wearer it usually means the cornea is not getting enough oxygen: a poorly fitting or poorly moving lens, or a lens with low oxygen permeability. It is more common with soft lenses than rigid gas-permeable lenses, and with extended wear. Infection, inflammation, sutures and wounds can also cause it. § TB 3 · p. 43 · Handout 3 · p. 4 · Slides 3 · #7
>
> If the cause is removed and oxygen returns, the vessels empty but remain as **ghost vessels**: fine white lines (Fig. 3.20C). Normally the healthy limbus is a barrier to vessel growth. § TB 3 · p. 43
>
> Your notes: "we do NOT want vessels in the cornea at all" · "approx. 1 mm in the limbus OK" · your sketch: a vessel loop at the limbus is "normal"; one reaching into the cornea is "NOT normal". Your notes

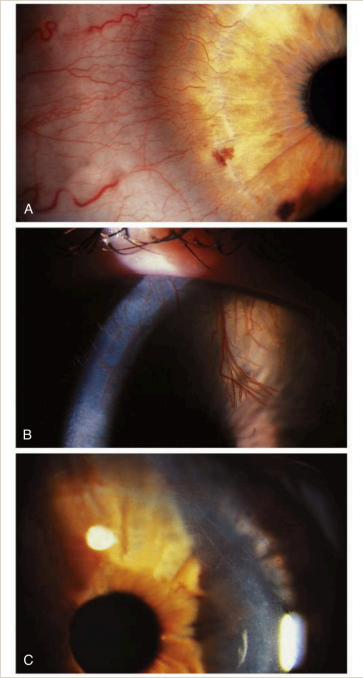


Fig. 3.20 A, early neovascularization from conjunctival loops · B, large vessels in the corneal stroma · C, ghost vessels after the vessels empty.The slide shows A. § TB 3 · p. 43 · Slides 3 · #7

ONLY IN YOUR NOTES · CHECK THEM

"Corneal neovascularization → posterior embryotoxon, arcus" Your notes. The textbook's cornea chapter does not mention *posterior embryotoxon*; ask in class or check chapter 6. It describes *corneal arcus* only as an aging change, not a vessel change: a yellow-white ring of cholesterol deposits in the peripheral stroma, separated from the limbus by clear cornea. Under age 40, suspect raised blood lipids. § TB 3 · p. 48

Review note: "2 und 3 resistant to damage" Your notes. The textbook calls **Bowman layer (2)** and **Descemet membrane (4)** very resistant, not the stroma (3). § TB 3 · p. 33, 38

5 · Sclera

The sclera is the posterior five-sixths of the outer connective tissue coat. § TB 4 · p. 53

FEATURE	WHAT IT IS	YOUR NOTES
Functions	**Maintains the shape of the globe · resists internal and external forces · attachment for the extraocular muscles** (the tendon collagen merges with scleral fibrils). § TB 4 · p. 53 · Handout 3 · p. 5 Exam	review: "sclera forms the eyeball"
Tissue	**Thick, dense connective tissue.** Collagen fibrils vary in diameter (25–230 nm) and lie in **irregular**, branching, interlacing bundles; this gives strength and flexibility. Thickness: 1 mm at the posterior pole, 0.3 mm just behind the rectus insertions. § TB 4 · p. 53	"irregular → gives the white colour / opaque"

Why opaque	Fewer glycosaminoglycans (one quarter of the cornea's), so less water (68%). Varied fibril sizes and irregular spacing scatter light. § TB 4 · p. 53–54	
Continuity	Continuous with the **corneal stroma at the limbus**. The innermost sclera merges with the choroid (suprachoroid). § TB 4 · p. 53 · Handout 3 · p. 5	
Scleral spur	A ring of **circularly oriented collagen** on the inner, anterior sclera; wedge-shaped in cross-section. Its posterior part is the **origin of the ciliary muscle** fibres. Most trabecular meshwork sheets attach to its anterior side. § TB 4 · p. 53 · Handout 3 · p. 5	
Lamina cribrosa	A sieve of scleral tissue that bridges the **posterior scleral foramen**; the optic nerve bundles pass through its holes. It is the **weakest part** of the outer coat, so high eye pressure can push it back: optic nerve **cupping** in glaucoma (Fig. 4.3). § TB 4 · p. 54 · Handout 3 · p. 5 · Slides 3 · #9	"see it in the optic nerve → bright"
Blood supply	Minimal; the sclera has **no capillary beds**, so it counts as avascular. Vessels only pass through it. It is nourished by small branches of the **episcleral and choroidal vessels** and of the long posterior ciliary arteries. § TB 4 · p. 54 · Handout 3 · p. 5	"avascular" · "episclera and choroidal blood vessels"



Fig. 4.3 The lamina cribrosa: the grey meshwork in the cup of the optic disc. § TB 4 · p. 56 · Slides 3 · #9

6 · Episclera and Tenon capsule

Near the limbus, from outside in: § TB 4 · p. 53, 59

bulbar conjunctiva → Tenon capsule → episclera → **sclera**

LAYER	STRUCTURE	FUNCTION	YOUR NOTES
Episclera	**Loose, vascularized connective tissue** just outer to the sclera. Its vessels come from the anterior ciliary arteries; the larger ones show through the conjunctiva. Joined to Tenon capsule by strands; thinner toward the back. § TB 4 · p. 55	**Blood supply to the sclera and the peripheral cornea**: capillary networks lie in front of the rectus insertions and around the peripheral cornea. § TB 4 · p. 55 · Handout 3 · p. 6	"very vascularized" · review: "blood supply, sending vessels to c…" (cut off)
Tenon capsule (fascia bulbi)	A thin, **fibrous sheet** of connective tissue outer to the episclera. Runs back from the limbus and fuses with the sclera at the optic nerve. § TB 4 · p. 56 · Handout 3 · p. 6	**Protects and supports the globe** and attaches it to the orbital connective tissue. The potential space between it and the episclera is a cavity in which the globe moves. § TB 4 · p. 56	"so thin; between episclera and conjunctiva" · "protects the globe – main purpose!"

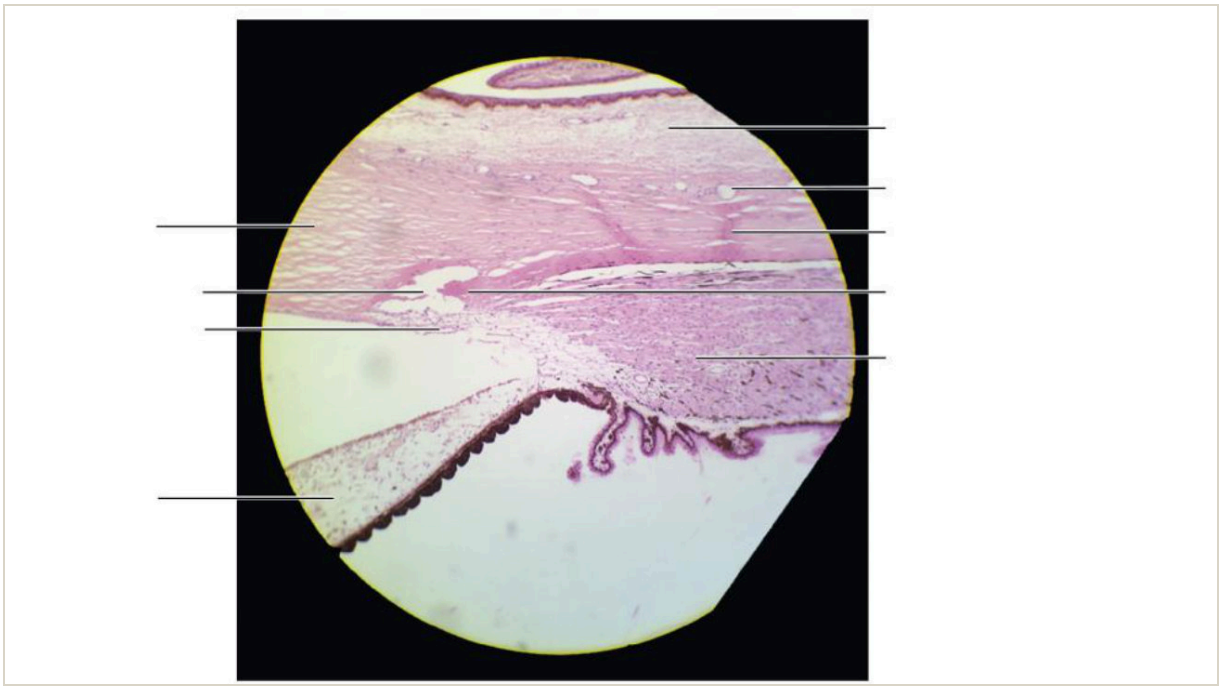
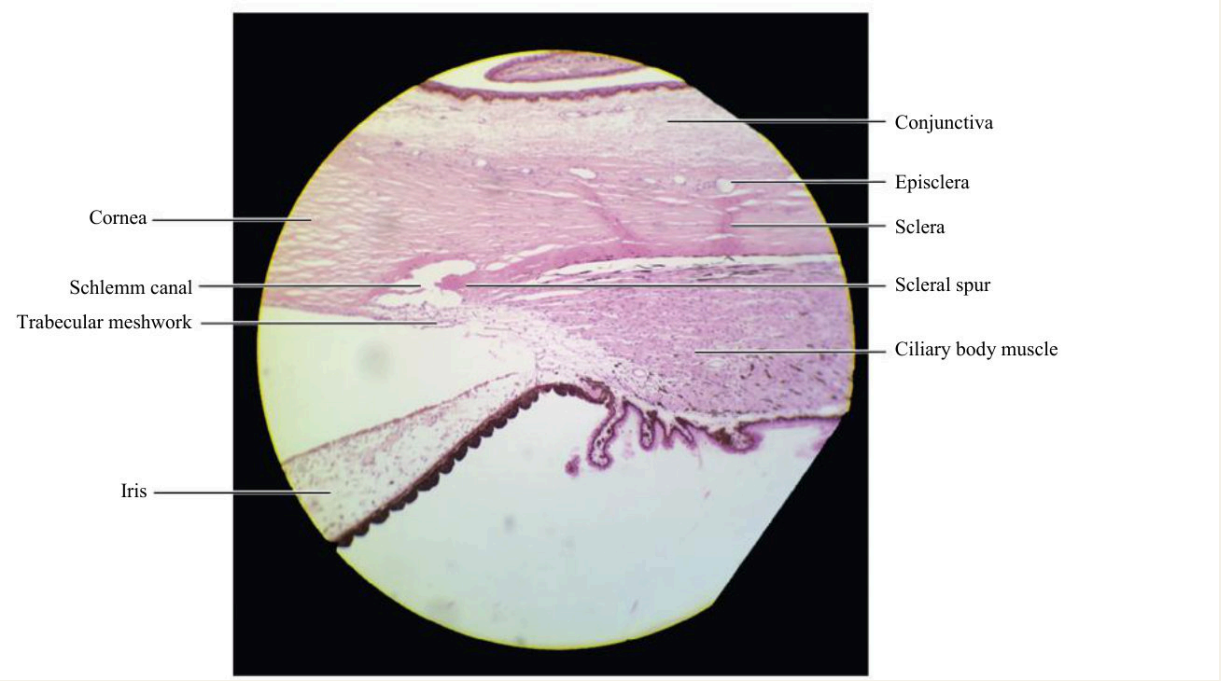


Fig. 4.1, labels removed. Name the 9 lines, then open the answer. § TB 4 · p. 54

▼ Answer: Fig. 4.1 with labels



Left: cornea · Schlemm canal · trabecular meshwork · iris. Right, top to bottom: conjunctiva · episclera · sclera · scleral spur · ciliary body muscle. Schlemm canal and the meshwork return in chapter 6.

7 · Conjunctiva

A thin, translucent **mucous membrane**. It runs from the limbus over the anterior sclera, folds at the fornices, and lines the eyelids. **Function: it lets the lids and the globe move smoothly over each other.** It has two layers: a stratified epithelium with **goblet cells**, which make the mucous layer of the tear film, and a connective tissue stroma (submucosa) with vessels, lymphatics and nerves. § TB 4 · p. 56–57

PART	WHERE	NOTES
Bulbar conjunctiva	covers the **sclera**	Translucent: the white sclera shows through. Loosely attached, except in the last 3 mm before the cornea, where it is tightly attached and merges with Tenon capsule and episclera. § TB 4 · p. 56
Fornix	the cul-de-sac joining bulbar and palpebral parts, superior and inferior	Loosely attached to the fascia of the levator, tarsal plate and extraocular muscles, so the conjunctiva moves with the globe and lids without stretching. § TB 4 · p. 56–57
Palpebral (tarsal) conjunctiva	lines the **eyelids**	Bright pink when the lid is everted. § TB 4 · p. 57

§ Handout 3 · p. 6 · TB 4 · p. 56–57 (Fig. 4.5). Your notes: bulbar "– sclera", palpebral "– lids"; review: "conj. smooth eye movement" Your notes. The picture of the three parts is in chapter 1, §5.

8 · Limbus

The **corneoscleral junction**: a band about 1.5–2 mm wide around the cornea. § TB 4 · p. 59 · Handout 3 · p. 6

- **Anterior boundary**: a plane joining the end of **Bowman layer** and the end of **Descemet membrane**. The handout says simply: "where Bowman layer and Descemet membrane end".
- **Posterior boundary**: a plane perpendicular to the globe surface through the posterior edge of the **scleral spur**.

§ TB 4 · p. 59 · Handout 3 · p. 6

CORNEA	AT THE LIMBUS BECOMES
Corneal epithelium (5 cells)	thicker conjunctival epithelium (10–15 cells)
Regular corneal stroma	irregular scleral stroma, gradually, with no visible line
Bowman layer · Descemet membrane	end at the anterior border (Descemet's end = **Schwalbe line**)
Endothelium	breaks up to cover the strands of the trabecular meshwork
(nothing)	conjunctival stroma, Tenon capsule and episclera **begin** here

§ TB 4 · p. 59

Also in the limbus: capillary loops of conjunctival and episcleral vessels that **feed the cornea** (§4), and the **limbal stem cells** that renew the corneal basal cells. § TB 4 · p. 60 · TB 3 · p. 46

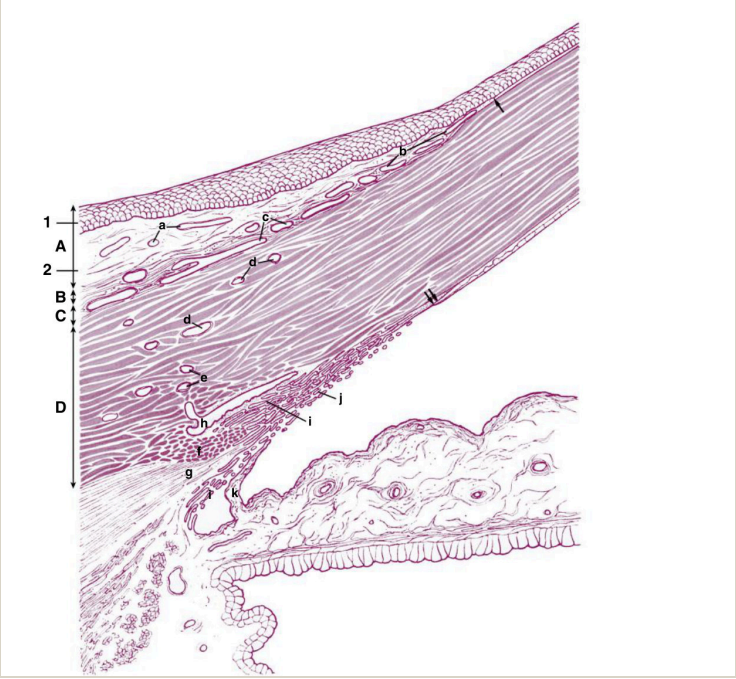


Fig. 4.2 Limbus. **A** limbal conjunctiva (1 epithelium, 2 stroma) · **B** Tenon capsule · **C** episclera · **D** limbal stroma (sclera and cornea merging) · **b** peripheral corneal arcades, reaching the end of Bowman layer (single arrow) · **f** scleral spur · **h** Schlemm canal · **i, j** trabecular meshwork · double arrow: end of Descemet membrane. § TB 4 · p. 55 · Slides 3 · #8

NOT TAUGHT FROM TEXTBOOK CHAPTERS 3 AND 4

Lecture 3 and the objectives do not cover these, so they are left out: corneal curvature, diameters and astigmatism; the named proteoglycans; keratoconus; aquaporins and the details of ion transport; glucose metabolism; the biochemistry of wound repair; UV absorption and photokeratitis; keratoplasty and refractive surgery; contact-lens effects on the endothelium; corneal aging lines; testing corneal sensitivity; treatments of recurrent erosion. From chapter 4: myopia and scleral ectasia, scleral colour, the scleral canals, scleral innervation and aging, scleritis/episcleritis, plica and caruncle, conjunctival vessels, lymphatics and nerves, pinguecula, pterygium, conjunctivitis, palisades of Vogt and limbal stem cell deficiency. Schlemm canal and the trabecular meshwork belong to chapter 6.

Key facts

535–555 µm central corneal thickness · 640–670 µm peripheral
5 layers: epithelium (5–7 cells, ~50 µm) · Bowman (8–19 µm) · stroma (450–500 µm, ~90%) · Descemet (5 → ~15 µm, doubles by 40) · endothelium (1 layer, 5 µm, 70–80% hexagonal)
Do not regenerate: **Bowman layer** and **endothelium** (cells spread instead) · Regenerates: **Descemet membrane**
Endothelial density: 3000–4000 cells/mm² (child) → 1000–2000 (age 80) · fails below ~400–500
<1% of light scattered by the cornea · stroma clearest at 75–80% water · sclera 68% water
Minor abrasion heals in 24–48 h · basement membrane doubles by 60
O_2: tears (main) · aqueous + limbal capillaries (minor) · Nutrients: aqueous (main) · limbal vessels (minor)
Nerves: V1 → nasociliary → long + short ciliary · 70–80 nerves · ~7000 nociceptors/mm²
Sclera: 1 mm (posterior pole) → 0.3 mm (behind rectus insertions) · limbus 1.5–2 mm wide

Self-check

▼ What is the average central corneal thickness, and how does the periphery compare?

535–555 μm centrally; the periphery is thicker, 640–670 μm. § TB 3 · p. 30

▼ Why does fluorescein stain an abrasion but not a healthy cornea?

The zonula occludens (tight junctions) between the surface epithelial cells stop the dye passing between cells. Where they are broken, the dye passes through Bowman layer into the anterior stroma and glows green under cobalt blue light. § TB 3 · p. 30

▼ Which layers and structures are involved in recurrent corneal erosion, and what are two causes?

The basal epithelial cells, their hemidesmosomes, the basement membrane, the anchoring fibrils through Bowman layer, and the anterior stroma. Abnormal hemidesmosomes or anchoring fibrils let the epithelium slough again and again. Causes: incomplete healing of a superficial abrasion, or epithelial basement membrane dystrophy; with age, a thickened basement membrane can outgrow the anchoring fibrils. § TB 3 · p. 31 · TB 1 · p. 8–9

▼ What happens when Bowman layer is injured? When Descemet membrane is?

Bowman layer does not regenerate; it is replaced by epithelium or stromal scar. Descemet membrane is very resistant and can be regenerated. § TB 3 · p. 33, 38, 46–47

▼ What three features of the stroma keep it transparent, and what causes it to opacify?

The very regular arrangement of the fibrils, their uniform diameter, and the restricted distance between them, which cancels scattered light by destructive interference. Edema (spacing disturbed) or scarring (wide, disordered fibrils) opacify it. § TB 3 · p. 37 · Handout 3 · p. 2

▼ How does the endothelium keep the cornea clear, and what happens when endothelial cells are lost?

Its ionic pumps move ions, and water follows, out of the stroma into the aqueous, balancing the slow leak in. The cells do not divide: neighbours enlarge and spread to cover the gap, density falls, and below about 400–500 cells / mm² the stroma swells. § TB 3 · p. 39, 44

▼ Where does the cornea get its oxygen and its nutrients?

Oxygen: mainly atmospheric oxygen dissolved in the tears; small amounts from the aqueous and limbal capillaries. Nutrients: mostly from the aqueous, through the leaky endothelium; some from conjunctival and episcleral vessels in the limbus. § TB 3 · p. 43, 45 · Handout 3 · p. 4

▼ What is corneal neovascularization, and what does it suggest in a contact lens wearer?

New vessels sprouting from the perilimbal capillaries into the cornea, in response to oxygen deprivation. In a lens wearer it usually means the cornea is not getting enough oxygen (poor fit or movement, low-permeability lens, extended wear). § TB 3 · p. 43

▼ Which nerves supply the cornea, and why do they matter beyond pain?

Branches of the long and short ciliary nerves (nasociliary ← ophthalmic V1 ← trigeminal). They are neurotrophic: without them the epithelium becomes permeable, adheres poorly and heals badly. § TB 3 · p. 42 · Handout 3 · p. 4

▼ Give the functions of the sclera, episclera, Tenon capsule and conjunctiva.

Sclera: keeps the globe's shape, resists forces, anchors the extraocular muscles. Episclera: blood supply to the sclera and peripheral cornea. Tenon capsule: protects and supports the globe; the globe moves within it. Conjunctiva: smooth movement of the lids over the globe; goblet cells add mucus to the tears. § TB 4 · p. 53, 55–57 · Handout 3 · p. 5–6

▼ What are the scleral spur and the lamina cribrosa, and why is the lamina cribrosa clinically important?

Scleral spur: a ring of circular collagen on the inner anterior sclera, origin of the ciliary muscle. Lamina cribrosa: the sieve of sclera bridging the posterior scleral foramen, through which the optic nerve passes. It is the weakest part of the outer coat, so raised eye pressure can cup it (glaucoma). § TB 4 · p. 53–54

▼ Where are the boundaries of the limbus?

Anterior: a plane joining the ends of Bowman layer and Descemet membrane. Posterior: a plane through the posterior edge of the scleral spur, perpendicular to the globe surface. § TB 4 · p. 59 · Handout 3 · p. 6

§ Sources for this chapter: Handout 3 "Cornea, Conjunctiva, and Sclera Anatomy" (with your notes; its "Images used" line lists chapter-2 figures by mistake) · Slides 3 (Figs. 3.3, 3.4, 3.5, 3.9, 3.20, 4.2, 4.3; slide 5 is an uncaptioned photo) · Review 14 · p. 1 (objectives and your typed notes) · Textbook ch. 3, Cornea, pp. 30–48 · ch. 4, Sclera, Conjunctiva, and Limbus, pp. 53–60 · ch. 1, pp. 8–9 (hemidesmosomes).

The crystalline lens

How the lens is built from the inside out, and how its layers tell you where a cataract sits.

WHAT THE EXAM ASKS

Prof. Goodwin's four objectives for the lens: § Guidelines · p. 2

- Exam "Describe how the embryonic nucleus forms." → section 2
- Exam "Describe how the lens fibers form around the embryonic nucleus." → section 3
- Exam "Describe how the Y sutures can be used to determine in which nucleus a cataract is located." → sections 4 and 6
- Exam "Identify the lens capsule, Y sutures, and embryonic nucleus." → the label drills in sections 4 and 7

Your typed notes beside them: posterior epithelium grows forward into the space · fibres start at the equator, go to the middle and form sutures · anterior Y, posterior inverted · between them = embryonic nucleus, along the suture = fetal · "just have a look at the picture what the parts look like". Your notes

Words. The lecture is titled "Intraocular Lens Anatomy", but in the textbook an *intraocular lens (IOL)* is the artificial lens put in after cataract surgery. This script says **crystalline lens** (or just "lens") for the eye's own lens, and IOL only for the implant. § Handout 4 · p. 1 · TB 7 · p. 108

1 · Terms and optics

The lens is **avascular** and transparent. It sits in the **posterior chamber**: behind the iris, in front of the vitreous, hung from the ciliary body by the zonules. § TB 7 · p. 97

TERM	MEANS	NUMBERS
Poles	the centres of the anterior and posterior surfaces; lens **thickness** = anterior pole to posterior pole	3.6–4.5 mm (mean 4 mm) unaccommodated; grows 0.02 mm a year all life
Equator	the largest circumference: the ring where the anterior and posterior parts meet	diameter 6.5 mm (infant) → 9.0 mm horizontal × 9.7 mm vertical (adult, reached in the teens)
Biconvex	both surfaces curve outward; the **posterior is steeper**	radius of curvature: anterior 8–14 mm · posterior 5–8 mm

§ TB 7 · p. 97 · Handout 4 · p. 1. You wrote "360° all around the lens!?" beside the equator: yes, it is a ring, not a point.

Optical properties

- Lecture Refractive power of the **unaccommodated** lens ≈ **20 D**. § Handout 4 · p. 1 · TB 7 · p. 97

- Power depends on surface curvature, refractive index, the index change to the surrounding fluids, and thickness. § TB 7 · p. 97
- **Gradient index**: protein concentration rises toward the centre, so the index rises from cortex (1.38) to nucleus (1.41). § TB 7 · p. 97
- Accommodation amplitude: maximum **14 D** at age 8–12, near **zero after 50** (presbyopia). § TB 7 · p. 97, 103 You wrote "around 50 you lose accommodation".

2 · How the embryonic nucleus forms

Exam The lens comes from **surface ectoderm**. Its first fibres are made from the vesicle's *posterior* wall: § TB 9 · p. 140, 142–145 · TB 7 · p. 97–98

surface ectoderm touches the optic vesicle → thickens: **lens placode** (~day 27) → invaginates: lens pit →
lens vesicle, a hollow sphere of one cell layer, separates (~day 33) →
posterior cells elongate forward = **primary lens fibres** → they fill the lumen → **embryonic nucleus**

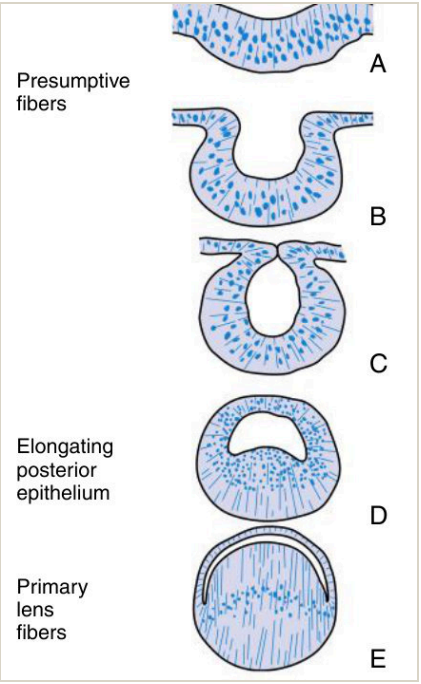


Fig. 7.2 Development of the embryonic nucleus. A placode · B invagination · C hollow vesicle lined with epithelium · D posterior cells elongate · E primary fibres fill the lumen; the anterior epithelium stays. § TB 7 · p. 98 · Slides 4 · #2

CONSEQUENCE	WHY
No posterior epithelium in the adult lens	it was used up to make the primary fibres § TB 7 · p. 97 · TB 9 · p. 145
Embryonic nucleus has no sutures	primary fibres simply fill the vesicle from back to front; sutures appear only with the secondary fibres § TB 9 · p. 145
Embryonic nucleus = the centre of the lens	every later fibre is laid down outside it § TB 7 · p. 100

Capsule present from the start	the vesicle's basal lamina becomes the lens capsule (evident at 5 weeks) § TB 9 · p. 144–145

§ Cell orientation in the vesicle: **apical** surfaces face the lumen, **basal** surfaces face outward. This is why, later, the anterior suture is made of apical fibre ends and the posterior suture of basal ends. TB 7 · p. 97, 100

3 · Epithelium, germinative zone and secondary fibres

PART	STRUCTURE	JOB
Anterior lens epithelium	a **single layer of cuboidal cells** under the anterior capsule only; basal side on the capsule, apical side facing the lens centre; joined by desmosomes and gap junctions, few if any tight junctions	secretes the anterior capsule all life · metabolic transport (Na^+/K^+-ATPase pumps) § TB 7 · p. 98, 103
Germinative zone	the band of epithelium in the **preequatorial** region, just anterior to the equator	the site of **mitosis**, which continues throughout life § TB 7 · p. 98
Secondary lens fibres	every fibre made from germinative-zone cells; in cross-section a flattened **hexagon** (3 × 9 μm); an outer fibre is up to 1 cm long; full of **crystallins**; interlocking (ball-and-socket) lateral membranes let fibres slide in accommodation	transparency and refraction § TB 7 · p. 98–99

How a secondary fibre forms around the embryonic nucleus

Exam

cell divides in the **germinative zone** → daughter cell migrates **posteriorly to the equator**, leaves the cell cycle → elongates: **apical** end toward the anterior pole (under the epithelium), **basal** end toward the posterior pole (along the posterior capsule) → meets fibres from the other side: a **suture** → loses its organelles (nucleus, mitochondria) → **mature fibre, laid down *outer* to all older fibres**

§ TB 7 · p. 98, 104 · TB 9 · p. 145

- This happens **all around the equator**, so each layer is a complete shell: the lens grows like an **onion**, youngest fibres outside, oldest in the centre. § TB 7 · p. 98 You wrote "through whole life they become fatter because they are like an onion".
- **Lens bow**: the fibre nuclei of one layer, joined by a line, form an arc bowed toward the front. § TB 7 · p. 98
- Timing: first layer of secondary fibres complete by **week 7**; Y sutures visible in the **third month**. § TB 9 · p. 145

4 · The nuclei and the Y sutures

The fibres are grouped by *when* they were made, so the zones are nested shells: § TB 7 · p. 100 · Fig. 7.7C

ZONE (INSIDE → OUT)	CONTAINS THE FIBRES MADE	SUTURES
Embryonic nucleus	primary fibres (from the posterior epithelium); the very centre; highest refractive index	**none**
Fetal nucleus	embryonic nucleus + secondary fibres formed **before birth**	**Y sutures**: anterior upright Y, posterior inverted Y
Adult nucleus	embryonic + fetal nuclei + fibres formed from birth to sexual maturation	more stellate, branching
Cortex	fibres formed after sexual maturation; lowest refractive index	complex stars

§ TB 7 · p. 100 · TB 9 · p. 145. The textbook adds a second scheme: fibres before sexual maturation = *juvenile* nucleus, then adult nucleus to middle age, the rest = cortex. Use the table's version unless asked.

Sutures

- A suture is where the fibre ends of one layer meet at a pole. **Anterior** suture = **apical** ends; **posterior** suture = **basal** ends. § TB 7 · p. 100
- The secondary fibres made in the embryo meet in three branches: the **Y sutures**. Anterior: **upright Y**. Posterior: **inverted Y**. § TB 7 · p. 100 Lecture § Handout 4 · p. 3–4
- A fibre that starts at the **tip** of a Y branch at one pole ends in a **fork** of the Y at the other pole: this conserves the lens shape (Fig. 7.7A). § TB 7 · p. 101
- Later sutures get more complex: after birth more stellate; early adulthood 6–9 branches; middle to old age 9–15 complex branching stars. Their offset, complex pattern helps transparency. § TB 7 · p. 100

Label the lens zones and sutures

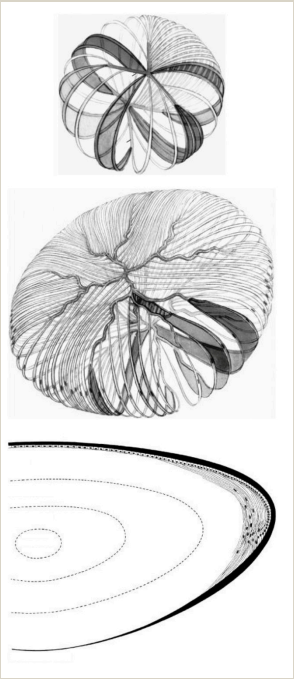
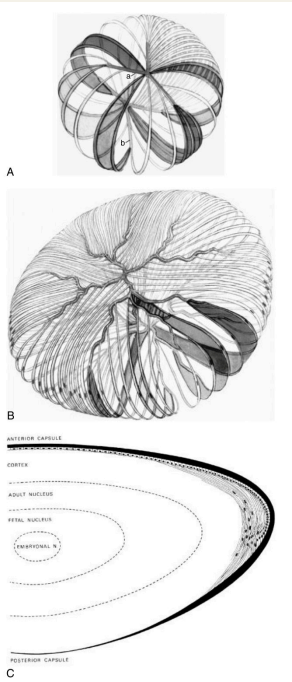


Fig. 7.7, labels removed. Top (A, fetal nucleus): name the two sutures at the leader lines. Bottom (C, adult lens): name the two capsule lines and the four zones from outside in. § TB 7 · p. 101 · Slides 4 · #3 Identify

▼ Answer: Fig. 7.7 with labels



A: (a) anterior Y suture · (b) posterior Y suture (inverted). B: the adult cortex sutures, branching stars. C: anterior capsule · cortex · adult nucleus · fetal nucleus · embryonic nucleus ("embryonal n.") · posterior capsule. The capsule is drawn thicker in front than behind.

5 · Lens capsule

FEATURE	DETAIL
What it is	a transparent **basement membrane** enclosing the whole lens; with time the **thickest basement membrane in the body** § TB 7 · p. 97
Made of	mainly **collagen**; no elastic fibres, yet **highly elastic** thanks to its lamellar arrangement § TB 7 · p. 98
Barrier	semipermeable: keeps out large molecules (albumin, haemoglobin), lets nutrients and antioxidants in; sealing the lens off early gives it **immune privilege** § TB 7 · p. 97, 103
Thickness	anterior capsule made by the anterior epithelium and **thickens with age** (anterior pole ~11 → 15 μm; the ring around it is thickest, ~13.5 → 16 μm) · equator ~7 μm · **posterior pole thinnest**, ~3.5 μm, barely changes § TB 7 · p. 98
Shape	the capsule would take a **more spherical** shape; the pull of the zonules, which insert into it (the **zonular lamella**), counteracts this. That is the basis of accommodation § TB 7 · p. 98
In cataract surgery	the anterior capsule is opened, epithelium and fibres removed, the rest of the capsule is **left in place** and the IOL goes inside it § TB 7 · p. 108 (you wrote "by cataract surgery they leave it")

6 · Clinical correlate: cataracts

Any lens opacity is a **cataract**; they are named by location or cause. Age is the major risk factor; others include disease (diabetes), genetics, trauma, radiation, nutrition and congenital factors. Cataracts are the leading cause of blindness worldwide. § TB 7 · p. 105–106 · Handout 4 · p. 3

TYPE	WHERE	LOOKS LIKE	CLINICAL NOTE
Nuclear	embryonic, fetal or adult nucleus	central opacity, yellow to brown (**brunescence**), Fig. 7.12	raises lens power: a **myopic shift** (a hyperope may briefly see better) § TB 7 · p. 106
Cortical	cortex	**spokes**, thick at the periphery, tapering inward, following the fibres, Fig. 7.13	starts near the equator, progresses slowly; hurts vision only when it reaches the pupil area § TB 7 · p. 106–107
Posterior subcapsular	just under the **posterior capsule**	seen in retroillumination, Fig. 7.14	affects vision **early** (on the visual axis, near the nodal point) · risk: long-term high-dose **steroids**, radiation § TB 7 · p. 106–108

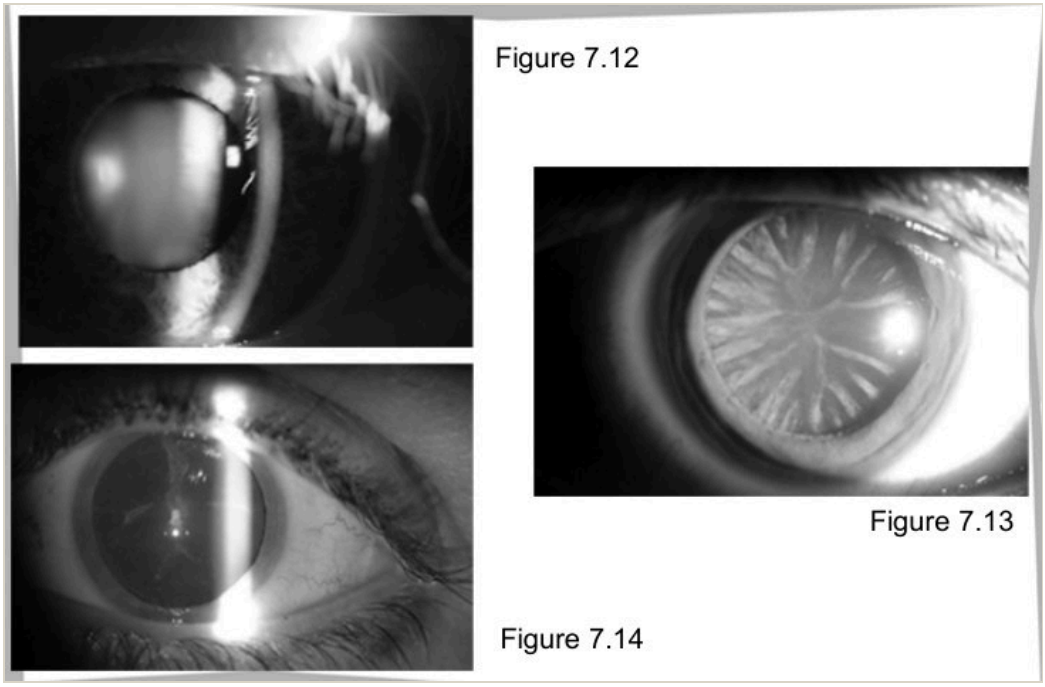


Fig. 7.12 nuclear cataract (optic section) · Fig. 7.13 spokes of a cortical cataract · Fig. 7.14 posterior subcapsular cataract (retroillumination). § Slides 4 · #4 · TB 7 · p. 106–107

You wrote "1. yellowish – age related" under cataracts: that is the nuclear type's brunescence. § TB 7 · p. 106

Using the Y sutures to place a cataract

Exam The Y sutures belong to the **fetal nucleus**, and the embryonic nucleus (no sutures) lies **between** the anterior Y and the posterior inverted Y. Everything formed after birth lies **outside** them. So, at the slit lamp: § TB 7 · p. 100 · TB 9 · p. 145 · Handout 4 · p. 4

Find the anterior upright Y and the posterior inverted Y. Where is the opacity?

> **Between the two Ys, in the centre → embryonic nucleus** (formed before the sutures existed)
>
> **On or along a Y suture → fetal nucleus** (the Y sutures are part of it)
>
> **Anterior to the anterior Y or posterior to the posterior Y → adult nucleus or cortex** (later shells; spokes = cortical)

§ The first two branches are exactly your lecture drawing ("red dot between the Ys = cataract in embryonic nucleus; purple dot on the Y = fetal nucleus; both you are born with it") and your typed review note. Your notes The third branch follows from the layering in TB 7 · p. 100.

> CLINICAL · CONGENITAL AND SUTURAL CATARACTS
>
> Both the embryonic and fetal nuclei are formed before birth, so an opacity there points to a developmental cause. If the primary fibres fail to elongate and pack in order, the result is a **cataract of the primary fibres** (embryonic nucleus); interference with the secondary fibres gives a **sutural cataract** along the Ys. Rubella in the mother hits the lens between weeks 4 and 7, while the primary fibres form. § TB 9 · p. 145

7 · Slit-lamp appearance

An **optic section** shows the lens **biconvex**, crossed by bright and dark **zones of discontinuity** (changes in light scatter). In front the zones are convex forward; behind the centre they are concave forward. In the handout's

order, front to back: § Handout 4 · p. 4 · TB 7 · p. 100

#	ZONE	APPEARANCE	WHAT IT IS
1	Anterior lens capsule	first **bright** line, convex forward	the capsule
2	Subcapsular clear zone	**dark** line	newly synthesised fibres
3	Remodelling zone (zone of disjunction)	next **bright** line	cortex losing transparency at an abrupt change in cell differentiation
4	Cortex and nuclei	grey zones	rest of the cortex, adult, fetal and embryonic nuclei
5	Anterior Y suture	upright Y, may be visible	belongs to the fetal nucleus
6	Embryonic nucleus	the centre	primary fibres
7	Posterior inverted Y suture	upside-down Y (Fig. 7.9)	in the posterior fetal nucleus
8	Posterior cortex	zones concave forward	cortex behind the nucleus
9	Posterior capsule	final zone	the capsule

§ TB 7 · p. 100, 102. Your notes for zones 1–3 ("in the picture" with a red, yellow and blue dot) match the dots on Fig. 7.8 below.

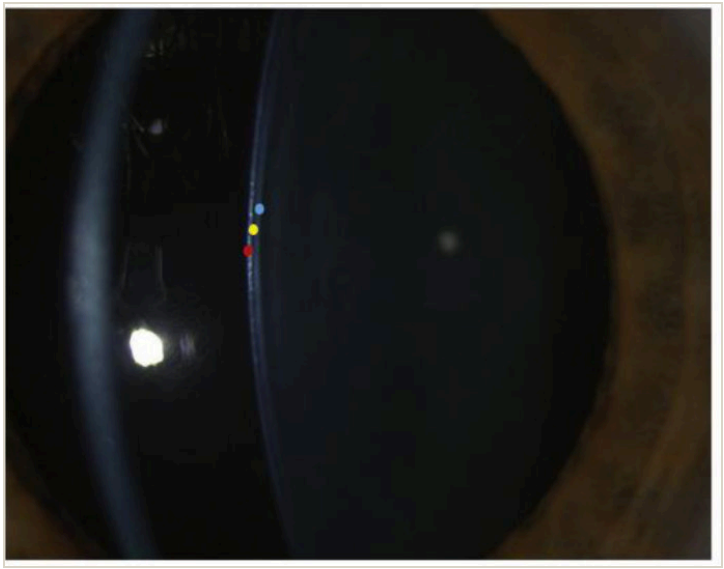


Fig. 7.8 Optic section of the lens. Name the zone at the red, yellow and blue dots, and say which are bright and which dark. § TB 7 · p. 102 · Slides 4 · #5

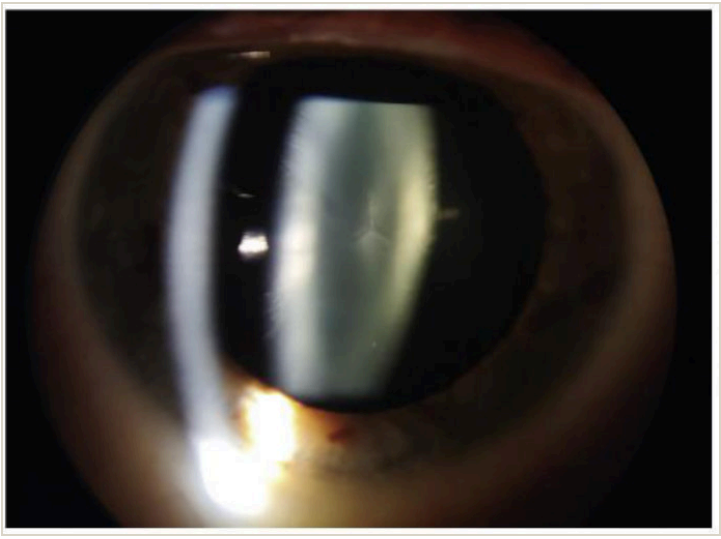
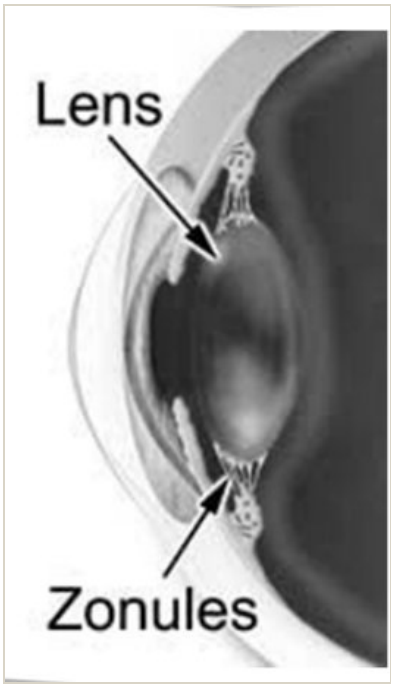


Fig. 7.9 Find the suture. Which one is it, and in which nucleus does it lie? § TB 7 · p. 102 · Slides 4 · #3 Identify

▼ Answer: Figs. 7.8 and 7.9

Red = **anterior capsule** (bright) · yellow = **subcapsular clear zone**, newly synthesised cortex (dark) · blue = **remodelling zone** (bright); behind them the grey zones of the rest of the cortex and the nuclei. Fig. 7.9: the **posterior Y suture**, seen as an upside-down Y, in the posterior part of the **fetal nucleus**. § TB 7 · p. 100, 102

8 · Zonules



The lens and its zonules. § Slides 4 · #6

FEATURE	DETAIL
Names	**zonules (of Zinn)** = **suspensory ligament** of the lens
Made of	threadlike **microfibrils** of extracellular matrix with fibrillin; very extensible, but **no true elastic fibres**
Origin	basement membrane of the **nonpigmented ciliary epithelium** in the **pars plana**, and the valleys between the ciliary processes in the **pars plicata**
Insertion	the lens capsule: most at the **preequatorial and postequatorial** regions, few directly at the equator; they merge with the capsule from the equator to near both poles (zonular lamella)
Job	hold the lens in place and transmit the ciliary muscle's pull: their tension keeps the lens flat (unaccommodated)

§ TB 7 · p. 98, 102. Your note "attach to the lens at the equator area, some go all the way to the poles · from the ciliary body to the lens capsule" fits; the textbook's refinement is that most fibres attach just in front of and just behind the equator.

9 · Accommodation

Distance (emmetropic eye): ciliary muscle relaxed → ciliary ring wide → zonules stretched → tension on the capsule holds the lens flat, ≈ 20 D. § TB 7 · p. 102

Near (classic Helmholtz theory): § TB 7 · p. 102–103

near target → **retinal blur** (cone-driven) → **ciliary muscle contracts** (parasympathetic) →

ciliary ring diameter **decreases** → zonular tension **falls** → capsule moulds the lens toward its preferred **spherical** shape → **lens power increases**

IN ACCOMMODATION	CHANGE
Thickness (pole to pole)	increases, in the **nucleus**; cortex thickness unchanged
Equatorial diameter	decreases
Anterior surface	curves much more steeply; anterior pole moves forward → **anterior chamber shallower**
Posterior surface	steepens only slightly (and stays steeper than the anterior)

§ TB 7 · p. 103. Relaxation reverses it: the ciliary body is pulled back by Bruch membrane's elastic tissue, the ring widens, zonular tension flattens the lens again. Your drawing (thin lens for "emmetrop", round lens for "accommodation") shows the same.

Presbyopia: accommodation nears zero by about 50. The main cause is the lens itself: older fibres resist deformation, the capsule becomes thicker and less elastic, and the growing lens changes the zonules' angle of pull; the ciliary muscle's force does *not* decrease. § TB 7 · p. 103

ONLY IN YOUR NOTES · CHECK WITH THE TEXTBOOK

Beside "posterior cortex" you wrote "fibers that have been already developed". Your notes The textbook says only that the zones behind the nucleus are concave forward and end at the posterior capsule (TB 7 · p. 100). Each fibre runs from the anterior suture to the posterior one (TB 7 · p. 98, 104), so the posterior cortex is made of the same fibres as the anterior cortex and is not older. Treat the note as "already-formed fibres of the cortex", not as a separate fibre type.

NOT TAUGHT FROM TEXTBOOK CHAPTER 7 (AND THE LENS PART OF CHAPTER 9)

Left out because no lecture point or objective asks for them: lens physiology (crystallin chemistry, gap-junction and epithelium–fibre interface details, metabolism, the lens's ionic current), ultraviolet absorption and oxidative stress (glutathione, ascorbate), the lens paradox, the molecular mechanisms of each cataract type (diabetic sorbitol, the cortex–nucleus barrier), the cataract grading system (Fig. 7.11), lens shagreen, vitreous attachments (hyaloid capsular ligament, Wieger ligament, retrolental space), minor zonule types, and from chapter 9 the induction signals, PAX6 and the hyaloid vessels around the fetal lens.

Key facts

≈ 20 D unaccommodated lens power · max accommodation **14 D** at 8–12 y → near **0** after 50 y
Thickness **3.6–4.5 mm** (mean 4), +0.02 mm/year · diameter 6.5 mm (infant) → **9.0 × 9.7 mm** (adult)
Radius of curvature: anterior **8–14 mm** · posterior **5–8 mm** (posterior steeper)
Refractive index: cortex **1.38** · nucleus **1.41**
Capsule: anterior pole 11 → 15 µm · equator ~7 µm · posterior pole **~3.5 µm** (thinnest)
Lens placode ~day 27 · lens vesicle ~day 33 · first secondary-fibre layer by week 7 · Y sutures visible month 3
Sutures: **3** branches (fetal Y) → 6–9 (early adulthood) → 9–15 (middle to old age)
Fibre cross-section: hexagon **3 × 9 µm**; outer fibre up to 1 cm long

Self-check

▼ Describe, step by step, how the embryonic nucleus forms.

Surface ectoderm thickens into the lens placode, which invaginates to form the lens vesicle, a hollow sphere of one cell layer. The posterior cells elongate forward as primary lens fibres until they fill the lumen; that core is the embryonic nucleus. The anterior epithelium stays. § TB 7 · p. 97–98 · TB 9 · p. 142–145

▼ Why does the adult lens have no posterior epithelium?

The posterior epithelium of the lens vesicle was used up to make the primary lens fibres. § TB 7 · p. 97

▼ Where are new lens fibres made, and how does a new fibre take its place around the embryonic nucleus?

In the germinative zone, the preequatorial epithelium. The daughter cell moves to the equator and elongates, apical end toward the anterior pole and basal end toward the posterior pole, until it meets fibres from the other side at a suture. It loses its organelles and lies outside all older fibres, like a new onion layer. § TB 7 · p. 98, 104

▼ Which fibre ends form the anterior suture, and which the posterior?

Anterior: the apical ends. Posterior: the basal ends. § TB 7 · p. 100

▼ Name the four lens zones from the centre out, with the fibres each contains.

Embryonic nucleus (primary fibres) · fetal nucleus (+ secondary fibres formed before birth) · adult nucleus (+ fibres from birth to sexual maturation) · cortex (fibres formed after sexual maturation). § TB 7 · p. 100

▼ A patient has an opacity between the anterior upright Y and the posterior inverted Y. Which nucleus is involved? And if it lies along a Y?

Between the Ys: the embryonic nucleus, which has no sutures. Along a Y: the fetal nucleus, which contains the Y sutures. Both formed before birth. § TB 7 · p. 100 · TB 9 · p. 145

▼ Where is the lens capsule thickest and thinnest, and what is it made of?

Thickest in the ring around the anterior pole (anterior capsule thickens with age); thinnest at the posterior pole (~3.5 μm). It is a basement membrane, mainly collagen, highly elastic though it has no elastic fibres. § TB 7 · p. 97–98

▼ Compare nuclear, cortical and posterior subcapsular cataracts: location and one clinical feature each.

Nuclear: in a nucleus; yellow-brown, myopic shift. Cortical: in the cortex; spokes following the fibres, affects vision late. Posterior subcapsular: under the posterior capsule; affects vision early, linked to steroids. § TB 7 · p. 106–108

▼ List, front to back, the slit-lamp zones of the lens.

Anterior capsule (bright) · subcapsular clear zone (dark) · remodelling zone (bright) · cortex and nuclei (grey) · anterior Y suture · embryonic nucleus · posterior inverted Y suture · posterior cortex · posterior capsule. § Handout 4 · p. 4 · TB 7 · p. 100

▼ Where do the zonules arise and insert?

From the basement membrane of the nonpigmented ciliary epithelium in the pars plana and the valleys of the pars plicata; they insert into the lens capsule, mostly just anterior and posterior to the equator. § TB 7 · p. 102

▼ What happens to the ciliary muscle, the zonules and the lens when you look at a near object?

The ciliary muscle contracts, the ciliary ring narrows, zonular tension falls and the capsule moulds the lens rounder: thicker, smaller equatorial diameter, steeper anterior surface, anterior chamber shallower. Power rises. § TB 7 · p. 102–103

§ Sources for this chapter: Handout 4 "Intraocular Lens Anatomy" (with your notes) · Slides 4 "Intraocular Lens Anatomy" · Exam guidelines (Review, p. 2, with your typed notes) · Textbook ch. 7, Crystalline Lens, pp. 97–108 · Textbook ch. 9, Ocular Embryology (lens), pp. 140–145.

Uvea: iris, ciliary body and choroid

The vascular middle coat: its landmarks, its vessels, its muscles, and the barrier that keeps the aqueous clear.

WHAT THE EXAM ASKS

The guidelines' objectives for this lecture, word for word. § Review 14 · p. 2, 5

- Exam "Identify and describe the landmarks present on the iris" → §2
- Exam "Name the iris and ciliary blood vessels" → §3
- Exam "Discuss how the iris and ciliary blood vessels and the ciliary body epithelium contribute to the blood aqueous barrier and uveitis" → §7
- Exam "Discuss the functions and innervation of the iris and ciliary body muscles" → §4, §6
- Exam "Name and recognize the location and characteristics of the two main parts of the ciliary body" → §5
- Exam "Discuss the vascular patterns of the arteries of the choroid and choriocapillaris" → §8
- Exam "Describe the location of vortex veins draining blood from the choroid" → §8
- Exam "Discuss how Bruch's membrane is involved in age-related macular degeneration" → §9
- Exam Circulation: "Describe the anastomosis that occurs to form the major circle of the iris" → §3 (only the part this lecture covers; the rest of the blood supply is the Circulation topic)

1 · Three regions, front to back

REGION	WHERE	MAIN JOB
Iris	in front of the lens; its centre opening is the pupil	diaphragm: regulates how much light reaches the retina
Ciliary body	ring from the iris root back to the ora serrata	makes the aqueous, part of the blood–aqueous barrier, accommodation
Choroid	ora serrata to the optic nerve, between sclera and retina	blood supply to the outer retina

The uvea is called the **vascular layer** because its largest part, the choroid, is mostly blood vessels. § Handout 5 · p. 1 · TB 5 · p. 62, 68, 75

2 · Iris landmarks

The handout's iris outline runs 1 root · 2 vessels · 4 pupil · 5 posterior epithelium; there is no point 3 in it. The landmarks the guidelines ask for come from the textbook. § Handout 5 · p. 1–4

LANDMARK	WHERE / WHAT	WHY IT MATTERS
Iris root	peripheral end of the iris, joining it to the front of the ciliary body; goes all the way round (360°). **Thinnest** part of the iris. § Handout 5 · p. 1 · TB 5 · p. 62	tears away in blunt trauma: **iridodialysis** (box below)
Collarette	circular, jagged, slightly raised ridge ~1.5 mm from the pupil margin; the **thickest** part of the iris. Attachment site of the fetal pupillary membrane. § TB 5 · p. 62	divides the iris into two zones; the minor circle lies behind it
Pupillary zone	from the pupil out to the collarette; holds the sphincter. § TB 5 · p. 62, 64	colour often differs from the ciliary zone
Ciliary zone	from the collarette out to the iris root. § TB 5 · p. 62	shows the circular contraction folds
Trabeculae	radial columns of collagen; white fibres in light irises. § TB 5 · p. 63, 65	hidden by pigment in brown irises
Crypts	oval openings where the anterior border layer is missing: **Fuchs crypts** either side of the collarette, **peripheral crypts** near the root. § TB 5 · p. 62, 65–66	let aqueous move quickly in and out of the stroma as the iris changes volume
Contraction folds	circular folds on the ciliary zone, made as tissue moves toward the root in dilation. § TB 5 · p. 66	
Pupillary ruff (frill)	the pigmented posterior epithelium curls round the pupil margin onto the front; dark serrated ring. § Handout 5 · p. 4 · TB 5 · p. 65	darker than the rest of the iris
Posterior surface	fairly smooth: radial contraction furrows (pupillary zone), structural folds of Schwalbe (ciliary zone, continuing as the valleys between ciliary processes), circular folds. § TB 5 · p. 66–67	

Pupil: slightly nasal and inferior to the iris centre. The iris is ~12 mm across and lies like a truncated cone: the pupil margin rests on the lens, in front of the root. § TB 5 · p. 62

Four layers, front to back: **anterior border layer · stroma + sphincter · anterior epithelium + dilator · posterior epithelium**. § TB 5 · p. 62

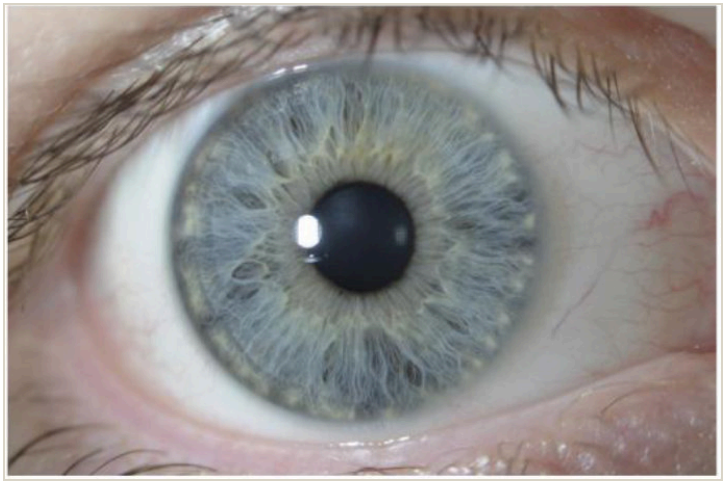


Fig. 5.1 Find the collarette, the pupillary and ciliary zones, the trabeculae and the crypts, then open the answer. § TB 5 · p. 62 · Slides 5 · #2

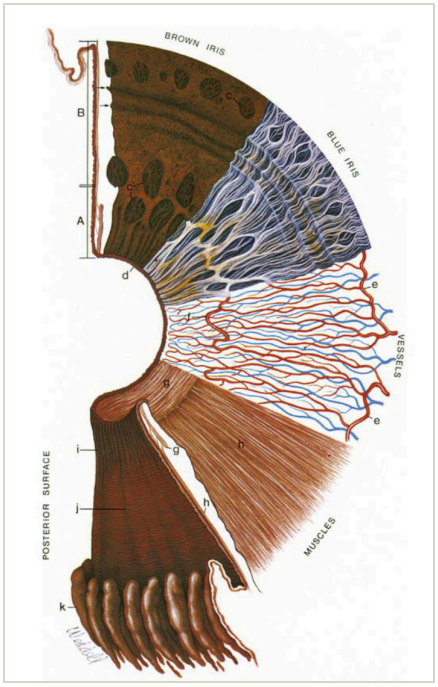


Fig. 5.17 Surfaces and layers of the iris. Name A, B and c–k. § TB 5 · p. 69 · Slides 5 · #4

▼ Answer: Fig. 5.1 and Fig. 5.17

Fig. 5.1: the yellowish jagged ring about a third of the way out is the collarette; inside it the pupillary zone, outside it the ciliary zone; the white radial strands are trabeculae; the dark oval holes beside the collarette are crypts.

Fig. 5.17: A pupillary portion · B ciliary portion · arrows: circular contraction furrows · c Fuchs crypts · d pigment ruff · e major arterial circle (in the ciliary body) · f incomplete minor arterial circle · g sphincter muscle · h dilator muscle · i radial contraction furrows · j structural folds of Schwalbe · k pars plicata. Brown iris: dense, matted anterior border layer; blue iris: thinner border layer, trabeculae stand out. § TB 5 · p. 69

CLINICAL CORRELATES · IRIS ROOT

Iridodialysis: a blow to the eye or head tears the thin iris root from the ciliary body (Fig. 5.5). Vessels and nerves can be damaged: bleeding into the anterior and/or posterior chamber, sector paralysis of the iris muscles. § Handout 5 · p. 1 · TB 5 · p. 62

Hyphema: the torn branches from the major circle bleed into the anterior chamber; the blood settles inferiorly (Fig. 5.27). § TB 5 · p. 75 · Slides 5 · #3

"So we have to look at the edges of the iris" after trauma. Your notes

3 · Iris and ciliary blood vessels

VESSEL	WHERE	CHARACTERISTICS

Major (arterial) circle of the iris	in the **ciliary body** stroma, in front of the circular ciliary muscle, near the iris root (pars plicata) § TB 5 · p. 63, 70 · TB 12 · p. 197	formed by the **long posterior ciliary arteries + anterior ciliary arteries**; supplies the iris and the ciliary body
Radial iris vessels	iris stroma, from the root to the pupil margin § Handout 5 · p. 2 · TB 5 · p. 63	branches of the major circle; wrapped in collagen so they do not kink as the iris moves
Minor circle of the iris	iris stroma, behind the collarette § Handout 5 · p. 2 · TB 5 · p. 64	**incomplete** circle, an embryological remnant; capillary arcades run from it toward the pupil
Iris capillaries	iris stroma	**not fenestrated**; endothelium joined by zonula occludens → part of the blood–aqueous barrier § Handout 5 · p. 2 · TB 5 · p. 64, 74
Ciliary body capillaries	ciliary stroma, mostly in the ciliary processes, near the pigmented epithelium	**fenestrated**, large, with wide lumens: leaky, for aqueous production § Handout 5 · p. 6 · TB 5 · p. 70

How the major circle forms

long posterior ciliary arteries (nasal + temporal) → run forward in the suprachoroidal space →
enter the ciliary body, branch up and down + anterior ciliary arteries (from the rectus muscle arteries) →
major circle of the iris

§ TB 5 · p. 70, 79 · TB 12 · p. 198 · Fig. 12.8. The branches anastomose with each other and with the anterior ciliary arteries. From the circle: radial vessels → iris; branches → ciliary body.

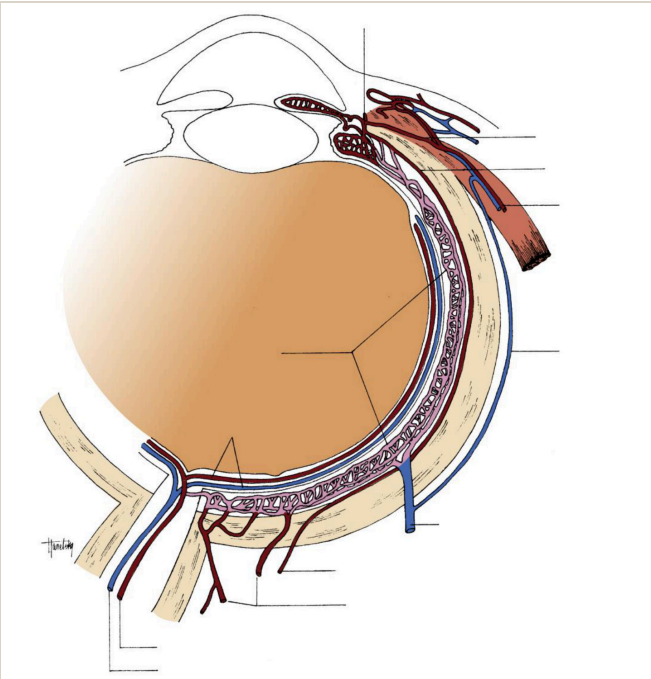
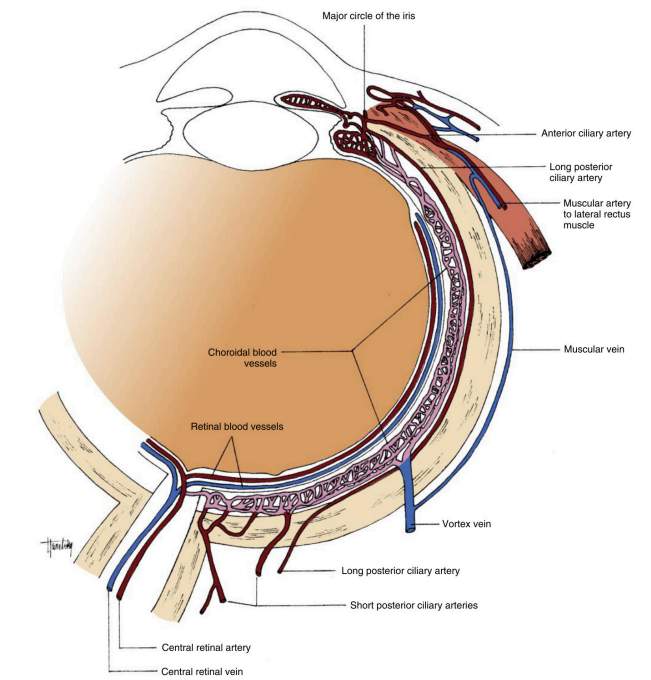


Fig. 12.8, labels removed. Name the 12 lines, then open the answer. § TB 12 · p. 199 · Slides 5 · #11

▼ Answer: Fig. 12.8 with labels



Top, then clockwise: major circle of the iris · anterior ciliary artery · long posterior ciliary artery · muscular artery to lateral rectus muscle · muscular vein · vortex vein · long posterior ciliary artery · short posterior ciliary arteries · central retinal vein · central retinal artery · retinal blood vessels · choroidal blood vessels.

4 · Pupil and iris muscles

The pupil regulates retinal illumination; its diameter is **1–9 mm** depending on the light: **miotic** (very small) in bright light, **mydriatic** (large) in dim light. § Handout 5 · p. 3 · TB 5 · p. 62

MUSCLE	LOCATION AND FIBRES	INNERVATION	ACTION
Sphincter	smooth muscle, a **circular** band 0.75–1 mm wide round the pupil, in the stroma of the pupillary zone; cells joined by tight junctions. Keeps working even if cut radially. § TB 5 · p. 64	**parasympathetic**	constricts the pupil: **miosis**
Dilator	smooth-muscle processes of the **anterior iris epithelium** (myoepithelial cells), 3–5 layers, **radial**, from the iris root to below the middle of the sphincter. § TB 5 · p. 64	**sympathetic**	pulls the pupillary portion toward the root: **mydriasis**

§ Handout 5 · p. 3 · TB 5 · p. 64, 68. In accommodation the sphincter also contracts, cutting spherical aberration (TB 5 · p. 72). The pupillary pathways themselves are chapter 11.

Posterior iris epithelium

A single layer of **heavily pigmented** columnar cells joined by tight junctions and desmosomes, lining the posterior chamber. The two iris epithelia lie apex to apex. Peripherally it loses its pigment and continues as the inner, nonpigmented ciliary epithelium; the anterior iris epithelium continues as the outer, pigmented one. § Handout 5 · p. 4 · TB 5 · p. 64–65, 68

CLINICAL CORRELATES · POSTERIOR IRIS EPITHELIUM

Pupillary ruff: the posterior epithelium seen at the pupil margin as a dark serrated ring (§2). § Handout 5 · p. 4 · TB 5 · p. 65

Pigmentary dispersion syndrome: the iris bows backward and its posterior surface rubs on the zonules → pigment is shed into the anterior chamber → deposits on the iris, lens, corneal endothelium (**Krukenberg spindle**, Fig. 5.18A) or in the trabecular meshwork, where it can block aqueous outflow. On transillumination the red fundus reflex shows through the depigmented spots (Fig. 5.18B). § Handout 5 · p. 4 · TB 5 · p. 67, 70 · Slides 5 · #6

"This can cause glaucoma." Your notes (the textbook says only that outflow may be compromised; glaucoma is chapter 6)

5 · Ciliary body: shape and two parts

A **ring** from the iris root to the ora serrata (≈5.9 mm wide nasally, 6.7 mm temporally). In sagittal section a **triangle**: base anterior (one corner at the scleral spur, the iris root from its middle), outer side against the sclera, inner side lining the posterior chamber and a little of the vitreous chamber, apex at the ora serrata. § Handout 5 · p. 5 · TB 5 · p. 68–69

	PARS PLICATA (CORONA CILIARIS)	PARS PLANA (ORBICULARIS CILIARIS)
Location	**anterior**	**posterior**, from the pars plicata to the ora serrata

Surface	wider; **70–80 ciliary processes** project into the posterior chamber; the valleys between them are the **valleys of Kuhnt**. A process ≈ 2 mm long, 0.5 mm wide, 1 mm high.	**flat**
Zonules	some attach in the valleys	most arise here (internal limiting membrane) and run forward through the valleys to the lens
Other	dense fenestrated capillaries in each process: the big surface for aqueous secretion	the vitreous base reaches ~2 mm over its posterior part

§ Handout 5 · p. 5 · TB 5 · p. 69, 73 · Slides 5 · #7–8 (Fig. 5.21A)

Ora serrata: the scalloped border between ciliary body and retina; forward-pointing *dentate processes* (teeth) with *oral bays* between. Here the pigmented ciliary epithelium becomes the RPE and the nonpigmented one becomes neural retina. § TB 5 · p. 69, 71

Layers, outer to inner: **supraciliaris** (lets the ciliary body slide on the sclera; fluid or trauma here → ciliary body detachment, Fig. 5.22) · **ciliary muscle** · **ciliary stroma** (holds the major circle) · **outer pigmented epithelium** · **inner nonpigmented epithelium**. § TB 5 · p. 69–70 · Slides 5 · #9

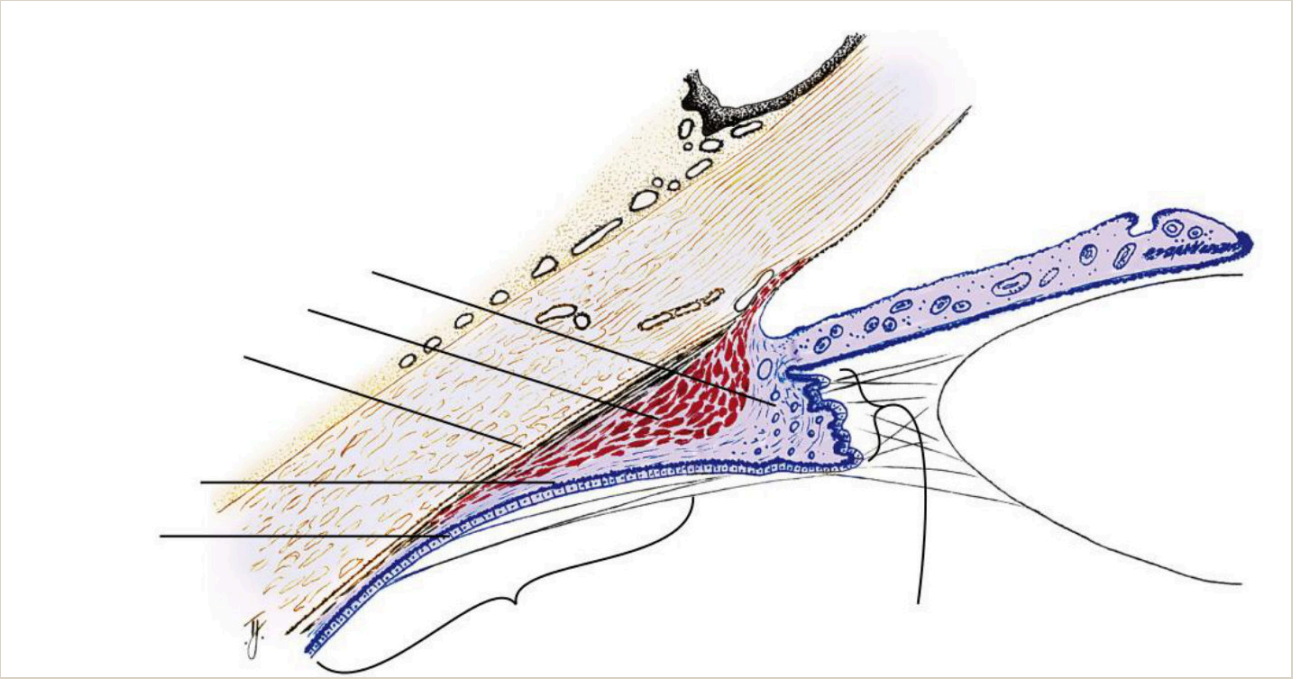
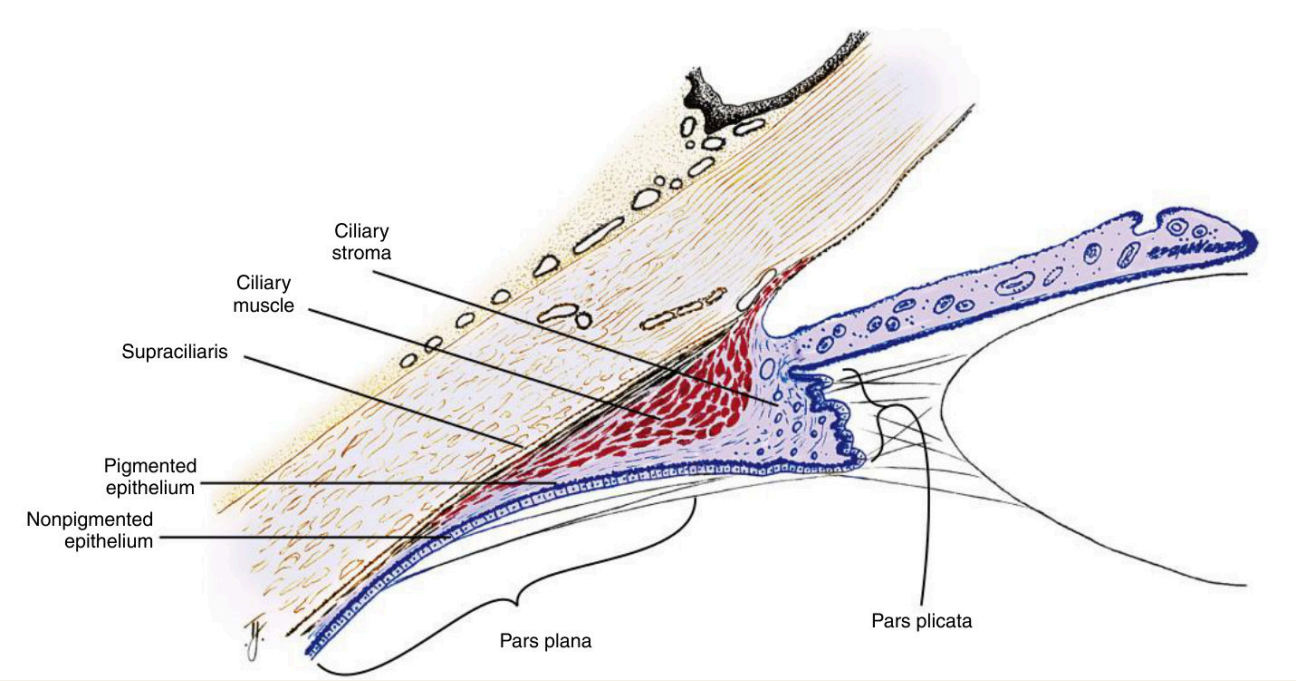


Fig. 5.20, labels removed. Name the 7 lines and brackets, then open the answer. § TB 5 · p. 71

▼ Answer: Fig. 5.20 with labels



Top to bottom on the left: ciliary stroma · ciliary muscle · supraciliaris · pigmented epithelium · nonpigmented epithelium. Brackets: pars plana (flat, posterior) · pars plicata (the processes, anterior).

6 · Ciliary muscle, accommodation and innervation

Smooth muscle in three fibre directions: **longitudinal** (outer, along the sclera; from the scleral spur to the anterior choroid), **radial**, and **circular** (innermost, sphincter-like, near the major circle). § Handout 5 · p. 6 · TB 5 · p. 70

Accommodation (near focus)

parasympathetic → ciliary muscle contracts →

longitudinal fibres pull the choroid forward; circular fibres shrink the ring toward the lens → zonule tension released →

lens rounder, thicker, front curve steeper → **more power: near focus**

§ TB 5 · p. 71–72 · Handout 5 · p. 7. Muscle relaxed → zonules taut, lens thinner → distance vision (the eye "at rest"). Your notes: "zonule more loose, thicker lens" / "lens thinner".

Innervation of the uveal muscles

FIBRES	ROUTE	SUPPLY
Parasympathetic	from the ciliary ganglion	ciliary muscle, iris sphincter, choroidal vessels

Sympathetic	superior cervical ganglion → long ciliary nerves	iris dilator and ciliary muscle
Sympathetic	superior cervical ganglion → ophthalmic and short ciliary nerves	choroidal vessels
Sensory	nasociliary nerve (ophthalmic division of V)	the whole uvea

§ TB 5 · p. 79

Sympathetic effect on the ciliary muscle: the sources differ. The handout: "no clinically significant effect on the ciliary body" Lecture. The textbook: it "likely has an inhibitory effect" TB. Know both; in the exam, lead with the handout's line. § Handout 5 · p. 6 · TB 5 · p. 70

7 · Blood–aqueous barrier and uveitis

GROUNDWORK FROM TEXTBOOK CHAPTER 1 · TIGHT JUNCTIONS

Zonula occludens is a belt of tight junctions round the apex of each cell; rows of ridges seal the space between cells. A substance cannot pass *between* such cells; it must go *through* them. These junctions help form the blood–aqueous and blood–retinal barriers, and disease can break them. § TB 1 · p. 8

The ciliary body epithelium **produces and secretes the aqueous**: diffusion, ultrafiltration and mainly (80–90%) active secretion, chiefly by the nonpigmented layer; ≈2.3 μL per minute. Aqueous outflow is chapter 6. § Handout 5 · p. 6–7 · TB 5 · p. 73–74

The barrier, step by step

blood in ciliary capillaries → fenestrations: large molecules and protein get out → ciliary stroma → pigmented epithelium → nonpigmented epithelium: zonula occludens block the gaps, so molecules must pass through the cells → **posterior chamber: selected aqueous, little protein**

§ TB 5 · p. 71, 74 · Handout 5 · p. 6

SITE	VESSELS	SEAL
Ciliary body	capillaries **fenestrated**: leaky	the seal is the **inner nonpigmented epithelium**, its cells joined by zonula occludens
Iris	capillaries **not fenestrated**; aqueous freely enters the iris stroma through the crypts	the seal is the **capillary endothelium**, joined by zonula occludens

Result: plasma has **~200×** more protein than aqueous. Low protein = little light scatter = maximum transmission.

§ TB 5 · p. 74

CLINICAL CORRELATE · UVEITIS

Inflammation breaks the zonula occludens of the nonpigmented ciliary epithelium, so the blood–aqueous barrier fails: immune factors and white cells enter the aqueous → **cells and flare** in the anterior chamber (Fig. 5.26). With the slit lamp a conical beam in normal aqueous is invisible ("optically empty"); particles scatter the light and make it visible: the **Tyndall phenomenon**. A lot of whitish material settles inferiorly as a **hypopyon**. § Handout 5 · p. 7 · TB 5 · p. 74 · Slides 5 · #10

Your notes: "protein in the aqueous, like dust in a cinema beam: it shouldn't be there"; uveitis is a painful, red, light-sensitive eye Your notes (the symptoms are not in the textbook chapter).

8 · Choroid: vessels, choriocapillaris, vortex veins

From the ora serrata to the optic nerve, between sclera and retina; mostly blood vessels; feeds the outer retina and takes its waste; ~300 μm thick under the fovea, its thickest point. § Handout 5 · p. 8 · TB 5 · p. 75, 77

LAYER (OUTER → INNER)	WHAT IT IS
Suprachoroid lamina (lamina fusca)	thin pigmented bands of connective tissue over the suprachoroidal space, which carries the long posterior ciliary arteries and nerves forward (and drains some aqueous)
Choroidal stroma	pigmented, vascular loose connective tissue: the large and medium vessels
Choriocapillaris	the capillary bed (below)
Bruch membrane	innermost, acellular, fused to the RPE (§9)

§ TB 5 · p. 75, 77

Vascular pattern of the arteries

short posterior ciliary arteries (ring round the optic nerve) → Haller layer: large vessels, outer → Sattler layer: medium vessels → **choriocapillaris, inner**

§ Handout 5 · p. 8 · TB 5 · p. 76, 79. The vessels get smaller as they branch and come closer to the retina.

The handout says the choroidal arteries are branches of the **short** posterior ciliary arteries. The textbook adds that the short ones supply the posterior pole, while branches of the **long** posterior ciliary arteries supply the anterior choroid and wedge-shaped nasal and temporal zones; the border between two supplies is a **watershed zone**, prone to ischaemia. § Handout 5 · p. 8 · TB 5 · p. 75, 77, 79

CHORIOCAPILLARIS

Structure	**single layer** of **anastomosing, fenestrated** capillaries; fenestrations mostly face the retina
Lumen	3–4× an ordinary capillary: 2–3 red cells side by side
Arrangement	lobules, each fed from the Sattler layer; the whole choroid fills at once
Macula	**densest** here, and the **sole blood supply** of the macular area
Limit	only in the choroid; it does not continue into the ciliary body

§ Handout 5 · p. 8 · TB 5 · p. 76–77

Vortex veins: where the choroid drains

choriocapillaris → venules → veins gather in a vortex (whirl) pattern in each quadrant →

4 or more vortex veins leave the eye

One vortex vein per **quadrant** at least; they exit the globe ~6 mm **posterior to the equator** and can be seen with a dilated pupil (Fig. 12.15, 5.32). Choroidal veins have no valves. They drain most of the uvea. § Handout 5 · p. 8 · TB 5 · p. 76, 79 · TB 12 · p. 204 · Slides 5 · #14–15

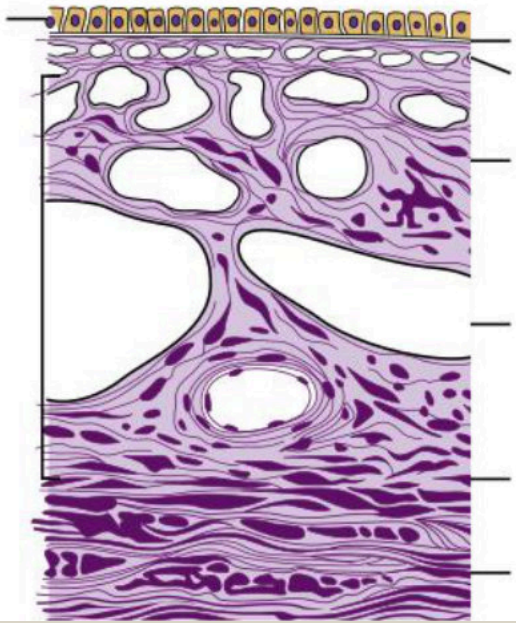
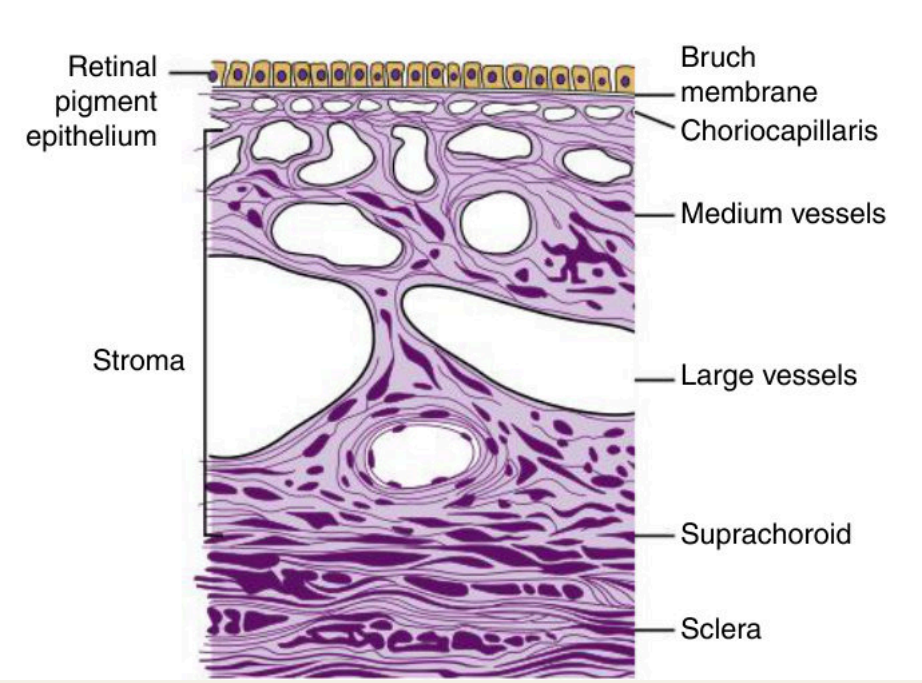


Fig. 5.29, labels removed. Name the 7 lines and the bracket, then open the answer. § TB 5 · p. 75 · Slides 5 · #12

▼ Answer: Fig. 5.29 with labels



Left: retinal pigment epithelium · stroma (bracket). Right, top to bottom: Bruch membrane · choriocapillaris · medium vessels (Sattler) · large vessels (Haller) · suprachoroid · sclera.

9 · Bruch membrane and age-related macular degeneration

Bruch membrane (the *basal lamina*) is the innermost layer of the choroid, from the optic nerve to the ora serrata. Five layers, outer to inner: **basement membrane of the choriocapillaris · outer collagenous zone · elastic layer · inner collagenous zone · basement membrane of the RPE**. Filaments of the RPE basement membrane merge with the inner collagenous zone: a **tight adhesion** of RPE to choroid. Everything going from choroid to retina, and waste going back, must cross it. § Handout 5 · p. 9 · TB 5 · p. 77–78

From drusen to vision loss

ageing: lipofuscin builds up in the RPE →

drusen: lipids, cholesterol, proteins between the RPE basement membrane and the inner collagenous zone →

Bruch membrane turns **hydrophobic**: blocks water and metabolites →

outer retina starved; water trapped under the RPE can detach it → **RPE atrophy, or new leaky vessels**

§ TB 5 · p. 77–79 · Fig. 5.38

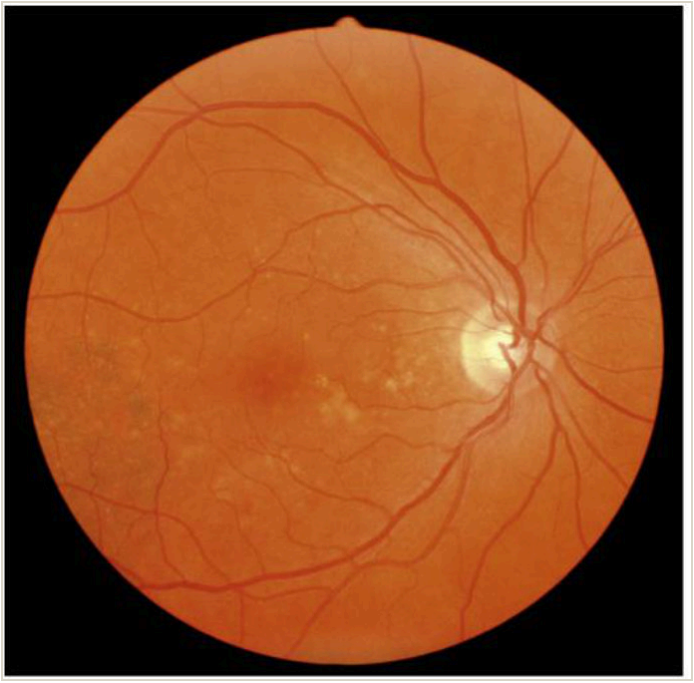


Fig. 5.35 Drusen: small yellow-white pinhead spots in the fundus. § TB 5 · p. 77 · Slides 5 · #16

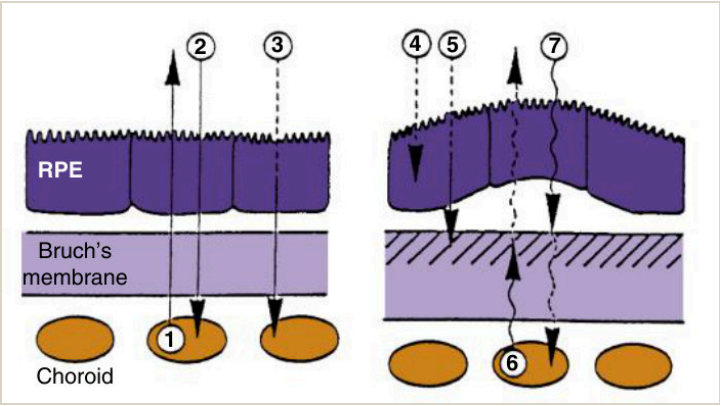


Fig. 5.38 Left, young: 1 metabolites from the choroid to the retina, 2 water from the retina to the choroid, 3 waste cleared to the choroid. Right, older: 4 lipofuscin in the RPE, 5 lipid waste in Bruch membrane, 6 metabolites blocked, 7 hydrophobic barrier traps water → RPE detachment. § TB 5 · p. 79 · Slides 5 · #17

CLINICAL CORRELATE · AGE-RELATED MACULAR DEGENERATION (AMD)

Degeneration of the choroid–retina interface in the macula; the most common cause of blindness in Western countries. Risks: age, genetics, smoking, diet low in antioxidants, Caucasian ethnicity, oxidative damage. § TB 5 · p. 78

- **Early**: drusen and RPE changes.
- **Advanced**: either **geographic atrophy** of choroid, RPE and photoreceptors, or **neovascularization**: drusen keep RPE-made VEGF from reaching the choriocapillaris → ischaemia → more VEGF → fragile new vessels from the choriocapillaris grow under the RPE or through Bruch membrane into the retina, leak and bleed.
- No cure; intravitreal **anti-VEGF** injections can prevent vision loss from neovascular membranes. No treatment for atrophy yet.

§ Handout 5 · p. 9 · TB 5 · p. 78–79. The macula depends on the choriocapillaris alone (§8), which is why this is where it shows.

ONLY IN YOUR NOTES · CHECK BEFORE YOU RELY ON THEM

- "Supply sphincter and Müller's: parasympathetic" Your notes: "Müller's" here means the circular ciliary muscle. The textbook chapter does not use that name; it just says the ciliary muscle and sphincter are parasympathetic.
- Vortex veins "minimum 4, up to 10–12" Your notes: the textbook says only "four or more".
- Drusen are "phospholipid build-up" Your notes: the textbook says lipids, cholesterol and proteins. One word on the drusen slide ("build up ___ lipids") cannot be read.
- Fig. 12.15: choroidal vessels show in the first (lightly pigmented) fundus, not in the second (darker) one Your notes. You wrote "cortex vessel" next to it; the figure shows vortex veins, so read it as "vortex".

NOT TAUGHT FROM TEXTBOOK CHAPTER 5

Left out because neither the handout nor the guidelines ask for it: iris colour and melanocyte studies, clump cells and the cells of the anterior border layer, Michel and Fuchs spurs, synechiae and iris bombé, iridectomy and iridotomy, heterochromia, iris and choroidal nevi, the fine structure of the ciliary muscle (V-shaped bundles, muscle stars), the ion-transport model of aqueous secretion (carbonic anhydrase, aquaporins) and its circadian rhythm, presbyopia research, choroidal lymphatics, AMD supplement trials, ageing changes of the uvea, and the reference list. Aqueous outflow and IOP are chapter 6; the full orbital blood supply is the Circulation topic.

Key facts

3 uveal regions: iris · ciliary body · choroid (front to back)
Pupil **1–9 mm** · iris ≈ **12 mm** across · collarette ≈ **1.5 mm** from the pupil margin · iris root = thinnest, collarette = thickest
Sphincter **0.75–1 mm** wide, circular, parasympathetic → miosis · dilator **3–5** layers, radial, sympathetic → mydriasis
70–80 ciliary processes (pars plicata) · valleys of Kuhnt between them · pars plana flat, ends at the ora serrata
Major circle = long posterior ciliary aa. + anterior ciliary aa., in the ciliary body stroma · minor circle incomplete, behind the collarette
Iris capillaries **not** fenestrated · ciliary and choriocapillaris capillaries fenestrated
Aqueous: active secretion **80–90%** · ≈ **2.3 µL/min** · plasma protein ≈ **200×** aqueous
Choroid ≈ **300 µm** under the fovea · Haller (large) → Sattler (medium) → choriocapillaris (single layer, lumen **3–4×**)
≥ 4 vortex veins, one per quadrant, exit ≈ **6 mm** behind the equator · Bruch membrane: **5** layers

Self-check

▼ Name the iris landmarks from the pupil to the root, and say which is the thinnest and which the thickest part.

Pupillary ruff at the margin · pupillary zone (sphincter) · collarette (~1.5 mm out, thickest; Fuchs crypts either side) · ciliary zone with contraction folds · peripheral crypts · iris root (thinnest). § TB 5 · p. 62, 65–66

▼ What is iridodialysis, why does it happen at that site, and what can follow?

The iris root tearing from the ciliary body after blunt trauma; the root is the thinnest part. Damaged vessels bleed (hyphema in the anterior chamber, settling inferiorly); nerve damage can paralyse a sector of the iris muscles. § Handout 5 · p. 1 · TB 5 · p. 62, 75

▼ Which arteries form the major circle of the iris, where does it lie, and what does it supply?

The long posterior ciliary arteries (after running forward in the suprachoroidal space) anastomose with each other and with the anterior ciliary arteries. It lies in the ciliary body stroma near the iris root, in front of the circular muscle, and supplies the iris (radial vessels) and the ciliary body. § TB 5 · p. 70, 79 · TB 12 · p. 198

▼ Where is the minor circle of the iris, and how is it different from the major circle?

In the iris stroma behind the collarette; incomplete; an embryological remnant. The major circle is complete and lies in the ciliary body, not the iris. § TB 5 · p. 63–64

▼ Give the location, fibre direction, innervation and action of the sphincter and the dilator.

Sphincter: circular smooth muscle in the stroma of the pupillary zone; parasympathetic; miosis. Dilator: radial processes of the anterior iris epithelium from the root to the mid-sphincter; sympathetic; mydriasis. § TB 5 · p. 64, 68

▼ How does the blood–aqueous barrier work in the ciliary body, and how in the iris?

Ciliary body: fenestrated capillaries let protein out, but the zonula occludens of the inner nonpigmented epithelium stop it passing between the cells into the posterior chamber. Iris: the capillaries are not fenestrated and their endothelium is sealed by zonula occludens. § TB 5 · p. 74 · TB 1 · p. 8

▼ What happens to the barrier in uveitis, and what do you see at the slit lamp?

The zonula occludens of the nonpigmented ciliary epithelium break down, so white cells and protein enter the aqueous: cells and flare (Tyndall phenomenon, the beam becomes visible); much material forms a hypopyon. § TB 5 · p. 74

▼ The ciliary muscle contracts. What happens to the zonules, the lens and the focus? Which nerves drive it?

Ring shrinks and choroid is pulled forward → zonules slacken → lens thickens, front curve steepens → more power, near focus (accommodation). Parasympathetic (ciliary ganglion). Sympathetic: "no clinically significant effect" (handout) / "likely inhibitory" (textbook). § Handout 5 · p. 6 · TB 5 · p. 70–72, 79

▼ Compare the pars plicata and the pars plana: position, surface, and what attaches.

Pars plicata: anterior, wider, 70–80 ciliary processes with valleys of Kuhnt; some zonules attach in the valleys. Pars plana: posterior, flat, runs to the ora serrata; most zonules arise here; the vitreous base sits over its back ~2 mm. § TB 5 · p. 69

▼ Describe the arterial pattern of the choroid from the short posterior ciliary arteries inward, and the choriocapillaris.

Short posterior ciliary arteries → large vessels (Haller, outer) → medium (Sattler) → choriocapillaris: a single layer of anastomosing, fenestrated, wide capillaries in lobules, densest in the macula, its only blood supply. Long posterior ciliary arteries feed the anterior choroid. § Handout 5 · p. 8 · TB 5 · p. 76, 79

▼ Where are the vortex veins?

At least one in each quadrant (four or more), where the choroidal veins gather in a vortex; they leave the globe ~6 mm behind the equator. § Handout 5 · p. 8 · TB 5 · p. 76 · TB 12 · p. 204

▼ How is Bruch membrane involved in AMD?

With age lipid-rich drusen collect between the RPE basement membrane and the inner collagenous zone; Bruch membrane becomes hydrophobic, blocking water and nutrients to the outer retina and waste back out. Result: RPE atrophy and photoreceptor loss (geographic atrophy) or VEGF-driven new vessels that cross Bruch membrane, leak and bleed. Water trapped under the RPE can detach it. § TB 5 · p. 77–79

§ Sources for this chapter: Handout 5 "Uveal Tract (Uvea)" (with your notes) · Slides 5 "Uvea: Iris, Ciliary Body, and Choroid" (with your notes) · Exam guidelines (Review 14 · p. 2, 5) · Textbook ch. 5, Uvea, pp. 62–79; ch. 12, pp. 197–199, 204 (major circle, Fig. 12.8, vortex veins); ch. 1, p. 8 (tight junctions).

IOP and the aqueous chambers

Three chambers, the drainage angle, and the two ways aqueous leaves the eye.

WHAT THE EXAM ASKS

- Exam "Compare the three chambers of the globe" → section 1. § Review 14 · p. 3
- Exam "Identify the anatomical appearance of the anterior chamber angle structures with gonioscopy" → sections 2 and 3, label drill. § Review 14 · p. 3
- Exam "Describe ways that aqueous exits the anterior chamber" → section 5. Your note beside it: conventional and unconventional flow "will be a flow chart!" Be able to draw both routes box by box. § Review 14 · p. 3

1 · The three chambers compared

The anterior and posterior chambers hold **aqueous humour**; the vitreous chamber holds the **vitreous gel**. § TB 6 · p. 82

	ANTERIOR CHAMBER	POSTERIOR CHAMBER	VITREOUS CHAMBER
Contains	aqueous	aqueous; zonule fibres cross it	vitreous body: transparent gel, 98.5–99.7% water, collagen + hyaluronic acid
In front	**corneal endothelium**	**posterior iris surface**	**posterior lens surface** and the retrozonular part of the posterior chamber
Behind	**anterior iris surface** + pupillary area of the **anterior lens**	**anterior face of the vitreous**	retina and optic disc
Periphery	**trabecular meshwork**, part of the **ciliary body**, **iris root** (= the angle)	**ciliary body** (ciliary processes project into it) and the **equatorial lens**	**pars plana** of the ciliary body, then retina
Shape · size	centre deeper than periphery	annular (ring) space behind the iris	largest: about **80**% of the eye's volume
Special parts	the **anterior chamber angle**, where aqueous leaves	canal of **Hannover** (among the zonules) · canal of **Petit** (behind the last zonules, a potential space)	**patellar fossa** (the dip the lens sits in) · strongest attachment: **vitreous base** at the ora serrata

Job	route of aqueous to the exit	receives the aqueous as it is secreted	holds the retina against the choroid, stores metabolites, absorbs shocks, transmits and refracts light

§ TB 6 · p. 82 (anterior) · p. 89 (posterior, Fig. 6.12) · p. 90–92 (vitreous) · Handout 6 · p. 1. Your notes for the boundaries match: corneal endothelium · iris + anterior lens · corneoscleral junction (TM, CB, iris root).

The **anterior chamber angle** lies at the periphery of the anterior chamber, where corneoscleral and uveal tissue meet. Aqueous exits through it. § TB 6 · p. 82 · Handout 6 · p. 2

2 · The angle structures

The aqueous exits through the **filtration apparatus**: trabecular meshwork + Schlemm canal. With the scleral spur they fill the **internal scleral sulcus**, a groove at the inner corneoscleral junction. § TB 6 · p. 82

STRUCTURE	LOCATION	MADE OF · FEATURES	FUNCTION
Scleral spur	posterior edge of the internal scleral sulcus	**dense, coarse collagen fibres** § Handout 6 · p. 2 · TB 6 · p. 83	posterior part: **attachment for the ciliary muscle** (longitudinal fibres' tendon); anterior part: TM sheets attach, which widens the trabecular spaces and keeps Schlemm canal open § TB 6 · p. 82
Trabecular meshwork (TM)	encircles the anterior chamber, on the inner side of the internal scleral sulcus	**avascular**; **triangular** in cross-section: apex at **Schwalbe line**, base at the scleral spur. Perforated sheets: 3–5 at the apex, branching to 15–20 at the spur. Covered by endothelium continuous with the **corneal endothelium**; these cells phagocytose debris.	filters aqueous toward Schlemm canal. Its cells are contractile, like smooth muscle; **ciliary muscle contraction widens its spaces** and lowers outflow resistance § TB 6 · p. 82–83
· **Uveal meshwork**	**inner** part, facing the anterior chamber	**cordlike**; attaches to ciliary stroma and longitudinal ciliary muscle, some to the iris root; **largest pores**	first filter; entry to *both* outflow routes § TB 6 · p. 82, 86
· **Corneoscleral meshwork**	**outer** part, toward Schlemm canal	**sheetlike**; attaches to the scleral spur; pores smaller nearer the canal	narrower filter of the conventional route § TB 6 · p. 82, 86

Juxtacanalicular tissue	between the last TM sheet and the endothelium of Schlemm canal	also called the cribriform layer: cells in a matrix of collagen, elastic-like fibres, ground substance	with the canal's inner wall, the **site of highest outflow resistance** § TB 6 · p. 83, 87
Schlemm canal	**outer** to the TM, **anterior** to the scleral spur; outer wall against limbal sclera	**circular vessel**, a venous channel that normally holds **aqueous, not blood**. Endothelium with tight junctions (like a blood vessel), incomplete basement membrane (like a lymph channel). Internal collector channels (of Sondermann) are blind pouches toward the TM.	lets aqueous pass **out of the anterior chamber**; aqueous crosses the inner-wall endothelium, mostly by passive diffusion, also through pores and giant vacuoles § Handout 6 · p. 3 · TB 6 · p. 83–84, 86–87
External collector channels	around the outer wall of Schlemm canal, about **25–37**	branch from the canal: these vessels are *not* Schlemm canal itself	empty into the **deep scleral plexus**, then the **intrascleral plexus**, then the **episcleral veins** § TB 6 · p. 87
Schwalbe line	anterior end of the TM	the **termination of Descemet membrane**	anterior landmark of the angle § TB 6 · p. 82

§ Where the collector channels drain: the handout says the deep scleral *or* intrascleral plexus "and then go to the episcleral *or conjunctival* veins" (Handout 6 · p. 3); the textbook gives the sequence deep scleral plexus → intrascleral plexus → episcleral veins (TB 6 · p. 87). Know both wordings.

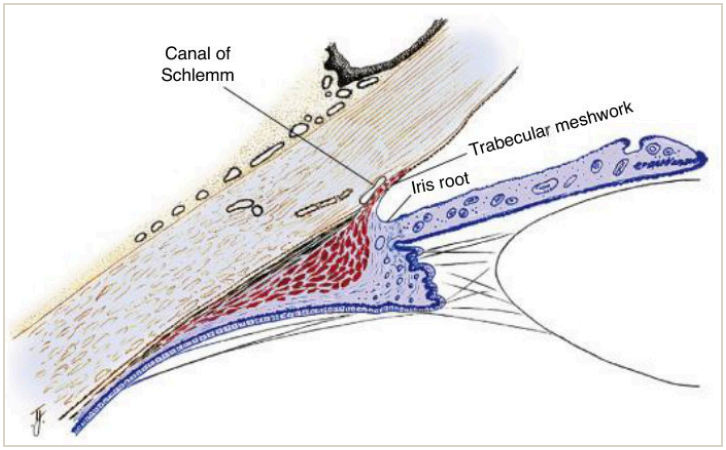


Fig. 6.1 Periphery of the anterior chamber: canal of Schlemm, trabecular meshwork, iris root. § TB 6 · p. 82

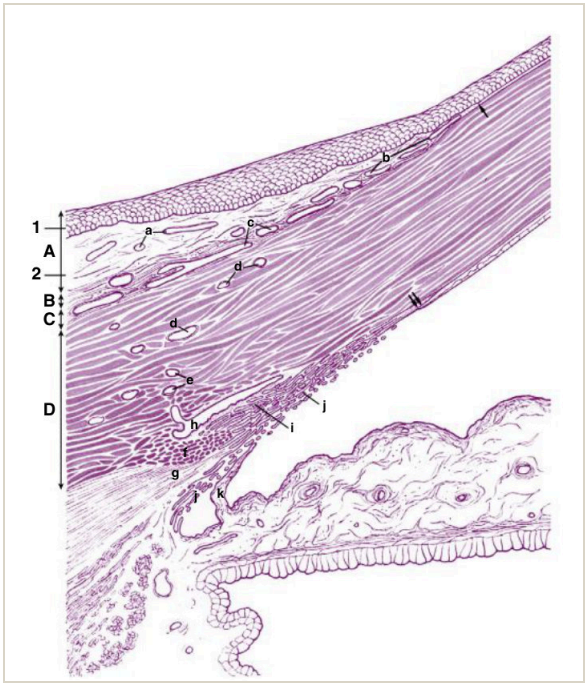


Fig. 6.2 The limbus. Key letters: c episcleral vessels · d intrascleral plexus · e deep scleral plexus · f scleral spur · g longitudinal ciliary muscle · h Schlemm canal · i corneal (corneoscleral) TM · j uveal meshwork · k iris process · double arrow: end of Descemet membrane. § TB 6 · p. 83 · Slides 6 · #2

Label the angle in histology

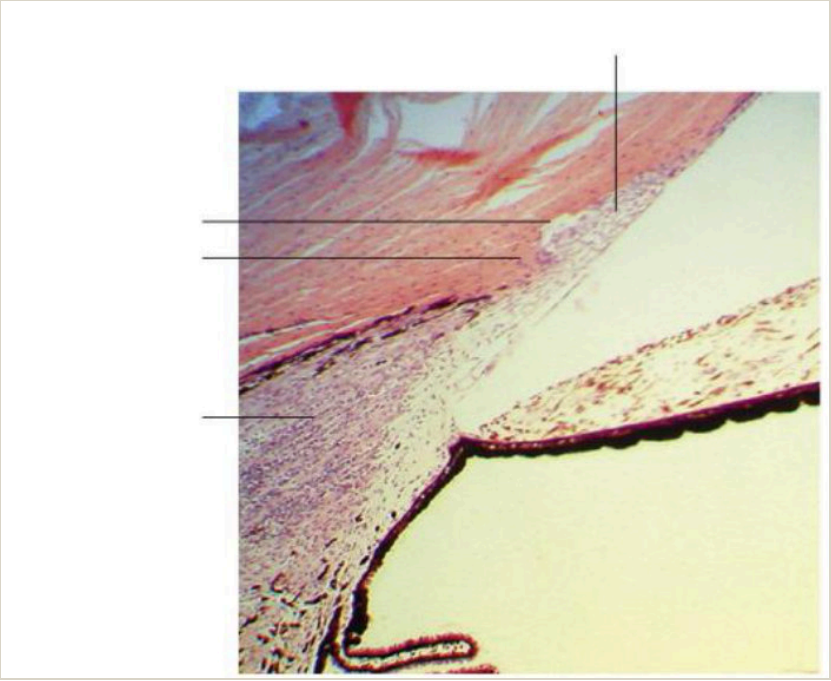
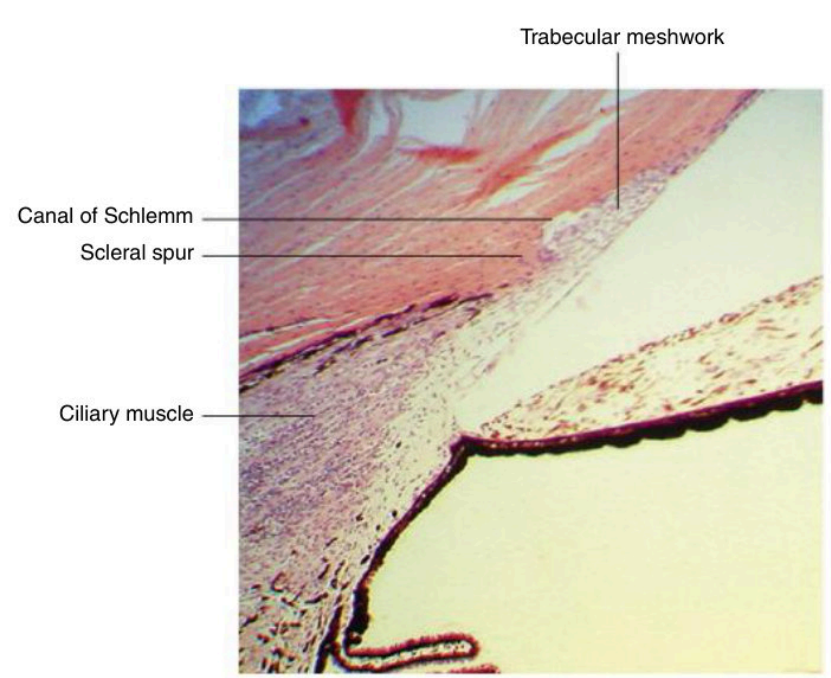


Fig. 6.3, labels removed. Name the four lines, then open the answer. § TB 6 · p. 83 · Slides 6 · #4

▼ Answer: Fig. 6.3 with labels



Top to bottom: trabecular meshwork · canal of Schlemm · scleral spur · ciliary muscle. The iris is the tissue at lower right with the dark pigmented edge; the cornea runs to the upper right.

3 · Gonioscopy: seeing the angle

CLINICAL: WHY GONIOSCOPY

The angle cannot be seen directly: the limbus is opaque, and light from the angle does not leave the cornea because of **total internal reflection**. A **gonioscopy lens** overcomes it with **mirrors** (Fig. 6.7), so the examiner can view the angle structures. The view is as if you stood on the lens and looked across the iris to the angle. Angle width is first estimated with the biomicroscope; gonioscopy follows if it looks narrow or outflow is in doubt. § Handout 6 · p. 3 · TB 6 · p. 84–85 · Slides 6 · #6–8

Seen with gonioscopy, from **posterior** (near the pupil) to **anterior** (toward the cornea). The letters are those of Figs. 6.8 and 6.9, in the same order: § TB 6 · p. 85–86

A · iris root → B · ciliary body band → C · scleral spur → D · trabecular meshwork → **E · Schwalbe line**

STRUCTURE	HOW IT LOOKS	YOUR NOTES
Iris root	most posterior; peripheral iris running into the angle	"once you find the pupil, then you can follow out" Your notes
Ciliary body	band just anterior to the iris root	darker than the iris Your notes
Scleral spur	line anterior to the ciliary body	the white line you see is the scleral spur Your notes
Trabecular meshwork	its **posterior part, where most aqueous is filtered, is mildly pigmented** § TB 6 · p. 86	"has fluid in it", more posterior part
Schwalbe line	most anterior: the end of Descemet membrane	hard to see Your notes
Schlemm canal	lies **behind the TM** and is generally **not visible**. If blood backs up into it, a **thin red line** appears in the TM area (Fig. 6.9). This happens when pressure on the gonioscopy lens compresses the episcleral veins so that episcleral venous pressure exceeds IOP. § TB 6 · p. 85–86	"we do not see"

In a **wide-open angle** the whole TM is visible. As peripheral iris approaches the TM the angle **narrows** and access to the trabecular openings decreases. Pigment or debris clogging the meshwork is visible with gonioscopy.

§ TB 6 · p. 85

Label the angle seen through the gonioscope

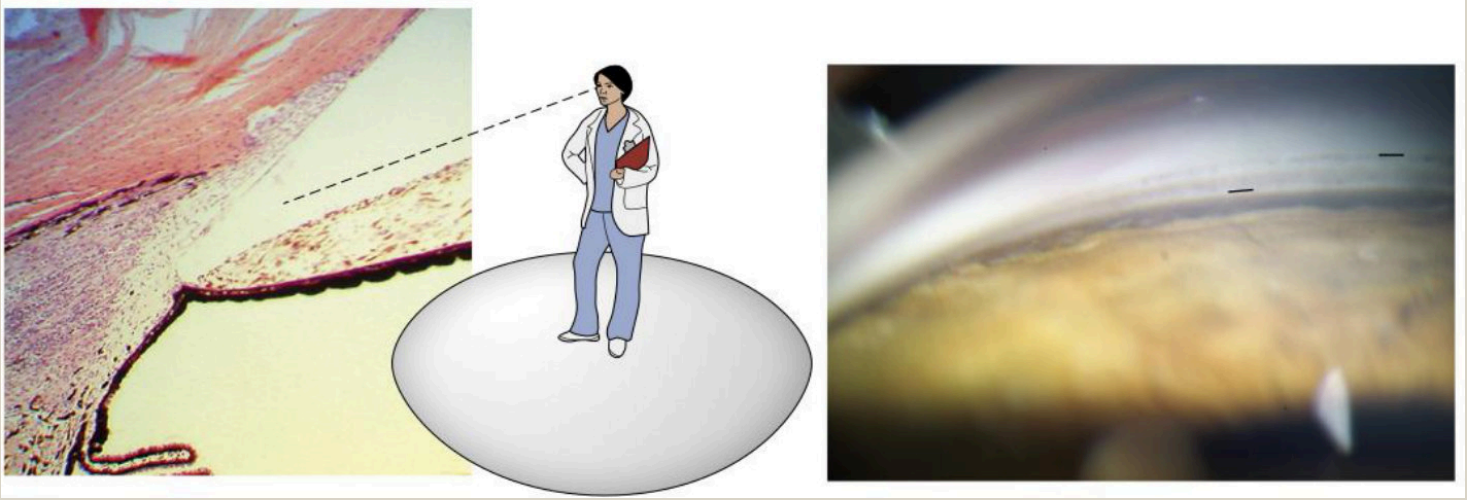
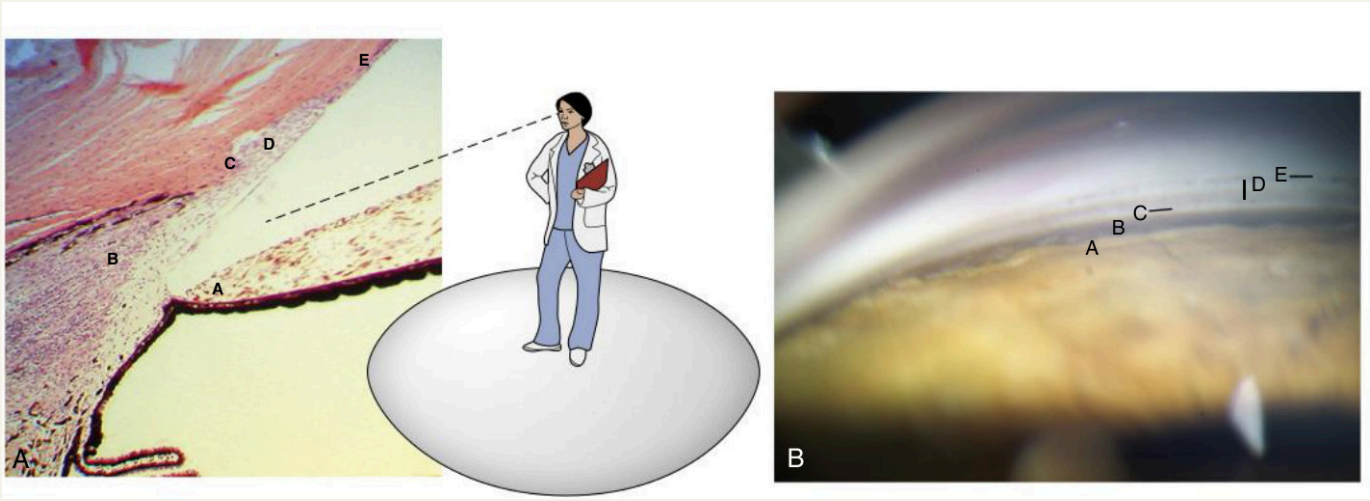


Fig. 6.8, letters removed. Left: histology of the angle. Right: the same angle through the gonioscope. On the right image, name the five lines from posterior to anterior, then find the same five structures on the histology. § TB 6 · p. 86 · Slides 6 · #7–8

▼ Answer: Fig. 6.8 with letters



A iris root · B ciliary body · C scleral spur · D trabecular meshwork · E Schwalbe line. The dashed line is the examiner's line of sight across the iris to the angle.

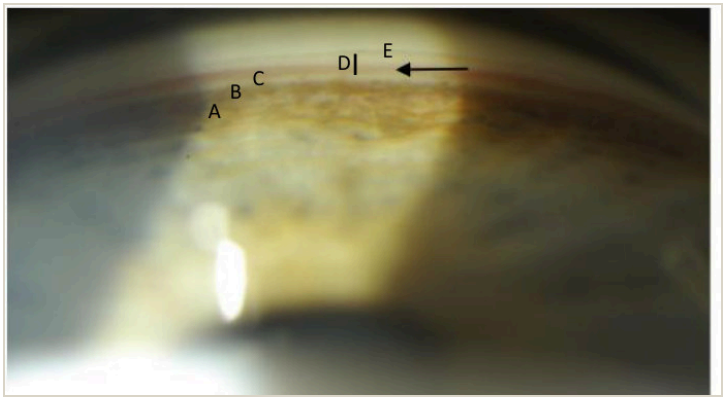


Fig. 6.9 Blood backed up into Schlemm canal (arrow) gives the posterior TM a pink line. A–E as in Fig. 6.8. § TB 6 · p. 86

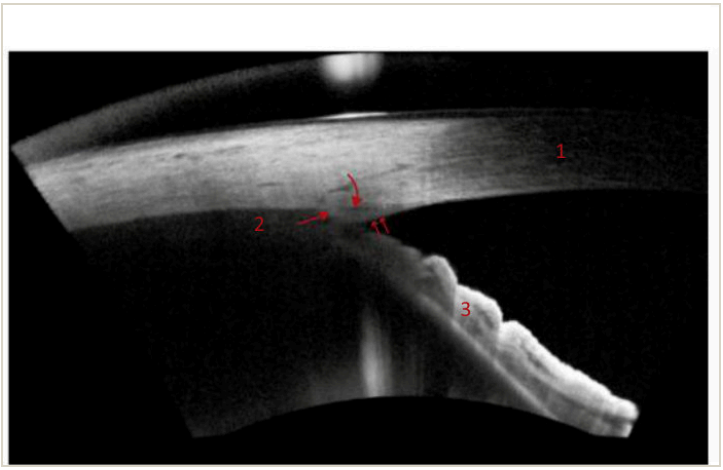


Fig. 6.5 OCT of the angle: 1 cornea · 2 ciliary body · 3 iris · curved arrow Schlemm canal · arrow scleral spur · double arrow trabecular meshwork. § TB 6 · p. 84 · Slides 6 · #5

4 · Aqueous: what it does and where it is made

POINT	WHAT THE SOURCES SAY
Function	supplies metabolites, mainly **oxygen and glucose**, to the **avascular cornea and lens**; **carries waste away**; its constant volume **maintains IOP**. § TB 6 · p. 85, 87
Production site	**pars plicata** of the ciliary body: secreted through the **epithelium covering the ciliary processes** into the **posterior chamber**. § TB 6 · p. 85
Mechanisms	diffusion, ultrafiltration and **active secretion** (80–90% of production, mostly by the nonpigmented ciliary epithelium). Detail in chapter 5. § TB 5 · p. 73
Circulation	between iris and lens, through the **pupil** into the anterior chamber. There it circulates in **convection currents**: down along the cooler cornea, up along the warmer iris, out at the periphery. § TB 6 · p. 85

§ Handout 6 · p. 4 (II A–B). Your notes agree: "to provide nutrients to the cornea and lens · removes/reduces waste products" · "by pars plicata (hills and valleys) in posterior chamber".

From production to the angle

ciliary processes (pars plicata) → posterior chamber → between iris and lens → pupil → anterior chamber → **anterior chamber angle**

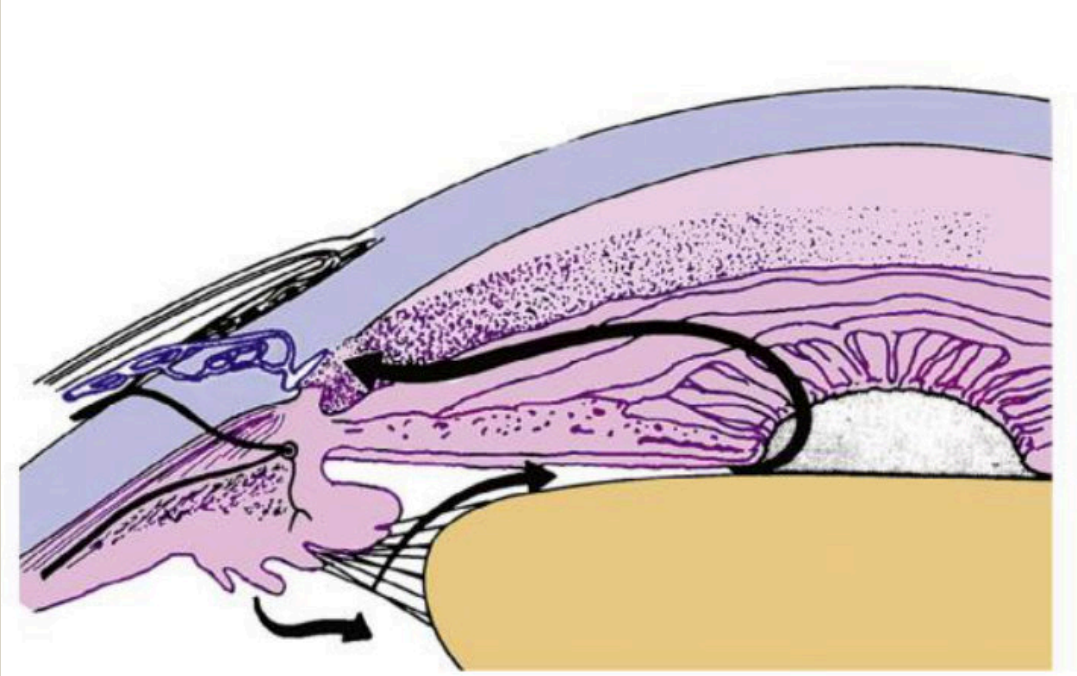


Fig. 6.10 Flow of aqueous: formed in the ciliary processes, around the lens, through the pupil, out through the TM into Schlemm canal and on to the episcleral veins. The lower arrow from the ciliary body is the unconventional route. § TB 6 · p. 86 · Slides 6 · #3, #10

5 · How aqueous exits the anterior chamber: the two routes

There are **two** routes; both start in the **uveal meshwork**. § TB 6 · p. 86 · Handout 6 · p. 4

Conventional outflow = trabecular flow

anterior chamber → uveal meshwork → corneoscleral meshwork → juxtacanalicular tissue → endothelium of Schlemm canal (inner wall) → Schlemm canal → external collector channels → deep scleral plexus → intrascleral plexus → **episcleral veins**

§ TB 6 · p. 86–87 · Handout 6 · p. 3–4. Your note: uveal TM → corneal TM → Schlemm's canal → external collector channels → episclera. Handout end point: episcleral *or conjunctival* veins.

Unconventional outflow = uveoscleral flow

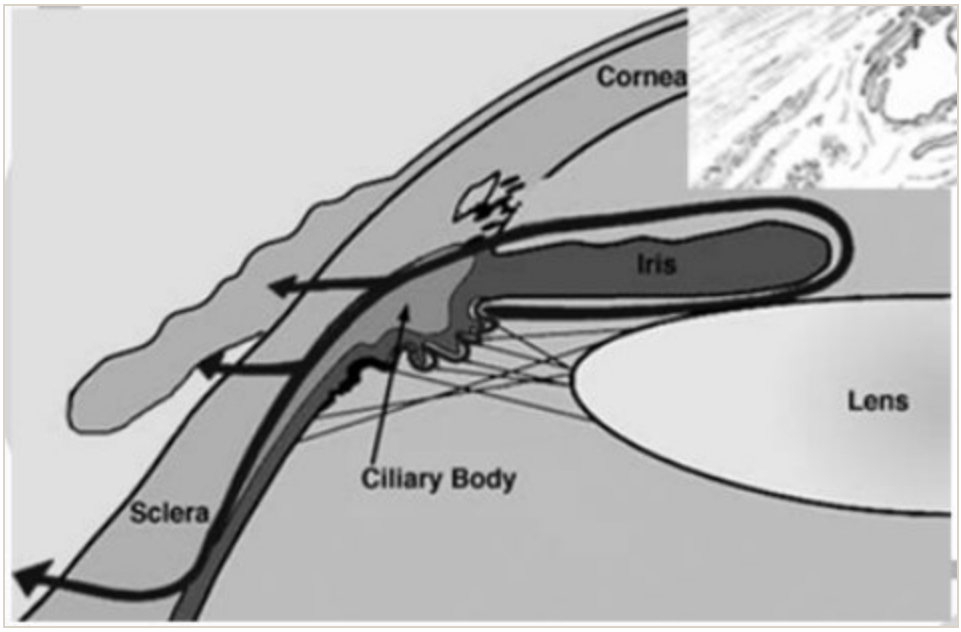
anterior chamber → uveal meshwork (or straight into the ciliary body face) → spaces between the ciliary muscle bundles → **supraciliary and suprachoroidal spaces**

From the suprachoroidal space, three exits: § TB 6 · p. 86

Supraciliary / suprachoroidal space

> **through the sclera** → absorbed into the orbital vessels
>
> **into the choroid** → anterior ciliary veins and vortex veins
>
> **lymphatic channels** in the ciliary stroma

§ TB 6 · p. 86–87. Slides 6 · #12 shows the route as arrows out through the ciliary body and sclera. Your notes list four exits (sclera · anterior ciliary veins · suprachoroidal collector channel · lymphatic channels); the textbook names three and does not mention a "suprachoroidal collector channel".



Unconventional route: arrows leave the angle through the ciliary body and pass out through the sclera. Top right corner: part of Fig. 6.2 overlapping on the slide. § Slides 6 · #12

The two routes side by side

	CONVENTIONAL (TRABECULAR)	UNCONVENTIONAL (UVEOSCLERAL)
Path	TM → Schlemm canal → collector channels → episcleral veins	uveal TM → ciliary muscle → suprachoroidal space → sclera / choroid / lymphatics
Share	the rest	**5–60**% of total outflow; **decreases with age**
Pressure	**pressure-driven**; keeps IOP steady; more variable	fairly **constant**, not affected by IOP
Resistance	mainly at the **juxtacanalicular tissue + inner wall of Schlemm canal**; rises with pigment or debris in the pores, fewer TM cells, collapse of the canal	little: no continuous epithelium covers the ciliary body where it borders the anterior chamber
Ciliary muscle contraction	**increases** outflow (spaces widen, canal enlarges)	**decreases** outflow (cycloplegics increase it)

§ TB 6 · p. 83, 86–88

6 · Intraocular pressure: the balance

- IOP stays in a narrow range because **production = exit** ($F_{in} = F_{out}$). § TB 6 · p. 87–88
- **Production stays fairly constant**, so most raised IOP comes from **decreased outflow**, i.e. raised resistance in the conventional route. § TB 6 · p. 87–88
- For understanding only: $F_{in} = (P_{\text{ciliary capillaries}} - IOP) \cdot C_{in} + S$ (active secretion) and $F_{out} = (IOP - P_{\text{episcleral veins}}) \cdot C_{out} + U$ (uveoscleral). C is **facility**, the reciprocal of resistance. § TB 6 · p. 88
- Normal IOP **10–21 mm Hg**; low to mid 20s suspect; upper 20s or higher need close monitoring. Measured with a tonometer (non-contact or Goldmann applanation). § TB 6 · p. 88

CLINICAL: GLAUCOMA, AND DIFFERENT DRUGS FOR DIFFERENT FLOWS

Raised IOP can damage the retinal nerve fibre layer, mechanically or by reducing perfusion; the loss shows at the optic disc as an enlarged, deeper cup. In glaucoma the raised IOP usually comes from resistance in the **conventional** route (juxtacanalicular tissue, inner wall of Schlemm canal). Normal-tension glaucoma damages the nerve fibres at normal IOP. § TB 6 · p. 88

- **Less production**: beta-blockers, alpha$_2$ agonists (brimonidine), carbonic anhydrase inhibitors. § TB 6 · p. 88
- **More conventional outflow**: pilocarpine (contracts the ciliary muscle, opens the TM sheets), Rho kinase inhibitors. § TB 6 · p. 88–89
- **More uveoscleral outflow**: **prostaglandins**, the most effective drugs (remodel the matrix between ciliary muscle bundles); brimonidine also. § TB 6 · p. 88–89

Your notes: "we have to understand it pretty well, because of glaucoma" · "different medication for different flow". Your notes

ONLY IN YOUR NOTES · CHECK

- Unconventional outflow is a "more recent discovery" and "goes backwards" Your notes: the textbook does not say either.
- Handout p. 1, next to the anterior chamber: "space that holds vitreous". The anterior chamber holds **aqueous**; the vitreous chamber holds the vitreous. § TB 6 · p. 82
- Handout p. 3, next to Schlemm canal: "doesn't have blood in it, but vitreous". It holds **aqueous**, not vitreous. § TB 6 · p. 83

NOT TAUGHT FROM TEXTBOOK CHAPTER 6

Left out because no objective or handout point needs them: TM stem cells; the fine structure of Schlemm canal pores and giant vacuoles; aqueous veins of Ascher; nasal–temporal differences; the Krukenberg spindle; steroid-induced IOP rise; glaucoma surgery; tonometry technique; age and ethnic changes in IOP and the angle. Also most of the vitreous: attachments other than the vitreous base, its zones, Cloquet canal, hyalocytes, collagen and hyaluronic acid distribution, ascorbate and oxygen, liquefaction, posterior vitreous detachment and vitreomacular traction.

Key facts

3 chambers: anterior + posterior (aqueous, joined at the pupil) · vitreous (gel, about **80%** of the eye's volume)
Angle order, posterior → anterior: **iris root · ciliary body · scleral spur · TM · Schwalbe line**
TM: triangle, apex Schwalbe line, base scleral spur · **3–5** sheets at the apex → **15–20** at the spur · uveal = inner, cordlike · corneoscleral = outer, sheetlike
25–37 external collector channels
2 outflow routes · uveoscleral = **5–60%** of outflow, pressure-independent
Unconventional exits: **3** (sclera · choroid → anterior ciliary + vortex veins · lymphatics)
Normal IOP **10–21 mm Hg** · highest resistance: juxtacanalicular tissue + inner wall of Schlemm canal

Self-check

▼ Give the anterior, posterior and peripheral boundaries of the anterior chamber.

Anterior: corneal endothelium. Posterior: anterior iris surface and the pupillary area of the anterior lens. Peripheral: trabecular meshwork, part of the ciliary body, iris root. § TB 6 · p. 82

▼ Compare the posterior and vitreous chambers: boundaries and contents.

Posterior: ring space bounded by the posterior iris, equatorial lens, anterior vitreous face and ciliary body; holds aqueous and the zonules (canals of Hannover and Petit). Vitreous: largest (about 80% of volume), bounded by the posterior lens, pars plana, retina and optic disc; holds the vitreous gel. § TB 6 · p. 89–90

▼ What is the scleral spur made of, and what attaches to it?

Dense, coarse collagen fibres. The ciliary muscle (longitudinal fibres' tendon) attaches posteriorly; trabecular sheets attach anteriorly. § Handout 6 · p. 2 · TB 6 · p. 82–83

▼ Name the two divisions of the trabecular meshwork and how they differ.

Uveal meshwork: inner, cordlike, largest pores, attaches to ciliary stroma and muscle. Corneoscleral meshwork: outer, sheetlike, attaches to the scleral spur, smaller pores toward Schlemm canal. § TB 6 · p. 82

▼ What is Schwalbe line?

The termination of Descemet membrane: the apex of the TM and the most anterior angle structure in gonioscopy. § TB 6 · p. 82, 85

▼ Why is a gonioscopy lens needed to see the angle?

The limbus is opaque, and light from the angle is trapped by total internal reflection at the cornea. The lens's mirrors overcome this. § Handout 6 · p. 3 · TB 6 · p. 84

▼ List the angle structures seen with gonioscopy, from posterior to anterior. Which one is usually not visible?

Iris root, ciliary body, scleral spur, trabecular meshwork, Schwalbe line. Schlemm canal lies behind the TM and is not visible unless blood backs up into it. § TB 6 · p. 85–86

▼ Why can a red line appear in the TM during gonioscopy?

Pressing on the lens compresses the episcleral veins; when their pressure exceeds IOP, blood backs up into Schlemm canal. § TB 6 · p. 85

▼ Draw the conventional outflow route as a flow chart.

Anterior chamber → uveal meshwork → corneoscleral meshwork → juxtacanalicular tissue → inner-wall endothelium → Schlemm canal → external collector channels → deep scleral plexus → intrascleral plexus → episcleral veins (handout: episcleral or conjunctival veins). § TB 6 · p. 86–87 · Handout 6 · p. 3

▼ Draw the unconventional outflow route, with its exits.

Anterior chamber → uveal meshwork → spaces between ciliary muscle bundles → supraciliary and suprachoroidal spaces → (1) through the sclera to orbital vessels, (2) into the choroid → anterior ciliary and vortex veins, (3) lymphatic channels in the ciliary stroma. § TB 6 · p. 86

▼ Where is outflow resistance highest, and why does IOP usually rise?

Juxtacanalicular tissue and the inner wall of Schlemm canal. Production stays fairly constant, so raised IOP usually comes from decreased conventional outflow. § TB 6 · p. 87–88

▼ Where is aqueous made, what does it do, and what is normal IOP?

Pars plicata ciliary processes, into the posterior chamber. It feeds the avascular cornea and lens (oxygen, glucose), removes waste and maintains IOP. Normal IOP: 10–21 mm Hg. § TB 6 · p. 85, 87–88

§ Sources for this chapter: Handout 6 "Intraocular Pressure (IOP) and Aqueous Chambers" (with your notes) · Slides 6 "IOP and Aqueous Chambers" (with your notes) · Review 14 · p. 3 (objectives and your typed notes) · Textbook ch. 6, Aqueous and Vitreous Humors, pp. 82–92 · TB 5 · p. 73 (aqueous production).

Retina

The neural coat: its cells, its ten layers as the microscope and the OCT show them, its regions, its two blood supplies and its barrier.

WHAT THE EXAM ASKS

The guidelines' objectives for this lecture, word for word. § Review 14 · p. 3–4

- Exam "Describe the functions of the RPE" → §2
- Exam "Contrast the types of photoreceptors and under what conditions they are stimulated" → §3
- Exam "Describe the portions of and functions the photoreceptor" → §3
- Exam "Identify the types, functions, and synapses of ganglion cells" → §4
- Exam "Identify each of the ten retinal layers on an OCT" → §5, §6
- Exam "Identify the photoreceptor parts on an OCT" → §6
- Exam "Describe the structures and/or connections found in each of the 10 retinal layers" → §5
- Exam "Describe the position of the macula relative to the disc" → §7
- Exam "Describe the anatomic differences between the fovea and the remainder of the retina, including the relative concentration of photoreceptors and blood vessels" → §7
- Exam "Describe the location, patterns, and structures supplied by the retinal and choroidal blood vessels" → §8
- Exam "Describe how the anatomy contributes to the blood retinal barrier" → §9
- Exam Circulation: "Describe the blood vessels that supply the anterior and posterior choroid (outer retina) and differentiate this from the blood vessels that supply the inner retina" → §8 (only the retinal-versus-choroidal part; the rest of the blood supply is the Circulation topic)

Your typed review notes stress "picture OCT": expect an OCT image to label. Your notes

1 · General retinal anatomy

The retina is the innermost coat, between the choroid and the vitreous, from the edge of the optic disc to the ora serrata. It has two parts from the two layers of the optic cup: the outer **retinal pigment epithelium** (RPE), stuck firmly to the choroid everywhere, and the **neural retina**, attached to the RPE only in a ring round the disc and at the ora serrata. It holds the first three neurons of the visual pathway. § TB 8 · p. 111

Terms

TERM	TEXTBOOK / HANDOUT	YOUR NOTES

Macula and fovea	Macula: darker area at the posterior pole for sharpest acuity and colour vision. Fovea: the depression at its centre. Histologists and clinicians use the words differently (§7). § TB 8 · p. 111, 128	central vision
Posterior pole	A clinical term for part of the fundus; "no universal agreement" on its definition. § TB 8 · p. 128	posterior portion of the eyeball: retinal blood vessels, macula and optic nerve Your notes
Peripheral retina	Most of the retinal area; rods dominate; made for gross form and motion. Central retina: rich in cones and ganglion cells, small, made for acuity. § TB 8 · p. 128	anterior
Fundus	The textbook uses the word (the fundus as seen with the ophthalmoscope) but does not define it.	= retina Your notes

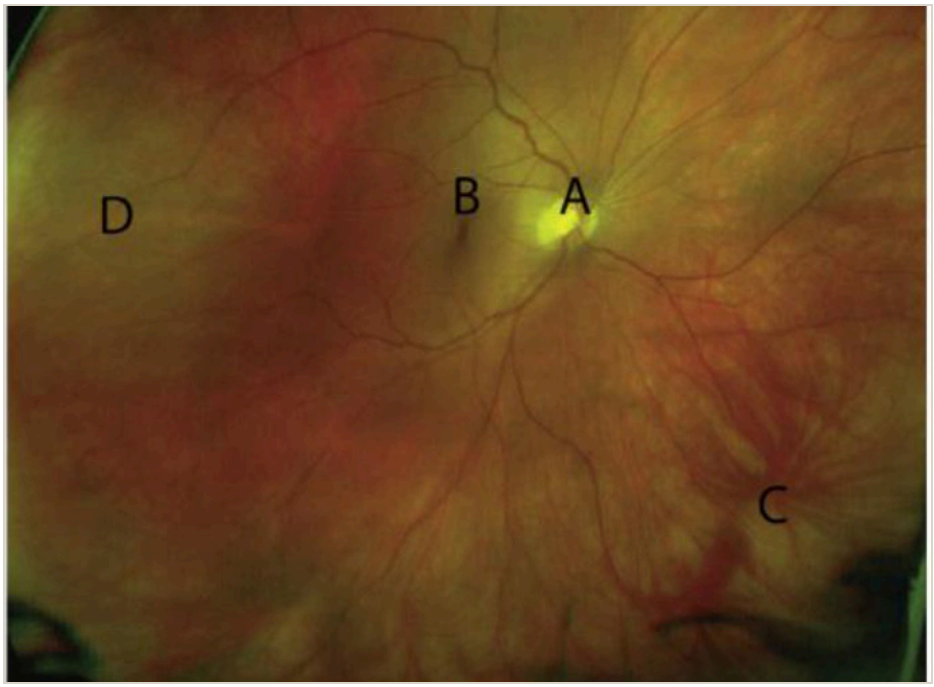


Fig. 8.20 Name A–D. *Answer:* A optic disc · B macula (both central retina) · C vortex vein · D ciliary nerve (both peripheral retina). § TB 8 · p. 128 · Slides 7 · #2

Overview of the retinal pathway

light passes through the inner layers → photoreceptor outer segment (1st neuron) → bipolar cell (2nd) → ganglion cell (3rd) → nerve fibre layer → **optic nerve → brain**

§ TB 8 · p. 111, 121. Light crosses almost the whole retina before it reaches the outer segments; the neural signal then runs back the opposite way. Horizontal and amacrine cells modify the signal sideways before it leaves the eye.

2 · Retinal pigment epithelium (RPE)

FEATURE	DETAIL
Layer	Outermost retinal layer, **one cell thick**, hexagonal pigmented cells; columnar at the posterior pole, **longer, narrower and more densely pigmented in the macula** (one reason the macula looks dark); cuboidal toward the ora serrata, where it continues as the pigmented ciliary epithelium. § Handout 7 · p. 2 · TB 8 · p. 111
Basal side	Faces the choroid; infolded; its basement membrane is part of **Bruch membrane**, so the RPE–choroid bond is strong. § TB 8 · p. 111
Apical side	**Microvilli** extend into the photoreceptor layer and envelop the outer segment tips (Fig. 8.2). **No junctions** join RPE and photoreceptors: only the subretinal (potential) space and the interphotoreceptor matrix hold them together. § Handout 7 · p. 2 · TB 8 · p. 111–112 · Slides 7 · #4
Between RPE cells	**Zonula occludens** + zonula adherens near the apex (terminal bars); desmosomes; gap junctions. § TB 8 · p. 111
Clinical	Retinal detachment separates the neural retina from the RPE, the weak, junction-free interface; subretinal fluid cuts the photoreceptors off from the choroid's nutrients. § TB 8 · p. 112

Functions of the RPE

FUNCTION	HOW	YOUR NOTES
Blood–retinal barrier	Zonula occludens join the RPE cells; the RPE selectively moves nutrients from the choriocapillaris into the retina (glucose transporters on both sides) and waste (lactate, water) back out. § Handout 7 · p. 3 · TB 8 · p. 121–122 Exam	zonula occludens: very tight junction · glucose into the retina, lactic acid + water out
Phagocytosis of photoreceptor discs	Shed outer-segment discs are engulfed; many lysosomes; up to 2000 discs per cell daily. Undigested remains build up as lipofuscin. § Handout 7 · p. 3 · TB 8 · p. 122	phagocytosis
Metabolises and stores vitamin A	Vitamin A is part of the photopigment; the RPE re-isomerises all-trans-retinol back to 11-cis-retinal for the rods. § Handout 7 · p. 3 · TB 8 · p. 122, 125	vitamin A restored in the RPE and taken up by the photoreceptors
Produces growth factors	Secretes **VEGF**, which keeps the choriocapillaris working; balanced by an antiangiogenic factor (PEDF), because too much VEGF causes neovascularisation. § Handout 7 · p. 3 · TB 8 · p. 122	growth factors help the vessels in the choroid

Absorbs light	Melanin granules absorb excess light and reduce scatter. § Handout 7 · p. 3 · TB 8 · p. 122	light absorbed by the pigment in the RPE
(textbook's sixth)	Helps form the interphotoreceptor matrix. § TB 8 · p. 122	

Your review notes count "5 main purposes", as the handout does. Your notes

3 · Photoreceptors: rods and cones

Both contain photopigments that are activated by light. What makes a cell a rod or a cone is less its shape than the **light level in which it works**. § Handout 7 · p. 4 · TB 8 · p. 111

	RODS	CONES
Stimulated in	**dim light**: scotopic vision; very sensitive, poor detail, no colour (shades of grey) Exam	**bright light**: photopic vision; sharp acuity and colour Exam
Photopigment	rhodopsin in free-floating discs	one of three: S (blue, 420 nm), M (green, 531 nm), L (red, 588 nm); ≥90% are L or M; S only 5–10%
Outer segment	long, uniform; discs separate from the plasma membrane; shed mostly early morning	shorter, often tapering; discs continuous with the membrane; shed at the end of the day
Inner segment, fibres	inner and outer segment same width; long thin outer fibre	wider ellipsoid with more mitochondria; short or absent outer fibre, so cone nuclei lie just inside the ELM
Synaptic terminal	**spherule** (round)	**pedicle** (broad, flat)
Number · where	80–110 million; none in the foveola; densest 20–25° from the fovea	4–5 million; concentrated in the macula, only cones in the foveal centre
Convergence	high: many rods → one ganglion cell (sensitivity to light and motion)	low: down to 1 cone : 1 ganglion cell in the fovea (detail)

§ TB 8 · p. 114–115, 121, 123–124, 128–129 · Handout 7 · p. 4. Your notes: rods → dim, cones → bright; cones 420 / 531 / 588 nm.

The six portions, outside in

PORTION	LIES IN LAYER	CONTAINS · FUNCTION	YOUR NOTES

1 Outer segment	photoreceptor layer, tip in RPE microvilli	stack of membranous discs (600–1000 per rod) holding the visual pigment: **converts light into a neural signal** § TB 8 · p. 113–114	membranous discs · rhodopsin
2 Cilium	photoreceptor layer	connecting stalk (9 pairs of tubules, no central pair) joining outer and inner segment; conduit for new disc material § TB 8 · p. 114	connecting stalk
3 Inner segment · ellipsoid	photoreceptor layer (outer part)	many **mitochondria**: energy § TB 8 · p. 114	outer → mitochondria
3 Inner segment · myoid	photoreceptor layer (inner part)	ER and Golgi: **protein synthesis** (not contractile in humans) § TB 8 · p. 114	closer to the body; ER and Golgi
4 Outer fibre	crosses the ELM	links inner segment to cell body § TB 8 · p. 114	
5 Cell body	outer nuclear layer	nucleus § TB 8 · p. 114	nucleus, DNA
6 Inner fibre	outer plexiform layer	axon ending in a spherule or pedicle that synapses with bipolar and horizontal cells; releases **glutamate** § Handout 7 · p. 5 · TB 8 · p. 114–115	axon → synapse

How it works: in the dark the photoreceptor is depolarised (≈ −40 mV) and releases glutamate continuously. Light activates the pigment → cGMP falls → Na^+ channels in the outer segment close → the cell hyperpolarises (≈ −75 mV) → **less glutamate**. § TB 8 · p. 124–125

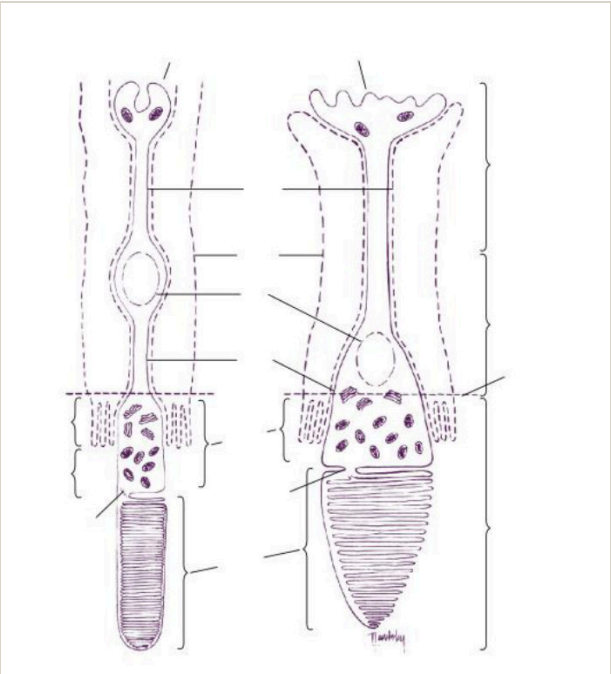
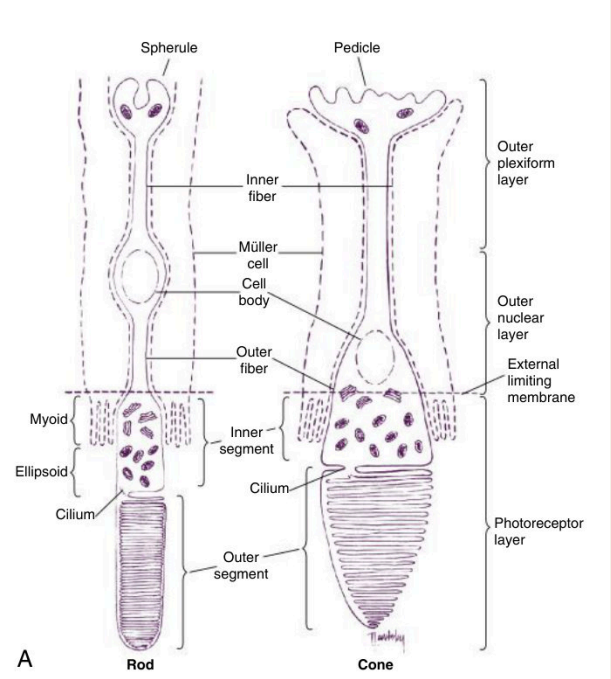


Fig. 8.4A, labels removed. Name the parts of the rod and cone and the layers on the right, then open the answer. § TB 8 · p. 114 · Slides 7 · #5

▼ Answer: Fig. 8.4A with labels



Top down: spherule (rod) · pedicle (cone) · inner fibre · Müller cell (dotted outline) · cell body · outer fibre · external limiting membrane · inner segment: myoid then ellipsoid · cilium · outer segment. Layers, right: outer plexiform · outer nuclear · external limiting membrane · photoreceptor layer.

4 · The other retinal cells

CELL	DENDRITE / INPUT SYNAPSES WITH	AXON / OUTPUT SYNAPSES WITH	FUNCTION · TRANSMITTER	YOUR NOTES
Bipolar (2nd neuron)	photoreceptors + horizontal cells	ganglion + amacrine cells	relays the signal photoreceptor → ganglion; ON and OFF types; glutamate § Handout 7 · p. 5 · TB 8 · p. 115, 125–126	second-order neuron · same synapses
Ganglion (3rd neuron)	bipolar + amacrine cells	brain: lateral geniculate nucleus and others (below)	first retinal cell to fire action potentials; axons form the nerve fibre layer and optic nerve; glutamate § Handout 7 · p. 6 · TB 8 · p. 117, 126	third order · bipolar + amacrine

Horizontal	photoreceptors, bipolar cells, other horizontal cells (gap junctions); all in the outer plexiform layer	**visual integration**: inhibitory feedback to photoreceptors, feed-forward to bipolars (centre–surround); **modulate cone responses**, not rods § Handout 7 · p. 7 · TB 8 · p. 118	photoreceptors, bipolar, horizontal
Amacrine	bipolar axons, ganglion dendrites and cell bodies, other amacrines; all in the inner plexiform layer	**modulate the information that reaches the ganglion cells**; mostly inhibitory (GABA, glycine); the AII amacrine carries the rod signal to ganglion cells § Handout 7 · p. 7 · TB 8 · p. 118, 124, 127	bipolar, ganglion, amacrine
Müller (glia)	not a signalling cell; spans from the photoreceptor layer (apex) to the inner surface (endfeet = ILM)	**structure**; absorbs K^+, neurotransmitters (glutamate, GABA) and metabolic waste; **metabolises, synthesises and stores glycogen**; wraps vessels and axons § Handout 7 · p. 7 · TB 8 · p. 118	retina wouldn't "work" without them · wraps all blood vessels

Two routes through the retina

cone → cone bipolar → **ganglion cell (3-neuron chain)**

rod → rod bipolar (always ON) → All amacrine → **ganglion cell (4-neuron chain)**

§ TB 8 · p. 123, 126–127. Rod bipolars rarely synapse on ganglion cells directly.

Ganglion cell types

TYPE	ENDS IN	CARRIES
P cells (P1 = midget)	**parvocellular** layers of the LGN	cone input, colour, fine detail, high contrast; concentrated centrally; 80% of ganglion cells § Handout 7 · p. 6 · TB 8 · p. 116, 123, 126
M cells (parasol)	**magnocellular** layers of the LGN	movement and flicker; large dendritic trees; poor wavelength sensitivity § Handout 7 · p. 6 · TB 8 · p. 116, 123, 126

Intrinsically photosensitive (ipRGC)	hypothalamus (suprachiasmatic nucleus), midbrain (olivary pretectal nucleus), LGN	contain **melanopsin**: respond to light without photoreceptors (peak ≈480 nm); non-image tasks: circadian rhythm / melatonin, pupillary light response; only 0.3–0.75% of ganglion cells § Handout 7 · p. 6 · TB 8 · p. 117

The textbook adds a fourth type, koniocellular (blue–yellow), which the handout does not list. § TB 8 · p. 116

Where ganglion axons synapse

TARGET	ROLE
Lateral geniculate nucleus (≈90%)	vision: parvo (detail, colour) and magno (motion) layers § TB 8 · p. 117, 123
Pretectal (olivary) nucleus, midbrain	pupillary reflexes § TB 8 · p. 117, 123
Suprachiasmatic nucleus, hypothalamus	circadian rhythm § TB 8 · p. 117, 123
Superior colliculus	turning the eyes, head and neck toward a target. The textbook says only "other centres" that drive eye, head and neck movements; the name is in your notes. § TB 8 · p. 123 Your notes

5 · The ten retinal layers

The ten "layers" are really one epithelium plus **three layers of cell bodies** (nuclear) with their **processes and synapses** (plexiform) between them. § Handout 7 · p. 8 · TB 8 · p. 111, 119

#	LAYER	STRUCTURES AND CONNECTIONS	VESSELS · CLINICAL
1	Retinal pigment epithelium (RPE)	one layer of pigmented cells (4–6 million; each serves 30–40 photoreceptors); zonula occludens § TB 8 · p. 119	outer blood–retinal barrier
2	Photoreceptor layer	outer and inner segments of rods and cones; Müller cell processes between the inner segments § TB 8 · p. 119	avascular
3	External limiting membrane (ELM)	not a true membrane: **zonula adherens** between photoreceptors and Müller cells at the inner segments; may restrict large molecules § TB 8 · p. 119	avascular
4	Outer nuclear layer (ONL)	**cell bodies of rods and cones** (cone nuclei in one row next to the ELM); thickest in the fovea (~10 rows of cone nuclei) § TB 8 · p. 119–120	avascular

5	Outer plexiform layer (OPL)	inner fibres of rods and cones; synapses of spherules and pedicles with **bipolar dendrites and horizontal cells** (cone triads); middle limiting membrane; Henle fibre layer at the fovea § TB 8 · p. 120, 129	fed by **both** retinal and choroidal circulation; the middle limiting membrane marks the edge of the retinal vessels
6	Inner nuclear layer (INL)	cell bodies of **horizontal, bipolar, amacrine and Müller cells** (and displaced ganglion cells) § TB 8 · p. 120	deep (and middle) capillary plexus at its outer and inner edges · **dot/blot haemorrhages**
7	Inner plexiform layer (IPL)	synapses of **bipolar axons with ganglion dendrites** (2nd → 3rd neuron) and with amacrine processes; amacrine–ganglion and amacrine–amacrine synapses § TB 8 · p. 120	
8	Ganglion cell layer (GCL)	ganglion cell bodies; usually one cell thick, 8–10 near the macula; displaced amacrines, glia § TB 8 · p. 120	superficial capillary plexus
9	Nerve fibre layer (NFL)	**ganglion cell axons** running parallel to the surface to the disc (unmyelinated); thickest at the disc margin; papillomacular bundle from the macula (chapter 12) § TB 8 · p. 120	superficial vessels; radial peripapillary capillaries · **flame haemorrhages**
10	Internal limiting membrane (ILM)	innermost boundary: **Müller cell endfeet** + basement membrane; vitreous fibres may fuse with it; astrocytes replace Müller cells at the disc § TB 8 · p. 120	**retinal sheen**

§ Handout 7 · p. 8–9 · Slides 7 · #3, #8, #13–14. Your notes match the table: photoreceptor layer = outer + inner segments, Müller cells · ONL thickest in fovea · INL: horizontal, bipolar, amacrine, Müller, deep vessels · GCL thickest around the fovea · NFL: superficial vessels, ganglion axons, thickest round the disc · ILM: innermost, Müller cells.

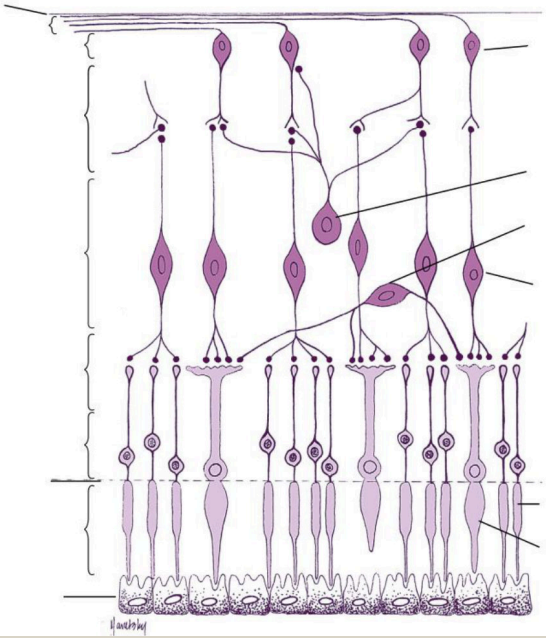
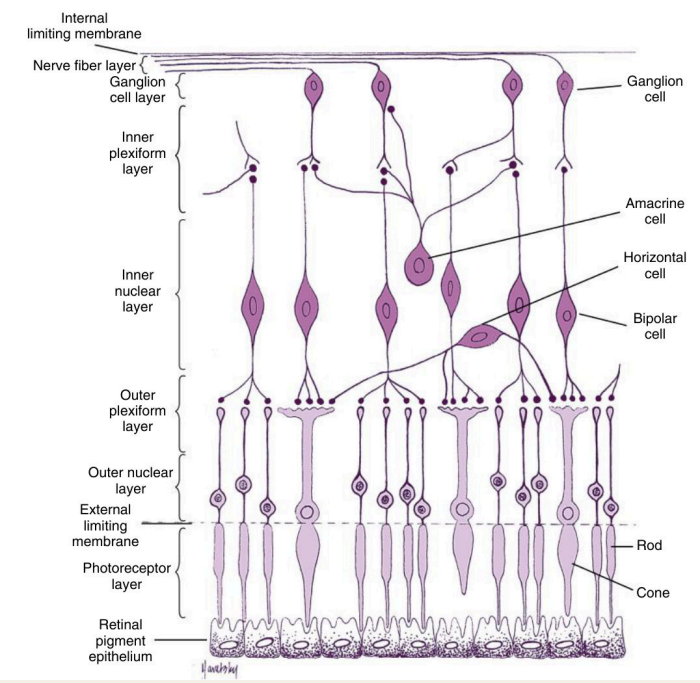


Fig. 8.8, labels removed. Name the ten layers (left) and the six cells (right), then open the answer. § TB 8 · p. 117 · Slides 7 · #6, #8

▼ Answer: Fig. 8.8 with labels



Left, top down: internal limiting membrane · nerve fibre layer · ganglion cell layer · inner plexiform · inner nuclear · outer plexiform · outer nuclear · external limiting membrane · photoreceptor layer · retinal pigment epithelium. Right: ganglion cell · amacrine cell · horizontal cell · bipolar cell · rod · cone.

CLINICAL CORRELATES · THE LAYERS ON THE FUNDUS

Retinal haemorrhages take the shape of the layer they bleed into. In the NFL, blood spreads along the parallel axons: a feathery **flame-shaped** haemorrhage (from the radial peripapillary or superficial plexus). In the INL, where cells are stacked vertically, it stays round: a **dot or blot** haemorrhage (middle or deep plexus). Both are seen in diabetic retinopathy (Fig. 8.32). § Handout 7 · p. 9 · TB 8 · p. 133–134 · Slides 7 · #11

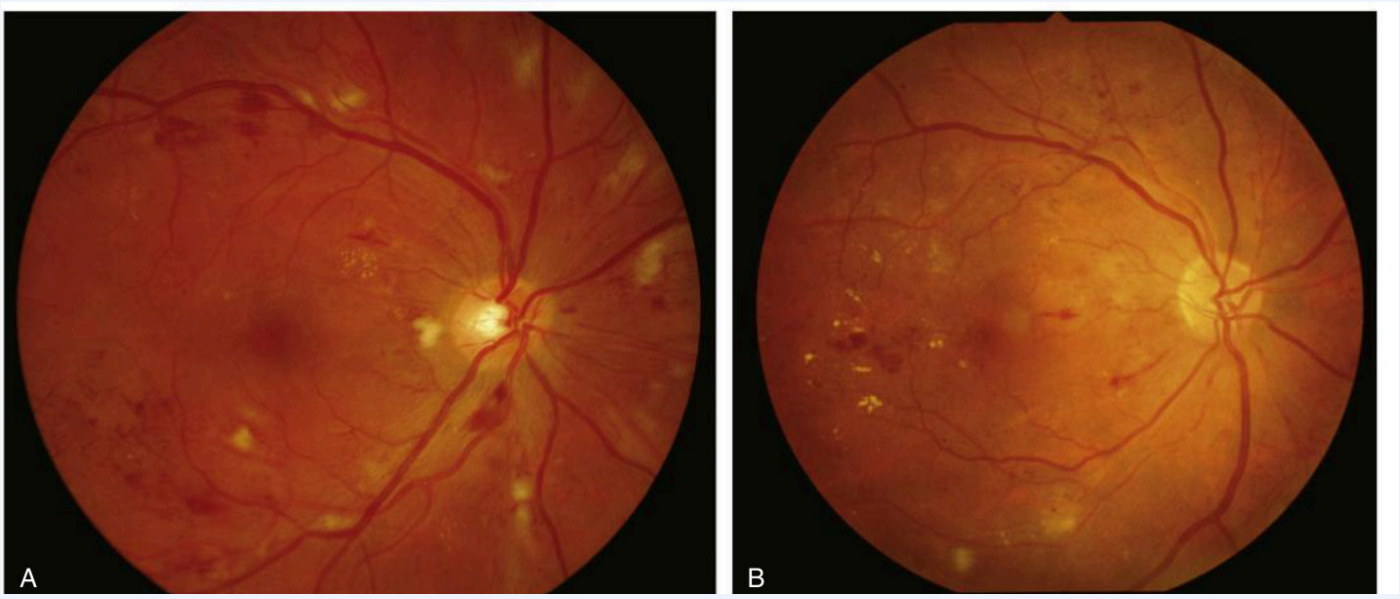


Fig. 8.32 Diabetic retinopathy: A flame-shaped, B dot and blot haemorrhages. § TB 8 · p. 134

Retinal sheen: reflections from the ILM, glistening round the macula and vessels in young people (Fig. 8.12), fading with age. § Handout 7 · p. 9 · TB 8 · p. 121 · Slides 7 · #12, #19

6 · The layers on OCT

GROUNDWORK FROM TEXTBOOK CHAPTER 1 · OCT

Optical coherence tomography uses light waves to take a non-invasive cross-sectional image of the eye's tissues: a 3-D map of the retina and optic nerve head that can measure the thickness of single retinal layers. **OCT angiography** detects moving blood and images the retinal and choroidal vessels without dye. § TB 1 · p. 4

Reading rule: layers of **axons and synapses are hyperreflective (bright)**; layers of **nuclei are hyporeflective (dark)**. The ILM is a thin line brighter than the NFL. § TB 8 · p. 121 Your notes: "synapses = lighter, nucleus = darker".

Your notes

HISTOLOGICAL LAYER	ON OCT (FIG. 8.13), VITREOUS AT THE TOP	BRIGHT / DARK
10 ILM	internal limiting membrane: first thin line under the vitreous	bright
9 NFL	nerve fibre layer, thicker toward the disc	bright

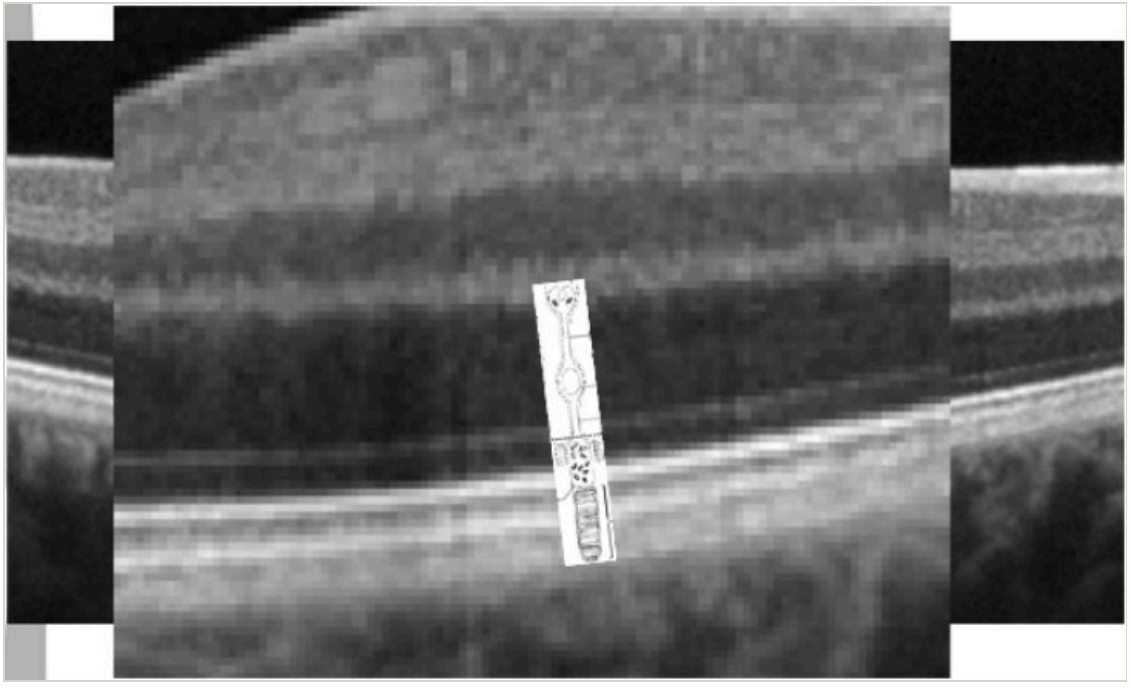
8 GCL	ganglion cell layer	dark
7 IPL	inner plexiform layer	bright
6 INL	inner nuclear layer	dark
5 OPL	outer plexiform layer	bright
4 ONL	outer nuclear layer: the thick dark band, thickest under the fovea	dark
3 ELM	external limiting membrane: the **dimmest** of the four outer lines	bright (faint)
2 Photoreceptor layer	**ellipsoid zone** (inner segment mitochondria) and **interdigitation zone** (outer segment tips meeting RPE microvilli)	bright, bright
1 RPE	**RPE/Bruch complex**: the most posterior bright band; choroid below	bright

§ TB 8 · p. 121 · Slides 7 · #15–16. The textbook says the exact histology of the outer bands is still somewhat uncertain.

Photoreceptor parts on OCT

ONL = photoreceptor cell bodies (dark) → ELM = zonula adherens at the inner segments (faint line) →
ellipsoid zone = inner segment mitochondria (bright) → interdigitation zone = outer segment tips (bright) →
RPE/Bruch complex (bright)

§ TB 8 · p. 121. Your review notes: "3 bright lines at the bottom: RPE, outer segment, inner segment (ellipsoid); dim line: external limiting membrane". At the fovea the elongated cone outer segments push the bands down: the **fovea externa** (TB 8 · p. 129).



A photoreceptor drawn over the outer OCT bands: cell body in the ONL, inner segment at the ellipsoid zone, outer segment down to the RPE. § Slides 7 · #16

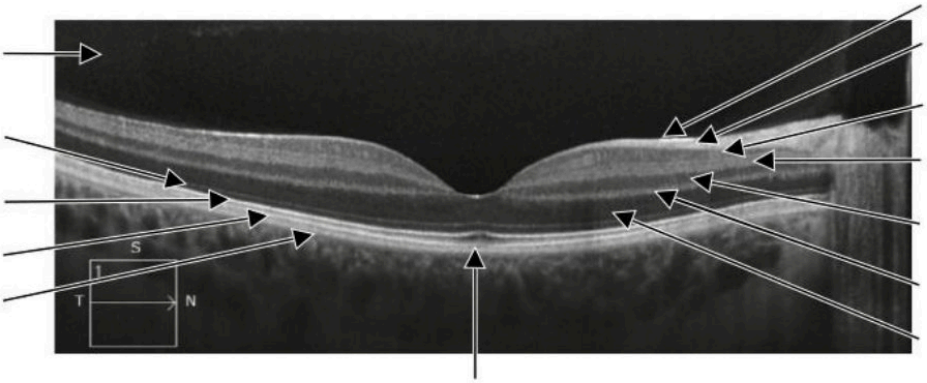
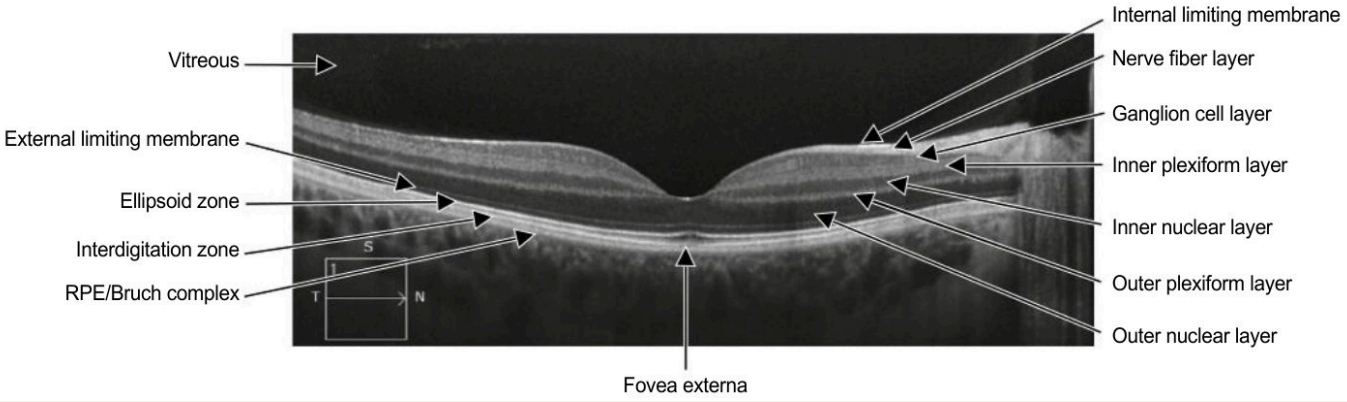


Fig. 8.13, labels removed. Name the 13 arrows, then open the answer. § TB 8 · p. 121 · Slides 7 · #15, #18 Exam

▼ Answer: Fig. 8.13 with labels



Left, top down: vitreous · external limiting membrane · ellipsoid zone · interdigitation zone · RPE/Bruch complex. Right, top down: internal limiting membrane · nerve fibre layer · ganglion cell layer · inner plexiform layer · inner nuclear layer · outer plexiform layer · outer nuclear layer. Bottom: fovea externa.

On the slides the same scan comes back with pathology: the vitreous pulling on the fovea (vitreomacular traction, "pulled away") and macular oedema (Fig. 8.27). § Slides 7 · #17–18 · TB 8 · p. 120–121, 129

7 · Retinal regions

REGION	FACTS
Macula (lutea)	Darker area of the central retina, ~5.5 mm across (includes fovea, parafovea, perifovea). Dark because its RPE cells are taller and more pigmented and the choroid / choroidal capillary bed is thicker under it. Yellow hue from **xanthophyll pigments, lutein and zeaxanthin**, most concentrated here; they filter short-wavelength light, reducing chromatic aberration, and may be antioxidant. § Handout 7 · p. 11 · TB 8 · p. 128

Fovea (centralis)	Shallow depression in the centre of the macula, ~1.5 mm across. Bipolar and ganglion cells displaced **laterally**; **rod-free**; **capillary-free**. Its wall is the clivus, its floor the foveola. § Handout 7 · p. 12 · TB 8 · p. 128–129
Ora serrata	Peripheral end of the retina, **~5 mm anterior to the equator**, scalloped (bays and dentate processes), ~2 mm wide; reaches further forward medially. **Firm retina–vitreous attachment** (the vitreous base). Neural retina continues as the nonpigmented, RPE as the pigmented ciliary epithelium. § Handout 7 · p. 12 · TB 8 · p. 130
Optic disc (optic nerve head)	Where ganglion axons gather and leave the eye; where the **central retinal artery enters** (just nasal of centre) and the **central retinal vein exits**. **1.7 mm horizontal × 1.9 mm vertical** (slightly tall). Only NFL + ILM: no photoreceptors, so it is the **physiologic blind spot**; paler because there is no RPE. The **physiologic cup** is a central depression that varies a lot in size and depth. § Handout 7 · p. 13 · TB 8 · p. 130, 132

Where the macula lies relative to the disc

- Textbook: the macula's centre is about **3.5 mm lateral (temporal) to the edge of the disc** and about **1 mm inferior to the centre of the disc**. § TB 8 · p. 128 Exam
- Handout: puts "1 mm inferior to the center of the disc" under the *fovea*. Same point, since the fovea is the macula's centre: know both wordings. § Handout 7 · p. 12
- "A little inferior and temporal, 17°" Your notes: the angle is not in the textbook chapter.

The fovea against the rest of the retina

	FOVEA / FOVEOLA	REST OF THE RETINA
Photoreceptors	**cones only** in the centre (rod-free zone 0.35–0.7 mm, ≈1° of field); highest cone density, 164,000–300,000 / mm²; thin, elongated, rod-like cone outer segments	rods outnumber cones; rods densest 20–25° out; both fall off toward the ora serrata; periphery rod-dominated
Blood vessels	**foveal avascular zone**, 0.4–0.7 mm: no retinal capillaries, so light reaches the photoreceptors unobstructed; fed from the choroid alone	retinal capillary plexuses in the inner layers; few vessels in the far periphery
Inner layers	INL and GCL pushed aside onto the foveal walls; cone axons run obliquely as the **Henle fibre layer**; foveola has only RPE, photoreceptors, ELM, ONL (~10 rows of cone nuclei), Henle fibres, ILM	all ten layers; parafovea has the most bipolar and ganglion cells (INL up to 12, GCL up to 9 cells thick)
Wiring	≈1 cone : 1 ganglion cell → detail and colour	high rod : ganglion ratio → sensitivity to light and motion, poor form

Thickness	foveola 0.13–0.23 mm	0.18 mm at the equator, 0.11 mm at the ora serrata

§ TB 8 · p. 121, 128–130 · Slides 7 · #20–22. Your notes: "only photoreceptors, no bipolar, ganglion or horizontal cells · not any blood vessels · squeezed photoreceptors" (on the OCT of Fig. 8.25).

> TERMINOLOGY: HISTOLOGIST VERSUS CLINICIAN
>
> What the histologist calls the **fovea** the clinician calls the **macula**; the histologist's **foveola** is the clinician's **fovea**. Clinically the macula is the darker area about the size of the disc, and the fovea its very centre. § TB 8 · p. 128

8 · Blood supply: retinal and choroidal

	INNER RETINA	OUTER RETINA (PHOTORECEPTORS)
Supplied by	**central retinal artery** (branch of the ophthalmic artery) § Handout 7 · p. 14 · TB 8 · p. 132 · TB 12 · p. 195	**choroid** (choriocapillaris); nutrients diffuse through Bruch membrane and the RPE § Handout 7 · p. 14 · TB 8 · p. 132
Layers	inner layers, ILM to INL; the OPL is fed by **both** § TB 8 · p. 132	everything **outer to the OPL is avascular** § Handout 7 · p. 14 · TB 8 · p. 132
Pattern	enters at the disc → superior and inferior branches → each nasal and temporal; nasal branches run straight to the ora, temporal ones **arch round the macula**; end in capillary arcades ~1 mm from the ora § TB 8 · p. 132–133	posterior choriocapillaris from the **short posterior ciliary arteries** (retina back to the equator); anterior choriocapillaris from recurrent branches of the **long posterior ciliary arteries** and the **anterior ciliary arteries** (retina from the equator forward) § TB 12 · p. 197 · TB 5 · p. 79
Vessel wall	**unfenestrated** endothelium with zonula occludens, pericytes; **end vessels** (no anastomoses); autoregulated, no autonomic control § TB 8 · p. 133	choriocapillaris **fenestrated**; choroid autonomically innervated § TB 8 · p. 133
Flow	≈60 mL/min/100 g	≈2000 mL/min/100 g: oxygen must diffuse to the inner segments § TB 8 · p. 128
Drains to	central retinal vein, exits at the disc § Handout 7 · p. 13	vortex veins (chapter 5 §8)

Three capillary networks of the retinal circulation

PLEXUS	LAYER	PATTERN

Radial peripapillary capillary plexus	nerve fibre layer	radial round the disc, parallel to the axons; densest superotemporal and inferotemporal, where the NFL is thickest
Superficial capillary plexus	ganglion cell layer	long vessels from the superior and inferior arcades
Deep capillary plexus	edges of the inner nuclear layer (textbook splits it into a middle plexus at the inner edge and a deep one at the outer edge)	lobular vortexes

§ Handout 7 · p. 14 · TB 8 · p. 132 · Slides 7 · #21, #25. A capillary-free zone 0.15 mm wide surrounds each retinal vessel (TB 8 · p. 133). Your notes put the superficial plexus in the GCL with "(AI check)": the textbook confirms it.

The two routes from the ophthalmic artery

ophthalmic artery → central retinal artery (in the optic nerve) → optic disc → retinal arteries and 3 capillary plexuses → **inner retina (ILM to OPL)**

ophthalmic artery → short posterior ciliary arteries (posterior) · long posterior + anterior ciliary arteries (anterior) → choriocapillaris → Bruch membrane + RPE → **outer retina (photoreceptors)**

§ TB 8 · p. 132 · TB 12 · p. 195, 197 · TB 5 · p. 79

Cilioretinal artery

Enters the retina from the **temporal edge of the disc** but comes from the **choroidal (ciliary) circulation**, not the central retinal artery; it supplies the macular area. Present in **15–20%** of people (handout and TB ch. 8); TB ch. 12 quotes studies giving 15–50%. § Handout 7 · p. 15 · TB 8 · p. 133 · TB 12 · p. 198 · Slides 7 · #23, #26

> CLINICAL CORRELATE · CENTRAL RETINAL ARTERY OCCLUSION
>
> When the central retinal artery is blocked the inner retina loses its supply. A cilioretinal artery, fed from the choroid, keeps the macula alive, so central vision can be spared. § Handout 7 · p. 15 · TB 8 · p. 133 · TB 12 · p. 198
>
> On slide 27 you circled the area between disc and macula and wrote "normal retina"; this is probably the zone a cilioretinal artery kept supplied. The slide has no caption, so check it with your teacher. Your notes § Slides 7 · #27

9 · Blood–retinal barrier

GROUNDWORK FROM TEXTBOOK CHAPTER 1 · TIGHT JUNCTIONS

Zonula occludens is a belt of tight junctions round the apex of each cell; rows of ridges seal the space between cells. A substance cannot pass *between* such cells; it must go *through* them. These junctions help form the blood–retinal and blood–aqueous barriers, and disease can break them. § TB 1 · p. 8

The barrier keeps plasma components that would block light out of the retina. § TB 8 · p. 133

Outer barrier (choroid side)

choriocapillaris: fenestrated, large molecules leak out → choroidal tissue → Bruch membrane: passes them easily → **RPE zonula occludens: stop**

Inner barrier (retinal vessels)

blood in retinal capillary → **unfenestrated endothelium sealed by zonula occludens: stop**

§ TB 8 · p. 133–134 · Handout 7 · p. 3. Supporting players: astrocytes round the retinal vessels are part of the barrier (TB 8 · p. 119); Müller cells wrap the vessels (TB 8 · p. 118). The ELM's zonula *adherens* is only a possible barrier to large molecules, not a tight seal (TB 8 · p. 119). Your review notes: "choroid vessels are fenestrated · RPE has tight junctions, zonula occludens".

ONLY IN YOUR NOTES · CHECK BEFORE YOU RELY ON THEM

- Ganglion axons synapse with the "optic radiation" Your notes: **not right**. Ganglion axons end in the lateral geniculate nucleus (and pretectal nucleus, hypothalamus, colliculus); the optic radiation is the next stage, leaving the LGN for the cortex (chapter 12). § TB 8 · p. 117
- Fig. 8.20: you labelled the vessel at C "short ciliary nerve" and the area at D "long ciliary nerve" Your notes. The textbook's caption says C is a **vortex vein** and D a ciliary nerve. § TB 8 · p. 128
- "2–80 photoreceptors for 1 horizontal cell" Your notes: the first number is hard to read and the textbook gives no such range for horizontal cells. Its numbers: an HIII horizontal cell contacts 9–12 cones in the macula and 20–25 in the periphery; a rod bipolar cell contacts 15–20 rods centrally and up to 80 peripherally. § TB 8 · p. 116, 118
- Cup-to-disc ratios "6/10" and "2/10" on the two discs of slide 23, "have to watch the cup is not getting bigger", "with glaucoma bigger cup" Your notes. The textbook: normal cups vary widely (Fig. 8.29); the cup enlarges with age, which must be considered when assessing the disc for glaucoma. § TB 8 · p. 132, 135 · Slides 7 · #23
- "A yellowish macula is normal, not like 'staining', that is not normal" Your notes: the textbook covers only the normal yellow pigment.
- Lamina cribrosa "little holes … just there, not good or bad" Your notes: the textbook says its openings can be seen through the nerve fibres in some people, especially with a deep cup (Fig. 8.28). § TB 8 · p. 130–131
- You wrote "(AI check)" beside three points; the textbook confirms all three: the myoid holds ER and Golgi for protein synthesis (TB 8 · p. 114); amacrine cells take bipolar input and pass it to ganglion cells (TB 8 · p. 118); the superficial plexus lies in the GCL (TB 8 · p. 132).
- "Kljukasta arterija" (hook-shaped artery) beside the cilioretinal artery Your notes: fits the textbook's picture of the vessel looping out of the temporal disc edge. § TB 12 · p. 198 (Fig. 12.7)

NOT TAUGHT FROM TEXTBOOK CHAPTER 8

Left out because neither the handout nor the guidelines ask for them: the RPE ion-transport model (Fig. 8.14), the many bipolar subtypes (midget, diffuse, blue, giant), horizontal subtypes HI–HIII, amacrine subtypes (A17, A18), the details of phototransduction, ON/OFF pathways and sublaminae, receptive fields, light and dark adaptation, retinal metabolism, the electroretinogram, inherited degenerations (retinitis pigmentosa, Stargardt), papilledema, parafoveal/perifoveal measurements, aging changes, Alzheimer disease and the reference list.

Key facts

10 layers: RPE · photoreceptor · ELM · ONL · OPL · INL · IPL · GCL · NFL · ILM (1 epithelium + 3 nuclear layers + plexiform layers between)
3-neuron chain: photoreceptor → bipolar → ganglion (rods: 4, via AII amacrine) · transmitter all three: glutamate
5 RPE functions: barrier · phagocytosis · vitamin A · growth factors (VEGF) · light absorption
6 photoreceptor portions: outer segment · cilium · inner segment (ellipsoid + myoid) · outer fibre · cell body · inner fibre
Cones **420 / 531 / 588 nm** (S / M / L) · ipRGC ≈480 nm · rods 80–110 million, cones 4–5 million
4 bright outer OCT bands: ELM (dimmest) · ellipsoid · interdigitation · RPE/Bruch
Macula centre **3.5 mm temporal** to the disc edge, **1 mm inferior** to the disc centre · macula ~5.5 mm · fovea ~1.5 mm · foveola ~0.35 mm · rod-free zone 0.35–0.7 mm · foveal avascular zone 0.4–0.7 mm
Disc **1.7 × 1.9 mm** (horizontal × vertical) · ora serrata **5 mm** anterior to the equator
3 retinal capillary networks: radial peripapillary (NFL) · superficial (GCL) · deep (INL) · avascular outer to the OPL
Cilioretinal artery: **15–20%**, temporal disc edge, choroidal origin · ≈**90%** of ganglion axons go to the LGN

Self-check

▼ List the functions of the RPE and give one structural fact behind each.

Blood–retinal barrier (zonula occludens between RPE cells; controls nutrient and waste transport) · phagocytosis of shed discs (lysosomes; lipofuscin builds up) · metabolises and stores vitamin A (re-forms 11-cis-retinal) · growth factors (VEGF keeps the choriocapillaris working) · absorbs light (melanin reduces scatter). § Handout 7 · p. 3 · TB 8 · p. 121–122

▼ Contrast rods and cones: light level, pigment, terminal, distribution and convergence.

Rods: dim light (scotopic), rhodopsin, spherule, absent from the foveola and densest 20–25° out, high convergence (sensitivity, motion). Cones: bright light (photopic), S/M/L pigments (420/531/588 nm), pedicle, only type in the foveal centre, down to 1:1 with ganglion cells (detail, colour). § TB 8 · p. 115, 121, 123

▼ Name the six portions of a photoreceptor from the RPE inward, with the layer each lies in and its job.

Outer segment (photoreceptor layer; discs with visual pigment, light → signal) · cilium (connecting stalk) · inner segment: ellipsoid (mitochondria) and myoid (ER, Golgi, protein synthesis) · outer fibre (across the ELM) · cell body (ONL, nucleus) · inner fibre (OPL; spherule/pedicle synapses with bipolar and horizontal cells, glutamate). § TB 8 · p. 113–114

▼ What are the three ganglion cell types in the handout, where does each project, and what does each carry?

P (midget) → parvocellular LGN: colour, fine detail, contrast. M (parasol) → magnocellular LGN: movement, flicker. ipRGC (melanopsin) → suprachiasmatic nucleus, olivary pretectal nucleus, LGN: circadian rhythm, pupillary light response. § Handout 7 · p. 6 · TB 8 · p. 116–117, 123

▼ With what does a ganglion cell's dendrite synapse, and where? Its axon?

Dendrite: bipolar axons and amacrine processes, in the inner plexiform layer. Axon: runs in the NFL, leaves as the optic nerve, ends mainly (~90%) in the LGN; the rest in the pretectal nucleus, suprachiasmatic nucleus and centres for eye and head movement. § TB 8 · p. 117, 120, 123

▼ Which cell bodies are in the ONL, the INL and the GCL? Which synapses are in the OPL and the IPL?

ONL: rods and cones. INL: horizontal, bipolar, amacrine, Müller. GCL: ganglion cells. OPL: photoreceptor terminals with bipolar dendrites and horizontal cells. IPL: bipolar axons with ganglion dendrites and amacrine processes. § TB 8 · p. 119–120

▼ What are the ELM and the ILM made of?

ELM: zonula adherens between photoreceptors and Müller cells at the inner segments (not a true membrane). ILM: Müller cell endfeet covered by a basement membrane (astrocytes at the disc). § TB 8 · p. 119–120

▼ On an OCT, which layers are bright and which dark, and what are the four bright outer bands from the RPE inward?

Axon and synapse layers (NFL, IPL, OPL) bright; nuclear layers (GCL, INL, ONL) dark; ILM a thin bright line. Outer bands: RPE/Bruch complex · interdigitation zone (outer segment tips) · ellipsoid zone (inner segment mitochondria) · ELM (dimmest). § TB 8 · p. 121

▼ Where is the macula relative to the optic disc?

Its centre is about 3.5 mm temporal (lateral) to the disc edge and about 1 mm inferior to the disc centre; the handout gives the 1 mm inferior figure for the fovea. § TB 8 · p. 128 · Handout 7 · p. 12

▼ How does the fovea differ from the rest of the retina in photoreceptors, inner cells and blood vessels?

Only cones in the centre (rod-free 0.35–0.7 mm), at the highest density; bipolar and ganglion cells displaced laterally (Henle fibres), leaving RPE, photoreceptors, ELM, ONL, Henle fibres and ILM; no retinal capillaries (foveal avascular zone 0.4–0.7 mm), so it depends on the choroid. Elsewhere rods dominate, all ten layers are present and retinal capillaries feed the inner layers. § TB 8 · p. 128–129

▼ Which vessels supply the inner and the outer retina, and where is the border? Name the three retinal capillary networks.

Inner: central retinal artery (from the ophthalmic artery) via the disc. Outer: choriocapillaris, from the short posterior ciliary arteries (posterior) and the long posterior + anterior ciliary arteries (anterior). Outer to the OPL is avascular; the OPL gets both; the middle limiting membrane marks the border. Networks: radial peripapillary (NFL), superficial (GCL), deep (INL). § Handout 7 · p. 14 · TB 8 · p. 132 · TB 12 · p. 197

▼ What is a cilioretinal artery and why does it matter in central retinal artery occlusion?

A vessel from the choroidal circulation entering the retina at the temporal disc edge (15–20%) and supplying the macula; if the central retinal artery is blocked it keeps the macula supplied. § Handout 7 · p. 15 · TB 8 · p. 133

▼ Explain the two parts of the blood–retinal barrier.

Outer: the fenestrated choriocapillaris leaks large molecules, which pass Bruch membrane, but the zonula occludens between RPE cells stop them entering the retina. Inner: retinal capillaries are unfenestrated and their endothelium is sealed by zonula occludens. Astrocytes round the vessels also contribute. § TB 8 · p. 119, 133–134 · TB 1 · p. 8

▼ A fundus shows a feathery haemorrhage near the disc and round spots elsewhere. In which layers is the blood?

Flame-shaped: nerve fibre layer (radial peripapillary or superficial plexus). Dot/blot: inner nuclear layer (middle or deep plexus). § Handout 7 · p. 9 · TB 8 · p. 133

§ Sources for this chapter: Handout 7 "Retina" (with your notes) · Slides 7 "Retina" (with your notes) · Exam guidelines (Review 14 · p. 3–4, with your typed notes) · Textbook ch. 8, Retina, pp. 111–135; ch. 12, pp. 195, 197–198 (central retinal artery, Fig. 12.5, cilioretinal artery); ch. 5, p. 79 (choroidal arteries); ch. 1, pp. 4 (OCT) and 8 (tight junctions).

The orbit: bones, walls and openings

Seven bones, four walls, and what passes through each hole between them.

WHAT THE EXAM ASKS

- Exam "Draw and label the bones that contribute to each of the orbital walls." Your note beside it: "pyramide zeichnen können" (be able to draw the pyramid). Your notes § Guidelines · p. 4
- Exam "Describe the landmarks associated with each orbital bone, as well as the structures that are associated with each landmark." § Guidelines · p. 4
- Exam "List the nerves and blood vessels that travel within each of the canals, foramen, and fissures of the orbit." § Guidelines · p. 4

Sections 2–3 answer the first, section 2's landmark table the second, section 4 the third.

1 · A four-sided pyramid of seven bones

- The orbit is a **four-sided pyramid**: the **base** is the anterior orbital margin, the **apex** is posterior, inside the skull. Four walls: **roof, floor, medial wall, lateral wall**. § TB 10 · p. 163
- The two **medial walls are roughly parallel**; the two **lateral walls**, extended backwards, meet at about **90°** (Fig. 10.6). § TB 10 · p. 163 · Slides 8 · #2
- Widest part about **1.5 cm** behind the margin ("pear shaped"). The **floor** reaches only about **two-thirds** of the depth; the other three walls reach the apex. § TB 10 · p. 163
- **Seven bones**: frontal, maxillary, zygomatic, sphenoid, ethmoid, palatine, lacrimal. The **frontal, sphenoid and ethmoid** are single bones shared by both orbits. § TB 10 · p. 163 · Handout 8 · p. 1

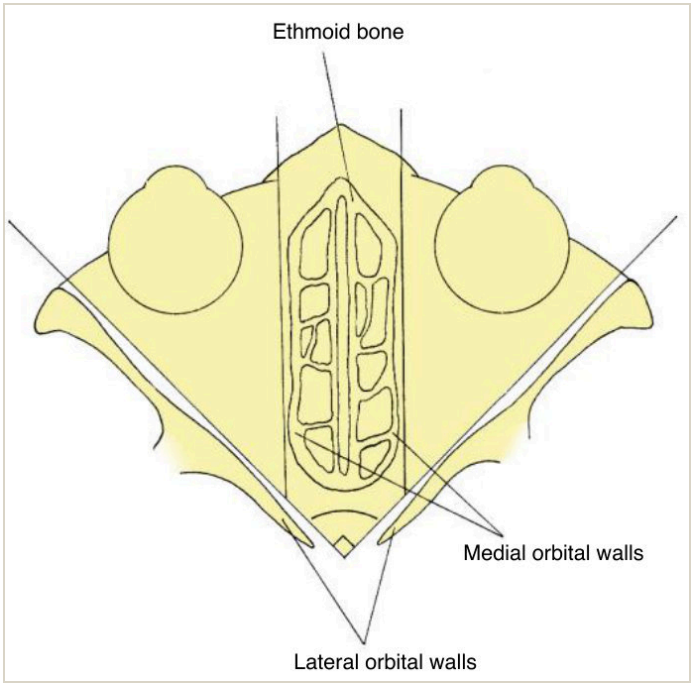


Fig. 10.6 Medial walls parallel, with the ethmoid between them; lateral walls at about a right angle. § TB 10 · p. 163 · Slides 8 · #2

The sphenoid: one bone, three orbital parts

PART	WHERE	IN WHICH WALL	OPENING IT MAKES
Body	midline, base of the cranium; holds the sphenoid sinus and the pituitary fossa (sella turcica)	medial wall, at the back	optic canal (with the lesser wing)
Lesser wing	from the front of the body; smaller and **higher** than the greater wing; joined to the body by roots (struts)	roof, at the back	optic canal (gap to the body) · superior orbital fissure (gap to the greater wing)
Greater wing	from the sides of the body	lateral wall, at the back	superior orbital fissure · foramen rotundum, ovale, spinosum lie in it

§ TB 10 · p. 159, 162, 168–169 · Fig. 10.3, 10.5 (Slides 8 · #3–4). Your note: the sphenoid "looks like a butterfly and the smaller one is the lesser wing". Your notes

How to draw it

The lecture drew the orbit as a pyramid seen from the front (your sketch, handout p. 1): Your notes

1. A large rectangle = the orbital margin (base). A small square in the middle = the apex.
2. Join the corners: four trapezoids. Top = roof, bottom = floor, outer side = lateral wall, nasal side = medial wall.
3. Fill in the bones from front (outer) to back (inner) with the table below; mark the openings: optic canal at the apex, superior orbital fissure between roof and lateral wall, inferior orbital fissure between floor and lateral wall.

§ Handout 8 · p. 1 · bones and openings from TB 10 · p. 163–166

2 · Wall by wall

WALL	SHAPE	BONES, FRONT → BACK	LANDMARKS	ON THE OTHER SIDE
Roof	triangle	**orbital plate of the frontal bone** (most) · **lesser wing of the sphenoid** (small, posterior)	fossa for the lacrimal gland · trochlea · supraorbital notch · supratrochlear notch	anterior cranial fossa (frontal lobe); the plate is thin here · frontal sinus in front
Floor	triangle	**orbital plate of the maxilla** (largest, medial) · **orbital plate of the zygomatic** (lateral) · **orbital process of the palatine** (small, at the very back)	inferior orbital fissure · infraorbital groove → canal → foramen · site of the blow-out fracture	maxillary sinus (its roof is the 0.5–1 mm orbital plate of the maxilla)
Lateral wall	roughly triangle	**zygomatic bone** (front) · **greater wing of the sphenoid** (back)	superior orbital fissure · inferior orbital fissure · zygomatic foramina · lateral orbital (Whitnall) tubercle	temporal fossa (zygomatic) · middle cranial fossa (greater wing)
Medial wall	rectangle	**frontal process of the maxilla** · **lacrimal bone** · **orbital plate of the ethmoid** (most) · **body of the sphenoid** (back)	fossa for the lacrimal sac (anterior + posterior lacrimal crests) · optic canal at the back · anterior and posterior ethmoidal canals (at the junction with the roof)	ethmoid air cells; the ethmoid plate is "paper thin" (lamina papyracea): the **thinnest wall**

§ TB 10 · p. 163, 165–166, 169 · Handout 8 · p. 3–6. Floor medial/lateral/posterior: your annotations, consistent with Fig. 10.7.

Where the walls meet: roof / lateral wall: superior orbital fissure (back), frontozygomatic and frontosphenoidal sutures (front) · floor / lateral wall: inferior orbital fissure · roof / medial wall: frontoethmoidal suture, with the ethmoidal canals. § TB 10 · p. 165–166

Handout vs textbook: the handout lists the optic canal under the medial wall's *body of the sphenoid*; the textbook describes it in the roof, between the lesser wing and the body. Both are right: it sits at the apex, where the two meet. The handout names the medial-wall bone "maxilla"; the textbook specifies its *frontal process*. § Handout 8 · p. 6 · TB 10 · p. 163, 165

Landmarks: which bone, what is there

LANDMARK	WALL · BONE	WHAT IT IS / WHAT IT HOLDS	YOUR NOTES
Fossa for the lacrimal gland	roof · frontal	indentation behind the **lateral** part of the superior margin; holds the lacrimal gland § TB 10 · p. 163	"deep on the bone"

Trochlea	roof · frontal	U-shaped **cartilage** (not bone) attached about 2 mm behind the superomedial margin; the tendon of the **superior oblique** runs through this pulley §TB 10 · p. 163	"super orbital muscle": the textbook says superior *oblique*
Supraorbital notch	roof · frontal (superior margin)	just medial to the centre of the superior margin; **supraorbital nerve and vessels**; closed into a foramen in 27–52% §TB 10 · p. 167	"supraorbital nerve + vessels"
Supratrochlear notch	roof · frontal (superomedial corner)	shallow groove; **supratrochlear nerve and vessels** §TB 10 · p. 167	"supratrochlear nerve + vessel"
Inferior orbital fissure	between floor and lateral wall	gap to the pterygopalatine and temporal fossae (section 4) §TB 10 · p. 163, 169	"separation between floor and lat. wall"
Infraorbital groove → canal → foramen	floor · maxilla	groove runs forward from the inferior orbital fissure, is roofed over by a thin plate to become the **canal** inside the maxilla, and opens on the face as the **foramen**, 6.3–8.8 mm below the inferior margin; **infraorbital nerve and vessels** §TB 10 · p. 163, 169	"the thing going from groove to canal to foramen (sensory nerve, upper cheek and lid)"
Superior orbital fissure	between roof and lateral wall (lesser / greater wing)	7–8 mm gap to the middle cranial fossa, widest medially; a small spur on its lower edge (lateral rectus spine) anchors the lateral rectus §TB 10 · p. 169	"between LW + GR, cranial nerve 3, 4, 5, 6"
Lateral orbital (Whitnall) tubercle	lateral wall · zygomatic	small bump 2–3 mm inside the lateral rim (in >60%); attachment for the levator aponeurosis, lateral canthal tendon, lateral check ligament, suspensory ligament of Lockwood §TB 10 · p. 166	

Fossa for the lacrimal sac	medial · frontal process of maxilla + lacrimal	between the **anterior lacrimal crest** (maxilla) and the **posterior lacrimal crest** (lacrimal bone); holds the lacrimal sac; continues down as the **nasolacrimal canal** §TB 10 · p. 165	lacrimal sac drawn on the medial wall
Optic canal	apex · lesser wing + body of sphenoid	oval opening at the apex, formed by the **optic strut** from the lesser wing to the body; to the middle cranial fossa §TB 10 · p. 163, 168	at the apex in your sketch

The orbital margin (the base of the pyramid)

MARGIN	FORMED BY	NOTE
Superior	frontal bone	supraorbital and supratrochlear notches
Lateral	zygomatic process of the frontal + frontal process of the zygomatic	most exposed, so the **strongest** part of the margin
Inferior	maxilla + zygomatic, about equally	zygomaticomaxillary suture can be felt through the skin
Medial	frontal process of the maxilla (anterior lacrimal crest) + lacrimal bone (posterior lacrimal crest)	not continuous: it runs as a spiral around the lacrimal sac fossa (Fig. 10.14)

§TB 10 · p. 163, 167. Your note on the lateral wall: "strongest one (trauma, hit)" matches the textbook's lateral margin.

3 · Label the bones

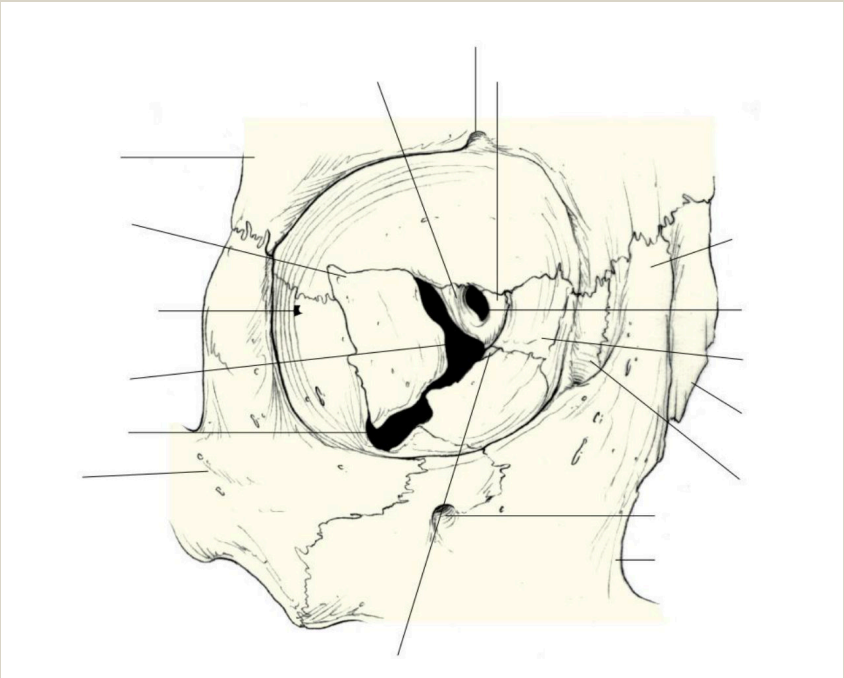
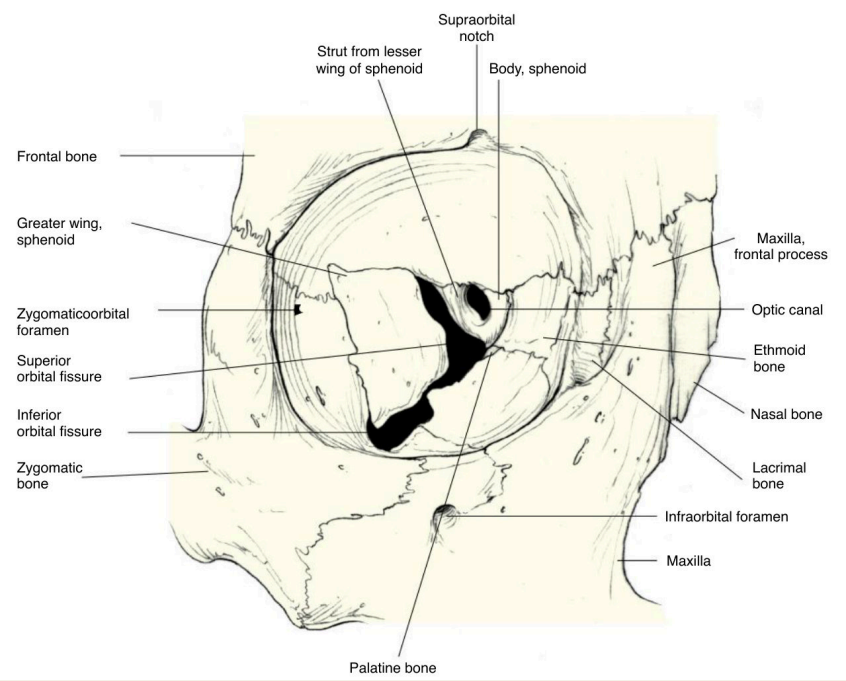


Fig. 10.7 Anterior view of the orbit, labels removed. Name all 17 lines, then open the answer. § TB 10 · p. 164 · Slides 8 · #5, #9 Label

▼ Answer: Fig. 10.7 with labels



Top: strut from lesser wing of sphenoid · supraorbital notch · body of sphenoid. Left, top to bottom: frontal bone · greater wing of sphenoid · zygomaticoorbital foramen · superior orbital fissure · inferior orbital fissure · zygomatic bone. Bottom: palatine bone. Right, top to bottom: frontal process of maxilla · optic canal · ethmoid bone · nasal bone · lacrimal bone · infraorbital foramen · maxilla.

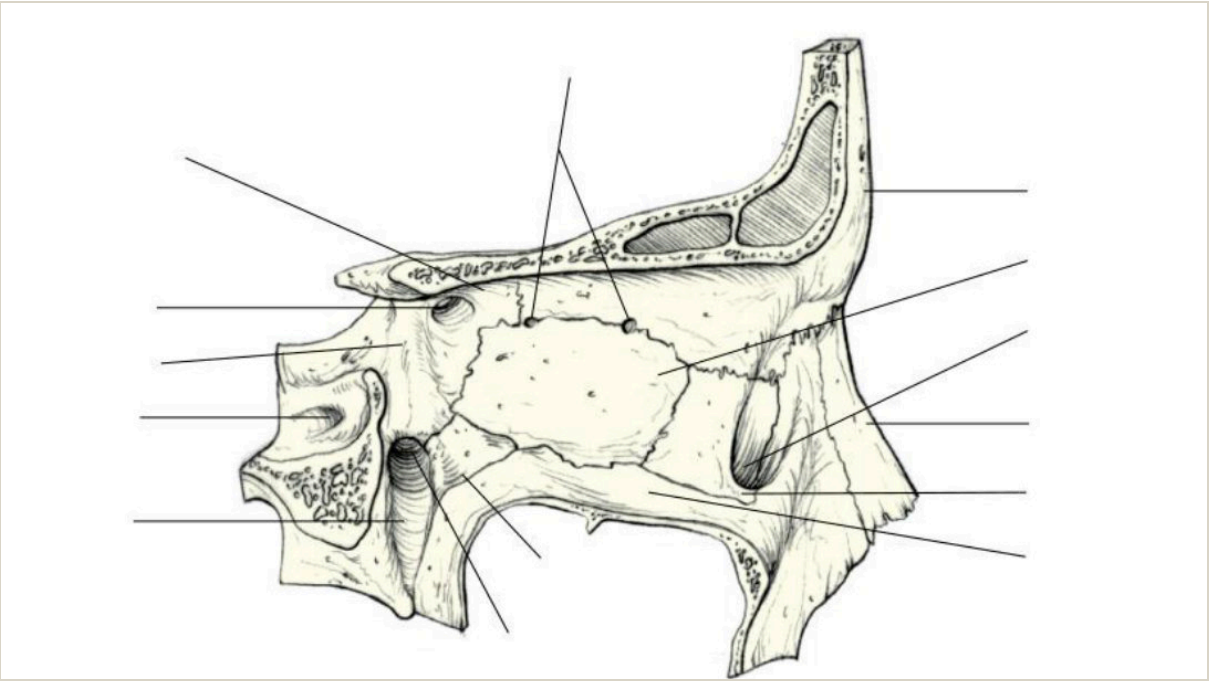
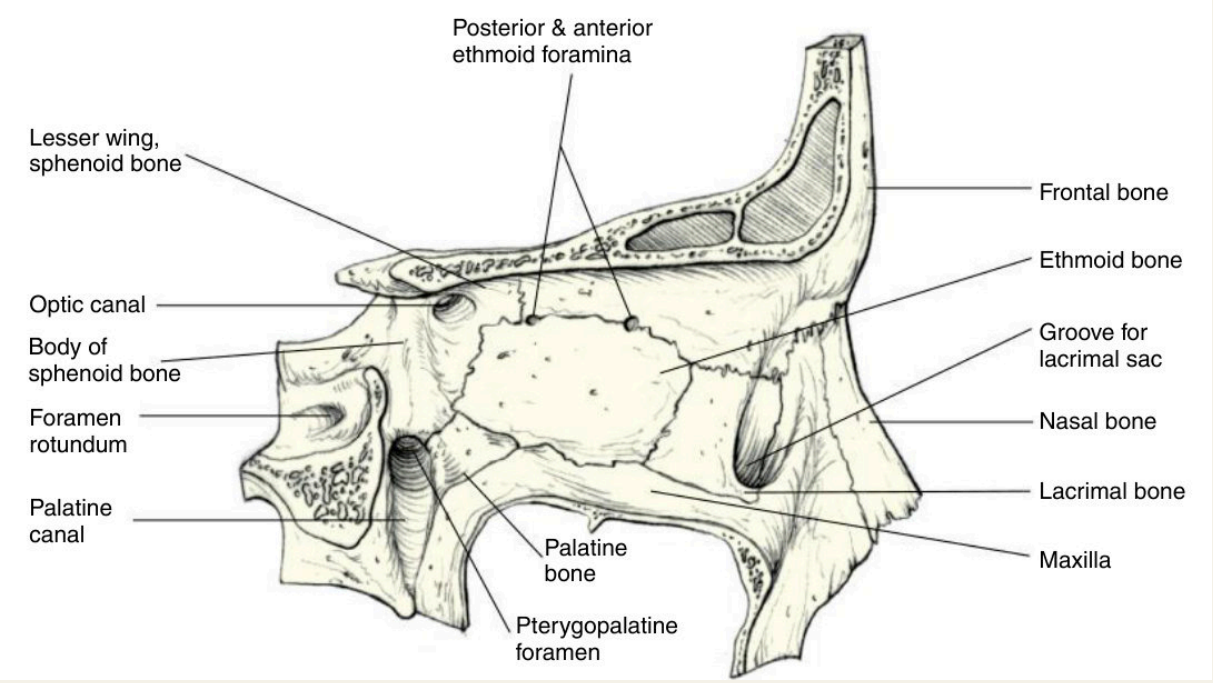


Fig. 10.11 Medial orbital wall (front is on the right), labels removed. Name all 13 labels, then open the answer. § TB 10 · p. 166 · Slides 8 · #8, #10 Label

▼ Answer: Fig. 10.11 with labels



Top: posterior and anterior ethmoid foramina (two lines). Left, top to bottom: lesser wing of sphenoid · optic canal · body of sphenoid · foramen rotundum · palatine canal. Bottom: palatine bone · pterygopalatine foramen. Right, top to bottom: frontal bone · ethmoid bone · groove for lacrimal sac · nasal bone · lacrimal bone · maxilla.

Read the medial wall front to back as the textbook lists it: maxilla (frontal process) → lacrimal → ethmoid → body of sphenoid. In your sketch these were the four bones of the medial wall ("4 bones"). § TB 10 · p. 165 · Handout 8 · p. 6

4 · What passes through each opening

OPENING	BETWEEN / IN	CONNECTS ORBIT WITH	NERVES	VESSELS
Optic canal	lesser wing + body of sphenoid (optic strut)	middle cranial fossa	**optic nerve** (exits)	**ophthalmic artery** (enter
Superior orbital fissure, *above* the tendinous ring	lesser + greater wing of sphenoid (roof / lateral wall)	middle cranial fossa	**lacrimal**, **frontal** (both V1), **trochlear** (CN IV)	**superior ophthalmic vein** (throug or above the ring)
Superior orbital fissure, *within* the ring	same	middle cranial fossa	**oculomotor** (CN III) superior + inferior divisions · **nasociliary** (V1) · **abducens** (CN VI)	inferior ophthalmic vein sometimes (more often drains into t superior one middle meningeal artery may enter to join the recurren meningeal branch of th lacrimal artery
Inferior orbital fissure	floor / lateral wall	pterygopalatine and temporal fossae	branches of the **maxillary nerve** (V2): **infraorbital** and **zygomatic** nerves	**infraorbital vessels** · a branch of th inferior ophthalmic vein may ex here
Infraorbital groove · canal · foramen	maxilla (floor → face)	face	**infraorbital nerve**	**infraorbital vessels**
Supraorbital notch (or foramen)	frontal, superior margin	forehead	supraorbital nerve	supraorbital vessels

Supratrochlear notch	frontal, superomedial corner	forehead	supratrochlear nerve	supratrochle vessels
Zygomaticofacial / zygomaticotemporal foramina	zygomatic bone	face, temporal fossa	"nerves and vessels" between orbit and face; the textbook names no more	
Anterior / posterior ethmoidal canals	frontoethmoidal suture (roof / medial wall)	ethmoid region	contents not named in textboo ch. 10	
Cranio-orbital (orbitomeningeal) foramen	above and lateral to the superior orbital fissure; in 46–55%	middle cranial fossa	none	where present, the middle meningeal / recurrent meningeal anastomosis runs here instead
Nasolacrimal canal	below the lacrimal sac fossa	nasal cavity (inferior meatus)	none	none: carries the **nasolacrima duct**

§ TB 10 · p. 163, 165–169 · Fig. 10.18. "V1" = ophthalmic division and "V2" = maxillary division of the trigeminal (CN V): the nerves themselves are chapter 10 (cranial nerves).

The common tendinous ring (annulus of Zinn) lies in front of the superior orbital fissure and the optic canal and is the origin of the **four recti**. Optic nerve + ophthalmic artery pass through the canal *and* the ring. Memory aid for the three nerves *above* the ring: **L-F-T** (lacrimal, frontal, trochlear). § TB 10 · p. 168–169

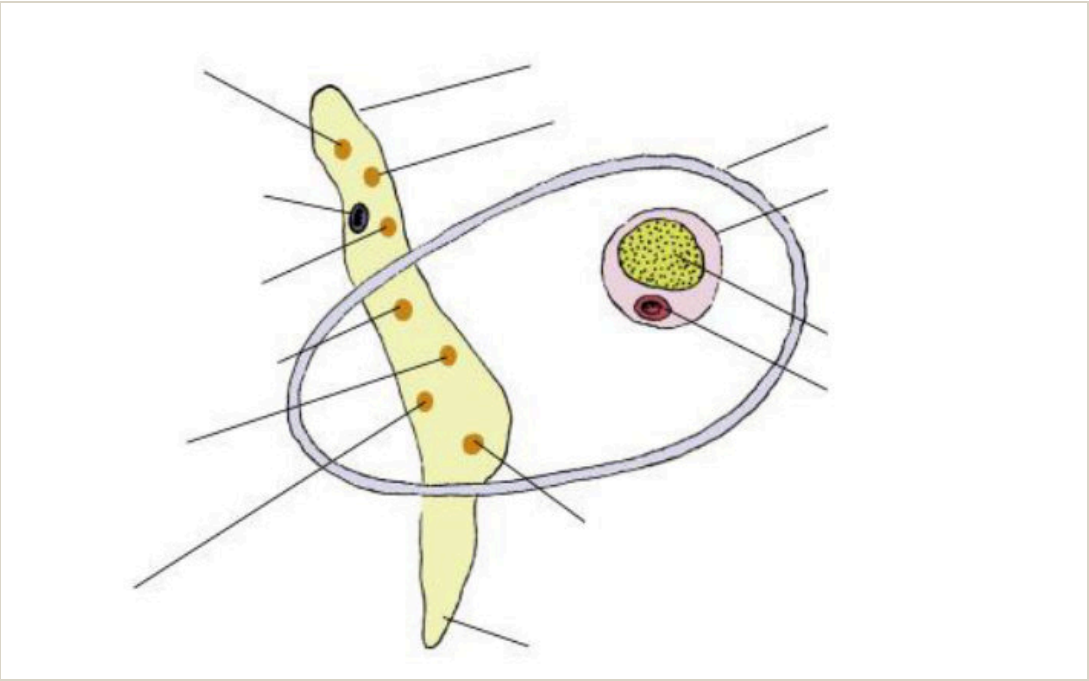
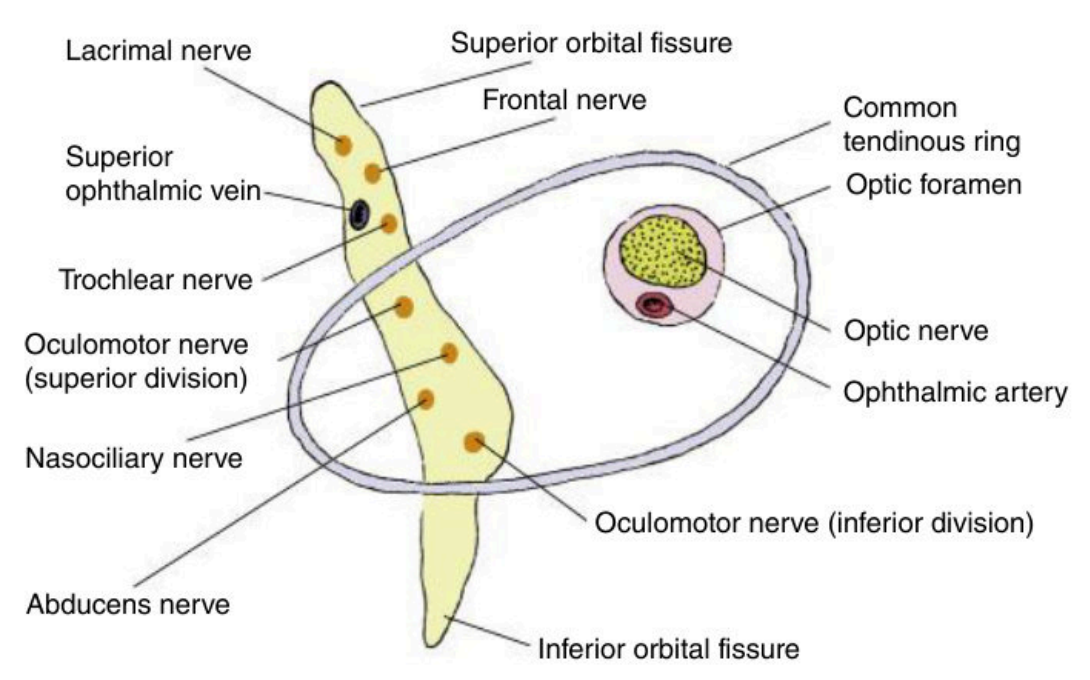


Fig. 10.18 The orbital apex: superior orbital fissure, common tendinous ring, optic foramen. Name all 14 lines, then open the answer. § TB 10 · p. 169 Label

▼ Answer: Fig. 10.18 with labels



In the fissure, top to bottom: lacrimal nerve · frontal nerve · superior ophthalmic vein · trochlear nerve (these above the ring) · oculomotor superior division · nasociliary nerve · abducens nerve · oculomotor inferior division (these within the ring). Also: superior orbital fissure · inferior orbital fissure · common tendinous ring · optic foramen · optic nerve · ophthalmic artery.

The infraorbital route

maxillary nerve (V2) → foramen rotundum → pterygopalatine fossa → inferior orbital fissure → infraorbital groove

→ infraorbital canal → infraorbital foramen → **face (cheek)**

§ TB 10 · p. 163, 169. The infraorbital and zygomatic nerves are branches of V2; they join the infraorbital vessels at the fissure.

Through the apex

middle cranial fossa → optic canal (inside the tendinous ring) → **orbit**

§ TB 10 · p. 168: the ophthalmic artery enters this way; the optic nerve travels the opposite direction, out of the orbit. The dura lining the canal sticks to the nerve's dura, so even a small lesion here can compress the nerve.

From the lacrimal sac fossa to the nose

fossa for the lacrimal sac → nasolacrimal canal (nasolacrimal duct) → **inferior meatus of the nose**

§ TB 10 · p. 165. The tear drainage itself is chapter 2.

5 · Blow-out fracture

CLINICAL CORRELATE · NAMED IN THE HANDOUT

- **How:** a blow to the orbital rim buckles the walls or suddenly raises the pressure inside the orbit. The **rim stays intact**; a wall breaks. § TB 10 · p. 163
- **Where:** the **floor**, in the thin bone along the **infraorbital canal**; the orbital contents can drop into the maxillary sinus below. § TB 10 · p. 163, 165 · Slides 8 · #7
- **Signs:** orbital swelling · ecchymosis (bruising) · **numbness** of the area of the infraorbital nerve (cheek) · **diplopia** (double vision) from restricted eye movement, especially in **upward gaze**. § TB 10 · p. 163, 165
- **Why the eye cannot move:** bruising or haematoma of the extraocular muscles, or the inferior muscles, fat and connective tissue **herniate and get trapped** in the fracture. § TB 10 · p. 163

§ Handout 8 · p. 4. Your notes: "Infraorbital canal is where the orbital bone breaks, if it breaks (trauma)" · "double vision".

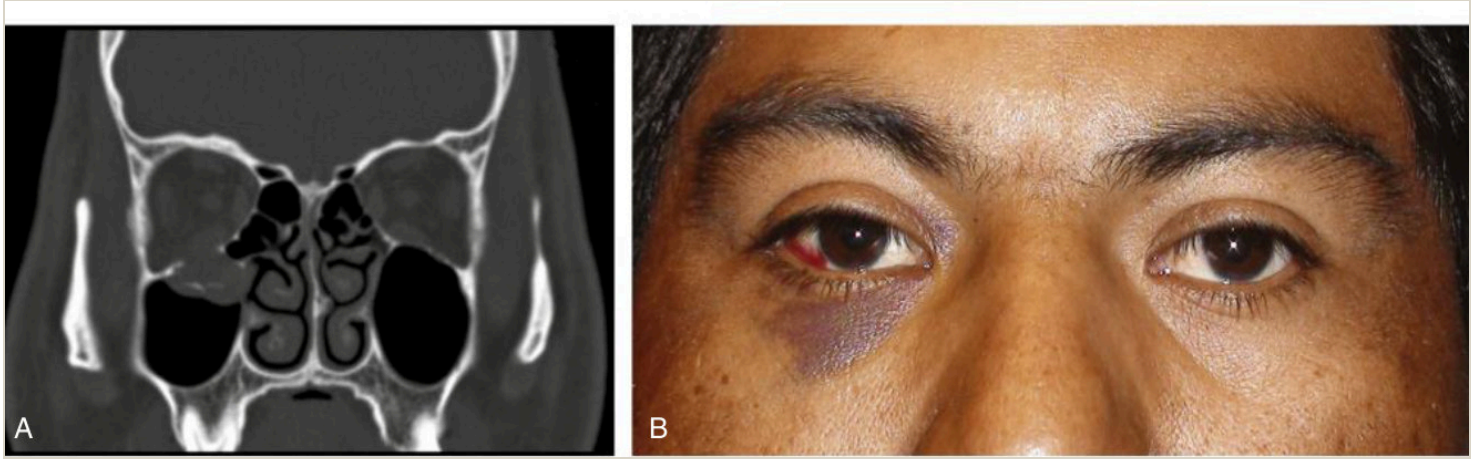


Fig. 10.10 Blow-out fracture of the right orbital floor. A: CT, orbital contents protruding into the right maxillary sinus. B: ecchymosis and subconjunctival haemorrhage; the patient had a numb right cheek and double vision. § TB 10 · p. 165 · Slides 8 · #7

ONLY IN YOUR NOTES · CHECK BEFORE YOU USE THEM

- Your typed note on the guidelines says the superior orbital fissure carries "C3, 4, 5 nur V1 und V2, 6" and "sympathetic fibres". The textbook: the fissure carries only **V1** branches (lacrimal, frontal, nasociliary); **V2** leaves through the foramen rotundum and reaches the orbit through the *inferior* fissure. Sympathetic fibres are not mentioned in textbook ch. 10: check them in chapter 10 (cranial nerves). Your notes § TB 10 · p. 162, 169
- "Optic canal: optic nerve, central ophthalmic artery": the textbook says the **ophthalmic artery**. Your notes § TB 10 · p. 168
- Infraorbital nerve to the "lower eyelid" / "upper cheek and lid": the textbook's chapter 10 only says the cheek goes numb when the nerve is hurt; check the lid in chapter 10 (cranial nerves, maxillary branches). Your notes
- "The thinnest bone of our orbita (not the …)" beside the ethmoid: the last word is unreadable. The textbook agrees the ethmoid (medial wall) is thinnest; the lacrimal is the smallest bone, "the size of a thumbnail". Your notes § TB 10 · p. 162, 165
- Two typed notes on the guidelines page are unclear: "inferior orbital fissure: same as ?" and "drying the flowsheet and add each of those bv (bild)". Ask whether the lecture wanted a drawn flow sheet of the vessels. Your notes

NOT TAUGHT FROM TEXTBOOK CHAPTER 10

Left out because lecture 8 and the exam guidelines do not cover them: the cranial bones and sutures in detail (parietal, occipital, temporal, inion, mandible, nasal bones, vomer, conchae), the foramina ovale and spinosum beyond naming them, orbital measurements beyond the key numbers, the paranasal sinuses beyond what borders each wall, the orbital connective tissue (periorbita, orbital septum, Tenon capsule, suspensory ligament of Lockwood, orbital muscle of Müller, septal system, orbital fat), the other clinical comments (inion for VEP, optic nerve damage, orbital and preseptal cellulitis, exophthalmos), aging changes and the reference list. Tenon capsule and the septum return with the eyelids and sclera (chapters 2–3).

Key facts

7 bones: frontal · maxillary · zygomatic · sphenoid · ethmoid · palatine · lacrimal (frontal, sphenoid, ethmoid single)
4 walls: roof 2 bones · floor 3 · lateral 2 · medial 4
90° between the two lateral walls · medial walls parallel
2/3: the floor reaches two-thirds of the depth · widest **1.5 cm** behind the margin
7–8 mm: superior orbital fissure · **6.3–8.8 mm**: infraorbital foramen below the inferior margin · **2 mm**: trochlea behind the superomedial margin
0.5–1 mm: orbital plate of the maxilla over the maxillary sinus
Thinnest wall: medial (lamina papyracea) · strongest margin: lateral · most fractured: floor

Self-check

▼ Name the bones of each orbital wall, front to back.

Roof: frontal (orbital plate), lesser wing of sphenoid. Floor: maxilla, zygomatic, palatine (orbital process). Lateral: zygomatic, greater wing of sphenoid. Medial: frontal process of maxilla, lacrimal, ethmoid (orbital plate), body of sphenoid. § TB 10 · p. 163, 165–166 · Handout 8 · p. 3–6

▼ Which three orbital bones are single, and which wall does not reach the apex?

Frontal, sphenoid, ethmoid. The floor, which reaches only about two-thirds of the depth. § TB 10 · p. 163

▼ Which bone holds the fossa for the lacrimal gland, and where exactly?

The frontal bone, behind the lateral part of the superior orbital margin (roof). § TB 10 · p. 163

▼ What is the trochlea made of, where is it, and what passes through it?

U-shaped cartilage on the frontal bone, about 2 mm behind the superomedial orbital margin; the tendon of the superior oblique muscle. § TB 10 · p. 163

▼ Which two bones form the fossa for the lacrimal sac, and which crests bound it?

Frontal process of the maxilla (anterior lacrimal crest) and the lacrimal bone (posterior lacrimal crest). Below, it continues as the nasolacrimal canal. § TB 10 · p. 165

▼ Between which bones do the optic canal, the superior and the inferior orbital fissures lie?

Optic canal: lesser wing and body of the sphenoid. Superior fissure: lesser and greater wings of the sphenoid (roof / lateral wall). Inferior fissure: floor and lateral wall. § TB 10 · p. 163, 168–169

▼ What passes through the optic canal?

The optic nerve (out) and the ophthalmic artery (in), both also through the common tendinous ring. § TB 10 · p. 168

▼ List what passes through the superior orbital fissure above the ring, and what within it.

Above: lacrimal, frontal, trochlear nerves; superior ophthalmic vein (through or above). Within: oculomotor superior and inferior divisions, nasociliary, abducens. § TB 10 · p. 169 · Fig. 10.18

▼ Trace the infraorbital nerve from the skull to the face. Which nerve is it a branch of?

Maxillary nerve (V2) → foramen rotundum → pterygopalatine fossa → inferior orbital fissure → infraorbital groove → canal → foramen (6.3–8.8 mm below the inferior margin). With the infraorbital vessels. § TB 10 · p. 163, 169

▼ What passes through the supraorbital and the supratrochlear notches?

The supraorbital nerve and vessels; the supratrochlear nerve and vessels. § TB 10 · p. 167

▼ A patient was hit in the eye: his cheek is numb and he sees double looking up. What broke, where, and why the symptoms?

A blow-out fracture of the orbital floor along the infraorbital canal, rim intact. Numb cheek: infraorbital nerve. Diplopia in upgaze: the inferior muscles, fat or connective tissue are trapped in the fracture, or the muscles are bruised. § TB 10 · p. 163, 165

▼ Which is the thinnest orbital wall, and which part of the margin is strongest?

The medial wall (lamina papyracea of the ethmoid). The lateral margin. § TB 10 · p. 165, 167

§ Sources for this chapter: Handout 8 "Orbital Bones" (with your notes) · Slides 8 "Orbital Bones" (with your notes) · Exam guidelines (Review 14, p. 4, with your typed notes) · Textbook ch. 10, Bones of the Skull and Orbit, pp. 159–169.

Extraocular muscles

Six muscles, three axes: where each muscle starts and ends, which nerve drives it, how it turns the eye, and which muscles work together.

WHAT THE EXAM ASKS

Two guideline topics, "Extraocular Muscles" and "Extraocular Muscles (cont.)": § Review handout 14 · p. 5

- Exam "Describe the location of the origin and insertion of each of the extraocular muscles." → sections 1–3
- Exam "List what nerves innervate each of the extraocular muscles" → section 3
- Exam "List the primary and secondary actions of each extraocular muscle." → section 5
- Exam "Describe muscle movements as the eye moves from a secondary position to a tertiary position, and use this to determine dysfunctional muscle." → sections 6–7
- Exam "Determine which extraocular muscles are agonist and antagonists when moving about each of Fick's axes." → section 8
- Exam "Determine the yoke muscles involved in each secondary and tertiary eye movement" → section 9

The handout also says **"Fick's axes DRAW"**: be able to draw the globe with its three axes (section 4). Your notes beside the objectives ask "which one is dysfunctional? X, Y, Z": expect cases. § Handout 9 · p. 7 · Review handout 14 · p. 5

1 · Six muscles and where the recti start

Four **recti** (medial MR, lateral LR, superior SR, inferior IR) and two **obliques** (superior SO, inferior IO). They are the extrinsic, voluntary muscles of the globe; the intrinsic ones (ciliary muscle, sphincter, dilator) are inside it. Recti from longest to shortest: SR, MR, LR, IR. § Handout 9 · p. 1 · TB 11 · p. 175, 179

Origin and insertion: the **origin** is the end fixed on bone, the **insertion** is the end on the globe that moves. Every movement is the insertion pulled toward the origin. § TB 11 · p. 183 · your notes, handout p. 1

STRUCTURE	WHAT IT IS	CLINICAL NOTE
Common tendinous ring (annulus of Zinn)	Oval band of connective tissue, continuous with the periorbita, at the orbital apex in front of the optic foramen and the medial part of the superior orbital fissure. Thickened upper and lower parts = upper and lower limbs (tendons). **Origin of all four recti.** § Handout 9 · p. 1 · TB 11 · p. 179	

Which rectus on which limb	MR and LR: both upper and lower limbs · SR: upper limb · IR: lower limb. § TB 11 · p. 179	
Dural sheath of the optic nerve	The origins of **MR and SR** also attach to it. § Handout 9 · p. 1 · TB 11 · p. 179	Why eye movement hurts in retrobulbar optic neuritis (box below)
Oculomotor foramen	The opening enclosed by the ring. Through it: optic nerve and ophthalmic artery (from the optic canal); superior and inferior divisions of CN III, CN VI and the nasociliary nerve (from the superior orbital fissure). These lie inside the **muscle cone** (the space the four recti enclose), so each rectus nerve enters the muscle on its cone-facing surface. § TB 11 · p. 180 · Slides 9 · #3 (Fig. 10.18)	CN IV, the frontal and lacrimal nerves and the superior ophthalmic vein pass **above** the ring, outside the cone. § TB 11 · p. 180

CLINICAL CORRELATE · RETROBULBAR OPTIC NEURITIS

Inflammation affecting the sheath of the optic nerve behind the globe. Usually **no visible fundus change** (slide 4 shows a normal-looking optic nerve), but **pain on extreme eye movement** can be an early sign. Why: the optic nerve sheath has a dense sensory nerve supply, and because the MR and SR origins are joined to it, moving the eye stretches the sheath. § Handout 9 · p. 1 · Slides 9 · #4 · TB 11 · p. 179

Your note: "when looking around and it is painful, it means the optic nerve is inflamed". § your notes, slides p. 2

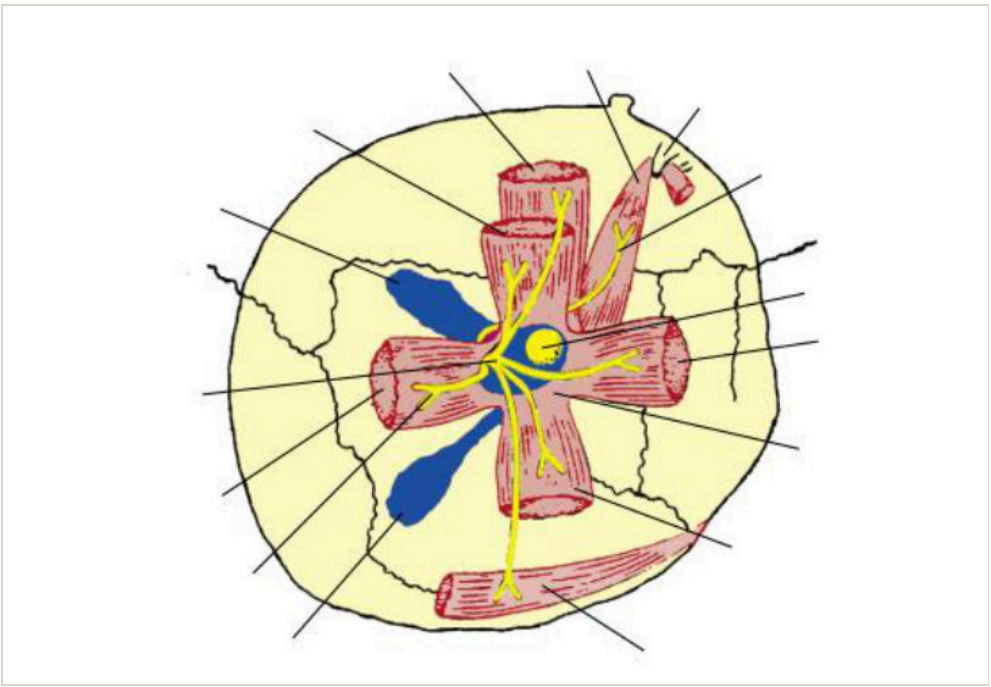
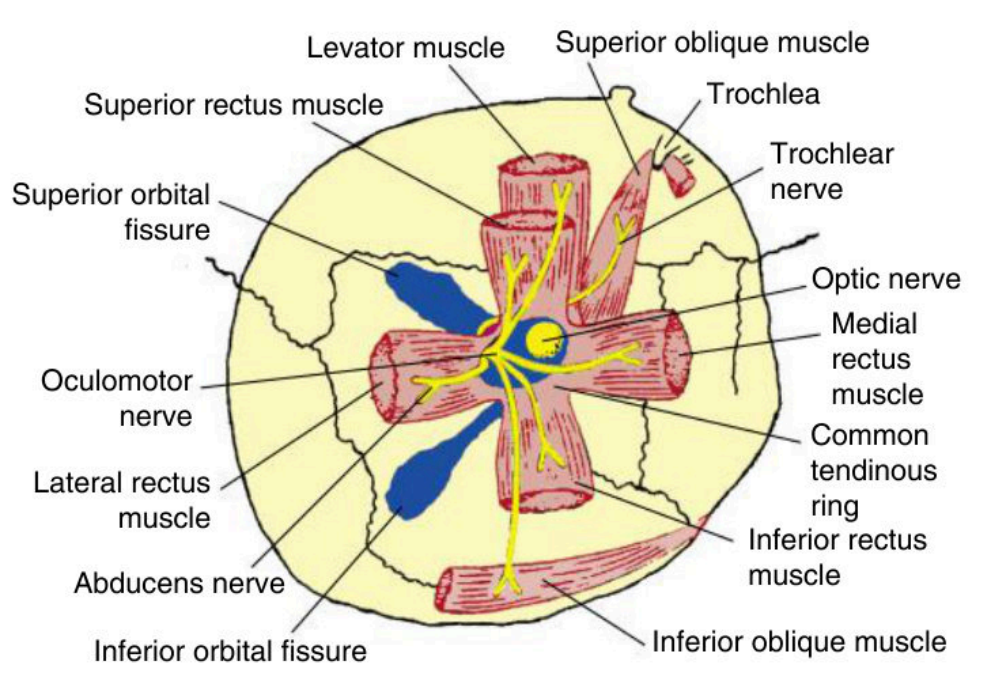


Fig. 11.11, labels removed: the orbital apex seen from the front, globe removed. Name the 15 lines (muscles, nerves, fissures, ring), then open the answer. § TB 11 · p. 180

▼ Answer: Fig. 11.11 with labels



Left side, top to bottom: superior rectus · superior orbital fissure · oculomotor nerve · lateral rectus · abducens nerve · inferior orbital fissure. Top: levator · superior oblique · trochlea · trochlear nerve (above the ring). Right side: optic nerve · medial rectus · common tendinous ring · inferior rectus · inferior oblique (the only muscle not near the apex).

2 · Where the recti insert

- All four recti insert on the **anterior globe**, in front of the equator. § Handout 9 · p. 2 · TB 11 · p. 180
- Their tendons **pierce Tenon capsule and merge with the scleral fibres**. A sleeve of the capsule covers the tendon for a short way, and the muscle slides freely inside it. § Handout 9 · p. 2 · TB 11 · p. 180
- **Spiral of Tillaux**: the line joining the four insertions is a spiral, starting closest to the limbus and moving away: **MR → IR → LR → SR**. § TB 11 · p. 180 · Fig. 11.12

RECTUS	INSERTION FROM LIMBUS	TENDON LENGTH	TENDON WIDTH	SHAPE OF INSERTION LINE
Medial	5.5 mm (5.2–5.7)	3.7 mm	10.3 mm	straight, vertical, bisected by the horizontal plane
Inferior	6.2 mm (6.0–6.3)	5.5 mm	9.8 mm	oblique, nasal end nearer the limbus
Lateral	6.6 mm (6.4–6.8)	8.8 mm	9.2 mm	parallel to the MR, straight or concave forward
Superior	7.2 mm (6.8–7.5)	5.8 mm	10.8 mm	oblique, nasal end nearer the limbus

§ TB 11 · p. 181–182, Table 11.1 (first number) and text (range in brackets). The rows are in Tillaux order.

3 · Muscle by muscle: the master table

The handout asks origin, insertion and innervation for each muscle, and a pathway for the two obliques. Cover the right-hand columns and recall them.

MUSCLE	ORIGIN	INSERTION	NERVE	PRIMARY ACTION	SECONDARY ACTIONS
Medial rectus	common tendinous ring (both limbs) + optic nerve sheath	anterior globe, medial	CN III, inferior division	adduction	none
Lateral rectus	common tendinous ring (both limbs) + spina recti lateralis on the greater wing of the sphenoid	anterior globe, lateral	CN VI (abducens)	abduction	none
Superior rectus	common tendinous ring (upper limb) + optic nerve sheath	superior, anterior globe	CN III, superior division	elevation	adduction, intorsion
Inferior rectus	common tendinous ring (lower limb)	inferior, anterior globe	CN III, inferior division	depression	adduction, extorsion
Superior oblique	anatomic: lesser wing of the sphenoid, medial to the optic canal · physiologic: **trochlea**	superior, posterior, lateral globe	CN IV (trochlear)	intorsion	depression, abduction
Inferior oblique	medial maxillary bone, at the front of the orbital floor	inferior, posterior, lateral globe	CN III, inferior division	extorsion	elevation, abduction

§ TB 11 · p. 184, Table 11.5 · TB 11 · p. 182, Table 11.2 · Handout 9 · p. 3–6. Shorthand for the nerves: **LR6 · SO4 · all the rest III**; of the four CN III muscles, only the SR takes the superior division. § TB 11 · p. 190

The details behind each row

MUSCLE	COURSE AND FEATURES	NERVE ENTERS ON	YOUR NOTES
MR	**Largest** EOM (probably from constant use in convergence). Runs along the medial orbital wall, through a connective tissue pulley just behind the equator, then follows the globe's curve. Its sheath forms the medial check ligament. § TB 11 · p. 181	lateral surface	"largest muscle (muscle we use the most)"

LR	Runs along the lateral orbital wall, pulley behind the equator. Longest rectus tendon (8.8 mm). Lacrimal artery and nerve on its upper border; ciliary ganglion, CN VI and ophthalmic artery between it and the optic nerve. Its sheath forms the lateral check ligament. § TB 11 · p. 181	medial surface	origin "CTR + greater wing of sphenoid"
SR	Runs forward under the levator along the orbital roof. The two sheaths are joined, so the **upper lid rises when you look up**. Muscle line makes **23°** with the sagittal (y) axis. § TB 11 · p. 181–182	inferior face (branches go on to the levator)	"CTR + optic nerve sheath"
IR	Runs along the orbital floor, parallel to the SR, also at **23°** to the sagittal axis. Its sheath and the IO's join in the suspensory ligament of Lockwood; the capsulopalpebral fascia goes to the lower tarsus, so the **lower lid drops when you look down**. § TB 11 · p. 182	superior surface	"anterior globe @ 23° angle towards nasal side"
SO	**Longest and thinnest** EOM, because of its 2.5 cm tendon. The tendon begins about 1 cm behind the trochlea (a U-shaped cartilage on the frontal bone), slides through it, turns back and passes under the SR to a fan-shaped insertion 12–18 mm from the limbus. The trochlea is the **effective origin**: the line from trochlea to insertion makes **55°** with the sagittal axis. § TB 11 · p. 182	posterior part (CN IV, from outside the cone)	"longest and thinnest muscle · trochlea · 55°"
IO	The **only EOM whose anatomic origin is in the anterior orbit**: maxillary bone, about 2 mm behind the inferomedial orbital rim, lateral to the nasolacrimal canal. Runs back and laterally below the IR. Shortest tendon (1 mm). Insertion posterior, lateral, mostly inferior, just outside the macula, about 15 mm from the limbus. Muscle line makes **51°** with the sagittal axis. § TB 11 · p. 182	upper surface	"51° · maxillary bone (stays still) · posterior lateral inferior globe"

The two oblique pathways

SO: lesser wing of sphenoid → forward along the upper medial orbit → trochlea (turns back) → under the SR → **superior posterior lateral globe**

IO: maxillary bone, front of orbital floor → backward and lateral, below the IR → **inferior posterior lateral globe**

§ Handout 9 · p. 5–6 · TB 11 · p. 182 · Fig. 11.9. Both obliques pull from the front-medial corner of the orbit toward the back of the globe; that is why they move the back of the eye and the front goes the other way.

4 · Fick's axes, ductions and positions of gaze

Every eye movement is a rotation about one or more of three axes that cross at the **centre of rotation**, about 13.5 mm behind the cornea (a little further back in myopia, further forward in hyperopia). The **anterior pole** is the

reference point: when the front moves up, the back moves down. § TB 11 · p. 182–183

AXIS	LIES	RUNS	ROTATION ABOUT IT (DUCTIONS)
X	horizontal (transverse)	nasal → temporal	**elevation** (supraduction) · **depression** (infraduction)
Y	horizontal (sagittal)	anterior pole → posterior pole	**intorsion** (incyclorotation): the 12 o'clock limbus point turns nasally · **extorsion** (excyclorotation): it turns temporally
Z	vertical	superior → inferior	**adduction** (anterior pole medially) · **abduction** (laterally)

§ Handout 9 · p. 7 · TB 11 · p. 183, Fig. 11.14, Fig. 11.15, Table 11.3. A duction moves one eye; movements of both eyes are versions (same direction) or vergences (opposite directions).

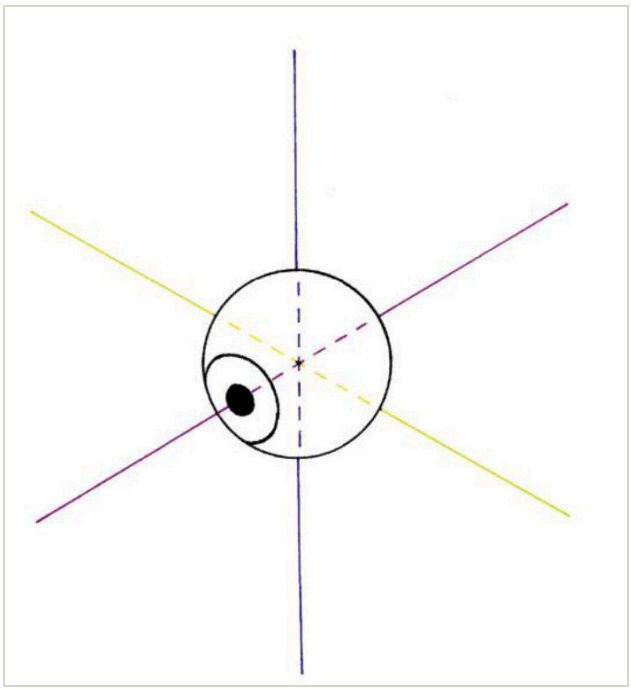
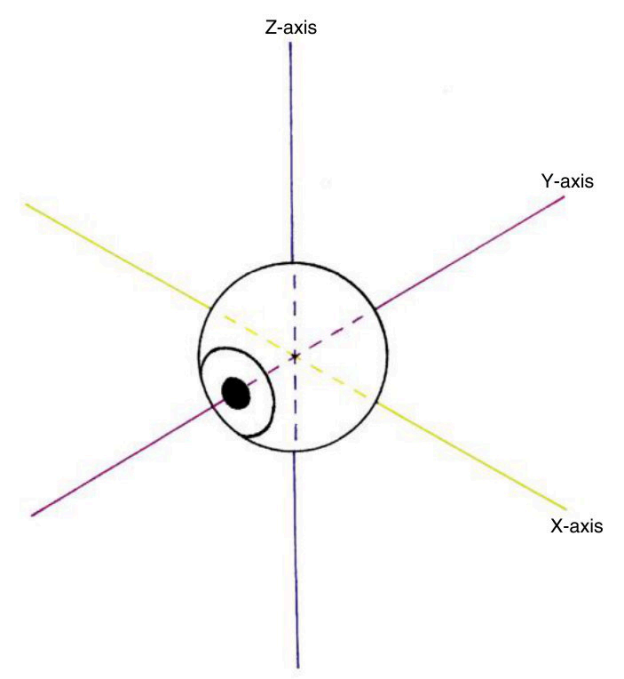


Fig. 11.14, labels removed. Name each axis and the two movements about it. Then draw it yourself from memory: the handout says DRAW. § TB 11 · p. 183 · Handout 9 · p. 7

▼ Answer: Fig. 11.14 with labels



Vertical line: Z-axis (adduction / abduction). Line through the cornea and the back of the eye: Y-axis (intorsion / extorsion). Side-to-side line: X-axis (elevation / depression).

Positions of gaze

POSITION	DEFINITION	YOUR NOTES
Primary	Head erect, eyes focused for infinity, looking at an object where the head's sagittal plane meets the horizontal plane through both centres of rotation.	"looking straight ahead, not accommodated"
Secondary	Rotation about **either** the vertical (z) **or** the horizontal (x) axis: straight right, left, up or down.	"vertical OR horizontal"
Tertiary	Rotation about **both** the vertical and the horizontal axis: the four oblique directions (up-right, up-left, down-right, down-left).	"vertical AND horizontal"

§ Handout 9 · p. 8 · TB 11 · p. 183

BOTH EYES MOVE	TERM	BOTH EYES MOVE	TERM
right	dextroversion	up and right	dextroelevation
left	levoversion	up and left	levoelevation
up	supraversion	down and right	dextrodepression
down	infraversion	down and left	levodepression

both adduct	convergence	both abduct	divergence

§ TB 11 · p. 183–184, Table 11.4

5 · Actions from primary position

Each action follows from where the insertion sits relative to the origin. The textbook warns this "one muscle contracts" model is hypothetical: all six muscles are always in some state of contraction. § TB 11 · p. 183–184

MUSCLE	ABOUT X (VERTICAL)	ABOUT Z (HORIZONTAL)	ABOUT Y (TORSION)	WHY
MR	—	**adduction**	—	lies parallel to the y-axis, perpendicular to x and z: one action only
LR	—	**abduction**	—	same, pulling temporally
SR	**elevation** (primary)	adduction	intorsion	insertion above the origin on the anterior globe → up · insertion lateral to the origin → in · oblique insertion on top of the globe → intorts
IR	**depression** (primary)	adduction	extorsion	insertion below the origin on the anterior globe → down · lateral to the origin → in · oblique insertion underneath → extorts
SO	depression	abduction	**intorsion** (primary)	insertion behind and below the trochlea: pulls the back of the eye up and medially, so the front goes *down* and *out*; the oblique pull on the top of the globe intorts
IO	elevation	abduction	**extorsion** (primary)	wraps under the globe; insertion behind and above the origin: pulls the back of the eye down and medially, so the front goes *up* and *out*

§ Handout 9 · p. 9–10 · TB 11 · p. 184–185 · Slides 9 · #5–15 (Figs. 11.15–11.18)

Memory hooks from the table: **the recti adduct** (SR, IR, MR), **the obliques abduct**. **Superior muscles intort** (SR, SO), **inferior muscles extort** (IR, IO). **Obliques do the opposite vertical job to their name**: SO depresses, IO elevates. § from TB 11 · p. 184, Table 11.5

Two views on the obliques. The textbook (and the lecture) call torsion the oblique muscles' primary action. Some authors call depression (SO) and elevation (IO) primary and torsion secondary. Know both; answer with torsion as primary. § TB 11 · p. 185

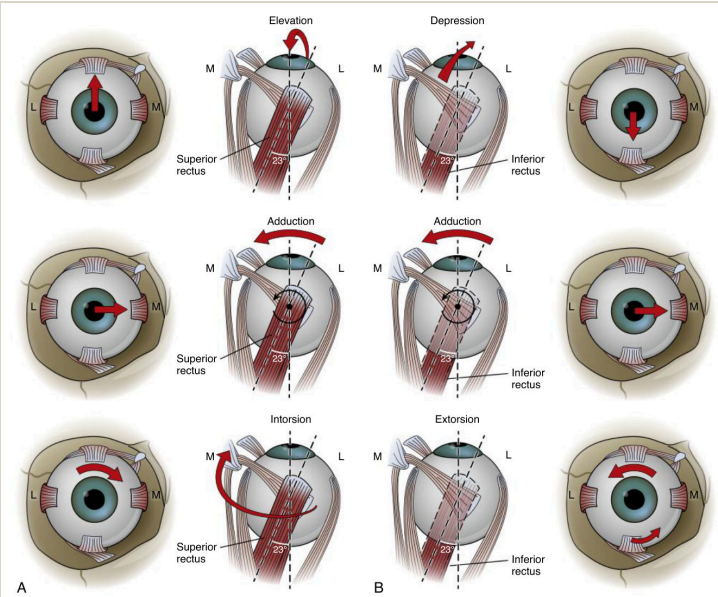


Fig. 11.17 Vertical recti from primary position. A, SR: elevation (x) · adduction (z) · intorsion (y). B, IR: depression · adduction · extorsion. Note the 23° angle to the sagittal axis. § TB 11 · p. 185 · Slides 9 · #5, 6, 12, 13

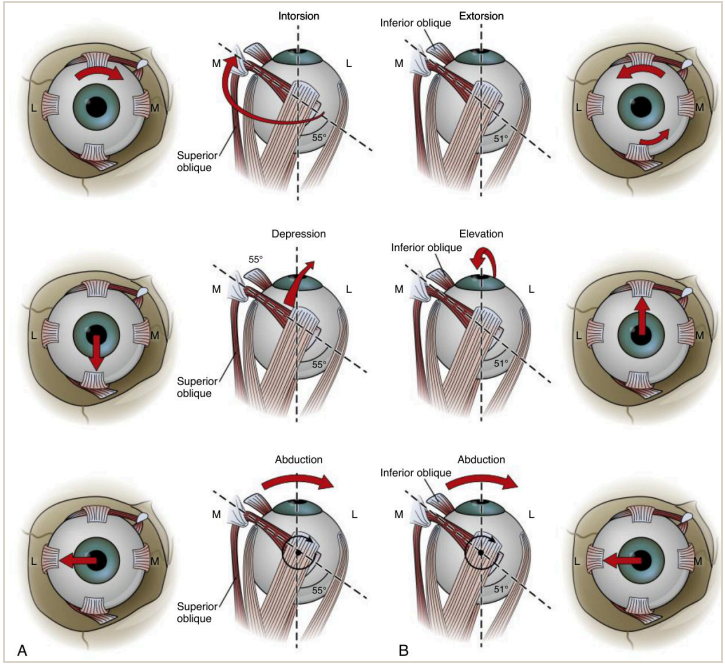


Fig. 11.18 Obliques from primary position. A, SO: intorsion (y) · depression (x) · abduction (z), line at 55°. B, IO: extorsion · elevation · abduction, line at 51°. § TB 11 · p. 186 · Slides 9 · #7, 8, 14, 15

6 · Actions from secondary positions

When the eye turns, the muscle's line of pull changes relative to Fick's axes. A muscle whose plane lies **parallel to the y-axis** acts purely vertically; one whose plane is **perpendicular to the y-axis** cannot act vertically at all. § TB 11 · p. 185–187 · Slides 9 · #16–17

MUSCLES	EYE ABDUCTED	EYE ADDUCTED
Vertical recti (SR, IR)	At about **23° abduction** their plane is parallel to the y-axis: **SR only elevates, IR only depresses**, no torsion. § Handout 9 · p. 11 · TB 11 · p. 185 · Fig. 11.19A	At about **67° adduction** (probably beyond what the orbit allows) their plane is perpendicular to the y-axis: SR and IR **cannot elevate or depress**. What remains is torsion. § TB 11 · p. 187 · Fig. 11.19B Your drawing beside it reads "torsional". Your notes
Obliques (SO, IO)	At **35–39° abduction** their plane is perpendicular to the y-axis: the obliques **cannot elevate or depress**. § TB 11 · p. 187 · Fig. 11.20B You wrote: SO → intorsion, IO → extorsion. Your notes	At **51–55° adduction** their plane is parallel to the y-axis: **SO only depresses, IO only elevates**. § Handout 9 · p. 12 · TB 11 · p. 187 · Fig. 11.20A
Horizontal recti (MR, LR)	With the eye elevated they also elevate slightly; with the eye depressed they depress it further. § TB 11 · p. 185	

The rule the whole clinical test rests on: the more the eye **abducts**, the more the **vertical recti** elevate and depress (and the less the obliques do); the more it **adducts**, the more the **obliques** elevate and depress. § TB 11 · p.

From a secondary to a tertiary position

Start in right or left gaze (secondary, about the z-axis), then look up or down (adding the x-axis): the eye is now in a tertiary position. The abducted eye now moves vertically with its **vertical rectus**, the adducted eye with its **oblique**. § TB 11 · p. 188

SECONDARY POSITION	THEN	TERTIARY POSITION	RIGHT EYE (MUSCLE TESTED)	LEFT EYE (MUSCLE TESTED)
Right gaze	up	up and right	abducted → **R SR**	adducted → **L IO**
Right gaze	down	down and right	abducted → **R IR**	adducted → **L SO**
Left gaze	up	up and left	adducted → **R IO**	abducted → **L SR**
Left gaze	down	down and left	adducted → **R SO**	abducted → **L IR**

§ TB 11 · p. 188 (ocular motility testing, steps 3–4)

7 · Finding the dysfunctional muscle

CLINICAL CORRELATE · EXTRAOCULAR MUSCLE ASSESSMENT

1. Look at primary position first. If one muscle is inactive, the eye is pulled in the direction of its *antagonist*. An eye turned toward the nose → underactive LR, unopposed MR. A paralysed MR → eye turned temporally. § TB 11 · p. 188

2. Then test motility with a small target (a bead), putting the eye where one muscle is the main actor:

follow the bead → far right and far left: horizontal → in left gaze: up, then down → **in right gaze: up, then down**

THE EYE CANNOT…	POSITION TESTED	DYSFUNCTIONAL MUSCLE
abduct	horizontal	**LR**
adduct	horizontal	**MR**
elevate	adducted	**IO**
depress	adducted	**SO**
elevate	abducted	**SR**
depress	abducted	**IR**

§ Handout 9 · p. 13 · TB 11 · p. 188, Fig. 11.22 · your notes on handout p. 13 match this table exactly

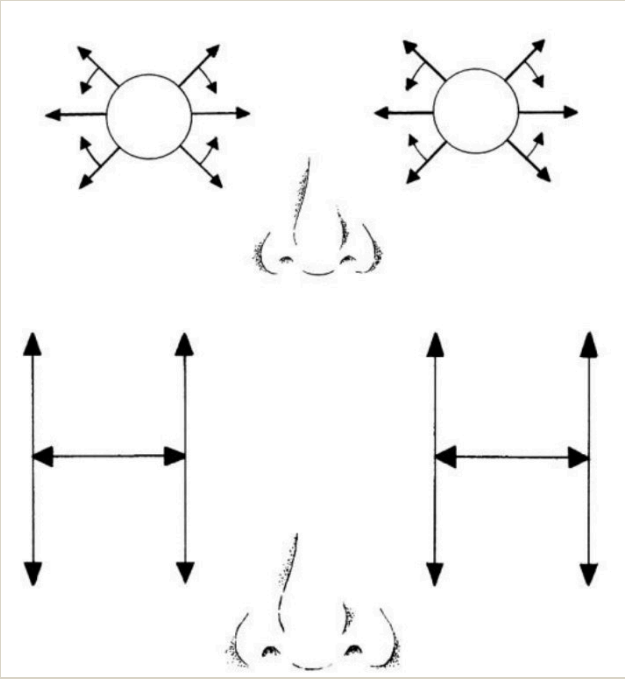
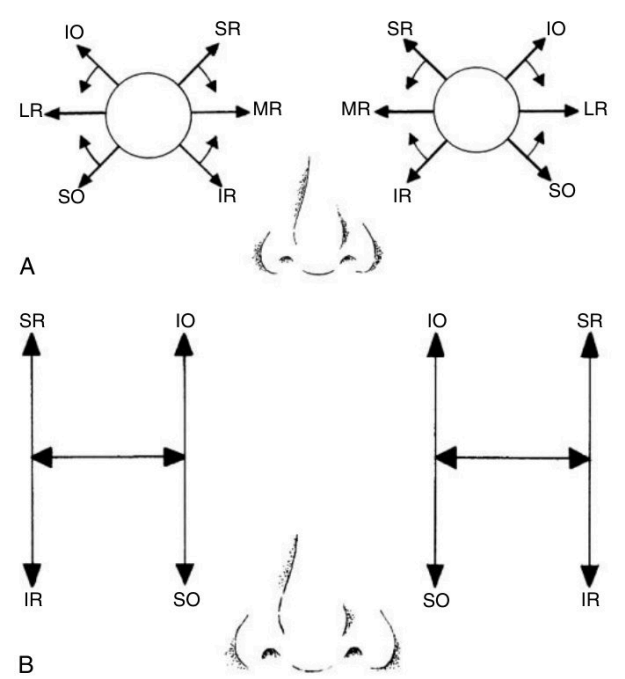


Fig. 11.22, labels removed. You face the patient: the left eye in the drawing is the patient's right eye. A: name the muscle at the end of each arrow (direction of pull from primary position). B: name the muscle at each end of the two "H"s. § TB 11 · p. 188

▼ Answer: Fig. 11.22 with labels



A, patient's right eye: IO up-out, SR up-in, LR out, MR in, SO down-out, IR down-in (curved arrows = torsion). The left eye mirrors it. B, the "H" for each eye: temporal side (abducted) SR above, IR below; nasal side (adducted) IO above, SO below.

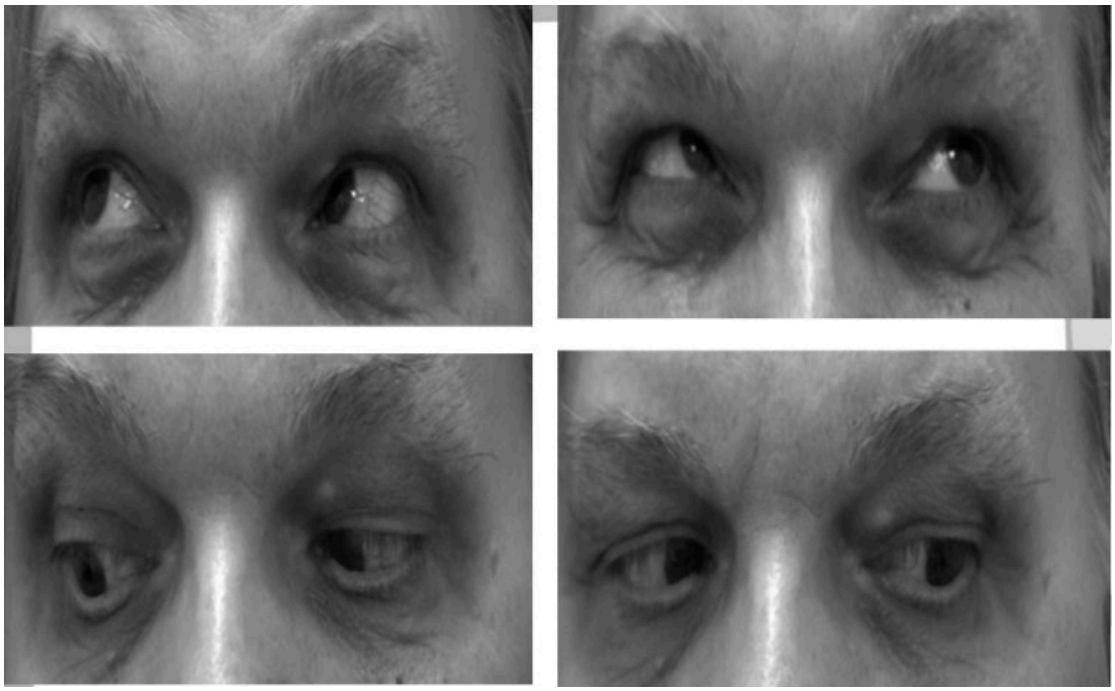


Fig. 13.14 Right fourth nerve palsy (CN IV → SO): the right eye cannot look down when it is adducted, and sits higher in primary gaze. Your note on the slide marks a panel "Right SO". The next slide (Fig. 13.17) shows a right CN VI palsy: slight esotropia in primary gaze and no abduction of the right eye in right gaze. § Slides 9 · #18–19 · TB 13 · p. 217–219

8 · Agonists and antagonists about each axis

- **Synergists / agonists** work together for the same action. **Antagonists** work in opposite directions. § Handout 9 · p. 14 · TB 11 · p. 188
- **Sherrington's law of reciprocal innervation**: contraction of a muscle comes with a simultaneous and proportional relaxation of its antagonist. Example: in adduction the MR contracts and the LR relaxes. § Handout 9 · p. 14 · TB 11 · p. 188 Exam

AXIS · MOVEMENT	AGONISTS (TOGETHER)	ANTAGONISTS (AGAINST THEM)
Z · adduction	MR (primary) + SR + IR	LR + SO + IO
Z · abduction	LR (primary) + SO + IO	MR + SR + IR
X · elevation	SR + IO	IR + SO
X · depression	IR + SO	SR + IO
Y · intorsion	SO (primary) + SR	IO + IR
Y · extorsion	IO (primary) + IR	SO + SR

§ Built from TB 11 · p. 184, Table 11.5, and the pairs the textbook names on p. 188

The same pair of muscles can be agonists about one axis and antagonists about another. That is the point of the objective:

PAIR	X (UP/DOWN)	Y (TORSION)	Z (IN/OUT)	NET EFFECT TOGETHER
MR + LR	—	—	antagonists	—
SR + IO	**agonists** (both elevate)	antagonists (in vs ex)	antagonists (in vs out)	torsion and horizontal cancel: **pure elevation** § TB 11 · p. 188
IR + SO	**agonists** (both depress)	antagonists	antagonists	**straight down** § TB 11 · p. 188
SO + IO	antagonists	antagonists	**agonists** (both abduct)	§ TB 11 · p. 188
SR + IR	antagonists	antagonists	**agonists** (both adduct)	
SR + SO	antagonists	**agonists** (both intort)	antagonists	
IR + IO	antagonists	**agonists** (both extort)	antagonists	

§ Your notes say the same for SR + IO: "elevate ⇒ x-axis ⇒ agonist · intorsion → extort ⇒ y-axis ⇒ antagonist · adduct → abduct ⇒ z-axis ⇒ antagonist", and "SR, SO agonist in torsional movement". § your notes, handout p. 14 · Review handout 14 · p. 5

9 · Yoke muscles

- **Yoke muscles**: the muscles of the *two* eyes that work together to make symmetric, binocular movements. § Handout 9 · p. 14 · TB 11 · p. 189
- **Hering's law of equal innervation**: innervation to the muscles of the two eyes is equal and simultaneous, so the two eyes normally move symmetrically. § Handout 9 · p. 14 · TB 11 · p. 189 Exam
- Don't mix the two laws up: Sherrington = agonist and antagonist in *one* eye · Hering = yoke muscles of *both* eyes.

GAZE	KIND	YOKE MUSCLES
Right (dextroversion)	secondary	R LR + L MR § TB 11 · p. 189
Left (levoversion)	secondary	L LR + R MR
Up (supraversion)	secondary	SR + IO in each eye (the elevating synergists) § from TB 11 · p. 188
Down (infraversion)	secondary	IR + SO in each eye (the depressing synergists) § from TB 11 · p. 188
Up and right	tertiary	**R SR + L IO**
Up and left	tertiary	**L SR + R IO**
Down and right	tertiary	**R IR + L SO**
Down and left	tertiary	**L IR + R SO**
Convergence	vergence	R MR + L MR § TB 11 · p. 189

§ TB 11 · p. 189, Fig. 11.23 (six cardinal positions). How to build any tertiary pair: the eye looking *out* uses its rectus, the eye looking *in* uses its oblique, and both do the same vertical job. Your note: "up left: R IO, L SR are yoke muscles". § your notes, Review handout 14 · p. 5

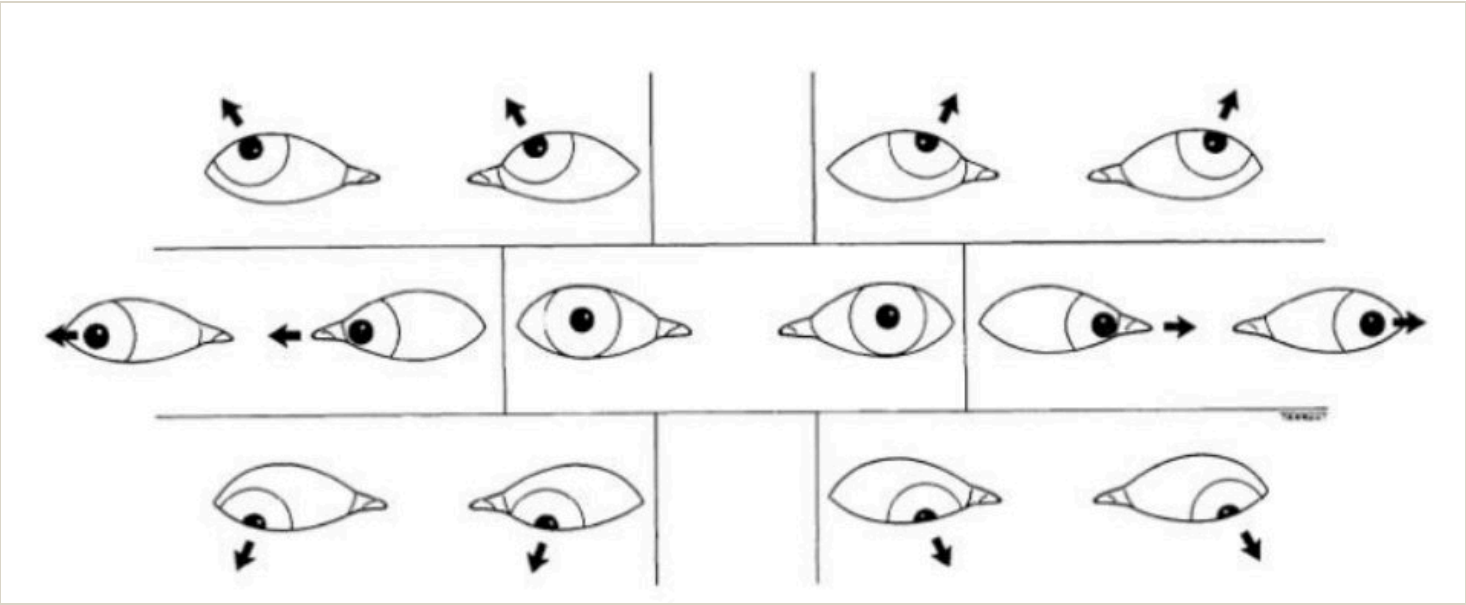
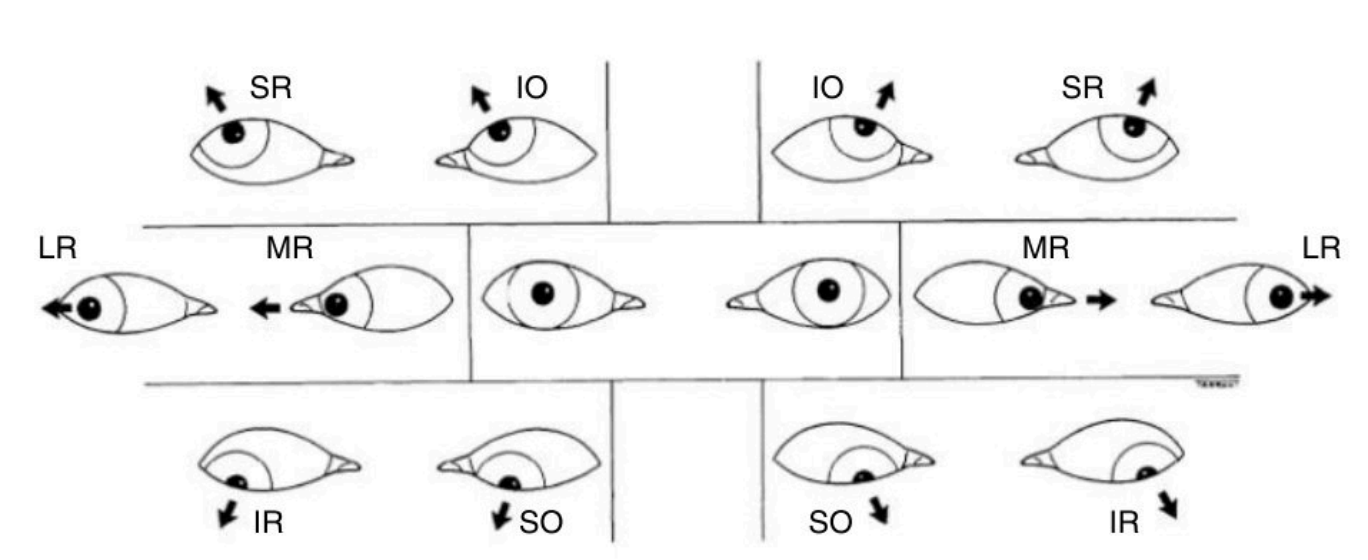


Fig. 11.23, labels removed: the six cardinal positions, as you face the patient (the left eye of each pair is the patient's right eye). Name the yoke muscle that moves each eye in the direction of its arrow. § TB 11 · p. 189

▼ Answer: Fig. 11.23 with labels



Patient's right gaze (pairs on the left of the figure): up R SR + L IO · across R LR + L MR · down R IR + L SO.
Patient's left gaze (right of the figure): up R IO + L SR · across R MR + L LR · down R SO + L IR.

ONLY IN YOUR NOTES · CHECK THESE

- Typed note beside the objectives: "IR: 1. depression, extorsion, **abduction**". The textbook gives IR **adduction** (its insertion is lateral to its origin), and your own handout note on p. 9 says adduction too. Learn adduction. §TB 11 · p. 184
- SO origin written as "Trochlear". The *anatomic* origin is the lesser wing of the sphenoid; the trochlea is the *physiologic* (effective) origin. Trochlear is the nerve (CN IV). §TB 11 · p. 182
- "IO: posterior lateral??" Confirmed: insertion inferior, posterior, lateral globe, near the macula. §TB 11 · p. 182
- "antagonist in lateral??" Depends on the pair: see the pair table in section 8 (SR + IO are antagonists horizontally; SO + IO are agonists, both abduct).
- Your yoke sketch on handout p. 14 labels an oblique in one eye with an MR in the other ("IO (R) / MR (L)", "SO / MR"). That is not a yoke pair in the textbook: yoke pairs are vertical with vertical (R IO + L SR) and horizontal with horizontal (R LR + L MR). Check the sketch against Fig. 11.23.
- SO intorts and IO extorts when the eye is abducted, and SR/IR are "torsional" when it is adducted: the textbook only says these muscles then cannot move the eye vertically. Your notes

NOT TAUGHT FROM TEXTBOOK CHAPTER 11

Left out because neither the objectives nor the handout ask for them: the microscopic anatomy of striated muscle (sarcomere, actin/myosin, sliding ratchet model), fibre types and the global/orbital layers, motor units, muscle pulleys, intermuscular septa and check-ligament detail, compartmentalization, the controversy over the IO's horizontal action, the muscles' blood supply (ophthalmic artery branches belong with circulation), ageing changes, and the clinical comments on myasthenia gravis, Brown syndrome, strabismus and Graves disease. The pathways of CN III, IV and VI and their palsies are taught in the cranial nerves chapter; here they appear only as the slides' two palsy photos.

Key facts

6 muscles: 4 recti (origin: common tendinous ring) + 2 obliques (SO from the lesser wing via the trochlea, IO from the maxillary bone)
Nerves: **LR6 · SO4 · the rest III** (SR = superior division, MR/IR/IO = inferior division)
Spiral of Tillaux: MR **5.5** → IR **6.2** → LR **6.6** → SR **7.2** mm from the limbus
Tendons: LR **8.8** mm (longest rectus) · MR **3.7** mm · SO **2.5 cm** · IO **1** mm
Angles to the sagittal axis: SR and IR **23°** · SO **55°** · IO **51°**
Pure vertical action: vertical recti at **23°** abduction · obliques at **51–55°** adduction
No vertical action: vertical recti at **67°** adduction · obliques at **35–39°** abduction
Centre of rotation **13.5** mm behind the cornea

Self-check

▼ Which muscles arise from the common tendinous ring, and which two also attach to the optic nerve sheath? Why does that matter?

All four recti. MR and SR also attach to the dural sheath of the optic nerve, so eye movement stretches the sheath: pain on eye movement in retrobulbar optic neuritis. § TB 11 · p. 179

▼ Give the anatomic and the physiologic origin of the superior oblique, and say why the physiologic one is used.

Anatomic: lesser wing of the sphenoid, medial to the optic canal. Physiologic: the trochlea, which acts as a pulley and changes the direction of pull, so the line of action runs from the trochlea to the insertion. § TB 11 · p. 182

▼ Which extraocular muscle has its origin in the anterior orbit, and where does it insert?

The inferior oblique: maxillary bone near the inferomedial orbital rim. Insertion: posterior, lateral, mostly inferior globe, just outside the macula. § TB 11 · p. 182

▼ Put the rectus insertions in order from nearest to farthest from the limbus. What is the line called?

MR → IR → LR → SR: the spiral of Tillaux. § TB 11 · p. 180

▼ List the nerve of each extraocular muscle.

MR, IR, IO: inferior division of CN III. SR: superior division of CN III. LR: CN VI. SO: CN IV. § TB 11 · p. 182, Table 11.2

▼ Give the primary and secondary actions of the SR, IR, SO and IO.

SR: elevation; adduction, intorsion. IR: depression; adduction, extorsion. SO: intorsion; depression, abduction. IO: extorsion; elevation, abduction. § TB 11 · p. 184, Table 11.5

▼ Name Fick's three axes, the direction each runs, and the movements about each.

X: horizontal, nasal to temporal: elevation/depression. Y: horizontal (sagittal), anterior to posterior pole: intorsion/extorsion. Z: vertical, superior to inferior: adduction/abduction. § Handout 9 · p. 7 · TB 11 · p. 183

▼ What is the difference between a secondary and a tertiary position of gaze?

Secondary: rotation about either the vertical or the horizontal axis. Tertiary: rotation about both. § TB 11 · p. 183

▼ The right eye is in right gaze and cannot look up. Which muscle? And if the left eye cannot look up in the same position?

Right eye is abducted, so its elevator is the vertical rectus: right SR. Left eye is adducted, so its elevator is the oblique: left IO. § TB 11 · p. 188

▼ A patient cannot depress the left eye when looking down and to the right. Which muscle and which nerve?

In right gaze the left eye is adducted, so its depressor is the left SO, supplied by CN IV (trochlear). § TB 11 · p. 188 · TB 11 · p. 182

▼ SR and IO: agonists or antagonists? Answer for each of Fick's axes.

X: agonists (both elevate). Y: antagonists (SR intorts, IO extorts). Z: antagonists (SR adducts, IO abducts). Together they produce pure elevation. § TB 11 · p. 188

▼ Name the yoke muscles for levoelevation and dextrodepression, and the law that governs them.

Up and left: left SR + right IO. Down and right: right IR + left SO. Hering's law of equal innervation: equal and simultaneous innervation to the yoke muscles. § TB 11 · p. 189, Fig. 11.23

§ Sources for this chapter: Handout 9 "Extraocular Muscles" (with your notes) · Slides 9 "Extraocular Muscles" (with your notes) · Review handout 14, p. 5 (objectives and typed notes) · Textbook ch. 11, Extraocular Muscles, pp. 175–190 · Textbook ch. 13, pp. 217–219 (Figs. 13.14, 13.17).

Cranial nerve innervation

The sensory nerves that carry the eye's feeling back to the brainstem, the two autonomic routes into the eye, and the motor nerves that move the eye and close the lids.

WHAT THE EXAM ASKS

Prof. Goodwin's guideline objectives for "Cranial Nerves" and the two "Cranial Nerves (cont.)" slides, word for word: § Review · p. 6

- Exam "List the branches, pathways, and functions of each branch of the ophthalmic nerve." → §2
- Exam "List the branches, pathways, and functions of the branches of the maxillary nerve that innervate structures surrounding the orbit" → §3
- Exam "Describe the central, preganglionic, and postganglionic sympathetic pathways to the eye and ocular structures." → §5 (and §7 for the lacrimal gland)
- Exam "Describe how the sympathetic system affects the eye and ocular structures." → §5
- Exam "Describe the central, preganglionic, and postganglionic parasympathetic pathways to the eye or ocular structures." → §6 (and §7)
- Exam "Describe the parasympathetic system affects the eye and ocular structures." → §6
- Exam "Describe the pathways of cranial nerves 3, 4, and 6." → §8–§11
- Exam "Describe the characteristic of a cranial nerve 3, 4, or 6 palsy." → §8–§11
- Exam "List the functions and pathway of the branches of cranial nerve 7 that innervate muscles surrounding the eye." → §12

Your typed notes beside these objectives: "nasociliary nerves, frontal 2 branches, lacrimal 0 branches · maxillary: infraorbital nerve · central: brain stem, superior ganglion · post. internal carotid, cavernous sinus, dilator and Müller · pupils sympathetic: dilate, Müller enlarge eye · hypothalamus, Edinger-Westphal, cranial 3 to ciliary ganglion, then short ciliary nerves · 4 cavernous sinus · 6 is in pons into the sinus cavernosus · CN7 lagophthalmos" Your notes. Read them as the skeleton of the answers: count the branches, learn each autonomic route as a **sequence of stations**, and link CN VII to lid closure. Every route below is drawn as a flow chart for that reason.

1 · Which cranial nerve does what

Six cranial nerves serve the orbit. CN V carries **sensation**; CN III, IV, VI and VII are **motor** to striated muscle; CN II (vision) belongs to the visual pathway chapter. § TB 13 · p. 208

NERVE	LEAVES THE BRAINSTEM AT	SUPPLIES	FUNCTION

III Oculomotor · superior division	midbrain	superior rectus · levator palpebrae superioris	elevation (with adduction, intorsion) · lifts the upper lid
III Oculomotor · inferior division	midbrain	medial rectus · inferior rectus · inferior oblique · ciliary ganglion	adduction · depression · elevation · parasympathetic to sphincter and ciliary muscle (miosis, accommodation)
IV Trochlear	midbrain	superior oblique	depression, abduction, intorsion
V Trigeminal	pons	skin, eye and orbit	sensation (mainly pain from cornea, iris, conjunctiva, sclera)
VI Abducens	pons	lateral rectus	abduction
VII Facial	pons	frontalis, procerus, corrugator, orbicularis · pterygopalatine (sphenopalatine) ganglion	facial expression, lid closure · secretomotor to the lacrimal gland

§ TB 13 · p. 208–209 (Table 13.1). Memory aid from this table: LR by VI, SO by IV, all the other extraocular muscles by III.

Two building rules you need for every pathway: a **somatic** motor fibre runs the whole way from the brainstem to the muscle (one neuron); an **autonomic** route has a **synapse in a ganglion** outside the CNS (two neurons: preganglionic, then postganglionic). A group of nerve cell bodies is a *nucleus* inside the CNS, a *ganglion* outside it. § TB 13 · p. 208 · TB 14 · p. 222

2 · Sensory: the ophthalmic division of the trigeminal nerve (V1)

The textbook follows sensory fibres **from the tissue back to the brain**, the way the signal travels: small nerves join into bigger ones, reach the trigeminal ganglion and then the pons. Learn them in that direction. § TB 13 · p. 208

Trigeminal nerve (CN V) · trigeminal ganglion

- **V1 Ophthalmic** = nasociliary + frontal + lacrimal
  - **Nasociliary** (5 branches): **infratrochlear** · **anterior ethmoid** · **posterior ethmoid** · **long ciliary** · **short ciliary** (via the sensory root of the ciliary ganglion)
  - **Frontal** (2 branches): **supratrochlear** · **supraorbital**
  - **Lacrimal** (no named branches)
- **V2 Maxillary**: **infraorbital** · **zygomatic** (= zygomaticotemporal + zygomaticofacial)
- **V3 Mandibular** (lower face; not an orbital nerve)

§ Handout 10 · p. 1–3 · TB 13 · p. 208–212 · your tree drawn on Handout 10 · p. 4 has exactly these names Your notes

The nasociliary nerve and its five branches

BRANCH	INNERVATES	PATHWAY	YOUR NOTES
Infratrochlear	medial canthus: **caruncle, canaliculi, lacrimal sac**, medial eyelids, skin at the side of the nose	pierces the orbital septum, enters the orbit **below the trochlea**, runs on the upper border of the medial rectus and becomes the nasociliary nerve	caruncle, canaliculi / lac. sac, med. lids + conj. · "under" (the trochlea)
Anterior ethmoid	skin along the **centre of the nose** (to the tip), nasal mucosa, ethmoid sinuses	enters the orbit with its artery through a foramen in the frontoethmoid suture, joins the nasociliary nerve on the medial wall	skin of medial nose, ethmoid sinus · "lesion on tip of nose"
Posterior ethmoid	ethmoid and **sphenoid** sinuses	same route as the anterior ethmoid, further back	ethmoid + sphenoid sinus
Long ciliary (2)	**cornea** and other anterior segment structures; also carry **sympathetic** fibres to the iris dilator	one medial, one lateral; run between choroid and sclera to the back of the globe and leave it about **3 mm on each side of the optic nerve**; join the nasociliary nerve. Seen with indirect ophthalmoscopy at **3 and 9 o'clock** (Fig. 13.4)	ant. segment: ant. sclera, iris, CB, cornea + TM
Short ciliary (6–10)	cornea, **iris, ciliary body**; they also carry sympathetic and parasympathetic fibres	through the choroid to the back of the globe; leave the sclera in a ring around the optic nerve with the short posterior ciliary arteries; enter the **ciliary ganglion**, pass through **without synapsing** and leave as its **sensory root**, which joins the nasociliary nerve	ant. segment: cornea + iris + ciliary body

§ Handout 10 · p. 1–2 · TB 13 · p. 208–210 · Slides 10 · #2, #4, #5

infratrochlear + anterior ethmoid + posterior ethmoid + 2 long ciliary + sensory root of the ciliary ganglion →

nasociliary nerve (medial orbit) → oculomotor foramen (inside the common tendinous ring) · superior orbital fissure →

joins frontal + lacrimal → **ophthalmic nerve (V1)**

§ TB 13 · p. 210

CLINICAL CORRELATE · HERPES ZOSTER AND HUTCHINSON SIGN

Herpes zoster is a CNS infection by the varicella-zoster virus: it lies dormant in a sensory ganglion and, when activated, travels down the sensory nerve to the skin, giving pain and a rash in that nerve's area. About 10% of cases involve V1. **Hutchinson sign**: zoster on the **tip of the nose** means the eye will often be involved too, because the tip of the nose (anterior ethmoid) and the globe (long and short ciliary nerves) are both nasociliary territory. § TB 13 · p. 210 · Handout 10 · p. 1 · Slides 10 · #3

The frontal and lacrimal nerves, and V1 itself

NERVE	INNERVATES	PATHWAY	YOUR NOTES
Supratrochlear	skin and muscles of the forehead · **medial upper eyelid**	supratrochlear notch (or foramen); pierces the superomedial corner of the orbital septum	forehead, medial upper eyelid
Supraorbital	forehead, **scalp**, upper eyelid	supraorbital notch (or foramen) with the supraorbital artery, lateral to the supratrochlear; the two join midway in the orbit → **frontal nerve**	forehead, central upper eyelid
Frontal	(the two above)	back through the orbit between the levator and the periorbita; leaves through the superior orbital fissure **above** the common tendinous ring	"frontal 2 branches"
Lacrimal	**lateral upper eyelid**, temple area, the lacrimal gland itself	fibres join in the gland; the nerve runs back along the upper border of the lateral rectus; may receive a branch from the zygomatic nerve carrying the gland's autonomic fibres; leaves through the superior orbital fissure above the muscle cone	lacrimal gland, lateral upper eyelid, temple area · "0 branches"

§ Handout 10 · p. 3 · TB 13 · p. 211 · Fig. 13.5 (Slides 10 · #6)

supratrochlear + supraorbital → frontal nerve → SOF, above the ring → **V1**

lateral upper lid + temple + gland → lacrimal nerve (on the lateral rectus) → SOF, above the cone → **V1**

nasociliary + frontal + lacrimal (behind the orbit) →

ophthalmic nerve in the **lateral wall of the cavernous sinus** (picks up proprioceptive fibres from III, IV, VI) →

trigeminal ganglion → **pons**

§ Handout 10 · p. 3 ("the nasociliary, frontal, and lacrimal nerves join to form the ophthalmic division") · TB 13 · p. 211–212

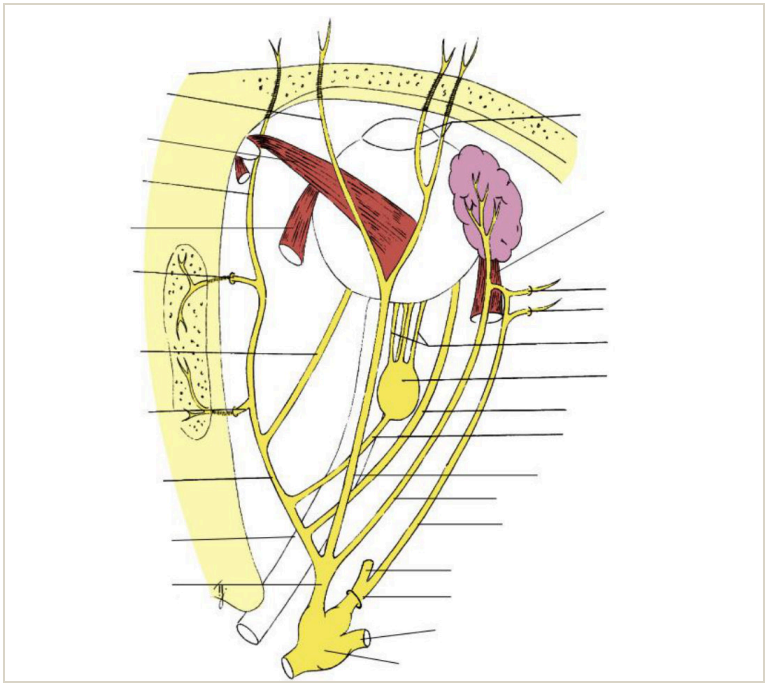
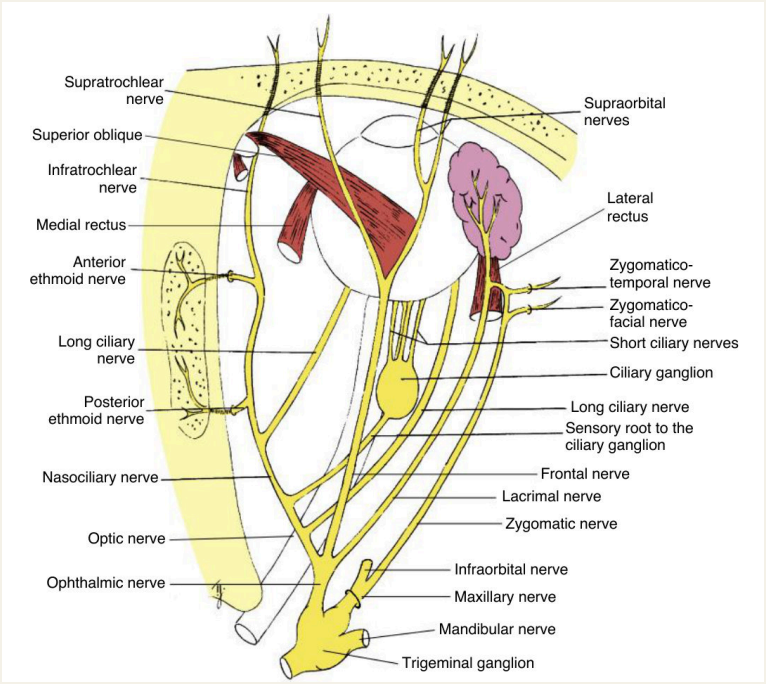


Fig. 13.1, labels removed. Name all 25 lines, then open the answer. § TB 13 · p. 209 · Slides 10 · #2, #5

▼ Answer: Fig. 13.1 with labels



Left, top to bottom: supratrochlear nerve · superior oblique · infratrochlear nerve · medial rectus · anterior ethmoid nerve · long ciliary nerve · posterior ethmoid nerve · nasociliary nerve · optic nerve · ophthalmic nerve. Right, top to bottom: supraorbital nerves · lateral rectus · zygomaticotemporal nerve · zygomaticofacial nerve · short ciliary nerves · ciliary ganglion · long ciliary nerve · sensory root to the ciliary ganglion · frontal nerve · lacrimal nerve · zygomatic nerve · infraorbital nerve · maxillary nerve · mandibular nerve · trigeminal ganglion.

3 · Sensory: the maxillary division (V2) and the way to the brainstem

NERVE	INNERVATES	PATHWAY	YOUR NOTES
Infraorbital	**lower eyelid**, cheek, upper lip; in its canal it also collects the **upper teeth** and maxillary sinus	infraorbital foramen → infraorbital canal and groove in the maxilla (with the infraorbital artery) → leaves the orbit through the **inferior orbital fissure** → maxillary nerve	lower eyelid, upper cheek, upper lip + teeth
Zygomatic	zygomaticotemporal: **temple** · zygomaticofacial: **lateral cheek and lower eyelid**; also carries the lacrimal gland's autonomic fibres (§7)	each enters the orbit through a foramen in the zygomatic bone; they join, run along the lateral orbital wall, leave through the **inferior orbital fissure** → maxillary nerve	lateral lower eyelid, lateral cheek, lateral forehead

§ Handout 10 · p. 3 · TB 13 · p. 211–212

infraorbital + zygomatic + palate, upper teeth, gums, cheek mucosa →

maxillary nerve, pterygopalatine fossa (picks up autonomic fibres from the pterygopalatine ganglion) → foramen rotundum →

lateral wall of the cavernous sinus → **trigeminal ganglion**

§ TB 13 · p. 212. The mandibular nerve (V3) enters through the foramen ovale and lies just below the cavernous sinus.

> CLINICAL · REFERRED PAIN
>
> Pain can be felt far from its source when both areas share a trigeminal network: an abscessed upper tooth (infraorbital nerve) can be described by the patient as **eye pain**. Suspect it when no orbital cause is found. § TB 13 · p. 211

Trigeminal ganglion and nuclei

STATION	WHERE	WHAT IT DOES
Trigeminal (gasserian, semilunar) ganglion	lateral to the internal carotid artery and the back of the cavernous sinus	holds the **cell bodies** of the sensory fibres of all three divisions (no synapse); fibres then enter the lateral pons
Principal sensory nucleus	pons	touch and pressure
Nucleus of the spinal tract	pons and medulla, down to C2–C4	pain and temperature
Mesencephalic nucleus	pons–midbrain junction	proprioception

§ TB 13 · p. 212 · Fig. 13.6 (Slides 10 · #7). From the nuclei the information goes on to the thalamus.

4 · The autonomic plan

The autonomic system supplies smooth muscle, glands and vessels. In the eye: the iris muscles, ciliary muscle, smooth muscle of the lids (Müller), choroidal and conjunctival vessels, and the lacrimal gland. The sympathetic side prepares for an emergency; the parasympathetic side keeps and restores the resting state. § TB 14 · p. 222

	SYMPATHETIC	PARASYMPATHETIC
Preganglionic cell bodies for the eye	spinal cord **T1–T3**	**midbrain** (Edinger-Westphal preganglionic cells) · **pons** (lacrimal / superior salivatory nucleus)
Ganglion	near the spine: **superior cervical ganglion**	near the target: **ciliary ganglion** · **pterygopalatine ganglion**
Preganglionic fibre	usually myelinated, ends in the ganglion with a synapse	
Postganglionic fibre	usually non-myelinated	to the iris and ciliary muscle: myelinated
Transmitter at the target	norepinephrine (adrenergic)	acetylcholine (cholinergic)
Eye targets	iris dilator, ciliary muscle, Müller muscle, lacrimal gland, choroidal and conjunctival vessels	iris sphincter, ciliary muscle, lacrimal gland, blood vessels

§ TB 14 · p. 222, 224, 227 · Handout 10 · p. 5–7

5 · The sympathetic pathway

Three neurons in a row: **central** (hypothalamus to spinal cord), **preganglionic** (spinal cord to superior cervical ganglion), **postganglionic** (ganglion to the eye). § Handout 10 · p. 5 · TB 14 · p. 222, 235 · Fig. 14.18

hypothalamus central neuron, through the brainstem → spinal cord (synapse; outflow T1–T3) preganglionic → ventral root → sympathetic chain (no synapse), over the apex of the lung, around the subclavian artery → ascends to the **superior cervical ganglion**, C2–C3, at the carotid bifurcation · synapse postganglionic → carotid plexus on the internal carotid artery → carotid canal into the skull → **cavernous sinus**

§ TB 14 · p. 222 (central pathway ends in the cervical spinal cord; preganglionic fibres leave in the first three thoracic nerves) · p. 235 (over the lung apex, around the subclavian artery). Your notes: "Hypothalamus ↓ spinal cord ↓ sup. cervical gang." and "LUNG" drawn beside Fig. 14.2 Your notes. The textbook names the first synapse in two ways: the "lateral horn" of T1–T3 (p. 222) and the "dorsal horn of the cervical spinal cord" (p. 235). Learn: upper spinal cord, outflow T1–T3.

In the cavernous sinus the fine postganglionic fibres leave the plexus and **hitch a ride on cranial nerves** to their targets: § TB 14 · p. 222

cavernous sinus → V1 → nasociliary → long ciliary nerves → **iris dilator (+ ciliary muscle)**

cavernous sinus → nasociliary (or straight from the plexus) → sympathetic root → ciliary ganglion, **no synapse** → short ciliary nerves → **choroidal blood vessels**

cavernous sinus → joins CN III → superior division (with the levator branch) → **Müller muscle, upper lid**

§ TB 14 · p. 222 · TB 13 · p. 214 (CN III receives sympathetic fibres in the sinus). Conjunctival vessels: via the long or the short ciliary nerves. An alternative route to Müller muscle through the infratrochlear or lacrimal nerve has been suggested. The lacrimal gland and lower-lid route is in §7.

TARGET (HANDOUT)	ARRIVES BY	EFFECT OF SYMPATHETIC STIMULATION
Iris dilator	long ciliary nerves	contracts → **pupil dilates** (mydriasis) → more light on the retina
Choroidal vasculature	short ciliary nerves	**vasoconstriction** (also conjunctival vessels)
Müller muscle	superior division of CN III	contracts → **widens the palpebral fissure** (raises the lid)
Ciliary muscle	long ciliary nerves	small inhibitory effect after sustained accommodation, in some people
Lacrimal gland	zygomatic → lacrimal nerve (§7)	constricts its vessels → indirectly **less secretion**

§ Handout 10 · p. 5 · TB 14 · p. 222, 226. Your notes beside the handout: "dilates / constrict / raises eyelid" and, on the guidelines, "Müller enlarge eye" Your notes. Sweat-gland fibres to the face mostly follow the external carotid artery instead; that matters for Horner syndrome (chapter 11).

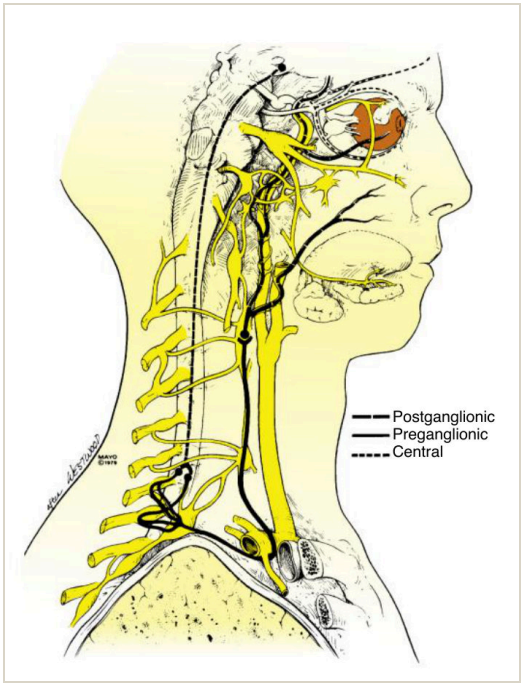


Fig. 14.18 The three sympathetic neurons: central (fine dashes, hypothalamus to spinal cord), preganglionic (solid, into the chest and up to the superior cervical ganglion), postganglionic (long dashes, along the internal carotid). § TB 14 · p. 236

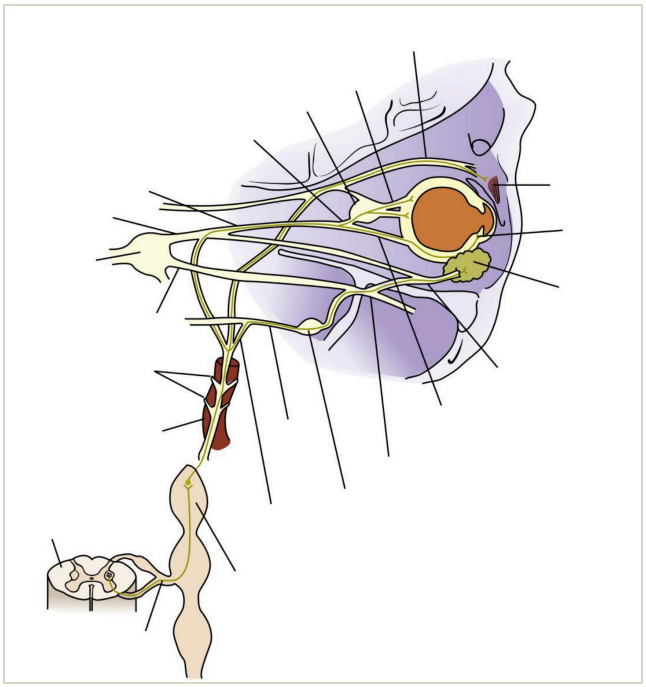
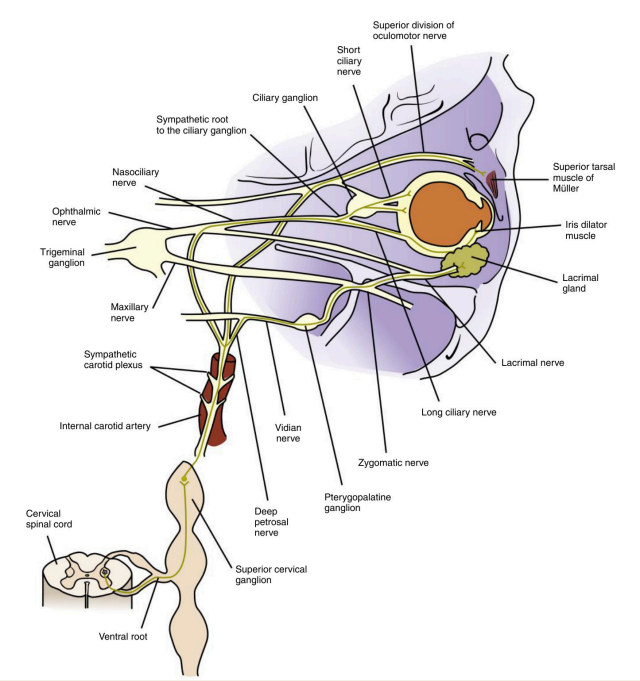


Fig. 14.2, labels removed. Name all 22 lines, then trace each route from the spinal cord to the dilator, Müller muscle, choroid and lacrimal gland. § TB 14 · p. 225 · Slides 10 · #9

▼ Answer: Fig. 14.2 with labels



Top, left to right: sympathetic root to the ciliary ganglion · ciliary ganglion · short ciliary nerve · superior division of oculomotor nerve. Left, top to bottom: nasociliary nerve · ophthalmic nerve · trigeminal ganglion · maxillary nerve · sympathetic carotid plexus · internal carotid artery · cervical spinal cord · ventral root. Bottom: superior cervical ganglion · deep petrosal nerve · vidian nerve · pterygopalatine ganglion · zygomatic nerve · long ciliary nerve. Right, top to bottom: superior tarsal muscle of Müller · iris dilator muscle · lacrimal gland · lacrimal nerve.

6 · The parasympathetic pathway

central input (see below) → **Edinger-Westphal preganglionic cells**, midbrain preganglionic → with the motor fibres of CN III → inferior division → parasympathetic root → **ciliary ganglion** · synapse postganglionic → short ciliary nerves → **ciliary muscle + iris sphincter**

§ Handout 10 · p. 6 · TB 14 · p. 223–224 · Fig. 14.3 (Slides 10 · #10)

- **Central pathway.** Your notes put the **hypothalamus** first ("hypothalamus (H) ↓ midbrain (Edinger-Westphal nucleus (CN3))", also on the guidelines) Your notes. The textbook does not name the hypothalamus for this route; it describes the inputs to the Edinger-Westphal cells from the pretectal nucleus (light) and from the cortex via the frontal eye fields (near). Those reflexes are chapter 11. § TB 14 · p. 229
- **Name trap.** In animals the Edinger-Westphal nucleus itself holds the preganglionic cells. In humans they lie just **dorsal** to it and are called the **Edinger-Westphal preganglionic cells**; the nucleus sits in the rostral part of the oculomotor nucleus. § TB 13 · p. 214 · TB 14 · p. 223
- **Inside CN III** the pupil fibres lie **superficially** as the nerve leaves the midbrain, move to the centre near the orbit, and run in the inferior division. This is why compression dilates the pupil (§8). § TB 13 · p. 215 · TB 14 · p. 229

The ciliary ganglion

FEATURE	FACT
Size and place	flat, about **2 mm long, 1 mm high**; inside the muscle cone between the **lateral rectus and the optic nerve**, about **1 cm** in front of the optic canal and common tendinous ring
Three roots (back edge)	**parasympathetic** (from the inferior division of CN III) · **sensory** (from the globe, to the nasociliary nerve) · **sympathetic** (to the vessels of the globe)
Synapse	**only the parasympathetic fibres**; sensory and sympathetic fibres pass straight through
Front edge	short ciliary nerves, each carrying all three fibre types
Where the fibres go	most to the **ciliary muscle**; only about **3%** to the iris sphincter

§ TB 14 · p. 224 · Fig. 14.3 inset

TARGET (HANDOUT)	EFFECT OF PARASYMPATHETIC STIMULATION
Iris sphincter	contracts → **miosis**: less retinal illumination, less chromatic and spherical aberration
Ciliary muscle	contracts → **accommodation** (focus on near objects)

§ Handout 10 · p. 6 · TB 14 · p. 224. Your notes: "miosis · accommodation"; guidelines: "more accommodation and miosis" Your notes. The iris is a balance of both systems: dilator (sympathetic) against sphincter (parasympathetic). § TB 14 · p. 225

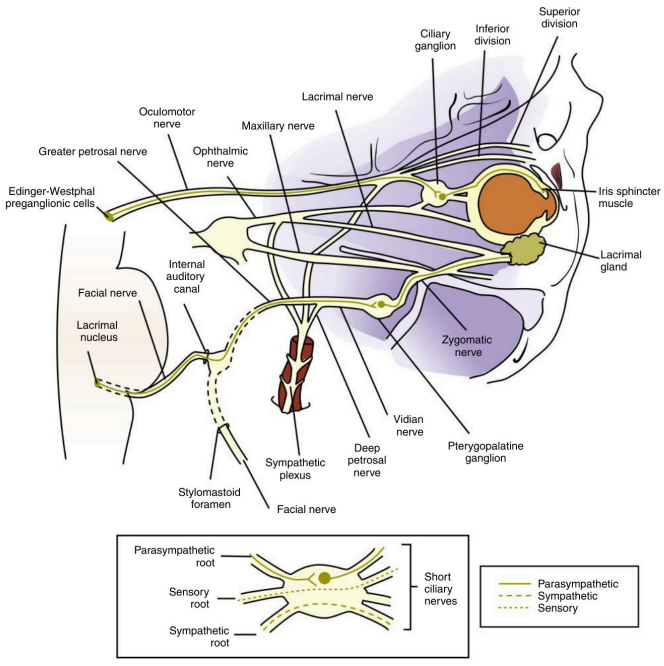


Fig. 14.3 Parasympathetic routes: Edinger-Westphal cells → CN III → ciliary ganglion → iris sphincter; lacrimal nucleus → facial nerve → greater petrosal → vidian → pterygopalatine ganglion → zygomatic → lacrimal gland. Inset: in the ciliary ganglion only the parasympathetic fibres synapse. § TB 14 · p. 226 · Slides 10 · #10

7 · The autonomic pathway to the lacrimal gland

Parasympathetic (secretomotor)

lacrimal (superior salivatory) nucleus, pons → preganglionic → with the motor fibres of CN VII → internal auditory canal → geniculate ganglion, **no synapse** → **greater petrosal nerve** → + deep petrosal = **vidian nerve** (nerve of the pterygoid canal) → **pterygopalatine ganglion** · synapse → postganglionic → maxillary nerve → zygomatic nerve → zygomaticotemporal branch (or a communicating branch to the lacrimal nerve) → **lacrimal gland**

Sympathetic (vasomotor)

hypothalamus → T1–T3 → superior cervical ganglion · synapse → postganglionic → carotid plexus → **deep petrosal nerve** → vidian nerve → pterygopalatine ganglion, **no synapse** → maxillary → zygomatic (→ lacrimal) nerve → **vessels of the lacrimal gland**

§ Handout 10 · p. 7 · TB 14 · p. 225–226 · Figs. 14.2, 14.3. The pterygopalatine ganglion is also called the sphenopalatine ganglion; it is a parasympathetic ganglion.

INNERVATION (HANDOUT)	ACTS ON	EFFECT
Parasympathetic	secretory cells	secretomotor → **more tears**
Sympathetic	blood vessels of the gland	less blood flow → indirectly **less secretion**; a zygomatic branch also reaches Müller muscle of the **lower** lid

§ TB 14 · p. 226 · TB 2 · p. 24

CLINICAL CORRELATE · THE CORNEAL REFLEX

Touching the cornea sets off three protective responses in both eyes: **blink, tearing and miosis**. The afferent arm is the trigeminal nerve; the blink is the facial nerve. A lesion of either CN V or CN VII weakens the reflex. § TB 14 · p. 227 · TB 13 · p. 220 · Handout 10 · p. 7

cornea → long (and short) ciliary nerves → nasociliary → V1 → trigeminal ganglion → **trigeminal nucleus, pons**

→ facial nucleus → CN VII, zygomatic branch → **orbicularis: blink**

→ lacrimal nucleus →

CN VII → greater petrosal → vidian → pterygopalatine ganglion → maxillary → zygomatic → lacrimal nerve → **lacrimal gland: tearing**

→ Edinger-Westphal preganglionic cells → CN III → ciliary ganglion → short ciliary nerves → **sphincter: miosis**

§ TB 14 · p. 227 (Fig. 14.4). Irritating other trigeminal branches also causes reflex tearing, for example plucking a nose hair.

8 · Oculomotor nerve (CN III)

	FACT	YOUR NOTES
Innervates	SR, MR, IR, IO, levator · carries the parasympathetic fibres (sphincter, ciliary muscle) and sympathetic fibres (smooth muscle of the lid) § TB 13 · p. 213	SR, MR, IO, IR, levator · "they go to sphincter, CB to accommodation"
Nucleus	**midbrain**, at the level of the **superior colliculus**, ventral to the cerebral aqueduct, dorsal to the medial longitudinal fasciculus. One subnucleus per muscle: the **SR** fibres cross (supply the contralateral eye); the single central **levator** nucleus supplies **both** levators; MR, IR, IO are ipsilateral § TB 13 · p. 213–214	"superior midbrain"

oculomotor nucleus, midbrain → fascicles through the midbrain → exits the **anterior midbrain** (interpeduncular fossa) → passes **between the superior cerebellar and posterior cerebral arteries** → runs forward lateral to, slightly below, the **posterior communicating artery** → pierces the roof of the **cavernous sinus**; lateral wall, above CN IV (receives sympathetic fibres) → **superior orbital fissure**, inside the common tendinous ring → **divides 2–3 mm behind the fissure**

superior division, above the optic nerve → **SR (underside) + levator**

inferior division, below the optic nerve → **MR · IR · parasympathetic root to the ciliary ganglion · IO**

§ Handout 10 · p. 8–9 · TB 13 · p. 214 · Fig. 13.10 (Slides 10 · #15). Your note: "Superior goes Müller through superior branch, everything else goes through inferior branch?" Your notes: yes for the autonomic fibres. Sympathetic fibres to Müller follow the superior division; parasympathetic fibres follow the inferior one. § TB 14 · p. 222–223

CLINICAL CORRELATE · CN III PALSY

CN III is **most often hit by an aneurysm of the posterior communicating artery**, but any of the neighbouring arteries (superior cerebellar, posterior cerebral) can compress it. § Handout 10 · p. 8 · TB 13 · p. 215

SIGN (HANDOUT LIST)	WHY
Ptosis	levator paralysed
Eye down and out	SO and LR work unopposed. The textbook: the eye is always **out**; **down** should follow from the SO but is not always visible clinically
Cannot adduct	MR
In abduction: cannot move up or down	SR, IR
In adduction: cannot move up	IO
Pupil dilated	sphincter paralysed
Decreased accommodation	ciliary muscle paralysed

§ Handout 10 · p. 9 · TB 13 · p. 215. Your notes on Slides 10 · #19: "levator muscle / SO LR still working · → MR not working" Your notes

Pupil involved or spared? The pupil fibres lie on the surface of the nerve as it leaves the midbrain. A **compressive** lesion (aneurysm) damages them first: a fixed, dilated pupil plus the muscle palsy. Being superficial, they are also closest to the nerve's own blood supply, so an **ischaemic** lesion (diabetes) usually **spares** the pupil. A CN III palsy with a dilated pupil is highly suspicious of a compressive intracranial lesion. The pupil detail is chapter 11. § TB 13 · p. 215 · TB 14 · p. 233



Fig. 13.12 CN III palsy. A: ptosis. B: primary gaze in the centre, arrows show the attempted gaze. The affected eye (with the ptosis) sits out and cannot adduct, elevate or depress. § TB 13 · p. 216 · Slides 10 · #19

9 · Trochlear nerve (CN IV)

	FACT	YOUR NOTES
Innervates	**superior oblique** only	superior oblique
Nucleus	**midbrain**, at the level of the **inferior colliculus**, below the oculomotor nucleus; its fibres cross, so each nucleus supplies the **contralateral** SO	"inferior midbrain"
Special	the **only cranial nerve leaving the dorsal (back) side** of the brainstem; the most slender cranial nerve	

trochlear nucleus, midbrain → exits the **posterior midbrain**, just below the inferior colliculus, and **crosses** to the other side → curves around the brainstem with the **superior cerebellar and posterior cerebral arteries**, lateral to CN III → **cavernous sinus**, lateral wall, between CN III and V1 → **superior orbital fissure**, *above* the ring (outside the muscle cone) → with the frontal nerve, medially over the levator and SR → **SO, orbital surface**

§ Handout 10 · p. 10 · TB 13 · p. 216 · Fig. 13.13 (Slides 10 · #20)

CLINICAL CORRELATE · CN IV PALSY

Eye deviation in primary position: the affected eye is **elevated** (hyperdeviation). **Restriction:** it **cannot move down when adducted**. Head posture: tilted toward the **opposite shoulder** (against the unopposed extorsion of the IO), chin down, face turned to the opposite side; a right SO palsy gives a head turned left, down, tilted to the left shoulder. Under 10 years the cause is usually congenital, at 21–40 years trauma. A nuclear lesion hits the **contralateral** SO; a nerve lesion the ipsilateral one. § Handout 10 · p. 10 · TB 13 · p. 217 · Slides 10 · #24 Your notes: "hyperdeviation · would not be able to depress in adduction" Your notes

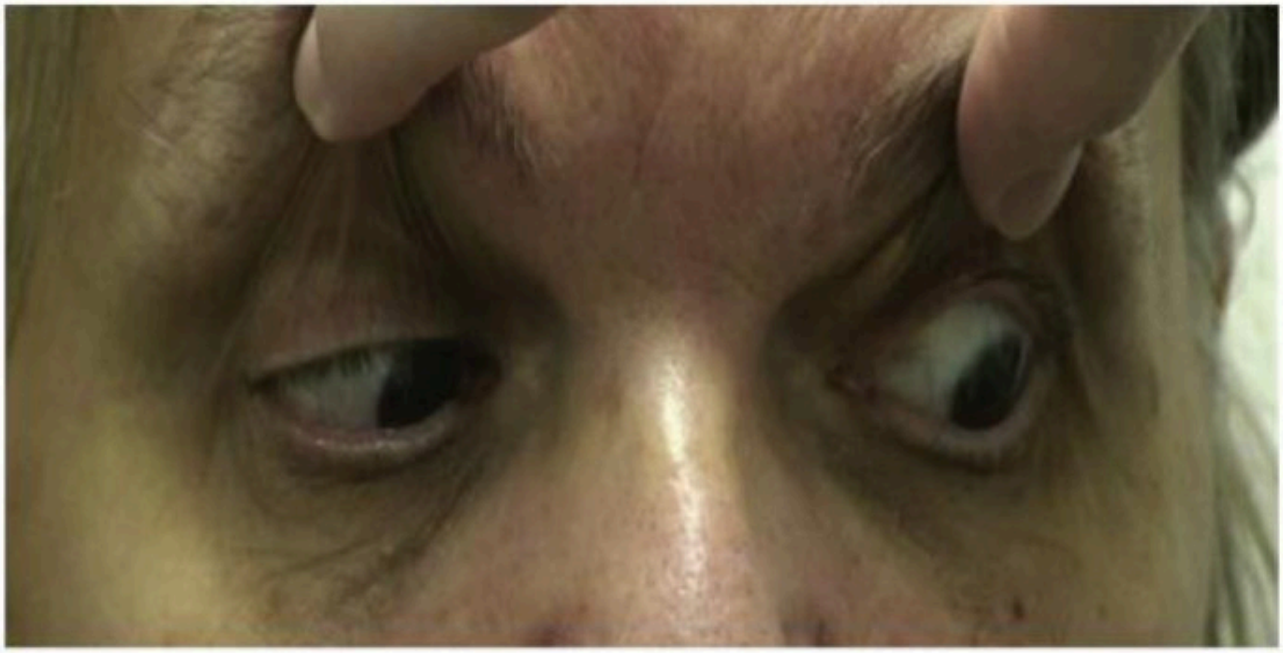


Fig. 13.14 Right CN IV palsy: looking down and to the left, the adducted right eye stays up. § TB 13 · p. 217 · Slides 10 · #24

10 · Abducens nerve (CN VI)

	FACT	YOUR NOTES
Innervates	**lateral rectus**	lateral rectus (LR)
Nucleus	**pons**, near the dorsal midline in the floor of the fourth ventricle. It also holds internuclear neurons that reach the **contralateral MR subnucleus** through the medial longitudinal fasciculus, so both eyes turn to the same side together	pons

abducens nucleus, pons → forward through the pons → exits the **groove between pons and medulla** → along the occipital bone at the skull base, up the posterior slope of the **petrous temporal bone** → **sharp bend over the petrous ridge** (tied down under the petrosphenoidal ligament) → **cavernous sinus**, beside the internal carotid artery (not in the wall) → **superior orbital fissure**, inside the ring → **LR, medial surface**

§ Handout 10 · p. 11 · TB 13 · p. 217–218 · Fig. 13.16 (Slides 10 · #25). In the sinus, sympathetic fibres travel with CN VI for a short way before joining V1. Your guidelines note: "6 is in pons into the sinus cavernosus" Your notes

CLINICAL CORRELATE · CN VI PALSY

The long course and the bend over the petrous ridge make CN VI **susceptible to trauma and raised intracranial pressure**: the brainstem is pushed back or down and the nerve is stretched over the bone. Skull-base fractures and basilar or carotid aneurysms also damage it. It is the **most common acquired isolated** extraocular-muscle nerve palsy in adults (in children under 18 the most common is CN IV). § Handout 10 · p. 11 · TB 13 · p. 218

Eye deviation in primary position: esotropia (eye turned in by the unopposed MR), worse at distance. **Restriction: cannot abduct**. The patient may turn the face toward the paralysed side. A lesion of the *nucleus* instead gives a gaze palsy: neither eye can look toward that side. § TB 13 · p. 218 · Slides 10 · #29 Your notes: "eso · abduction defect" Your notes



Fig. 13.17 Right CN VI palsy. Left image: right gaze, the right eye cannot abduct. Centre: slight esotropia in primary gaze. Right image: left gaze is normal. § TB 13 · p. 219 · Slides 10 · #29

11 · Where the nerves run together: cavernous sinus and superior orbital fissure

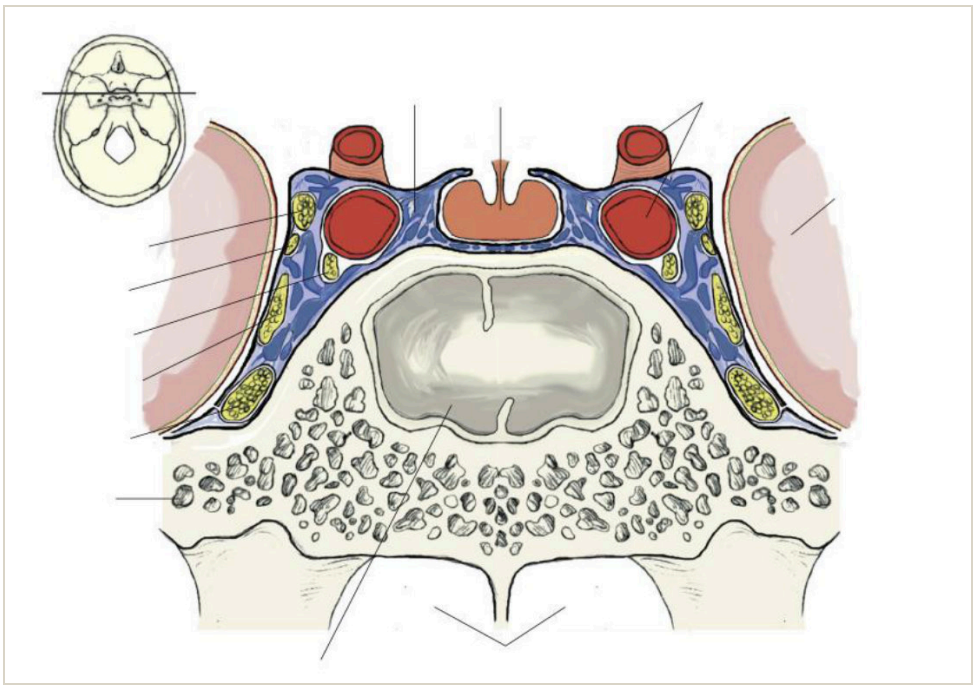
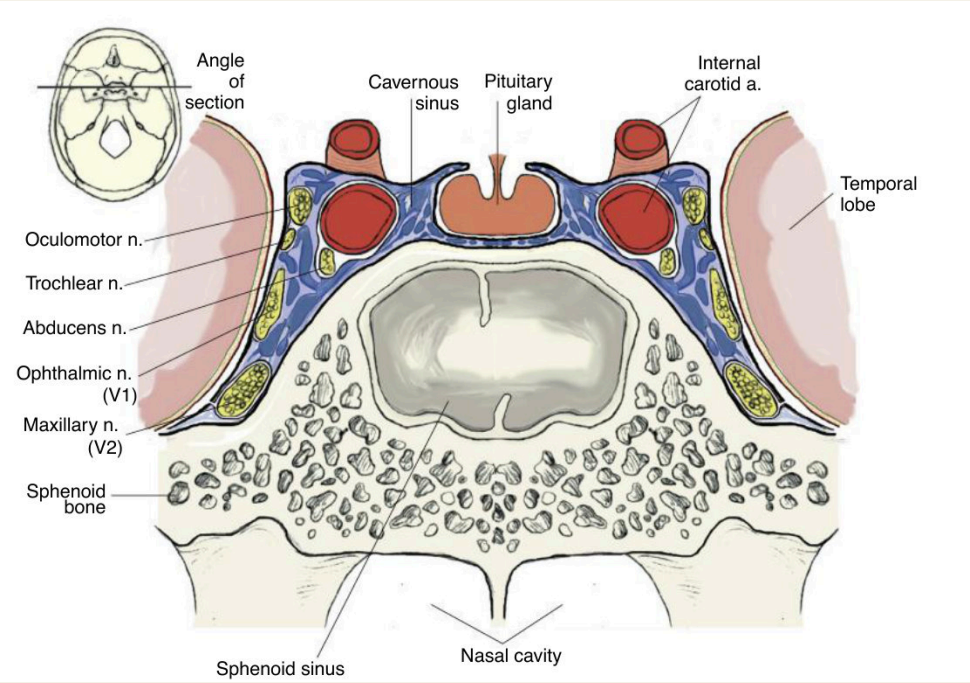


Fig. 13.7, labels removed. Name the five nerves on the left, and the structures in the centre. § TB 13 · p. 213 · Slides 10 · #16, #21, #26

▼ Answer: Fig. 13.7 with labels



In the lateral wall, top to bottom: **oculomotor n. · trochlear n. · ophthalmic n. (V1) · maxillary n. (V2)**. Free in the sinus beside the **internal carotid a.**: **abducens n.** Also: angle of section · cavernous sinus · pituitary gland · temporal lobe · sphenoid bone · sphenoid sinus · nasal cavity. You labelled the same on Slides 10 · #21: "ICA / CN6 / CN3 / CN4 / V1 / V2" Your notes.

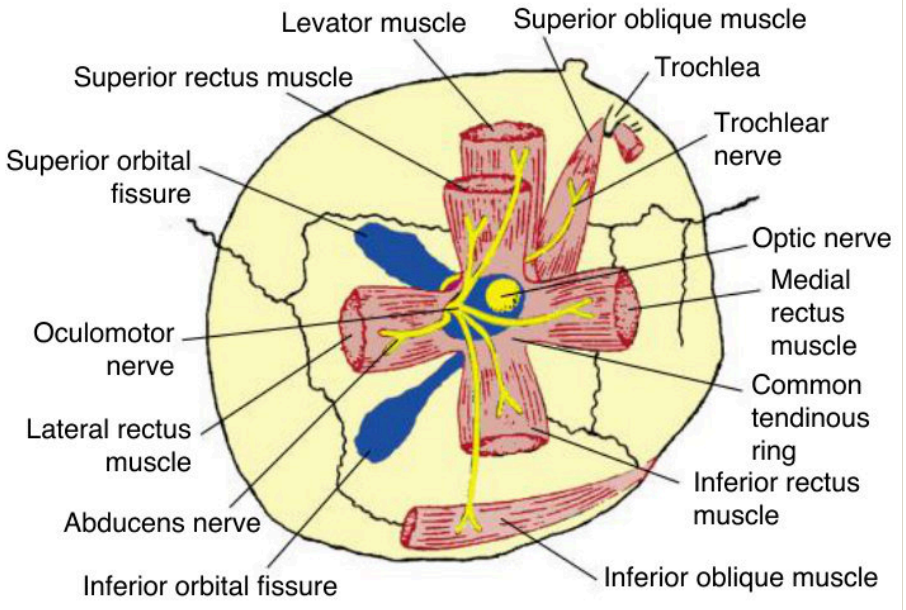


Fig. 13.11 Orbital apex. Through the common tendinous ring: CN III (both divisions), CN VI and the nasociliary nerve (with the optic nerve in the optic canal). Above the ring: CN IV, frontal and lacrimal nerves, superior ophthalmic vein. § TB 13 · p. 215, 218 · Slides 10 · #18 · see also Fig. 10.18 in chapter 8

LESION SITE	WHAT IT DOES

CN III nucleus	ipsilateral MR, IR, IO, **both** levators (bilateral ptosis), **both** SR; ipsilateral dilated pupil and loss of accommodation § TB 13 · p. 215
Cavernous sinus	III, IV, V1, V2 in the wall, VI and the sympathetic fibres around the carotid: **total ophthalmoplegia**, pupil and accommodation may be affected, **numb** V1/V2 skin. CN VI is often first hit by a carotid aneurysm here; **LR palsy + Horner syndrome on the same side = cavernous sinus lesion** § TB 13 · p. 215, 217, 218
Orbit, inside the cone	III, VI and nasociliary together: only the SO still works, eye slightly down and out and nearly immobile, reduced corneal sensation; usually vision loss from the optic nerve § TB 13 · p. 215, 218
Orbit, above the cone	CN IV with the frontal nerve: SO weakness plus numb forehead and scalp § TB 13 · p. 217

§ Your note beside "Cavernous sinus" on Handout 10 · p. 8: "if something happens here, eye stays straight and we don't feel anything" Your notes: that is the textbook's total ophthalmoplegia plus V1/V2 anaesthesia.

The three palsies side by side

	CN III	CN IV	CN VI
Muscles lost	SR, MR, IR, IO, levator + sphincter, ciliary muscle	SO	LR
Primary position	ptosis · eye out (and down)	eye up (hyperdeviation)	eye in (esotropia), worse at distance
Cannot	adduct · elevate or depress in abduction · elevate in adduction	depress in adduction	abduct
Pupil, accommodation	dilated · reduced	normal	normal
Head posture		tilt to the opposite shoulder, chin down, face to the opposite side	face turned toward the paralysed side
Typical cause (handout)	posterior communicating artery aneurysm	congenital (children), trauma	trauma, raised intracranial pressure

§ Handout 10 · p. 8–11 · TB 13 · p. 215–218

12 · Facial nerve (CN VII) to the muscles around the eye

Not in the handout's outline, but a guideline objective. CN VII has a large **motor** root (facial muscles) and a smaller root with taste and **parasympathetic** fibres (lacrimal gland, §7). § TB 13 · p. 219 · Review · p. 6

facial nucleus, pons → fibres arch around the abducens nucleus → exit at the **lower border of the pons** → **internal auditory meatus** → facial canal in the petrous temporal bone (greater petrosal nerve leaves here) →

stylomastoid foramen → below the external auditory canal, over the mandibular ramus → **temporal · zygomatic · buccal branches**

§ TB 13 · p. 219 · Fig. 13.18

BRANCH	MUSCLE	FUNCTION OF THE MUSCLE
Temporal (its frontal branch)	frontalis · **upper orbicularis oculi**	frontalis raises the brow (surprise, attention) · orbicularis closes the lid
Temporal, zygomatic, buccal (together)	procerus · corrugator superciliaris	procerus pulls the medial brow down (horizontal furrows over the nose) · corrugator pulls the brow down and medially (vertical furrows between the brows)
Zygomatic	**lateral** lower orbicularis	lid closure; orbicularis also lowers the brow
Buccal	**medial** lower orbicularis	lid closure

§ TB 13 · p. 219 · TB 2 · p. 10, 13. The eyelid chapter puts it more simply: orbicularis by the **temporal and zygomatic** branches (TB 2 · p. 22); the cranial-nerve chapter adds the buccal branch for the medial lower lid. Know both.

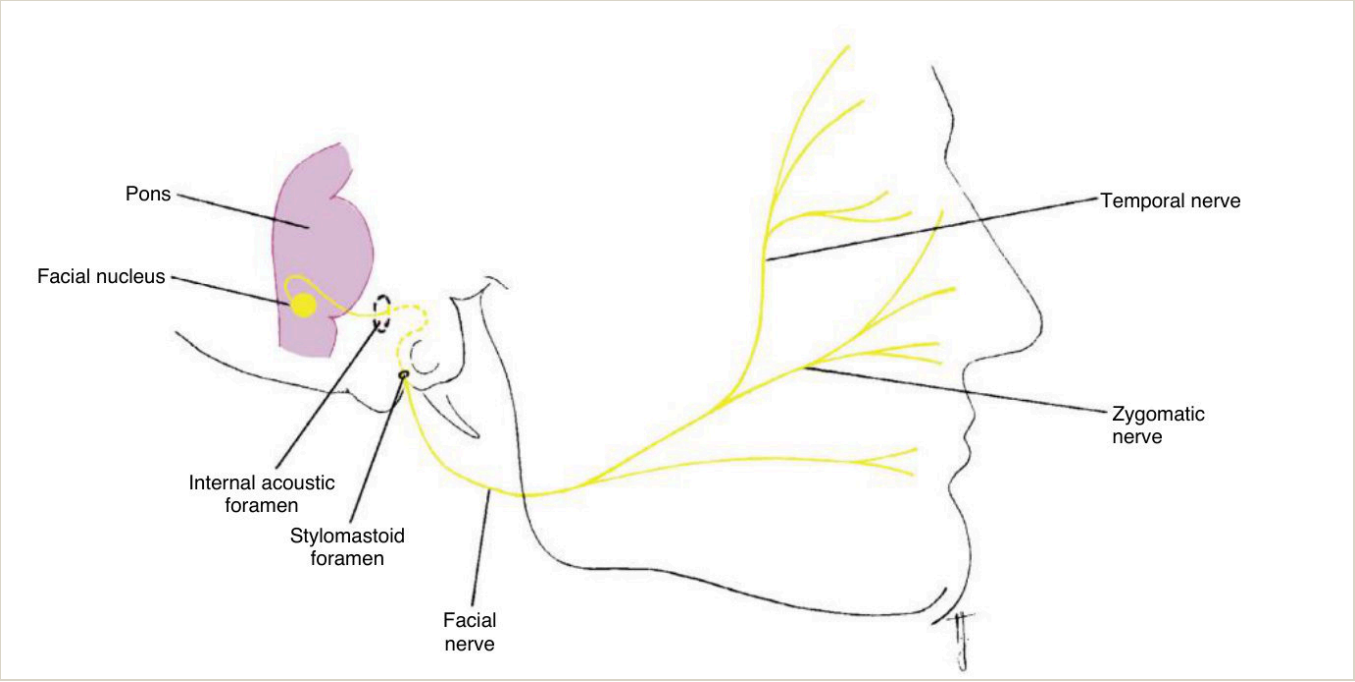


Fig. 13.18 Facial nerve: facial nucleus in the pons → internal acoustic foramen → stylomastoid foramen → temporal and zygomatic nerves to the muscles of the orbit. § TB 13 · p. 219

CLINICAL · FACIAL NERVE LESIONS AND LAGOPHTHALMOS

The upper part of the facial nucleus (frontalis, procerus, corrugator, orbicularis) gets input from **both** motor cortices, the lower part only from the opposite one. So a one-sided **cortex** lesion spares the forehead and lids, while a lesion of the **nucleus or the nerve** paralyses the **whole face** on that side. § TB 13 · p. 219 A paralysed orbicularis cannot close the lid: **paralytic lagophthalmos**, with drying of the inferior cornea, worst during sleep. § TB 2 · p. 11 Your guidelines note: "CN7 lagophthalmos" Your notes. CN VII is also the efferent arm of the blink in the corneal reflex (§7).

ONLY IN YOUR NOTES · CHECK

- Long ciliary nerves from the "ant. sclera, iris, CB, cornea + **TM**" (you added "AI shorts check"). The textbook says only "cornea and other anterior segment structures"; the trabecular meshwork is not named. § TB 13 · p. 208
- Infratrochlear nerve to the medial lids "+ conj.": the textbook names the medial eyelids, caruncle, canaliculi, lacrimal sac and side of the nose, not the conjunctiva. § TB 13 · p. 208
- Parasympathetic central pathway starting in the **hypothalamus**: not in the textbook (see §6).
- You crossed out the lacrimal-gland page of the handout (Handout 10 · p. 7) and wrote "is innervated by sympat. system". If the lecture skipped it, §7 is lower priority, but know that the textbook gives the gland **both** systems and that the **parasympathetic** one makes the tears.

NOT TAUGHT FROM TEXTBOOK CHAPTERS 13 AND 14

Left out because neither the handout nor the guidelines ask for them: the three corneal nerve plexuses and their numbers (chapter 3 territory), nerve loops of Axenfeld, the oculocardiac reflex, the exact layout of the subnuclei inside the CN III nucleus, fascicular CN III lesions (red nucleus, cerebral peduncle), aberrant regeneration of CN III, the tracts that coordinate eye movements, intrinsic choroidal neurons and parasympathetic vasodilation of the uvea, receptor types and eye-drop pharmacology. The pupillary light and near pathways, RAPD, tonic pupil, Horner syndrome and its drug tests are taught in chapter 11.

Key facts

5 branches form the nasociliary nerve (infratrochlear, anterior ethmoid, posterior ethmoid, long ciliary, sensory root/short ciliary) · **2** form the frontal (supratrochlear, supraorbital) · lacrimal has none · **3** nerves form V1
2 long ciliary nerves, **~3 mm** each side of the optic nerve, seen at **3 and 9 o'clock** · **6–10** short ciliary nerves
V2 orbital branches: **infraorbital** and **zygomatic**, both leave through the inferior orbital fissure; V2 enters the skull through the foramen rotundum
Sympathetic outflow **T1–T3** · synapse in the **superior cervical ganglion** at C2–C3 · 3 neurons: central, preganglionic, postganglionic
Ciliary ganglion **2 × 1 mm**, between LR and optic nerve, **~1 cm** in front of the optic canal · 3 roots · only parasympathetic fibres synapse · **~3%** of them go to the sphincter
Lacrimal: parasympathetic synapse in the **pterygopalatine** ganglion · greater petrosal + deep petrosal = **vidian** nerve
CN III divides **2–3 mm** behind the SOF · nucleus at the superior colliculus · CN IV at the inferior colliculus, the only dorsal exit · CN VI and CN VII nuclei in the pons
Through the common tendinous ring: **III (both divisions), VI, nasociliary** · above it: **IV, frontal, lacrimal**

Self-check

▼ Name the five branches that form the nasociliary nerve and one structure each one serves.

Infratrochlear (caruncle, canaliculi, lacrimal sac, medial lids) · anterior ethmoid (centre and tip of the nose, nasal mucosa, ethmoid sinuses) · posterior ethmoid (ethmoid and sphenoid sinuses) · long ciliary nerves (cornea, anterior segment) · short ciliary nerves via the sensory root of the ciliary ganglion (cornea, iris, ciliary body). § TB 13 · p. 208–210 · Handout 10 · p. 1–2

▼ What is Hutchinson sign, and why does it work?

Herpes zoster on the tip of the nose predicts eye involvement: the tip (anterior ethmoid) and the globe (ciliary nerves) are both supplied by the nasociliary nerve. § TB 13 · p. 210

▼ Which nerves form the frontal nerve, what do they supply, and where does it leave the orbit?

Supratrochlear (forehead, medial upper lid) and supraorbital (forehead, scalp, upper lid). It runs between the levator and the periorbita and leaves through the superior orbital fissure above the common tendinous ring. § TB 13 · p. 211

▼ Trace a pain signal from the lower eyelid to the brainstem.

Infraorbital nerve → infraorbital foramen, canal and groove → inferior orbital fissure → maxillary nerve → pterygopalatine fossa → foramen rotundum → lateral wall of the cavernous sinus → trigeminal ganglion → pons (nucleus of the spinal tract for pain). Lateral lower lid: zygomaticofacial → zygomatic nerve → inferior orbital fissure → maxillary nerve. § TB 13 · p. 211–212

▼ Give the central, preganglionic and postganglionic parts of the sympathetic route to the iris dilator.

Central: hypothalamus → brainstem → upper spinal cord. Preganglionic: T1–T3 ventral root → sympathetic chain, over the lung apex → superior cervical ganglion (synapse). Postganglionic: carotid plexus on the internal carotid → cavernous sinus → V1 → nasociliary → long ciliary nerves → dilator. § TB 14 · p. 222, 235

▼ Name three sympathetic targets in and around the eye, the route of each from the cavernous sinus, and the effect.

Iris dilator via the long ciliary nerves: mydriasis. Choroidal vessels via the sympathetic root, through the ciliary ganglion, then the short ciliary nerves: vasoconstriction. Müller muscle via the superior division of CN III: wider palpebral fissure. (Also the lacrimal gland vessels via deep petrosal, vidian and zygomatic nerves: less secretion.) § TB 14 · p. 222, 226

▼ What are the three roots of the ciliary ganglion, and which fibres synapse there?

Parasympathetic (from the inferior division of CN III), sensory (to the nasociliary nerve), sympathetic (to the globe's vessels). Only the parasympathetic fibres synapse; the short ciliary nerves carry all three types. § TB 14 · p. 224

▼ Describe the parasympathetic route to the eye and what it does.

Edinger-Westphal preganglionic cells (midbrain) → CN III → inferior division → parasympathetic root → ciliary ganglion (synapse) → short ciliary nerves → ciliary muscle (accommodation, most fibres) and iris sphincter (miosis, ~3%). § TB 14 · p. 223–224

▼ Where do the parasympathetic fibres to the lacrimal gland synapse, and which nerves carry them before and after?

In the pterygopalatine (sphenopalatine) ganglion. Before: lacrimal nucleus (pons) → CN VII → geniculate ganglion (no synapse) → greater petrosal → vidian nerve. After: maxillary → zygomatic → zygomaticotemporal branch or a communicating branch to the lacrimal nerve → gland. § TB 14 · p. 225–226

▼ A patient has ptosis, the right eye turned out, cannot adduct, and the right pupil is dilated. Which nerve, which likely cause, and why the pupil?

Right CN III. Likely a compressive lesion, typically a posterior communicating artery aneurysm. The pupil fibres run on the surface of the nerve as it leaves the midbrain, so compression hits them first; ischaemic (diabetic) palsies usually spare the pupil. § TB 13 · p. 215 · Handout 10 · p. 8–9

▼ Compare the deviation and restriction of a CN IV and a CN VI palsy, and say why CN VI is so vulnerable.

CN IV: eye elevated in primary position, cannot depress in adduction, head tilted to the opposite shoulder. CN VI: esotropia, worse at distance, cannot abduct. CN VI has a long course with a sharp bend over the petrous ridge, so trauma and raised intracranial pressure stretch it. § TB 13 · p. 217–218 · Handout 10 · p. 10–11

▼ Trace CN VII to the orbicularis oculi and name which branch supplies which part of the muscle.

Facial nucleus (pons) → around the abducens nucleus → lower border of the pons → internal auditory meatus → facial canal → stylomastoid foramen → over the mandibular ramus → branches. Upper orbicularis: frontal branch of the temporal nerve; lateral lower orbicularis: zygomatic; medial lower orbicularis: buccal. Temporal, zygomatic and buccal branches also supply frontalis, procerus and corrugator. § TB 13 · p. 219

§ Sources for this chapter: Review slide "Cranial Nerves" and "Cranial Nerves (cont.)" objectives (with your typed notes) · Handout 10 "Cranial Nerve Innervation to Ocular Tissues" (with your notes) · Slides 10 (with your notes) · Textbook ch. 13, Cranial Nerve Innervation of Ocular Structures, pp. 208–220 · Textbook ch. 14, Autonomic Innervation of Ocular Structures, pp. 222–236 · Textbook ch. 2, pp. 10–24 (eyelid muscles, lagophthalmos).

The pupillary pathways

How light and a near target make the pupil constrict, and how to tell from the pupils where a lesion is.

WHAT THE EXAM ASKS

Four guideline slides, "Pupils" and "Pupils (continued)" ×3, list these objectives: § Review handout · p. 7–8

- Exam "Describe the afferent, central, and efferent structures involved in the pupil reflex" → §1
- Exam "Describe the characteristics of a relative afferent pupillary defect and how the pupils would respond when there is an afferent pupillary defect." → §2
- Exam "List the causes of a relative afferent pupillary defect and describe how these conditions would present clinically" → §2
- Exam "Describe the pathway for the near pupillary response and how that differs from the light pupillary pathway." → §3
- Exam "Describe the pupil characteristics when there is a lesion in the area of the central pupillary fibers." → §3
- Exam "Describe the pupillary responses and general characteristics when there is a lesion affecting any part of the parasympathetic pupillary pathway" → §4
- Exam "Describe the pupillary responses and general characteristics when there is a lesion affecting the sympathetic pupillary pathway" → §5
- Exam "Describe the pharmacologic testing used to diagnose Horner syndrome" → §5
- Exam "Describe the characteristics of physiologic anisocoria" → §6
- Exam "Describe how lesions in the afferent, central, and efferent pupillary pathways will affect the direct, consensual, and near responses, as well as normal and abnormal pupillary reactions that occur during the swinging flashlight test" → §7, the summary table

The general autonomic routes (sympathetic and parasympathetic supply of the eye, ciliary ganglion) are in chapter 10. This chapter adds only what the reflexes and lesions need.

1 · The pupillary light pathway

Light into one eye makes *both* pupils constrict. The fibres that carry this signal are called **pupillary fibres**, as distinct from the visual fibres that carry the image. § TB 14 · p. 229 Afferent = sensory, carrying information in; efferent = motor, carrying it away. Your notes

LIMB	STRUCTURES, IN ORDER	DETAIL TO KNOW

Afferent	retina → optic nerve → chiasm → optic tract → **brachium of the superior colliculus** → **pretectal (olivary) nucleus** of the midbrain	Signal starts in intrinsically photosensitive retinal ganglion cells, with input from rods and cones. Pupillary fibres run with the visual fibres (nasal fibres cross in the chiasm), then leave the *posterior third* of the optic tract before the LGN. Synapse in the pretectal nucleus. § Handout 11 · p. 1 · TB 14 · p. 229
Central	pretectal nucleus → **Edinger-Westphal preganglionic cells** of *both* sides	Fibres go about equally to both sides; the crossing ones travel in the **posterior commissure**. § Handout 11 · p. 1 · TB 14 · p. 229
Efferent	Edinger-Westphal cells → **oculomotor nerve (CN III)**, inferior division → **ciliary ganglion** (synapse) → **short ciliary nerves** → **iris sphincter**	The parasympathetic route of chapter 10. In CN III the pupil fibres lie **superficially** as the nerve leaves the midbrain, then move centrally and inferiorly in the orbit. § Handout 11 · p. 1 · TB 14 · p. 223–224, 229

retina → optic nerve → chiasm → optic tract → brachium of sup. colliculus → pretectal nucleus → posterior commissure (to both sides) → Edinger-Westphal cells → CN III → ciliary ganglion → short ciliary nerves → **sphincter: miosis**

§ Handout 11 · p. 1 · TB 14 · p. 229 · Fig. 14.10. The handout says "pretectal nucleus / pretectal area", the textbook "pretectal olivary nucleus": the same structure.

CLINICAL CORRELATE · DIRECT AND CONSENSUAL RESPONSE

Direct response: the illuminated eye constricts. **Consensual response**: the other eye constricts too. The consensual response exists because the pathway crosses **twice**: nasal fibres in the chiasm, and about half the fibres of each pretectal nucleus in the posterior commissure. § Handout 11 · p. 2 · TB 14 · p. 231

Label the light pathway

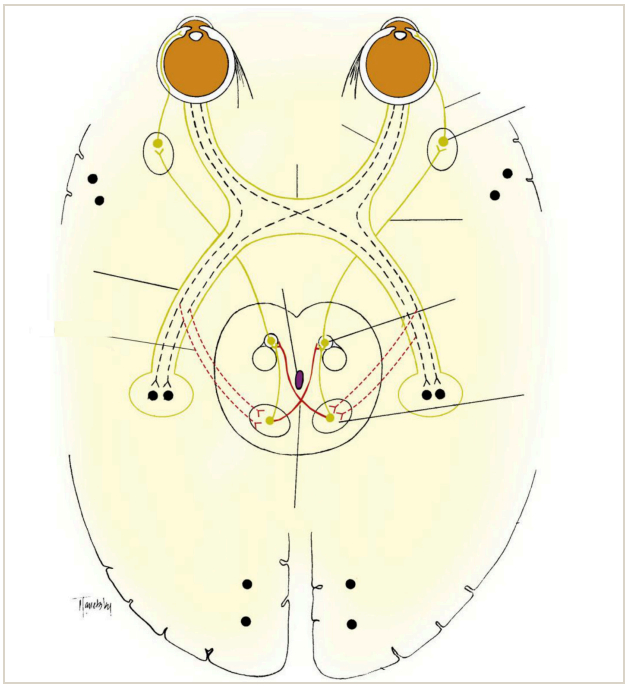
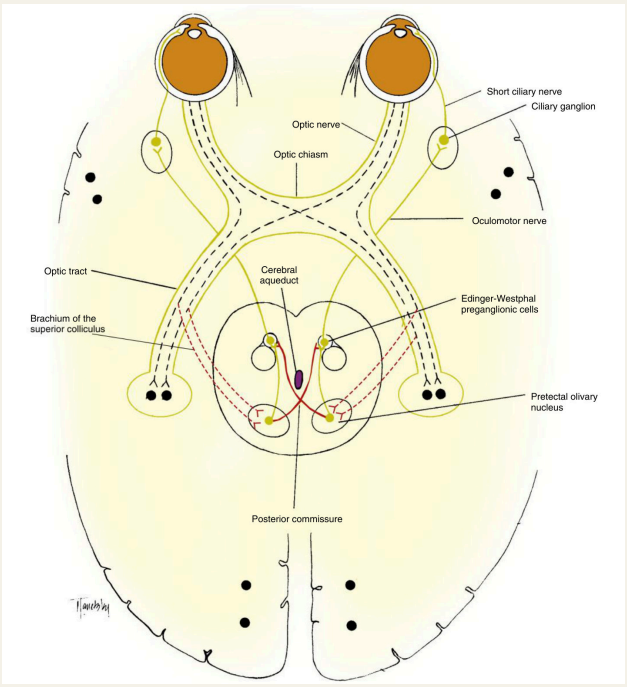


Fig. 14.10, labels removed. Name the 11 lines. Dotted lines: afferent; solid: efferent. § TB 14 · p. 231

▼ Answer: Fig. 14.10 with labels



Short ciliary nerve · ciliary ganglion · optic nerve · optic chiasm · oculomotor nerve · optic tract · cerebral aqueduct · Edinger-Westphal preganglionic cells · brachium of the superior colliculus · pretectal olivary nucleus · posterior commissure.

2 · Disruption of the afferent pathway: the RAPD

FEATURE	WHAT YOU SEE
Rule	An afferent lesion weakens the **direct response of the same eye** and the **consensual response of the other eye**. Light into the healthy eye gives normal responses in both. § Handout 11 · p. 2 · TB 14 · p. 231
Complete lesion	All fibres of one eye lost: light into that eye gives **no direct and no consensual** response. § TB 14 · p. 231–232
Partial lesion = RAPD	More often only some fibres are damaged; the defect shows only when one eye is *compared* with the other, hence **relative** afferent pupillary defect. § TB 14 · p. 232
Pupil size	Pupils stay equal, with no anisocoria: the weakened signal still reaches both Edinger-Westphal nuclei equally. Your notes § follows from TB 14 · p. 229, 232
Near response	Normal: the near pathway does not use the pretectal route (§3).

Swinging flashlight test

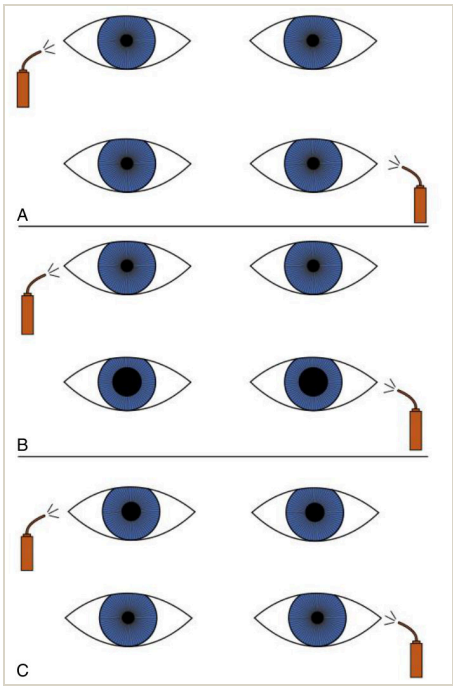


Fig. 14.11 **A** normal: equal constriction whichever eye is lit. **B** both pupils enlarge when the light moves to the left eye: severe RAPD OS. **C** both enlarge when the light moves to the right eye: mild RAPD OD. § TB 14 · p. 232 · Slides 11 · #3

1. Patient fixates a **distance target** (so no near response).
2. Swing the light from eye to eye, several times, rhythmically.
3. Light each pupil **equally**: same brightness, same time (about 2–3 s).
4. **Normal**: little or no change in pupil size as the light switches.
5. **RAPD**: when the light swings to the affected eye, **both pupils dilate** (or dilate quicker); when it swings back to the healthy eye, both constrict.

§ Handout 11 · p. 2 · TB 14 · p. 232. Brighter light shows subtle defects; record the light level, because the RAPD measured depends on it. § TB 14 · p. 232

Causes of an RAPD and how they present

SITE	WHICH EYE HAS THE RAPD	CLINICAL PICTURE

Retina	same eye	Must involve a large amount of retina to cause an RAPD. Your notes § Handout 11 · p. 3
Optic nerve	same eye	The **most common site**; e.g. **optic neuritis**: light into the affected eye → neither pupil constricts well; light into the other eye → both constrict normally (Fig. 14.12). A small nerve lesion can give a large RAPD. Your notes § Handout 11 · p. 3 · TB 14 · p. 232 · Slides 11 · #4
Chiasm	depends on the fibres lost	Listed by the textbook, not by the handout. § TB 14 · p. 232
Optic tract	**contralateral** eye	Behind the chiasm each tract carries fibres from both eyes, so an RAPD shows only if many more fibres of one eye are lost. The tract holds more crossed than uncrossed fibres, so a complete tract lesion gives an RAPD in the eye **opposite** the lesion. Visual field: homonymous hemianopia, the same side in both eyes (chapter 12). Your notes § Handout 11 · p. 3 · TB 14 · p. 232
Superior brachium	contralateral eye Your notes	Pupillary fibres only, after the visual fibres have left for the LGN, so an RAPD with a **normal visual field**. Your notes § Handout 11 · p. 3 · TB 14 · p. 232

CLINICAL CORRELATE · MEDIA OPACITIES DO NOT CAUSE AN RAPD

A cataract or other media opacity does **not** cause an RAPD in its own eye. A dense cataract may even cause an RAPD in the *other* eye: light scattered across the retina probably gives a stronger response on the cataract side. An RAPD in a cataract eye therefore means a retinal or optic nerve problem as well: you cannot blame the poor vision on the cataract. § Handout 11 · p. 3 · TB 14 · p. 232 Lecture: "important to know!"

3 · Disruption in the central pathway, and the near response

MIDBRAIN LESION OF …	RESULT
one pretectal nucleus	Possibly **no pupil defect**: the other pretectal nucleus still supplies both Edinger-Westphal nuclei. § TB 14 · p. 232
fibres pretectal → Edinger-Westphal (dorsal midbrain)	Both pupils show a **poor direct and consensual** response to light but a **brisk near** response: **light-near dissociation**. § Handout 11 · p. 4 · TB 14 · p. 232

The near response (accommodation-convergence reaction)

Not a true reflex but a **synkinesis** of three actions when a target comes near along the midline: **convergence** (medial recti), **accommodation** (ciliary muscle) and **miosis** (sphincter, for depth of field). Each can occur without

the others: plus lenses → miosis + convergence without accommodation; base-in prisms → miosis + accommodation without convergence; bright light → miosis alone. § TB 14 · p. 228–229

retina → optic nerve, chiasm, tract → LGN → optic radiations → striate cortex → frontal eye fields → oculomotor nucleus (medial rectus subnucleus) + Edinger-Westphal cells → CN III → **medial rectus · ciliary muscle · sphincter**

§ Handout 11 · p. 4 (afferent · CNS · efferent) · TB 14 · p. 229 · Fig. 14.9. The parasympathetic part runs via the ciliary ganglion and short ciliary nerves, as for light.

	LIGHT RESPONSE	NEAR RESPONSE
Afferent	leaves the optic tract **before** the LGN → brachium → pretectal nucleus	full visual pathway: LGN → optic radiations → **striate cortex**
Central	pretectal nucleus → Edinger-Westphal, both sides, through the dorsal midbrain (posterior commissure)	striate cortex → **frontal eye fields** → Edinger-Westphal + medial rectus subnucleus; reaches Edinger-Westphal from a more **ventral** direction, **bypassing** the pretectal route
Efferent	CN III → ciliary ganglion → short ciliary nerves → sphincter	the same, plus the ciliary muscle and, through CN III, the medial rectus
Result	miosis only	miosis + accommodation + convergence

§ Handout 11 · p. 4 · TB 14 · p. 229, 232

Label the near pathway

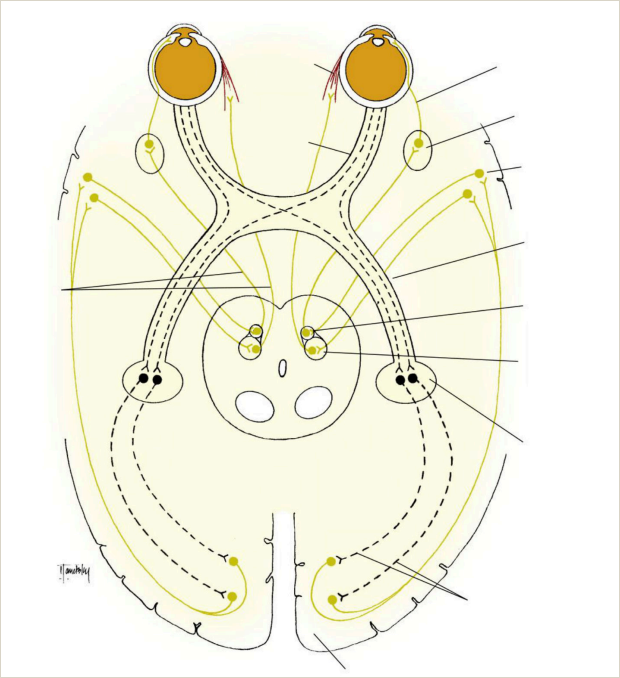
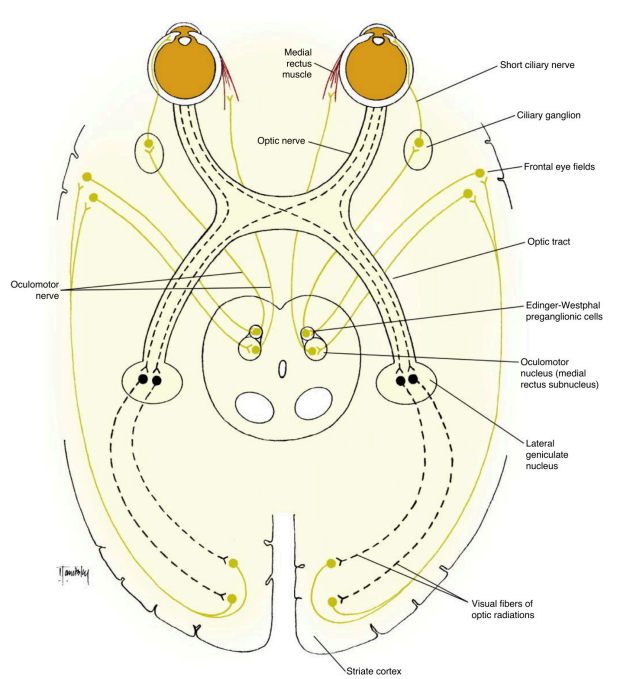


Fig. 14.9, labels removed. Name the 12 lines. Dotted: visual fibres to the cortex; solid: cortex → frontal eye fields → oculomotor nucleus → muscles. § TB 14 · p. 230

▼ Answer: Fig. 14.9 with labels



Medial rectus muscle · short ciliary nerve · ciliary ganglion · optic nerve · frontal eye fields · optic tract · oculomotor nerve · Edinger-Westphal preganglionic cells · oculomotor nucleus (medial rectus subnucleus) · lateral geniculate nucleus · visual fibres of the optic radiations · striate cortex.

Light-near dissociation

The near response is retained and **exceeds the best light response**; looking back to distance, the pupils redilate briskly. A near response greater than the light response is **always pathological**. § TB 14 · p. 232 · Fig. 14.13 · Slides 11 · #7–8

CLINICAL CORRELATES · TWO BILATERAL LIGHT-NEAR DISSOCIATIONS

	ARGYLL ROBERTSON PUPIL	DORSAL MIDBRAIN SYNDROME
Pupils	**very small** without a light stimulus; irregular (TB)	**normal to mid-dilated** (handout) · "fixed, mid-dilated" (TB)
Light / near	bilateral light-near dissociation	
Cause	most commonly **neurosyphilis**; also reported with diabetes, multiple sclerosis, stroke, Wernicke encephalopathy	most commonly a **pineal gland** abnormality: a tumour pressing on the back of the midbrain and the crossing pupillary fibres. Slide 9: "Parinaud's syndrome – pineal GBM"

§ Handout 11 · p. 6 · TB 14 · p. 233 · Slides 11 · #9. Other causes of light-near dissociation (TB): blindness from bilateral afferent damage, aberrant regeneration of medial rectus fibres to the sphincter, tonic pupil.

4 · Disruption of the efferent parasympathetic pathway

A parasympathetic lesion makes the **affected eye** show poor **direct, consensual and near** responses: the sphincter cannot be driven. Result: **ipsilateral mydriasis** and **anisocoria worse in bright light** (the healthy pupil constricts, the affected one cannot). The **larger** pupil is the abnormal one. § Handout 11 · p. 7 · TB 14 · p. 233

CAUSE	SITE	CHARACTERISTICS
CN III palsy	oculomotor nucleus or nerve	Large pupil, poor light and near responses, **plus** other CN III signs: ptosis (levator), weak superior, medial and inferior rectus and inferior oblique: check motility. Pupil fibres lie superficially, so **compressive** lesions (aneurysm) involve the pupil early, while **ischaemic** ones (diabetes) often spare it. A third nerve palsy with a dilated pupil is highly suspicious of a compressive lesion. § Handout 11 · p. 7 · TB 14 · p. 229, 233
Tonic pupil	**ciliary ganglion** or **short ciliary nerves** (postganglionic)	Dilated pupil, poor light response, loss of accommodation; near response **kept but slow**, with slow, long redilation (tonic). **Segmental constriction**: only sections of the sphincter work, so the border is irregular (Fig. 14.15, Slides 11 · #10–11). **Decreased corneal sensitivity** (corneal sensory fibres pass through the ganglion). **Cholinergic denervation supersensitivity** of the sphincter. Late: the pupil becomes small. § Handout 11 · p. 7–8 · TB 14 · p. 233–235
· causes		(1) autonomic dysfunction, e.g. **diabetes**; (2) orbital mass, trauma or surgery; (3) idiopathic = **Adie tonic pupil**: typically a woman of 20–40, 90% with diminished tendon reflexes. § Handout 11 · p. 7 · TB 14 · p. 235
Pharmacologically dilated pupil	sphincter receptors blocked by a drug	Recent fixed, dilated pupil; ask about the job (pharmacist, nurse, farmer, exterminator). Some "get the red out" drops dilate by stimulating the dilator. § Handout 11 · p. 8 · TB 14 · p. 235

Pilocarpine testing

Pilocarpine is a **direct-acting cholinergic agonist**: it acts on the sphincter itself, not on the nerve. § TB 14 · p. 228

DROP	NORMAL PUPIL	TONIC PUPIL	CN III PALSY	PHARMACOLOGIC DILATION
0.125% pilocarpine (dilute)	little or no effect	**constricts much more** than the normal eye (supersensitive)	—	no constriction
1% pilocarpine	constricts	constricts	constricts	**no constriction**: the receptors are blocked

§ Handout 11 · p. 8 (1% for the pharmacologically dilated pupil) · TB 14 · p. 234–235 (0.125% for tonic; drug-dilated pupil responds to neither) · Fig. 14.16. The handout lists only the 1% test; the 0.125% test is the textbook's.

5 · Disruption of the efferent sympathetic pathway: Horner syndrome

SIGN	WHY
Miosis	No dilator tone against the sphincter. § Handout 11 · p. 9 · TB 14 · p. 235
Anisocoria greater in dim light	In the dark the healthy pupil dilates, the Horner pupil hardly; the **smaller** pupil is the abnormal one. § Handout 11 · p. 9 · TB 14 · p. 233, 235
Normal reaction to light	The parasympathetic pathway is intact: brisk direct, consensual and near responses; swinging flashlight test normal. § Handout 11 · p. 9 · TB 14 · p. 235
Dilation lag	Normal pupil dilates within **5 s** of lights off (dilator active + sphincter relaxed). Horner pupil dilates only by sphincter relaxation: **10–20 s**. So anisocoria is greater at 5 s than at 15 s (Fig. 14.19). Distinguishes Horner from physiologic anisocoria. § Handout 11 · p. 9 · TB 14 · p. 236
Ptosis	Mild: loss of the smooth **Müller muscle**, not the levator. The lower lid rises slightly, so the fissure narrows (pseudo-enophthalmos). § Handout 11 · p. 9 · TB 14 · p. 235
Facial anhidrosis	No sweating on that side, with central and preganglionic lesions. Postganglionic: only the **medial forehead**, because most facial sweat fibres leave with the external carotid artery. § Handout 11 · p. 9 · TB 14 · p. 222, 235

§ Congenital Horner also gives iris heterochromia (Fig. 14.17, Slides 11 · #15). § TB 14 · p. 236

The three-neuron pathway and where it breaks

hypothalamus → brainstem → cervical spinal cord ⇒ apex of the lung → superior cervical ganglion ⇒ internal carotid artery → cavernous sinus → nasociliary nerve → long ciliary nerves → **iris dilator**

§ Handout 11 · p. 10 (⇒ marks a synapse) · TB 14 · p. 222, 235 · Fig. 14.18. Müller muscle is reached via the oculomotor nerve's superior division (chapter 10).

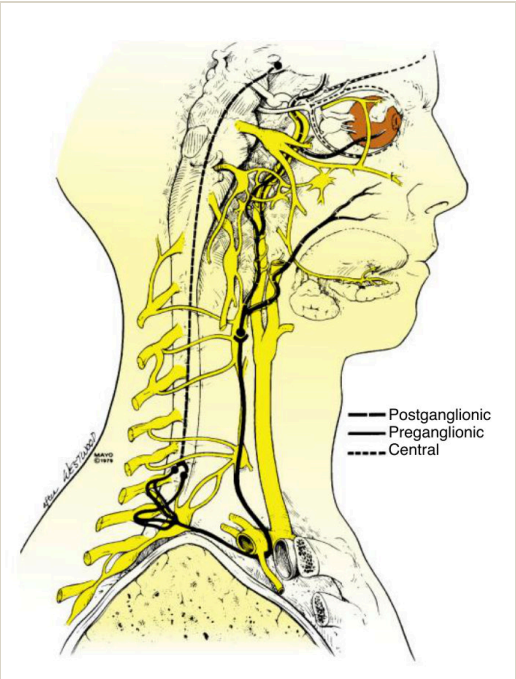


Fig. 14.18 Central neuron from the hypothalamus down to the cord; preganglionic into the chest and up to the superior cervical ganglion; postganglionic along the internal carotid to the cavernous sinus and orbit. § TB 14 · p. 236 · Slides 11 · #16
§ Handout 11 · p. 10 · TB 14 · p. 235

NEURON	CAUSES
1st · central	stroke, multiple sclerosis
2nd · preganglionic	thoracic injury or surgery, metastatic disease in the chest (the fibres cross the lung apex and loop around the subclavian artery)
3rd · postganglionic	skull base fracture, internal carotid injury; **painful Horner = internal carotid dissection**: an emergency · Horner + CN VI paresis = cavernous sinus lesion

SOURCES DIFFER · WHERE THE CENTRAL NEURON ENDS

The handout and the textbook's Horner section say the first neuron ends in the **cervical spinal cord**. The textbook's general pathway places the ocular preganglionic cells in the lateral horn of **T1–T3**, leaving in the first three thoracic nerves (chapter 10). Know both: "cervical cord" for the Horner chain, T1–T3 for the segments. § Handout 11 · p. 10 · TB 14 · p. 222, 235

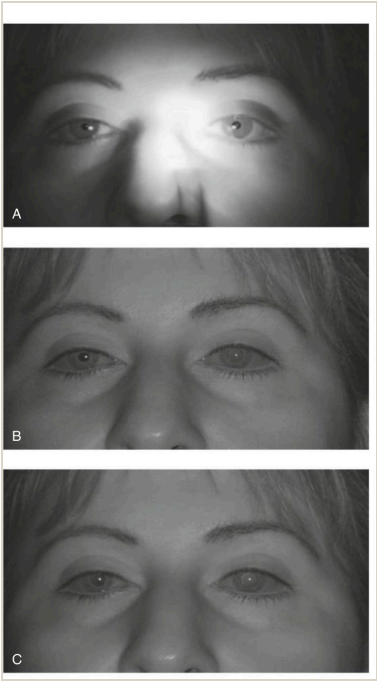


Fig. 14.19 Right Horner, dilation lag. **A** bright light · **B** 5 s after the light is removed: large anisocoria · **C** 15 s: smaller, as the right pupil slowly catches up. § TB 14 · p. 237 · Slides 11 · #14

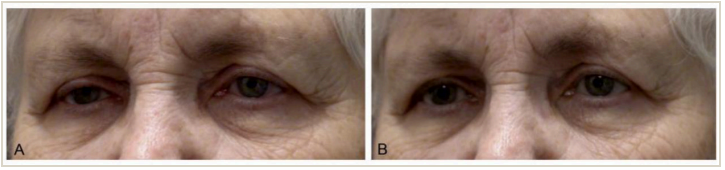


Fig. 14.21 Right Horner. **A** before drops: right miosis and ptosis · **B** after 1% apraclonidine in both eyes: the right pupil dilates and the ptosis improves. § TB 14 · p. 237 · Slides 11 · #17

Pharmacologic testing for Horner syndrome

DRUG	TYPE	NORMAL EYE	HORNER EYE	TELLS YOU
Apraclonidine 0.5 or 1% Lecture	direct-acting alpha-adrenergic agonist: mostly alpha 2, weak alpha 1	no change or slight miosis	**dilates**, and the ptosis improves: denervation up-regulates alpha 1 receptors on the dilator, which become supersensitive. Needs time: usually 1–2 weeks (reported from 36 h)	Horner: yes / no
Cocaine 2–10% TB	indirect agonist: blocks norepinephrine reuptake	dilates in 30–60 min	little or no dilation (no norepinephrine being released)	Horner: yes / no
Hydroxyamphetamine 1% TB	indirect agonist: releases stored norepinephrine	dilates	**preganglionic** lesion: dilates (3rd neuron intact) · **postganglionic**: no dilation	where the lesion is; give it 24–48 h after cocaine

§ Handout 11 · p. 11 (apraclonidine only) · TB 14 · p. 228, 236–237 · Fig. 14.20, 14.21. The lecture taught apraclonidine; the textbook adds cocaine and hydroxyamphetamine.

6 · Physiologic anisocoria

FEATURE	PHYSIOLOGIC (SIMPLE, BENIGN) ANISOCORIA
How common	about **20%** of people
Size	difference usually **< 1 mm**; more apparent in dim than bright light
Side	may switch sides; fluctuates, e.g. with stress Your notes
Reactions	both pupils round, react well to all stimuli, **dilate equally in the dark: no dilation lag**
Why	asymmetric sympathetic/parasympathetic balance; possibly asymmetric supranuclear inhibition of the Edinger-Westphal cells

§ Handout 11 · p. 11 · TB 14 · p. 233

Which pupil is abnormal? Sorting anisocoria

Anisocoria greater in dim light? The **smaller** pupil is abnormal.

> dilation lag, ptosis, anhidrosis, apraclonidine dilates it → **Horner syndrome**
>
> < 1 mm, no dilation lag, equal dilation in the dark → **physiologic anisocoria**

Anisocoria greater in bright light? The **larger** pupil is abnormal: a parasympathetic defect.

> ptosis + motility problems → **CN III palsy**
>
> slow tonic near response, segmental constriction, 0.125% pilocarpine constricts → **tonic pupil**
>
> no response even to 1% pilocarpine → **pharmacologically dilated pupil**

§ TB 14 · p. 233–236 · Handout 11 · p. 7–11

7 · Putting it together: lesion → responses

Example lesions on the **right** (OD). "Light OD" = light shone into the right eye.

LESION	LIGHT OD: DIRECT OD / CONSENSUAL OS	LIGHT OS: DIRECT OS / CONSENSUAL OD	NEAR	SWI FLA
Normal	brisk / brisk	brisk / brisk	normal	no c swit
Afferent, partial: R optic nerve or retina	**weak / weak**	normal / normal	normal	both whe light OD: **OD**

Afferent, complete: R optic nerve	**none / none**	normal / normal	normal	both fully
R optic tract or R superior brachium	normal / normal	**weaker / weaker**	normal	**RAP** (con
Central: dorsal midbrain (pretectal → Edinger-Westphal fibres)	**poor / poor**	**poor / poor**	**brisk** in both: light-near dissociation	no r diffe
Efferent parasympathetic: R CN III	**poor** / normal	normal / **poor**	**poor OD**	not u an R R pu hard mov pupi the a inpu
Efferent parasympathetic: R tonic pupil	**poor** / normal	normal / **poor**	OD **slow**, tonic, slow redilation	as ab
R pharmacologic dilation	**none** / normal	normal / **none**	**none OD**	as ab
Efferent sympathetic: R Horner	brisk / brisk	brisk / brisk	normal	norm
Physiologic anisocoria	brisk / brisk	brisk / brisk	normal	norm

§ Built from Handout 11 · p. 2–11 · TB 14 · p. 231–236. The swinging-flashlight column for efferent lesions is reasoned from the rules above, not stated in either source: an efferent lesion does not cause an RAPD.

ONLY IN YOUR NOTES · CHECK WITH THE LECTURER

- "Hippus should be the same in both eyes (AI check)". The textbook defines **hippus** as the normal, constant, rhythmic change of pupil size that reflects the autonomic balance, independent of light; it does not say whether it is equal in both eyes. During sleep the pupils are small (parasympathetic predominates). § TB 14 · p. 225
- Retinal detachment surgery is "one of the most common" causes of tonic pupil. The textbook says only "orbital surgeries". Your notes
- Pancoast tumour as a preganglionic Horner cause ("starting in liver"). The textbook says only "metastatic disease involving the chest": the "liver" part is not in the sources, so check it. Your notes
- "Vasculopathic" as a postganglionic cause; patients rarely notice the anhidrosis themselves; a Horner with headache or neck pain should go to the emergency room, "hours away from a stroke". Your notes (the emergency is in TB 14 · p. 235)
- Scopolamine patches (motion sickness) and farm chemicals as causes of a pharmacologically dilated pupil. The textbook names farmers, crop dusters, exterminators. Your notes
- "Anisocoria is a sign of Horner syndrome, but you do not have to have it." Your notes
- Review notes: "dilation lag is normally not tested; do it after the swinging flashlight test by taking the light away", then compare at 5 s and 15 s. Your notes

NOT TAUGHT FROM TEXTBOOK CHAPTER 14

Left out because no objective or handout point needs it: the autonomic supply of the lacrimal gland and of the choroidal and conjunctival vessels, the corneal reflex (Fig. 14.4), neurotransmitter and receptor details beyond what the drug tests need, drug-induced mydriasis for fundus examination, the inhibitory pathway to the dilator through the accessory optic system, theories of why the tonic pupil keeps a near response, and the 1% phenylephrine alternative for localising Horner. The general sympathetic and parasympathetic routes and the ciliary ganglion are taught in chapter 10.

Key facts

2 crossings make the consensual response: nasal fibres in the chiasm · about half the pretectal fibres in the posterior commissure
Pupillary fibres leave the optic tract's **posterior third**, before the LGN
Swinging flashlight: about **2–3 s** per eye, distance fixation, equal light
Normal pupil dilates within **5 s** in the dark; Horner pupil takes **10–20 s**; compare at 5 s and 15 s
Physiologic anisocoria: about **20%** of people, **< 1 mm**, no dilation lag
Pilocarpine **0.125%** → tonic pupil constricts · **1%** → every pupil except a drug-dilated one constricts
Apraclonidine **0.5–1%**: Horner pupil dilates; needs about 1–2 weeks of denervation · Cocaine **2–10%**, 30–60 min · Hydroxyamphetamine **1%**, 24–48 h after cocaine
Adie tonic pupil: woman **20–40** years, **90%** with diminished tendon reflexes
Sympathetic chain: **3** neurons: central (hypothalamus → cord) · preganglionic (cord → superior cervical ganglion) · postganglionic (→ dilator)

Self-check

▼ Trace the pupillary light pathway from the retina to the sphincter, naming the afferent, central and efferent parts.

Afferent: retina → optic nerve → chiasm → optic tract → brachium of the superior colliculus → pretectal (olivary) nucleus. Central: pretectal nucleus → Edinger-Westphal preganglionic cells of both sides (crossing in the posterior commissure). Efferent: CN III (inferior division) → ciliary ganglion (synapse) → short ciliary nerves → iris sphincter. § Handout 11 · p. 1 · TB 14 · p. 229

▼ Why does light into one eye constrict the other pupil?

Two crossings: nasal retinal fibres in the chiasm, and about half the fibres from each pretectal nucleus in the posterior commissure, so each Edinger-Westphal nucleus receives input from both eyes. § TB 14 · p. 231

▼ Right optic neuritis: what happens to each pupil when light is shone into the right eye, then into the left, and during the swinging flashlight test?

Light OD: weak direct OD and weak consensual OS. Light OS: normal direct and consensual. Swinging: both pupils dilate when the light moves to OD, so there is an RAPD OD. Pupils are equal in size and the near response is normal. § TB 14 · p. 231–232 · Fig. 14.12

▼ List the sites that can cause an RAPD. Which eye has the RAPD with a complete right optic tract lesion, and why?

Retina, optic nerve (the most common), chiasm (TB), optic tract, superior brachium. Right tract: **left** eye RAPD, because the tract carries more crossed (contralateral) than uncrossed fibres. § Handout 11 · p. 3 · TB 14 · p. 232

▼ A patient has a dense cataract in the right eye and an RAPD in the right eye. Can the cataract explain it?

No. Media opacities do not cause an RAPD in their own eye (a dense cataract may even cause one in the other eye), so look for retinal or optic nerve disease. § Handout 11 · p. 3 · TB 14 · p. 232

▼ How does the near pupillary pathway differ from the light pathway, and why does that allow light-near dissociation?

Near: the full visual pathway to the striate cortex → frontal eye fields → Edinger-Westphal and medial rectus subnucleus, approaching Edinger-Westphal ventrally. Light: optic tract → brachium → pretectal nucleus → Edinger-Westphal through the dorsal midbrain. A dorsal midbrain lesion cuts the pretectal fibres but spares the ventral near fibres: poor light response, brisk near response. § Handout 11 · p. 4 · TB 14 · p. 229, 232

▼ Compare Argyll Robertson pupils with dorsal midbrain syndrome.

Both: bilateral light-near dissociation. Argyll Robertson: very small (irregular) pupils, most commonly neurosyphilis. Dorsal midbrain: normal to mid-dilated pupils, most commonly a pineal gland abnormality (Parinaud syndrome on the slide). § Handout 11 · p. 6 · TB 14 · p. 233 · Slides 11 · #9

▼ A left pupil is large, reacts poorly to light and slowly to near, and the iris border constricts only in segments. Name the condition, its site, and the drop test.

Left tonic pupil: damage to the ciliary ganglion or short ciliary nerves (diabetes, orbital trauma, surgery or mass, or idiopathic Adie). 0.125% pilocarpine constricts it much more than the normal pupil (cholinergic denervation supersensitivity). Corneal sensitivity is often reduced. § Handout 11 · p. 7–8 · TB 14 · p. 233–235

▼ Why is a dilated pupil with a third nerve palsy more worrying than a pupil-sparing one?

The pupil fibres lie superficially in CN III as it leaves the midbrain, so compressive lesions such as an aneurysm involve them; ischaemic (diabetic) palsies often spare them. § TB 14 · p. 229, 233

▼ Give the six features of Horner syndrome, and the three neurons of the pathway with one cause each.

Miosis, anisocoria greater in dim light, normal light reaction, dilation lag, ptosis (Müller), facial anhidrosis. Central (hypothalamus → brainstem → cervical cord): stroke, MS. Preganglionic (cord → lung apex → superior cervical ganglion): thoracic surgery or injury, chest metastases. Postganglionic (internal carotid → cavernous sinus → nasociliary → long ciliary → dilator): carotid dissection (painful, emergency), skull base fracture. § Handout 11 · p. 9–10 · TB 14 · p. 235

▼ What does apraclonidine do in a normal eye and in a Horner eye, and why?

Normal: no change or slight miosis (mostly alpha 2 action). Horner: the pupil dilates and the ptosis improves, because denervation up-regulates alpha 1 receptors on the dilator, which respond to apraclonidine's weak alpha 1 action. It needs about 1–2 weeks to become positive. § Handout 11 · p. 11 · TB 14 · p. 236

▼ How do you tell physiologic anisocoria from Horner syndrome?

Both show a smaller pupil, worse in dim light. Physiologic: about 20% of people, < 1 mm, may switch sides, no dilation lag, no ptosis; apraclonidine and cocaine behave normally. Horner: dilation lag (more anisocoria at 5 s than 15 s), ptosis, anhidrosis, positive drop tests. § Handout 11 · p. 9–11 · TB 14 · p. 233, 236

§ Sources for this chapter: Handout 11 "Pupillary Pathway" (with your notes) · Slides 11 "Pupillary Pathways" · Review handout p. 7–8 (exam guidelines, with your typed notes) · Textbook ch. 14, Autonomic Innervation of Ocular Structures, pp. 222–237.

The visual pathway

From the retina to the occipital cortex: where the fibres run at every level, and how to read the lesion site from the visual field.

WHAT THE EXAM ASKS

Two guideline slides, "Visual Pathway" and "Visual Pathway (cont.)", list these objectives: § Review handout · p. 8

- Exam "Describe how areas of the visual field are represented on the retina" → §8
- Exam "Describe how the visual fibers exit the eye through the optic disc and how this would affect the visual field" → §9, §11
- Exam "Describe where the visual fibers are located in the optic nerve" → §9
- Exam "Describe how the fibers travel through the chiasm and how this affects the visual field" → §3, §9, §11
- Exam "Describe the location of inferior, superior, and macular fibers in the optic tract, lateral geniculate nucleus, optic radiations, and occipital lobe." → §10, the summary table
- Exam "Describe the location of the macular and peripheral vision representation on the visual cortex and how this affects the visual field." → §7, §11
- Exam "Correlate lesions in areas of the visual pathway with the visual field and vice versa" → §11, the lesion table and the three questions

Your note on the first page of the handout: "one of the most important one!" Your notes Retina and optic disc anatomy are in chapter 7; the pupillary fibres that leave the tract are in chapter 11.

1 · Overview: the retinogeniculocortical pathway

The name gives the three stations: **retino-** (retina) · **geniculo-** (lateral geniculate nucleus) · **cortical** (visual cortex). § Handout 12 · p. 1 Photoreceptor → bipolar (and amacrine) cell → ganglion cell: all these synapses lie in the retina. The ganglion cell axons are the fibres of the optic nerve, chiasm and tract; the next synapse is in the LGN. § TB 15 · p. 239

(optical system) → retina: photoreceptor → bipolar → ganglion cell → optic nerve → chiasm: nasal fibres cross → optic tract → LGN (synapse) → optic radiations → **visual cortex, occipital lobe**

§ Handout 12 · p. 1–2 · TB 15 · p. 239 · Fig. 15.1

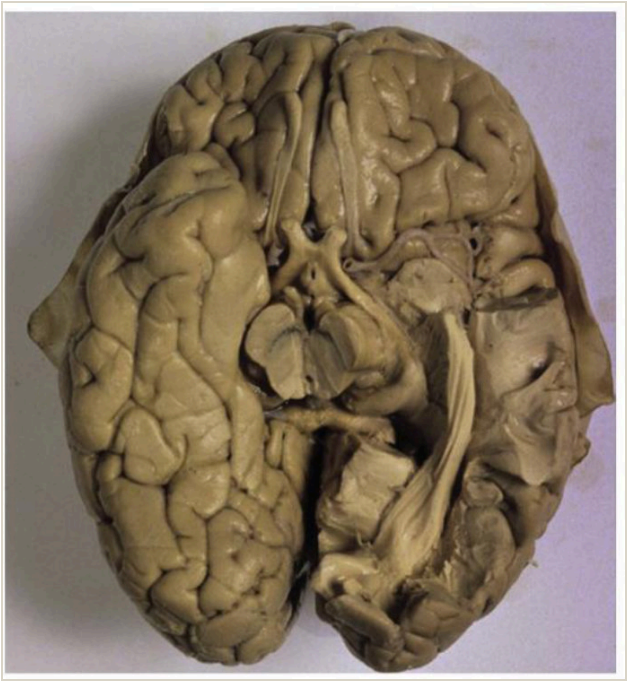


Fig. 15.8 The pathway from the optic nerves to the occipital lobe, seen from below. Find nerve, chiasm, tract and the radiations fanning back. § TB 15 · p. 244 · Slides 12 · #2, #12, #14

2 · Optic nerve (cranial nerve II)

Visual fibres: about **90**% end in the LGN; the other ~10% go to centres for the pupil, circadian rhythm and orienting head and eyes. 1 to 2.22 million fibres. Length **5–6 cm**, in four segments. § TB 15 · p. 240

SEGMENT	LENGTH	WHAT TO KNOW
1 · Intraocular	0.7–1 mm	= optic disc / optic nerve head. Ganglion cell axons from all over the retina **converge on the disc** and turn 90° to leave the globe. Prelaminar and laminar parts, named by the lamina cribrosa. § Handout 12 · p. 3 · TB 15 · p. 240 **Physiologic cup**: a depression in the disc surface, size varies widely. § TB 8 · p. 132 It enlarges when nerve fibre tissue is lost in glaucoma. § TB 4 · p. 54 **Blind spot**: the disc has no photoreceptors, so it gives the physiologic blind spot. § TB 8 · p. 132
2 · Intraorbital	30 mm	Longer than the globe-to-apex distance: a slight S-curve lets the eye move without stretching it. Surrounded by the rectus muscles; the **superior and medial rectus** sheaths adhere to it (pain on eye movement in optic neuritis). Three meningeal sheaths: **dura** (outer) · **arachnoid** · **pia** (inner); the subarachnoid space holds CSF and is continuous with the intracranial one. § Handout 12 · p. 4 · TB 15 · p. 240 **Myelination**: behind the lamina cribrosa, by **oligodendrocytes** (the CNS myelin cells); the lamina is thought to bar them, so the retina is unmyelinated. Diameter 1.5–1.8 mm at the retina → 3 mm after leaving the globe → 4–5 mm with sheaths. § Handout 12 · p. 4 · TB 15 · p. 240

3 · Intracanalicular	6–10 mm	In the **optic canal**, between the lesser wing and the body of the sphenoid (chapter 8). The ophthalmic artery enters the nerve's dural sheath here; the arachnoid merges with the pia in the canal. § Handout 12 · p. 5 · TB 10 · p. 163 · TB 15 · p. 240
4 · Intracranial	10–16 mm	Only the pia continues, through the subarachnoid space to the chiasm. Above: anterior perforated substance, olfactory tract root, anterior cerebral artery · medial: sphenoid sinus (thin bone) · below, then lateral: internal carotid artery. § Handout 12 · p. 5 · TB 15 · p. 240

CLINICAL CORRELATE · PAPILLEDEMA

Raised intracranial CSF pressure is passed into the optic nerve sheaths, compresses the nerve and stops axoplasmic flow in the prelaminar fibres: **bilateral swelling of the optic disc**. Disc swelling caused by raised intracranial pressure is called **papilledema** (Fig. 15.3). § Handout 12 · p. 4 · TB 15 · p. 240 · Slides 12 · #9 Your notes: the disc margins blur because of the fluid pressure. Your notes

3 · Optic chiasm

Roughly rectangular, ~15 mm wide × 8 mm front-to-back × 4 mm high, in the subarachnoid space bathed in CSF. § TB 15 · p. 240 "Important to know what is around it" Your notes: the pituitary gland below it is the commonest source of a bitemporal defect (§11). § TB 15 · p. 250

DIRECTION	NEIGHBOUR
Around	lies **within the circle of Willis** (a common site of aneurysms)
Anterior and superior	anterior cerebral and anterior communicating arteries
Each lateral side	internal carotid artery
Above	hypothalamus and the floor of the third ventricle
Below (~1 cm)	**pituitary gland** in the sella turcica
Immediately posterior	infundibulum (pituitary stalk)

§ Handout 12 · p. 6 · TB 15 · p. 241 · Figs. 15.4, 15.5 (Slides 12 · #10). Position over the sella: directly above in 75%; *prefixed* (short nerves, over the front of the pituitary) or *postfixed* (long nerves, over its back). TB 15 · p. 241–242

Arrangement of fibres: **nasal fibres cross** (decussate) to the contralateral tract; **temporal fibres stay ipsilateral**. Crossed : uncrossed ≈ **53 : 47**. Result: everything in the *left* visual field ends up in the *right* brain, and the right field in the left brain. § Handout 12 · p. 6 · TB 15 · p. 247, 249 Fibre details are in §9.

4 · Optic tract

A flattened band (~3.5 mm high, 5.1 mm long) from the posterolateral corner of the chiasm to the LGN, along the cerebral peduncle. Behind the chiasm the pathway runs on both sides of the brain. § TB 15 · p. 242–243

DESTINATION	FUNCTION
LGN (most fibres)	conscious vision: the visual pathway proper § Handout 12 · p. 7 · TB 15 · p. 242
Superior colliculus	visual orientation, foveation, control of saccades (does not analyse for perception) § TB 15 · p. 244 · your notes: "reflex head + saccadic movements" Your notes
Suprachiasmatic nucleus (hypothalamus)	synchronises the **circadian rhythm** § TB 15 · p. 248
Pretectum (pretectal nucleus)	afferent limb of the **pupillary light reflex**, via the superior brachium (chapter 11) § TB 15 · p. 242

§ The handout lists all three side branches as leaving the optic tract (Handout 12 · p. 7). The textbook says the same for the tract (TB 15 · p. 242) but also describes the suprachiasmatic fibres as leaving the back of the *chiasm* (TB 15 · p. 248). Know both.

5 · Lateral geniculate nucleus (LGN)

On the dorsolateral thalamus; all senses except smell pass through the thalamus. The ganglion cell axons end here, and most LGN output goes to the visual cortex. It is *not* a simple relay: it also gets cortical feedback and regulates what is sent on. § TB 15 · p. 243

PATHWAY	LGN LAYERS	CELLS	CARRIES
M · magnocellular	**1, 2** (inferior)	large	movement detection, low-spatial-frequency contrast § TB 15 · p. 244 · "flicker of light" Your notes
P · parvocellular	**3, 4, 5, 6**	medium	colour, high-spatial-frequency contrast § TB 15 · p. 244 · "visual acuity" Your notes
K · koniocellular	a thin layer below each of the six	small	input from short-wavelength (blue) cones § TB 15 · p. 243 · "blue and yellow colour" Your notes

§ Handout 12 · p. 8 names only P and M cells; the koniocellular layers are from the textbook and your notes.

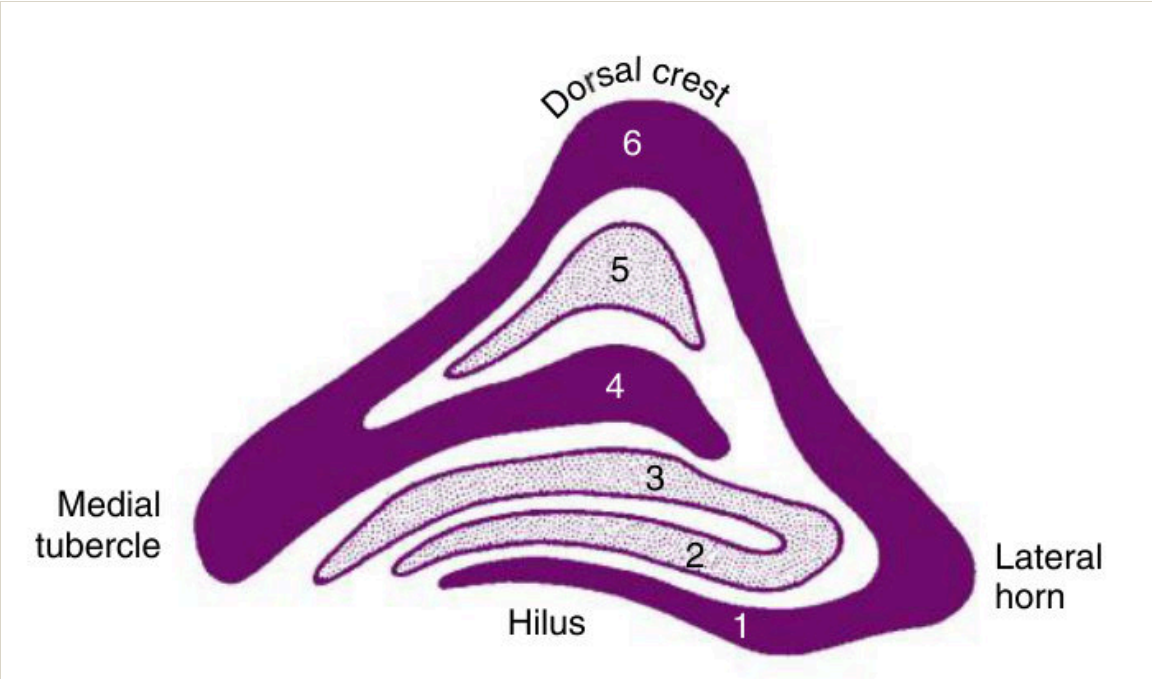


Fig. 15.6 Right LGN. **Crossed** (contralateral nasal) fibres end in layers **1, 4, 6**; **uncrossed** (ipsilateral temporal) in **2, 3, 5**. Each layer gets one eye only. § TB 15 · p. 243, 248

Each layer holds a **retinotopic map** of the contralateral hemifield, and the maps are stacked: a line through all six layers hits cells for the *same point* of the field. The fibres leave through the posterior limb of the internal capsule as the optic radiations. § TB 15 · p. 248

6 · Optic radiations (geniculocalcarine tract)

They **fan out** from the LGN deep in the white matter to the striate cortex. § TB 15 · p. 243

BUNDLE	ROUTE	CARRIES
Inferior (anterior bundle)	from the **lateral** LGN, sweeps forward around the tip of the temporal horn of the lateral ventricle = **Meyer loop**, through the **temporal lobe**	**inferior retina** → superior field
Superior (posterior bundle)	from the **medial** LGN, through the **parietal lobe**, lateral to the occipital horn	**superior retina** → inferior field
Macular	between the two	macula

§ Handout 12 · p. 9 · TB 15 · p. 243, 249 · Figs. 15.8, 15.17

7 · Visual cortex

Also called the **primary visual cortex, V1, or striate cortex** (from the white myelinated *stria of Gennari*). It lies almost entirely on the **medial surface of the occipital lobe**, about 1 cm wrapping round the posterior pole onto the lateral surface; most of it is buried in the calcarine fissure. § Handout 12 · p. 9 · TB 15 · p. 243–244

PART	WHERE	RECEIVES	LESION GIVES
Calcarine fissure	from the parieto-occipital sulcus to the posterior pole; divides V1	the horizontal meridian runs along its base	
Cuneus gyrus	**above** the calcarine fissure	superior radiations = **superior retina**	**inferior** field defect
Lingual gyrus	**below** the calcarine fissure	inferior radiations (Meyer loop) = **inferior retina**	**superior** field defect
Most posterior part (occipital pole)	tip of the lobe	**macula** (superior macula in the cuneus, inferior in the lingual gyrus); a large share of V1 for a small retinal area	central (macular) homonymous defect
Anterior part	toward the parietal lobe	peripheral retina; the most anterior strip = periphery of the nasal retina = the **temporal crescent**	peripheral defect, macula spared

§ Handout 12 · p. 9–10 · TB 15 · p. 244, 249, 252, 254 · Fig. 15.9 (Slides 12 · #15, #23)

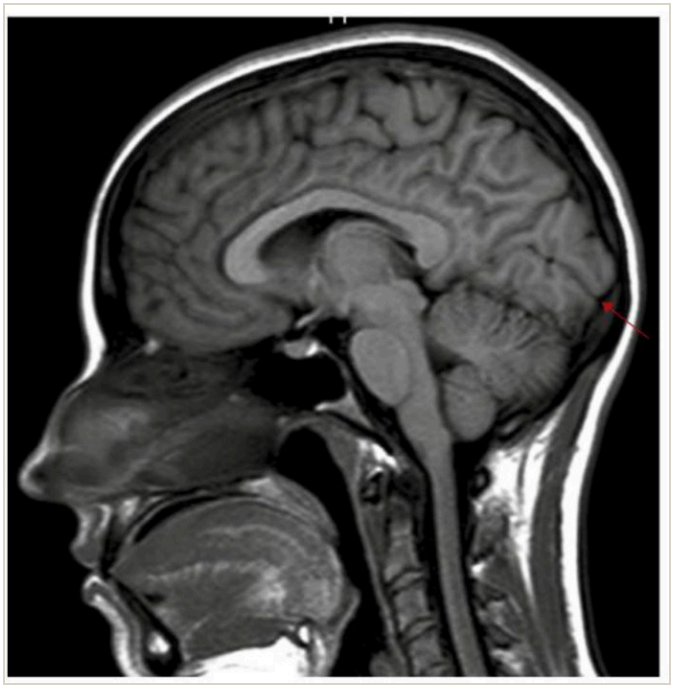
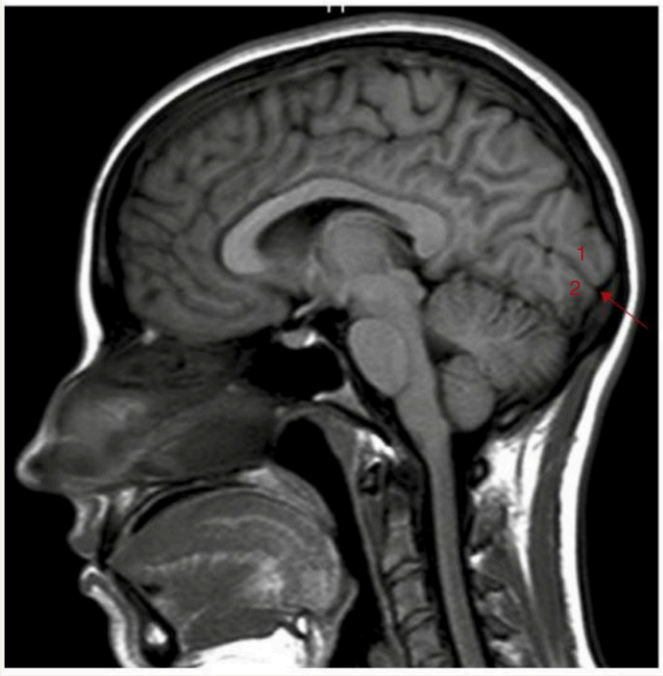


Fig. 15.9, labels removed. Name the fissure at the red arrow and the gyri just above and below it. § TB 15 · p. 244 · Slides 12 · #15

▼ Answer: Fig. 15.9 with labels



Arrow: calcarine fissure · 1 cuneus gyrus (above) · 2 lingual gyrus (below).

CLINICAL CORRELATE · MACULAR SPARING

A homonymous defect from an occipital lesion may leave central vision intact: **macular sparing**, accepted only if at least 3° is spared (fixation wobbles 1–2°). Explanations: the occipital pole may have a **dual blood supply** (posterior and middle cerebral arteries), and the macular projection covers a very large area. §TB 15 · p. 246, 254

8 · The visual field on the retina

The eye's optics **invert and reverse** the image: §TB 15 · p. 249 · Fig. 15.18

PART OF THE FIELD	IS IMAGED ON	REACHES THE CORTEX
Superior field	inferior retina	below the calcarine fissure (lingual gyrus)
Inferior field	superior retina	above the calcarine fissure (cuneus)
Nasal field	temporal retina	ipsilateral hemisphere (temporal fibres do not cross)
Temporal field	nasal retina	contralateral hemisphere (nasal fibres cross)
Fixation point	fovea	occipital pole

§ TB 15 · p. 249. Your notes: "Gesichtsfeld vs. Netzhautfeld": fibres and fields both use nasal/temporal/superior/inferior, so always say which one you mean. TB 15 · p. 249 gives the same warning.

- **The disc is always nasal to the macula** on the retina, so the **blind spot lies in the temporal field** of each eye. § Handout 12 · p. 12 · TB 15 · p. 249 · Fig. 15.19
- **Aligning the fields**: each eye is tested alone, looking at a fixation point; the field is split into four quadrants by a vertical and a horizontal line through fixation. § TB 15 · p. 249 On the printouts the left eye's field sits on the left and the right eye's on the right, so each blind spot is on the outer (temporal) side: "right eye blind spot on the right, left eye blind spot on the left". § Slides 12 · #6, #18 Your notes
- The temporal field is slightly larger than the nasal. The nasal field of one eye overlaps the temporal field of the other, except the far-temporal **temporal crescent**, seen by one eye only (nose and orbit block it for the other). § TB 15 · p. 249
- "Right visual field" means the right *side* of the field (of both eyes), not the right eye's field. The right side of the field is seen by the **left** cortex. § Handout 12 · p. 11 · TB 15 · p. 249 Exam

9 · Fibres at the disc, in the optic nerve and in the chiasm

Retina and disc

FIBRE GROUP (FIG. 15.12 COLOUR)	COURSE IN THE RETINA	ENTERS THE DISC AT
Papillomacular bundle (blue)	straight from the macula	**temporal** side; about ⅓ **of the disc** for only 1/20 of the retinal area
Temporal fibres (yellow)	cannot cross the **horizontal raphe** (a line through the fovea): they **arch** above and below the macula	**superior and inferior poles** (ST above, IT below); about ⅓ of the disc
Nasal to macula, temporal to disc (green)	radiate to the disc	superior and inferior disc, nasal to the temporal fibres (the temporal fibres pass through them)
Nasal to the disc (red)	radiate straight	**nasal** side; about ⅓ of the disc

§ Handout 12 · p. 13 · TB 15 · p. 246 · Fig. 15.12 (Slides 12 · #4, #19). Nasal and temporal retina are split by a vertical line through the fovea. Peripheral fibres lie more superficial (vitread) than central ones. TB 15 · p. 246–247

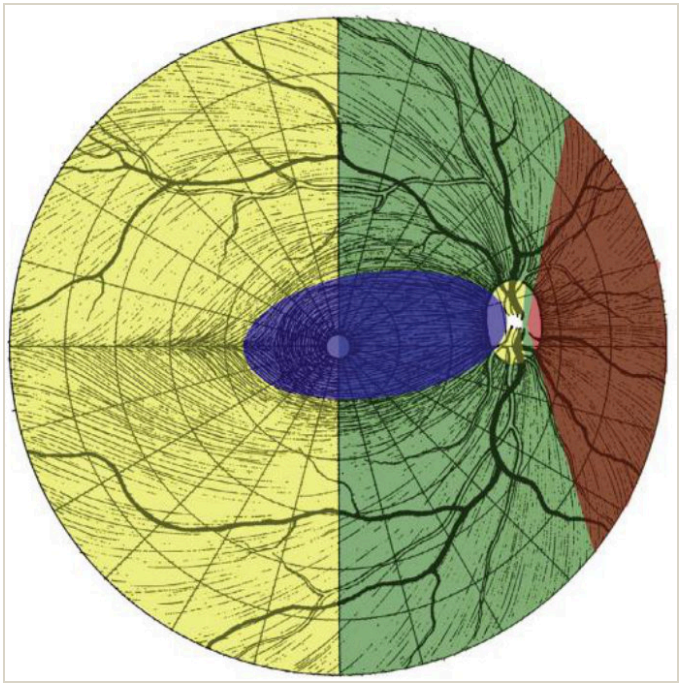


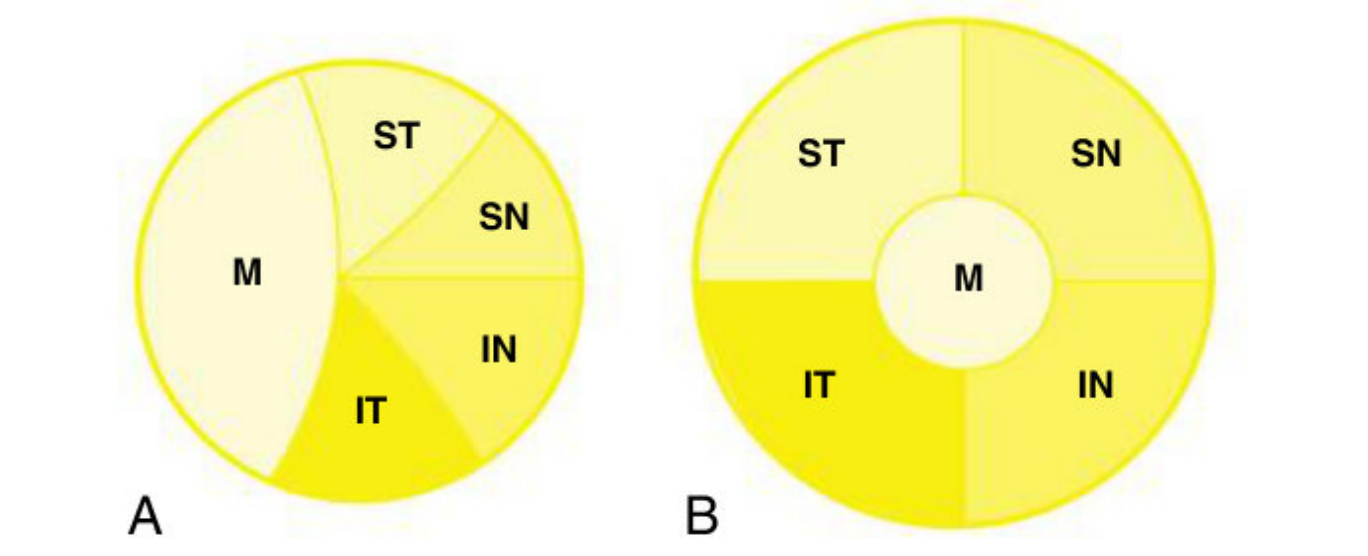
Fig. 15.12 Right retina: papillomacular bundle (blue), temporal arcuate fibres (yellow) meeting at the horizontal raphe, nasal fibres temporal to the disc (green) and nasal to the disc (red). § TB 15 · p. 246

Optic nerve

Just behind the lamina cribrosa the fibres keep the disc arrangement; within a short distance the **macular fibres move to the centre**, and the rest take their logical quadrants: ST superotemporal, IT inferotemporal, SN superonasal, IN inferonasal. § Handout 12 · p. 14 · TB 15 · p. 247 · Fig. 15.13B (Slides 12 · #8, #20)

Fig. 15.13, labels removed. Right eye, seen from the front (temporal is on the left). Write M, ST, SN, IN, IT in each sector of A (disc surface) and B (nerve near the chiasm). § TB 15 · p. 247 · Slides 12 · #4, #8, #20

▼ Answer: Fig. 15.13 with labels



A (disc): M on the temporal third · ST superior pole · IT inferior pole · SN and IN on the nasal side. B (nerve): M in the centre, ST · SN · IN · IT in the four quadrants.

Chiasm

FIBRES	COURSE THROUGH THE CHIASM
Temporal (ST, IT)	**do not cross**; run straight back on the **lateral** side into the ipsilateral tract
Inferior nasal	cross **anteriorly**, at a large almost 90° angle (most easily compressed). They were said to loop 1–2 mm into the opposite optic nerve (*Wilbrand knee*), but that is **controversial**
Superior nasal	cross more **posteriorly**, at a shallower angle
Nasal macular	cross; spread through most of the chiasm

§ Handout 12 · p. 6 · TB 15 · p. 247–248 · Fig. 15.14 (Slides 12 · #11, #21). Nasal fibres stay central even after crossing; most cross paracentrally.

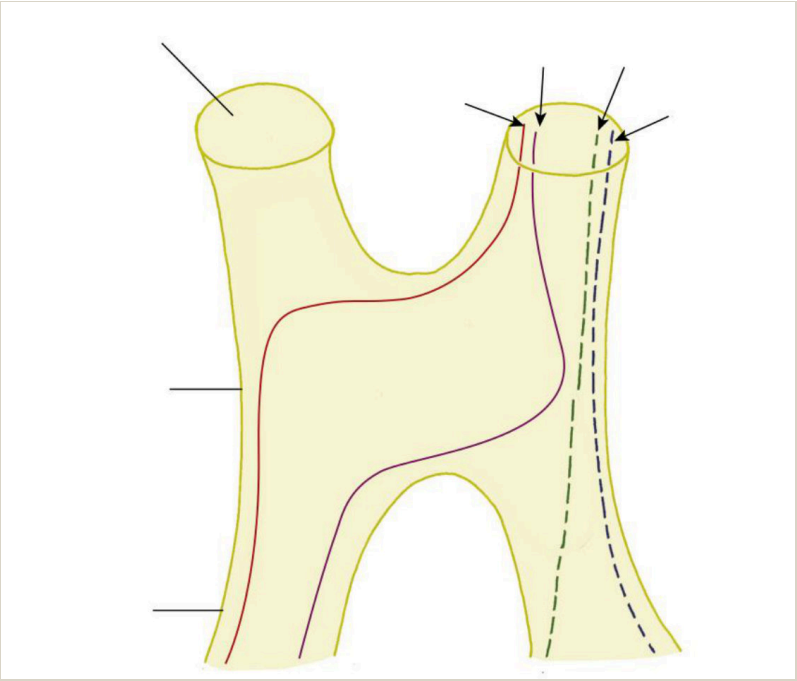
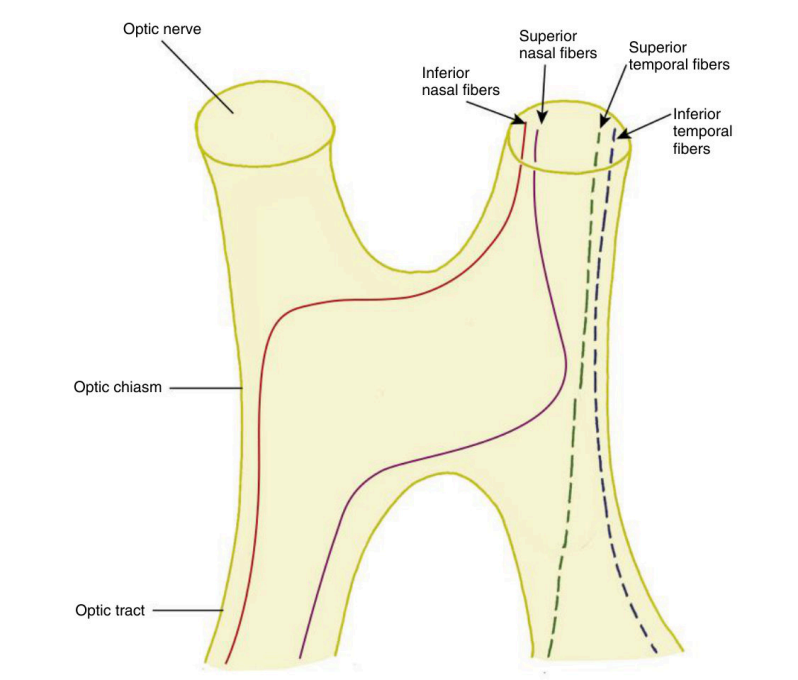


Fig. 15.14, labels removed. Name the three structures and the four fibre groups (solid lines cross, dashed lines do not). § TB 15 · p. 247 · Slides 12 · #11, #21

▼ Answer: Fig. 15.14 with labels



Optic nerve · optic chiasm · optic tract. Arrows, left to right: inferior nasal fibres (red, cross) · superior nasal fibres (purple, cross) · superior temporal fibres (green dashed, stay) · inferior temporal fibres (blue dashed, stay).

10 · Superior, inferior and macular fibres behind the chiasm

Behind the chiasm, crossed and uncrossed fibres from the same part of the field intermingle. The rule all the way to the cortex: **superior retina medial/upper, inferior retina lateral/lower, macula in between**. § TB 15 · p. 248–249

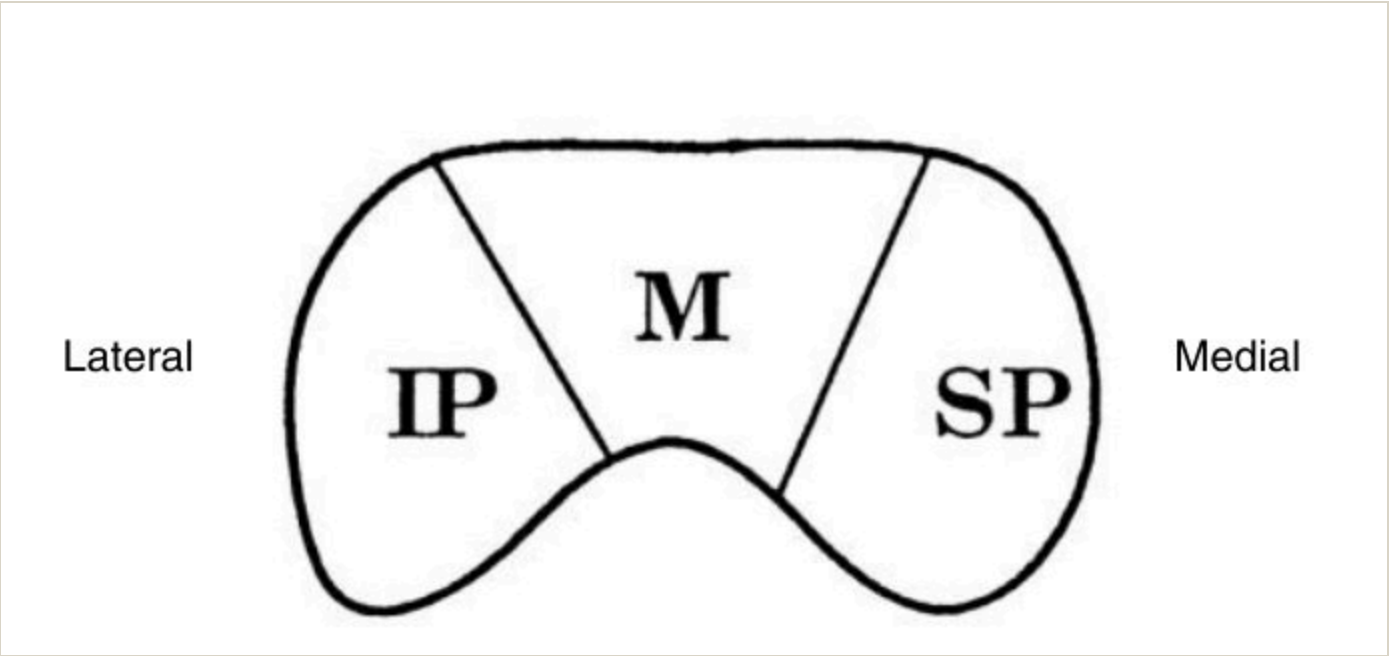


Fig. 15.15 Optic tract, coronal section. IP inferior peripheral (lateral) · M macular (middle) · SP superior peripheral (medial). § TB 15 · p. 248

LEVEL	SUPERIOR RETINAL FIBRES	INFERIOR RETINAL FIBRES	MACULAR FIBRES
Disc	superior pole (ST) + superior nasal	inferior pole (IT) + inferior nasal	temporal third
Optic nerve	superior quadrants (ST, SN)	inferior quadrants (IT, IN)	centre
Chiasm	SN cross posteriorly	IN cross anteriorly	nasal macular cross, spread throughout
Optic tract	**medial** (ipsilateral ST + contralateral SN)	**lateral** (ipsilateral IT + contralateral IN)	between
LGN	**medial**	**lateral**	**dorsal wedge**, ⅔–¾ of the LGN
Optic radiations	superior, through the **parietal lobe**	inferior, **Meyer loop** through the **temporal lobe**	between
Occipital cortex	**cuneus**, above the calcarine fissure	**lingual gyrus**, below it	**most posterior** (occipital pole); periphery anterior

§ TB 15 · p. 246–249 · Handout 12 · p. 9–10 Exam

11 · Lesion ↔ visual-field defect

A lesion anywhere along the pathway leaves a field defect whose shape follows the fibre pattern at that site. § Handout 12 · p. 13–17 · TB 15 · p. 246, 250

LESION SITE	FIELD DEFECT	CLUES

Choroid / outer retina	same shape as the lesion, in the matching place (inferotemporal retina → superonasal field)	one eye; may cross the horizontal and vertical midlines (Fig. 15.21) § TB 15 · p. 250
Nerve fibre layer, temporal fibres at the superior or inferior disc (glaucoma)	**arcuate defect**: curves round fixation from the blind spot to the horizontal nasal meridian, ending in a **nasal step**	one eye; **respects the horizontal line** (the raphe). Inferior disc → superior field (Fig. 15.22) § TB 15 · p. 250, 252 · Handout 12 · p. 13
Nerve fibre layer, a nasal bundle	wedge from the blind spot into the **temporal** field	one eye; less common § TB 15 · p. 250
Optic nerve	defect in the field of that eye only (complete lesion: blind eye, Fig. 15.20 #1)	**RAPD** on that side; later optic **atrophy** (disc pallor). Does not necessarily respect the horizontal line Your notes § TB 15 · p. 250–251
Junction of optic nerve and chiasm	**junctional scotoma**: depressed field of the eye on the lesion side + **superior temporal** defect of the other eye (#3)	the contralateral inferior nasal fibres cross anteriorly § TB 15 · p. 250–251

Chiasm (midline)	**bitemporal hemianopia** (#2)	both eyes; **respects the vertical line**. Commonest cause: **pituitary tumour**; the field defect is often the first sign. Crossed fibres suffer first (large crossing angle, maybe weak blood supply) (Fig. 15.23) § TB 15 · p. 250
Optic tract	**contralateral homonymous hemianopia**, often complete (#5); a partial one is **incongruent** (#4)	optic atrophy (pallor); complete tract lesion: RAPD in the **contralateral** eye (crossed fibres outnumber uncrossed) § TB 15 · p. 250–251
LGN	contralateral homonymous defect, moderate to complete, **congruent**	optic atrophy eventually § TB 15 · p. 250
Radiations, temporal lobe (Meyer loop)	contralateral homonymous defect, **superior**, incongruent (#6)	fibres spread out → incongruent, often one quadrant; **no atrophy, no RAPD** from here on § TB 15 · p. 250–252
Radiations, parietal lobe	contralateral homonymous defect, **inferior**, incongruent (#7); all radiations: total hemianopia (#8)	as above (Fig. 15.24) § TB 15 · p. 251–252

Occipital cortex	contralateral homonymous defect, **congruent**. Lingual gyrus → superior; cuneus → inferior. Anterior → peripheral, **macular sparing** (#9, #10) or a temporal crescent defect (#12); posterior → **macular** homonymous defect (#11)	corresponding fibres of both eyes lie side by side → congruent (Fig. 15.25) § TB 15 · p. 251–252

Behind the chiasm, visual acuity is usually normal: half the fovea is enough for 20/20. § TB 15 · p. 250 Homonymous = the defect is on the **same side** of the field in both eyes (nasal field of one eye + temporal field of the other); hemianopia = the whole half. Congruent = the two eyes' defects have the same shape. § TB 15 · p. 249–250

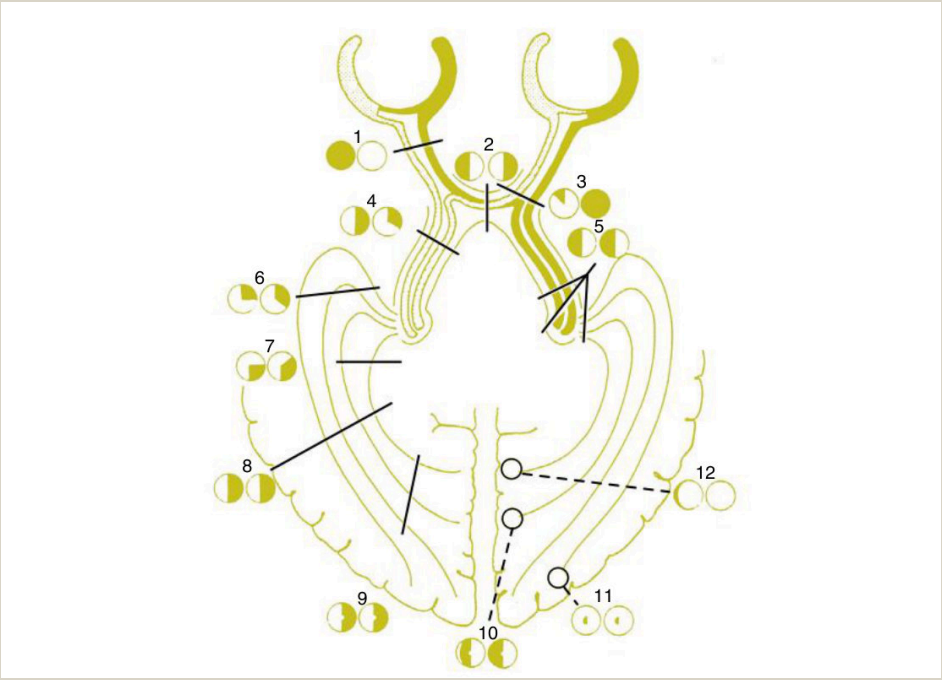


Fig. 15.20 Twelve lesion sites and their fields (left eye's field drawn on the left). Cover the table above and name each defect from the picture. § TB 15 · p. 251

From the field back to the lesion: three questions

1 · One eye or both? → one eye: in front of the chiasm · both: chiasm or behind (or two lesions) →

2 · Respects the horizontal or the vertical line? → horizontal: retina / disc / nerve · vertical: chiasm or behind →

3 · If vertical: bitemporal, or homonymous incongruent / congruent? → **chiasm · tract or radiations · LGN or cortex**

§ Your notes, Handout 12 · p. 17 Your notes .The steps rest on TB 15 · p. 249–252: a one-eye defect must lie in front of the chiasm; postchiasmal defects are homonymous; radiations give incongruent, occipital congruent defects.

ONLY IN YOUR NOTES · CHECK BEFORE YOU RELY ON IT

- **Altitudinal defect**: the whole upper or lower half of one eye's field, respecting the horizontal line; "blood vessel problem", nasal + temporal + macular fibres hit. Your notes Textbook ch. 15 does not describe it. Your typed guideline notes list it with the arcuate defect and nasal step as the three disc defects.
- **Central / cecocentral defect** from a lesion of the papillomacular bundle only. Your notes Not in textbook ch. 15.
- "Heteronymous hemianopia" for the bitemporal defect, marked "preveri" in your notes. Your notes The textbook only says *bitemporal*. It defines *homonymous* as nasal field of one eye + temporal field of the other, so a bitemporal defect is not homonymous.
- Blind spot about **17° temporal** to fixation and slightly below it. Your notes Textbook ch. 15 places it in the temporal field but gives no number.
- MS attacks oligodendrocytes → optic neuritis. Your notes Not in textbook ch. 15, which mentions optic neuritis only for the painful eye movements.
- Disc 1.9 mm vertical × 1.7 mm horizontal: confirmed by TB 8 · p. 130 (chapter 7).

NOT TAUGHT FROM TEXTBOOK CHAPTER 15

Left out because neither the handout nor the guidelines ask for them: the border tissues of Elschnig, Jacoby and Kuhnt and the fascicle counts of the optic nerve head; the layers and columns of V1 (ocular dominance and orientation columns); the extrastriate areas V2–V5 and the dorsal and ventral streams; agnosia, apraxia, aphasia and blindsight; the history of striate-cortex maps (Holmes, Horton) and the Wilbrand-knee studies; the blood supply of each part of the pathway (no objective covers it; only the occipital pole's dual supply is kept, for macular sparing); ageing of the pathway.

Key facts

Optic nerve **5–6 cm**: intraocular 0.7–1 mm · intraorbital 30 mm · intracanalicular 6–10 mm · intracranial 10–16 mm
Fibres: 1–2.22 million; **90%** to the LGN, ~10% to pupil, circadian and orienting centres
Nerve diameter: 1.5–1.8 mm at the retina → 3 mm behind the globe → 4–5 mm with sheaths
Disc thirds: macula ⅓ (from 1/20 of the retina) · temporal ⅓ · nasal ⅓
Chiasm 15 × 8 × 4 mm · pituitary **~1 cm below** · crossed : uncrossed **53 : 47**
LGN: magno **1, 2** · parvo **3–6** · crossed **1, 4, 6** · uncrossed **2, 3, 5** · macula ⅔–¾ (dorsal wedge)
Macular sparing: at least **3°** spared

Self-check

▼ Where on the retina is the superior nasal visual field imaged?

On the inferior temporal retina: the optics invert (superior ↔ inferior) and reverse (nasal ↔ temporal) the field. § TB 15 · p. 249

▼ Why is the blind spot in the temporal field of each eye?

The disc, which has no photoreceptors, is always nasal to the macula; nasal retina sees the temporal field. § Handout 12 · p. 12 · TB 15 · p. 249

▼ Where do the papillomacular bundle, the temporal fibres and the nasal fibres enter the disc?

Papillomacular bundle: the temporal side (⅓ of the disc). Temporal fibres arch round the macula to the superior and inferior poles. Nasal fibres radiate to the nasal side. § TB 15 · p. 246

▼ Glaucoma damages the temporal fibres at the inferior disc. Which defect, and where?

A superior arcuate defect, from the blind spot round fixation to the horizontal nasal meridian, ending in a nasal step; it respects the horizontal line (the temporal raphe). § TB 15 · p. 250

▼ Where are the macular fibres in the optic nerve, close behind the globe and further back?

Close to the lamina: as at the disc, on the temporal side. Within a short distance they move to the centre; ST, IT, SN and IN take their own quadrants. § TB 15 · p. 247

▼ Which fibres cross in the chiasm, and which of them cross most anteriorly?

The nasal fibres (about 53% of all). Inferior nasal fibres cross anteriorly (the disputed Wilbrand knee), superior nasal fibres posteriorly; temporal fibres stay lateral and uncrossed. § TB 15 · p. 247–248

▼ A pituitary tumour presses on the chiasm from below. Which field defect, and why?

Bitemporal hemianopia: the crossing nasal fibres of both eyes, which see the temporal fields, are hit first. § TB 15 · p. 250

▼ Where do superior, inferior and macular fibres lie in the optic tract and the LGN?

Tract: superior medial, inferior lateral, macular between. LGN: superior medial, inferior lateral, macula in a dorsal wedge of ⅔–¾. § TB 15 · p. 248

▼ A left temporal lobe lesion involves Meyer loop. Describe the defect.

A right homonymous defect, worse superiorly, incongruent: Meyer loop carries inferior retinal fibres, and the radiations are spread out. § TB 15 · p. 251–252

▼ Lesion in the right posterior lingual gyrus: which defect?

Left superior homonymous macular (central) defect, congruent: lingual gyrus → superior field; posterior → macula; right cortex → left field. § TB 15 · p. 249, 252 (Your guideline notes give this example.)

▼ A right homonymous hemianopia: name four findings that separate an optic tract lesion from an occipital one.

Tract: often incongruent if partial, optic atrophy (disc pallor), RAPD possible in the contralateral eye. Occipital: congruent, often macular sparing, no atrophy, no RAPD. § TB 15 · p. 250–252

▼ Why can a patient with a complete homonymous hemianopia still read 20/20?

Half of the fovea is enough for 20/20 acuity, so postchiasmal lesions usually spare acuity. § TB 15 · p. 250

§ Sources for this chapter: Handout 12 "Visual System" (with your notes) · Slides 12 "The Visual System" · Textbook ch. 15, Visual Pathway, pp. 239–254 (with TB 4 · p. 54, TB 8 · p. 130–132, TB 10 · p. 163) · Review handout · p. 8 (guidelines and typed notes).